

ity on the one hand and evolutionary flexibility on the other may be the consequence of the network or mesh-like nature of developmental and homeostatic controls.

CONCLUSION

In the preceding overview of the ways DNA encodes various kinds of biological information, the relationships between genotypes and phenotypes, the multiple functions in cells, and the various classes of genes that have evolved to serve these functions, we have tried to outline a general picture of the working tools with which biological traits are made. There are important broad generalizations that are seen repeatedly.

Biological processes use a limited number of genetic regulatory factors and their interactions. The diversity of mechanisms has been elaborated by gene duplication from a few ancestral starting points and by multiple uses of the same factors and pathways. This generic logic rests on overall principles of evolution and/or biochemistry, which can perhaps be attributed in a general way to the modular nature of coding, the fact that so many biological reactions involve the interaction among proteins (including ligands and their receptors), and the nature and ubiquity of cell membranes. All of this is constrained by the core of basic biochemical reactions from which so many of the constituents are derived and upon which it all depends.

A few of the “classic” kinds of genes, like single enzymes or structural proteins, are used only in one context for a direct function. Hemoglobin and tooth-enamel proteins are examples. But we need a different kind of explanation of specificity for the broader set of complex functions. Are signaling factors *Fgf4* or *Shh*, which we know are expressed in limbs and teeth (and many other structures), genes “for” teeth or limbs? Is the distal-less homeobox TF an insect wing spot or leg or tooth gene? Is the answer simply “yes,” or are there better ways to think of it? In a meaningful sense, many aspects of a biological trait are perhaps better defined not by the specific genes that bring it about but by their interactions. Many if not most of the signaling pathways have both positive and negative feedback components, which can be viewed as providing a stabilized interaction tool (Niehrs and Meinhardt 2002) that can function under a variety of circumstances.

There is degeneracy in RE sequences for a given TF. There is also similar regulatory degeneracy (sometimes referred to as promiscuity) in developmental signaling. The regulatory toolkit comprises sets of gene family members that are able to substitute for each other experimentally or evolutionarily—and this probably means that there is considerable cross-reaction among SFs and their receptors under normal conditions. For many traits, this leads to the many-to-many phenogenetic relationship described earlier.

The genes in the regulatory tool kit are also arbitrary relative to traits they control. This is reminiscent of the arbitrary coding nature of DNA. The complex nature of gene interactions is such that we can typically no longer associate a gene with a single function (Greenspan 2001). The action is in the specifics of interactions, and this also provides malleability and robustness. Just as a given type of tRNA transports the same amino acids to any protein whose message has the appropriate codon-anticodon match, a Hox or Fgf protein can regulate any gene that has the appropriate flanking RE sequences. In each case, there is binding specificity but functional arbitrariness.

Thus, the biological traits that have been the focus of evolutionary biology (limbs, flight, and the like) are the very specific end-stages of complex developmental processes, but much of how the traits get here is genetically arbitrary. This is very different from the view of evolution that has predominated since Darwin and of molecular biology since the modern synthesis and Central Dogma. Throughout the remainder of this book, we will see these principles in action.

PART III

An Internal Awareness of Self

Communication within Organisms

The purpose of this section is to review how some fundamental challenges in information transfer within a multicellular organism have been met, and how the mechanisms involved today evolved from simpler precursor states. First, we will look at the ways in which multicellular life has come about. We will look at various ways a modest number of basic “strategies” that are used have evolved, and how genes are used to achieve them. These strategies have been employed throughout the biosphere.

A major—if not the major—stage in the evolution of life as we know it was the evolution of the cell. Cells are more than gene-translating factories and are the essence of more complex organisms, constituting a fundamental organizing aspect of life. For an organism to function as a coherent whole, there must be an organized division of labor and communication among cells. We will look at how that occurs among nearby cells and then at the processes that allow long-distance communication among cells within an organism.

Finally, an organism develops generally from a rudimentary beginning, such as a single cell, through a highly orchestrated process of development, and we are beginning to learn how that works. A major aspect of that is the vital problem of reproduction, which itself has many meanings. Similarly, many if not most organisms are able to maintain viable states by responding to changes in their environment. We will describe general aspects of development as well as genes and the regulation of their expression that makes this responsiveness possible.

An important point is that within an organism communication largely involves specific signals that are received by cells specifically looking for them. Another problem of dealing with the internal world actually derives from the external. Organisms are always subject to being invaded by microorganisms that could do them harm. They have developed a variety of ways to cope.

Chapter 8

Making More of Life: The Many Aspects of Reproduction

Reproduction is so fundamental to life that it is one of the characteristics by which we *define* life. As well as the generation of new life in the short term, in the long term, reproduction provides an opportunity for heritable change of various kinds, and hence for evolution. A key view of 20th century biology was that life is basically a nucleic acid information phenomenon, in the dual senses that biological information flows from DNA to RNA to protein (the Central Dogma), but not the other way, both in an individual and across generations. Thus an important primary form of reproduction, *replication*, is molecular copying, of RNA for individual genes and of DNA for chromosomes. Replication of other molecules of life, such as amino acids, steroids, lipids, carbohydrates, and nucleic acid components themselves and reproduction of organisms follow as a result. We've seen in earlier chapters that this view can be tempered by other ways in which inherited change can come about, but they all involve reproduction in some fashion, and an important element is that in each case the heritable aspect of the new iteration is sequestered from direct modification by its parent. A new generation has a life of its own.

We usually associate the term "reproduction" with the production of new organisms from a parental generation. But this same process is essentially the way a single cell differentiates and divides into a multicelled organism. Among those cells, genetic information flow is also a unidirectional evolutionary microcosm. This includes somatic mutation that induces gene structure change and the various mechanisms that affect gene expression within each, partially sequestered, lineage of cells. There are many interesting subtleties in the processes of reproduction, which we discuss in this chapter.

THE PHENOMENA OF REPLICATION

Multicellular organisms have various ways of producing a next generation. Some, like bacterial biofilms, are essentially aggregates of otherwise free-standing cells that

reproduce by individual cell division. The biofilm does not reproduce as a single entity. Slime molds are another class of cells that normally live as individual single-celled organisms, although under some circumstances (to be described in Chapter 12), the cells are induced to come together to form multicellular aggregates, taking on many properties of multicellular organisms. At this *slug* stage, they have internal organization, signaling, various sensory behavior, cells that undergo apoptosis, and the like (e.g., Bonner 2000), and these reproduce basically as integrated organisms in which not all cells contribute directly to the reproductive process. Some simple multicellular animals, like sponges, can produce small aggregates of cells that shed as primitive larvae to become new organisms. A specific cell type is generally the precursor, but these are located throughout the parent's body. In plants, many or even most cells have the potential to produce an entirely new plant on their own, as even small pieces can regenerate complete new plants.

Most complex animals and plants have a specialized form of reproduction that we tend to treat as a standard for reproduction, although it is by no means the only or "best" form. In these, a type of cell, the *gamete*, is used only for reproduction. The organisms begin life as a single cell and grow through mitotic cell division. In many organisms, a lineage of *germ line* cells is set aside, usually early in development, and reserved for gamete production, while the rest of the organism's functions are carried out by other lineages of *somatic* (body) cells. Somatic cells may be replaced, lost, or even actively killed by apoptosis during life, and they can undergo natural selection that can even take the life of the organism (that is what cancer is). But although they are the product of a form of reproduction, and "beings" within the higher-order being known as an organism, somatic cells make no contribution to the next generation. Genetically, gametes are produced by meiosis and have half the chromosome complement of the somatic cells (e.g., are haploid vs. diploid). Typically, reproduction occurs by the union of gametes from different individuals, but self-fertilization can occur and some species reproduce by *parthenogenesis*, in which the organism's genome is fully represented without a separate fertilization stage; there is wide variation in the ways this kind of reproduction occurs in nature.

Somatic cells experience mutation just as germ cells do; however, although mutations accumulating in somatic cells may affect the organism (and cancer and other changes associated with aging are examples), only those in the germ line affect the next generation. Hence, in organisms reproducing via single gamete cells, this is where the evolutionary memory lies. Of course, somatic mutations may have an evolutionary impact if they determine whether a particular organism's gametes are transmitted.

One theory explaining the early and active sequestration of primordial germ cells is that they can escape the chromosomal modification by methylation that is used to control differential gene expression in somatic cells. Before transmission, gamete genomes in some species (including humans) are systematically imprinted, sometimes differently in males and females, by methylation at specified sites on the chromosomes or by other means, affecting how and/or when genes transmitted from that parent are used in the offspring. Another common characteristic of germ line lineages is to undergo fewer cell divisions during the life of the individual (Buss 1987); we may think of this as protecting the patrimony from mutational damage, but it is not always true. Sperm-producing cells in mammals undergo many cell divisions, with evidence showing that this results in correspondingly more mutations occurring through the male lineage (e.g., Crow 2000).

However, there is no one rule about this. Plants and many other organisms do not even sequester a special germ line; flowers, for example, reproduce via sexual reproduction but can be produced on hundreds of different stems. Some species, including many plants, are hermaphroditic, that is, the individual contains, and functions as, both sexes. And not all reproduction occurs through single gametes.

Most cells in multicellular organisms toil their lives away at the evolutionary service of the few cells that will contribute to the next generation. However, in sexually reproducing organisms, only half the genome is transmitted to a given offspring anyway. The prevailing view is that, for evolution to produce individuals who make this genetic sacrifice, the cell and organism must gain some selfish advantage in exchange. A general explanation is that somatic cells are helping to advance the reproductive cause of their genetically identical germline kin. This is only partially true because accumulating mutations make each somatic cell different. We might thus expect furious natural selection among the varying cells in our bodies all the time, but, although there is much machinery to interfere with aberrantly behaving cells, there is no master genome controlling this. In regard to reproductive self-sacrifice, germ cells from the same organism are, at least, a somatic cell's closest genetic relative but no way has evolved for a really superior somatically mutated cell to make a germline copy of itself (though in principle something like this could occur in plants).

We can take a similar view of the colonies of certain organisms like ants and wasps, in which most individuals do not even contribute gametes. This has been seen as a theoretically instructive example of how sterility (e.g., in drones or workers) could evolve: their reproductive self-sacrifice leads to the high reproductivity of the queen they protect, to whom they are closely related by a special chromosomal system called *haplodiploidy* (see below). But if this is compatible with evolution it is not inevitable: termites and naked mole rats are similarly eusocial, forming colonies with one breeding female, a few breeding males, and the rest nonbreeding, but these species are diploid, not haplodiploid.

Because of their locked-in reproductive fate, some biologists have suggested that it is meaningful to consider such colonies as organisms themselves. The idea has been around for a long time (Bateson 1894; 1913). Indeed, in many ways, our own somatic cells can be likened to the drones in a beehive, and our germ line to the queen. What an ant, or the set of drones, are to a hive may not be so different from what a liver is to a human. A major difference is that the individual ants are not stuck together, whereas, for example, individual nephrons in a kidney are. We could even consider a *species*, like humans, as a single large superorganism, with various sets of cells, aggregated in hierarchical ways and exchanging genetic variation over time. Genetically, this is not far-fetched at all because the responsible mechanisms are similar in nature.

Sexual reproduction is a nearly ubiquitous part of life and has evolved countless forms, many presumably having arisen independently. The most plausible evolutionary advantage leading to this is provided by the added variation produced by recombination of homologous chromosomes from different individuals. The partial sequestration made possible by sexual reproduction makes a species more able to respond to changes in the environment, while keeping the species from devolving into a set of totally sequestered lineages.

That so many multicellular organisms reduce their reproductive activity to the shedding of only single cells raises the question of whether sexual reproduction is

still a remnant of single-celled life. After all, reproduction could in principle occur by fusing many cells rather than one, and among more than two organisms, taking advantage of somatic mutations and natural selection within each organism's life history. This would allow a kind of semi-lamarckian evolution to take place: somatic cells used in reproductive fusions would be screened by selection for those cells that behaved well, including those carrying favorable somatic mutations. That can't happen now in lineages long committed to meiosis-based sexual reproduction, and problems could arise in terms of making a unified cellular tree of differentiation and development, or of self/nonself immune recognition. But different mechanisms might have evolved, and plants (for example) don't have a simple unified tree of development (see below).

Of course, some organisms do reproduce by shedding sets of somatic cells, and the rather fluid, context-specific nature of aggregation in otherwise single-celled organisms forming biofilms and slime molds shows that something of this kind can occur (Bonner 1998). Still, for whatever reason, the single-cell route has been repeatedly successful; whether it is "best" or not, it works and may be easiest to evolve. These are facts of importance to the definition of an individual from evolutionary perspectives (Buss 1987) and affect the tempo of evolutionary genetic change and perhaps also some aspects of its mode of action.

Viruses contain nucleic acids that have normal protein-coding function, but they are not alive and replicating when free-standing and must infect cellular organisms to reproduce. There, they shed their protein coat, allowing their RNA or DNA (depending on the type of virus) to interact with relevant enzymes in the host cell and to be copied. New viral coat proteins are coded for by the viral genome and translated into protein by the host cell. Multiple copies of viral DNA/RNA, and viral coat proteins, can circulate within the cell; thus a given virus can be assembled from components that did not all derive from a single parental ancestor. Viruses can exchange genes by recombination within the cell before new particles are formed, further adding to their repertoire of variation. Another way viruses reproduce is by integrating their DNA (or DNA copies of their RNA) into the host genome. If this occurs in somatic cells, the viral genome is inherited somatically; in a germ cell, the viral genome is transmitted to the next host generation via the horizontal transfer referred to in Chapter 2.

Bacterial plasmids, chloroplasts, mitochondria, and many chromosomally integrated genes are examples of the past or present infectious transfer of parasitic DNA that can be replicated as units within a cell and inherited through mitosis and meiosis.

THE SPECIAL PHENOMENON OF SEX DETERMINATION

We have described sexual reproduction from its general genetic point of view, which is a source of shuffling of chromosomes (and recombination during meiosis). But sexual reproduction requires much more than mechanisms for the production of male and female reproductive cells (sperm, egg, pollen, ovule, and correspondingly different cell structures in yeast and bacteria). Making gametes is just the beginning: they (that is, the organisms that carry them) have to find each other. This burden clearly reflects the importance of whatever advantage sexual reproduction must have. It has led to so many differences in behavior and morphology that sexual

dimorphism can be the greatest single source of variation among individuals in a species.

There is extensive variation in the genetic basis of sex determination. Even bacteria and yeast occasionally reproduce sexually, often induced to do so by nutritional or other environmental changes that we tend to characterize as stress. Morphological changes occur to denote “male” and “female” cells, which fuse to form a temporarily diploid organism, in which recombination can occur. In the case of bacteria, a small chromosomal element known as the fertility factor (F) can transfer between bacteria. F⁺ cells form short projections called *pili* on their surface, in a process called conjugation that joins F⁺ and F[−] cells; genetic information can then transfer through chromosomal recombination (see panels A–D in Figure 8-1A) (e.g., Griffiths et al. 1996; Lewin 2000; Suzuki et al. 1998). On rare occasions, the F element is incorporated directly into the bacterial genome and can thereafter be transferred to other bacteria in subsequent conjugation.

There are many variations among species, but the basic mating systems of yeast involves a “cassette” system. Figure 8-1B shows this for the brewer’s yeast *Saccharomyces cerevisiae* (e.g., Haber 1998). A gene in a chromosome III location called MAT codes for mating-related proteins. Depending on the mating type, one of two flanking genes, *HMLα* or *HMRa*, replaces the gene currently at MAT. Chromatin structures symbolized by gray ovals repress these genes, but at the appropriate time in the life cycle, a cut is made at MAT. Then, controlled in part by flanking

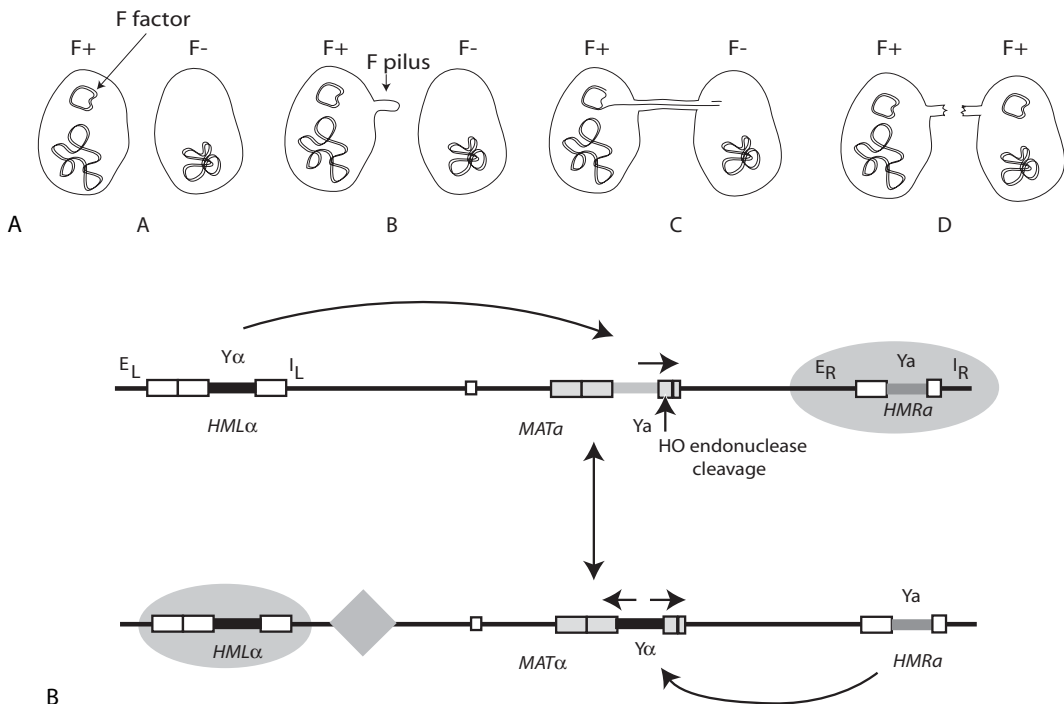


Figure 8-1. Some different kinds of mating systems in single-celled organisms. (A) Mating in bacteria (e.g., *E. coli*), stages A–D; (B) mating in yeast (*S. cerevisiae*). Separate lines show switching from each of two states. For explanations see text. (B) redrawn from (Haber 1998).

regulatory sequences (E_L , I_L , E_R , and I_R) recognized by DNA-binding regulatory proteins, and a repressor switch (symbolized in the figure by a gray diamond), a copy is made of the derepressed donor gene, and the copy is inserted at MAT replacing what was there. A subsequence within *HML α* or *HMR α* (Y_α or Y_a) codes for genes determining the different mating type. Haploid yeast of complementary mating types unite to form a diploid reproductive cell. The genetic mechanism for this switching, and the conditions under which diploid and haploid states occur or persist under different environmental circumstances, are highly variable among yeast species.

Many animals have sex determination in which males and females have different chromosomal complements. In mammals, females are homogametic and have two structurally identical X chromosomes, whereas males are heterogametic, with one X and one Y. The remaining chromosome sets, called *autosomes*, are structurally identical in each sex, although they may be differently modified in their respective somatic or gametic cells (see below), and of course may bear different alleles at any locus. Birds and nematodes have reversed male/female heterogamy.

Some species (e.g., some plants) produce diploid hermaphrodites, but closely related species may have obligatory sex-specific morphology and go through a haploid stage. In the haplodiploidy of social hymenopterans (ants and bees), males are haploid and come from unfertilized eggs, whereas workers and queens come from fertilized (and hence diploid) eggs. Queen development is environmentally induced by feeding larva on a special substance known as “royal jelly.”

Factors in the environment can determine sex in some vertebrates. In many reptile species, such as turtles, sex determination is temperature-dependent. However, these environmental effects appear to work via genetic pathways similar to those used in descendant vertebrates, so that experimental sex-hormone manipulation can override the effect of temperature. From their presence in diverse living reptiles, we might infer that sex hormones already existed ancestrally for some purpose and may have been recruited to take over the role previously played by temperature, becoming regulated by genes linked to sex chromosomes.

Many plants can be male, female, or hermaphroditic. Flowering plants do not sequester a separate germ line, and cells at the shoot meristems (tips) can differentiate into male and female flowers, or flowers can be produced that bear both stamens and carpels, depending on patterns of regulatory gene expression. There can be location-specific sex determination, such as male tassels atop and female flowers along the axis of corn. Once committed, the designated cells undergo meiosis that is similar, although not identical, to meiosis in animals. Although some plants can self-fertilize, others have mechanisms that force them to crosspollinate using, for example, insects, birds, or wind. Darwin wrote an entire monograph on the devious ways this is achieved by orchids (Darwin 1862).

Some species form two sexes without a specific set of sex chromosomes. This can be achieved by quantitative polygenic contribution of genes on several chromosomes or (as in some reptiles) through environmental effects.

SOME GENERAL GENETIC ISSUES

It is very interesting that, as important and pervasive as sexual reproduction is, its mechanisms are so highly variable and rapidly evolving. The mechanisms for sex

specification differ greatly at the molecular level. Some species have sex chromosomes, but they work in different ways in different species. The genes on the sex chromosomes are responsible for some aspects of sex determination, but they typically induce developmental cascades using genes all over the genome to effect the wealth of differences in size, morphology, and behavior between males and females that go well beyond their respective gonadal differences.

Some of these effects are qualitative, but others are quantitative (e.g., Griffiths, Miller et al. 1996; Raff 1996). In mammals, the mechanism is dichotomous and sex-linked. The *SRY* regulatory gene, a member of the *HMG/Sox* chromatin-associated DNA binding protein family (see Table 7-6), binds and bends DNA in a critical region of the Y chromosome, presumably providing access to transcription factor (TF) activity that initiates a cascade to turn a gonad into a testis. Downstream activation includes a related gene, *Sox9*, which is involved in male determination in mammals and other tetrapods (but also having unrelated developmental functions). The sex-determining cascade also involves X-linked genes. However, germ cell differentiation and gonadal and morphological sex differences are not inherently connected (and there are developmental genetic differences, even between humans and mice).

By contrast, sex determination is more quantitative than dichotomous in flies (Schutt and Nothiger 2000). In a zygote with two X chromosomes and two sets of autosomes (A), the ratio of X- and A-linked gene products is roughly 1.0, a state that activates a TF, *Sex lethal* (*Sxl*), to initiate a female-producing downstream cascade of differentiation. The latter activates a gene known as *Transformer* (*Tra*) that activates the gene *Doublesex* (*Dsx*), which leads to female development. These appear to involve sex-specific splicing variants (e.g., Graham et al. 2003). In XY males, the X-to-A dosage ratio is less than 1.0, inactivating *Sxl* and producing a male. However, not all flies use the same control system for activating *Dsx* (Dubendorfer et al. 2002; Graham, Penn et al. 2003), suggesting that the lower-down mechanism was established initially and its control assumed by various genes during arthropod evolution. In many fly species tested, *Sxl* is not even involved in sex determination (it is not true that once you've seen one fly you've seen them all, nor that all flies are like *Drosophila*). In the species *Ceratitis capitata* the X-to-A ratio appears not to be involved either, but a separate Y-linked factor called *M* is.

The widely studied animal model nematode *Caenorhabditis elegans* also uses a dosage-sensing mechanism, with activity of the autosomal gene controlled by the sex-chromosome dosage. However, the controlling sex-determining gene, *Xol*, is not homologous to *Sxl*, nor does the downstream cascade of genes that are on or off differently in male and female development bear much if any homology with the fly mechanism. Indeed, recent work suggests that even the clear-cut "mammalian" mechanism is not so clear-cut because mice and humans use *SRY*, and some other genes thought to be fundamental for sex determination, in different ways (Ostrer 2001; 2001). Furthermore, although they have a similar *SRY* mechanism, marsupials can be hormonally induced to form either sex, and gonadal development is determined by mechanisms separate from the sex-determining ones (Renfree and Shaw 2001).

The diverse sex-determining mechanisms appear to have evolved independently many times. However, some organisms, vertebrates and yeast for example, seem certainly to have had sexual reproduction ever since they shared a common ancestor;

therefore, the differences in mechanism seen today have probably arisen by phenogenetic drift rather than via independent evolution from nonsexual forms.

SEX-SPECIFIC CHROMOSOME MODIFICATION

Functions in the early zygote, from the single-cell stage onward, may be dominated by conditions inherited from the mother. This is especially true in species whose fertilized egg receives little besides DNA from the sperm. The nutrients and mRNA required for early development are produced by the mother. In mammals, maternally and paternally derived chromosomes arrive differently marked by *imprinting*, by methylation of the DNA (e.g., the Cs in CpG pairs) flanking the 5' start site of many vertebrate genes, which prevents promoter binding by TFs and thus represses transcription. In the formation of gametes, the parent resets its respective imprinting pattern.

How has this curious arrangement evolved? One explanation offered from a strongly darwinian behavioral-evolution perspective is that the fetus is a kind of parasite on the mother. It is related to the mother genetically but only by half, and she and the fetus compete for resources. The fetus could sap her of so much energy that she would not be able to reproduce again, which is against her interests, and conflicting with this is her clear interest in the health of the fetus.

Genes expressed in the trophoblast cells that form the extraembryonic tissues including the placenta are predominantly maternally derived, leading to a different possibility, that she is less likely to detect the implanted fetus as a foreign invader and destroy it. Sperm-derived chromosomes are generally more heavily imprinted and not expressed until somewhat later, and then in the cells that will form the embryo itself. Some imprinting signal does persist in the zygote, but differential imprinting cannot affect most autosomal genes, or at least not for long, because individuals typically show the expected equal inheritance proportion and effect of their autosomal alleles from both parents; for example, offspring from $AA \times aa$ matings typically manifest both A and a allelic effects. The known exceptions are notable for their rarity.

GENE DOSAGE AND RELATED ISSUES

Diploid organisms must be able to function genetically in a haploid state, during early development as just mentioned, and the haploid germ cells must also be viable. However, heterogamic sexual reproduction raises an additional *gene dosage* problem. To the extent that gene products coded by autosomal and X-linked genes interact in a particular cell type, a female mammal has a double dose of autosomal and X-specific genes but a male has only one-half dose of the latter. In fact, this kind of dosage difference determines sex in flies, as described earlier.

In some species, there is upregulation (increased expression) of genes on the single chromosome in heterogametes, whereas in others, including mammals, there is downregulation. This is achieved by random X-inactivation in females. During embryogenesis, in each cell in females, one of the X chromosomes is inactivated so that the cell only expresses genes from a single of its two X chromosomes. This is achieved by an interesting mechanism. A gene called *Xist* is transcribed from an X-inactivation center, and the resulting *Xist* RNA "paints" (covers) that X chromo-

some (but not the other X chromosome in the cell) to block its genes from transcription at positions that may be signaled by the location of dispersed repeat elements (Bailey et al. 2000; Brockdorff 2002; Plath et al. 2002). The result is that, in males and females, protein interactions work at the same dose level. The genetic mechanisms used to inactivate the entire X in this process were already available as part of the general mechanisms used in selective gene silencing in differentiating cells.

Because of the random and differently timed nature of inactivation, a female is somatically mosaic, with approximately half of each tissue using the maternal and half the paternal X-linked genes. Furthermore, because the inactivation pattern is mitotically inherited, the time at which a given inactivation event occurred in development determines the nature and size of the downstream clone—the developmental lineage—of cells expressing the particular active chromosome. This long-known principle has been shown to be incomplete, however, in that some genes do appear to escape inactivation. Interestingly, humans and mice differ considerably in the details of their *Xist* systems (e.g., Daniels et al. 1997). In addition, both the X- and Y-linked copies of some genes are active in males; the gene *Amelogenin* that is involved in dental enamel production is an example, with an active copy of the gene on both X and Y chromosomes.

Sex-linked genes in some organisms, one example being *Bombyx mori* (silkworm), appear not to be dosage compensated, showing that, clearly, interactions among gene products can be adjusted. Both male and female gametes are similar and in some species essentially identical. Some parent-specific marking persists in life so that alleles identical in sequence function differently if they were inherited from the maternal or paternal line. A few instances of this have been discovered in inherited disease, when the genes were examined carefully. Angelman syndrome is a set of conditions including developmental, speech, and other behavioral disturbances that arises when maternally imprinted information in a region of chromosome 15 is lost through various mechanisms such as aberrant meiosis. Alternatively, if paternal imprinting pattern in that chromosome region is absent a different disorder, Prader-Willi syndrome results; Prader-Willi has very different symptoms, predominated by morbid obesity. Several other known diseases are preferentially inherited through one parental line, again for presumed imprinting reasons.

VARIATION EVERYWHERE

Variation is everywhere in the sexual reproduction arena. And an arena it is, because of the behavioral complexities required for the two sexes to find each other. Add to this the variation involved in making different kinds of gametes, not to mention their delivery by various mechanisms as well as the diversity of genital organs, gestational systems, and the huge repertoire of associated appearance, behavior, and morphological differences between males and females. Indeed, it is too simple to think of sex as dichotomous (males and females). There are, for example, plants that make both sexes and may be self-fertilizing, reptiles that can be either sex depending on how hot they are (or were as eggs), and fish that can respond to environmental conditions by changing their sex. Many if not most aspects of sexual morphology are variable among closely related species (e.g., Renfree and Shaw 2001). We also have culturally based ideas about *gender*, that is male and female

behavior, but we know that there are exceptions, both in our species and many others.

More importantly perhaps is that as cultural beings who monitor our own societies closely, we know that there are all sorts of morphological as well as behavioral gradations related to plumbing as well as sex-related morphology and gender. These are not all highly correlated and hence are likely to be genetically unlinked. Not all this variation is dysfunctional in reproductive terms, and while in many or most species mating behavior involves stereotypical components, wide ranges of subtleties in morphology and behavior are compatible with successful reproduction. This probably indicates the kind of variation with which evolution works as sexual mechanisms change over time.

Finally, all evolutionary explanations have a Kiplingesque Just So element in them. Bdelloid rotifers illustrate this in regard to sex. These are microscopic aquatic animals that have been around for a long time and are phylogenetically diverse. They reproduce entirely by parthenogenesis (either by laying eggs or “hatching” already formed young) (Judson and Normark 2000; Welch and Meselson 2000). How this “violation” of so important an evolutionary rule as having sexual reproduction (and yet surviving quite well) occurred is not known, but of course even to propose this as a conundrum is to build in assumptions about the importance of sexual reproduction.

THE PACE OF REPLACEMENT

Organisms approach reproduction in various ways. Evolutionary biologists have sometimes characterized these in two general categories as r and K “strategies.” This nomenclature derives from an equation for the dynamics of population size, relating the rate of change in size of a population now of size N , to its intrinsic growth capability of the population (r), and the carrying capacity (maximum sustainable population) (K) of the local environment:

$$\Delta N = rN(K - N)/K$$

This is the logistic growth equation. At low relative population size, $K - N$ is large, $(K - N)/K$ is nearly equal to 1 so that growth occurs nearly at the maximum capacity (a factor of r per generation); but as population size nears the carrying capacity, K nearly equals N , so that $(K - N)/K$ is nearly zero, and growth tapers off.

In this conceptualization, a species that reproduces rapidly is said to be using an r strategy. Its offspring mature quickly and shed large numbers of gametes. Insects and fish that produce thousands of eggs are examples. They can tolerate massive die-off of their young and still have net successful reproduction during their lifetime. The parent typically does not invest much care or effort into the maturation of the offspring, but there are exceptions (for example, among mammals, mice are rapid reproducers, but they also nurse their young until self-sufficiency). The race for success is run by producing many offspring, increasing the odds that at least some will survive. Animals lower in the food chain, whose lives tend to be risky and short, often reproduce rapidly and can be more adaptable. Any individual offspring is, more or less, expendable in terms of the parents’ evolutionary interests. On average, in a stable environment, there is no net population growth after all this struggle, but an r strategy is able to recover more quickly from losses in numbers.

In contrast are the lumbering *K* strategists. Elephants, albatrosses, and some trees would be examples. So are humans and other primates. These organisms are, so to speak, near their carrying capacity or at least their ability to respond in terms of growth. They reproduce only irregularly, after a long juvenile time period, or only produce a small number of slow-growing offspring. In some cases, there is a need for longterm parental care so offspring can survive to reach reproductive age. Carnivores high in the food chain are fewer in number than their prey and may need more training for life. Because their reproduction is slow, maximizing the probability of survival of each individual offspring is important. A lost offspring is devastating to its parents' net reproductive output.

Mathematical theory has been developed to show how these reproductive behaviors might evolve genetically through the action of natural selection. Clearly, however, there is no one way to succeed in reproducing, nor is any way restricted to particular taxa, nor is a clade of taxa necessarily restricted to one way of doing business. But we should remember that while populations may fluctuate in size, no matter what their "strategy," in the long term the net reproduction on average for members of a population is just at the replacement level: one offspring produced per individual. In Chapter 3 we described various aspects of the way selection might act in regard to life history, to differentiate success among genotypes. The only invariant and merciless principle is that if it works, it works.

A QUICK AND INCOMPLETE TOUR OF ANIMAL AND PLANT DEVELOPMENT

A major hallmark of multicellular life is differentiation. Cells reproduce, and a single cell becomes a complete organism. Much of the work done on this subject before and even since our recently increased ability to document genetic mechanisms experimentally has concerned the determination of overall body plans. Multicellular organisms are quite diverse, but in fact there are only about 35 body plans associated with the major animal phyla (e.g., Raff 1996) and an additional set of basic plant forms. There is a similarly limited repertoire of basic patterns of structural organization of systems within an organism. We will see in the next chapter that a limited repertoire of developmental processes brings this about, helping to explain how the great diversity of life has been achieved.

Once a basic system has become highly integrated, early in development, it can be *canalized* or its general features retained, a term introduced by CH Waddington (Waddington 1942; 1957; Wilkins 2002) that we have several times mentioned. A view fully compatible with classical darwinian theory is that, once entrenched, so much of later development depends on the early establishment of a body plan that the latter is difficult to change. This leads to evolutionary and developmental stability, although at the expense of flexibility (e.g., Siegal and Bergman 2002). Other ways of buffering include heat shock (or chaperone) proteins that can serve as a "capacitor" against harmful environmental or mutational effects, epistatic feedback (Rutherford and Henikoff 2003; Wagner 2000), and redundancy. In Chapter 7, we discussed the way that tolerant enhancer elements can contribute to these kinds of stability over individual as well as evolutionary time.

Balancing selection, that is, selection against all extreme phenotypes (e.g., too big, too small) can facilitate the evolution of such stability. Of course, no foresight is involved in any case. Transmitted chromatin modification that affects gene expres-

sion is a way in which genetic variation can exist in a population but not be expressed, perhaps easily “reawakened” under stress (if the chromatin modification is itself variable or released by fortunate mutation). Selection need merely maintain the suitable forms in their respective environments. Later, crosses between what appear to be invariant strains, or other kinds of selection, can release the constraints on this variation so that it becomes expressed. Various effects of this kind have been observed in animals and plants (e.g., Dun and Fraser 1959; Lauter and Doebley 2002; Sollars et al. 2003).

Here, we will attempt a cursory survey of the basic aspects of multicellular organization that these processes have produced. Examples, illustrations, and even animations are plentifully available on the internet, and there are several excellent texts concerning morphological development and its genetics (Gilbert 2003; Raff 1996; Wilkins 2002; Wolpert et al. 1998). Tudge (Tudge 2000) has provided an excellent compendium of animal form, and there is an excellent resource on known evolutionary relationships among the creatures on Earth (Maddison 2003).

Animal body plans include various types of symmetry from essentially none (as in hydra and sponges) to radial (coelenterates), to pentameral (starfish and other echinoderms), to bilateral (ourselves). Plants establish radial symmetries in their roots, shoots and flowers, and branching symmetries in the arrangement of leaves on stems or veins within leaves. These symmetries define or reflect anatomic axes, most notably the anterior-posterior (AP) axis from head to tail and the dorsal-ventral (or dorsoventral, DV) axis from back to front and lateral or left-right or proximal-distal (or proximodistal, PD) when viewed relative to the midline; in plants, correspond to the main vertical and radial axes and the longitudinal axes of shoots and roots.

Evolutionary explanations of body plans usually relate to their current function. For example, the segmentation of the vertebral column enables animals to become long and flexible, and yet rigid from head to tail, and to have separate structures in different places along the way. The bilateral symmetry of our eyes and ears, the forked tongue in snakes, and the antennae in insects allow perception of the world in stereo. Of course, keeping in mind the step-by-step nature of complex trait evolution, these may or may not be the original selective advantages.

Not all body plans are as organized or simply explained. Bacterial aggregates form from free-floating single-celled organisms that, under environmental conditions that favor their congregation, can emit chemical signals that induce others nearby to change behavior and join together with no fixed kind of plan (Bonner 2000; Shapiro JA 1998). By aggregating on a surface, to which they adhere through the secretion of various polysaccharide or other substances in which they also become covered, the bacteria act in a sense as a single entity. The secreted adherent film can serve to concentrate trace nutrients, attract commensal organisms to process waste, or provide or recycle nutrients. The adherence may cause the walls to thicken, protecting the individual cells.

Different microstructures in the aggregates affect microenvironments, the flow of and access to water, response to antibacterial agents, and the like. In a sense, by organizing fluid flow in their environment, bacterial colonies become vascularized. More remarkably, it has been discovered that bacteria use different sets of genes when living in biofilm aggregates than when planktonic (free-floating), with major changes in the protein composition of their cell walls. More complex and diverse are ways that bacteria aggregate and accomplish things they could not do alone

(Armitage 1999; Hoyle and Costerton 1991; Miller and Bassler 2001; Parsek and Greenberg 2000; Pratt and Kolter 1999; Prigent-Combaret et al. 1999); sometimes, different species cooperate in this regard (we are anthropomorphizing in discussing these things as if they were done on purpose).

ANIMALS

Here we consider general characteristics of multicellular organization, and in Chapter 9 we will consider the genetic mechanisms that bring this structure about. First, we should repeat what has been noted by countless other biologists: the range of multicellular organization cannot be arrayed in a hierarchical Great Chain of Being graded from worst to best. The simplest organisms are still with us and, if anything, seem far less likely to become extinct than most complex organisms (who may, in fact, be at an overall disadvantage and more vulnerable to extinction in the long run). If there is one multicellular structure that is “better” in this sense, it might be the human brain because it is the only structure that seems to make it possible to exterminate so many others, around the globe, at least in the short run.

Starting Simple

Very simple organisms that can live as isolated single cells but can also form associations in various ways appear to have multiple evolutionary origins (Bonner 1988; 1998; 2000; Buss 1987). The simplest organisms that we think of as truly multicellular, such as sponges, have little organized structure or cellular differentiation. Most are referred to as *diploblast*, because they have an organized outer and inner cell layer but little cellular material or organization in between. Choanoflagellates are single-celled organisms consisting of a cell body and a collar of microvilli through which water is filtered, surrounding a single flagellum used for propulsion. Some choanoflagellates aggregate to form what may resemble an early form of metazoan life.

Sponges have only a few distinct cell types: a protective external layer; a ciliated internal layer that moves water through the organism; and a middle layer composed of freely migrating archeocytes that scavenge food particles and can differentiate into all the other cell types, as well as cells that lay down an extracellular matrix of spicules (Sponges 2002). A sponge is a hollow structure pierced by numerous incurrent and excurrent pores through which water flows. There is species-specific variation in shape, which implies that there is also underlying developmental programming, and specialized functions such as contractile ability and color variation. However, sponges do not develop from a single cell in the highly orchestrated hierarchical way we will see is typical of more complex animals. Their lesser level of tissue commitment and organization can be seen by the ability of disaggregated cells to reaggregate into a normal whole, something “higher”. Cells in vertebrate embryos have similar reaggregation properties but only early in embryogenesis (in a way that may be informative about evolution and mechanisms of early development) (Steinberg 1998). Environmental conditions can affect sponge morphology, reflecting their rather loose level of genetic programming.

Sponges can reproduce asexually by shedding buds or “gemmules,” small groups of cells that float away to anchor at a new location (Darwin used the term to hypothesize ubiquitous rudimentary and rather lamarckian units of inheritance). Even the

simple sponge can also differentiate into haploid sperm and eggs and undergo fertilization and the shedding of ciliated larvae. Reproduction can even be seasonal. Thus, without much real histology or organized form, some developmental complexity occurs.

Coelenterates (or Cnideria) include corals, jellyfish, and hydra. These species are diploblasts but make only a few specialized cell types, including the stinging nematocysts. They are typically bag-shaped with a single oral opening but can have internal septa, and they have more species-specific body plans than sponges. Coelenterates reproduce asexually as sedentary polyps or via sexual, free-swimming forms known as medusae; their life cycle often includes both forms. Corals are aggregates of polyps, whose association is so highly constrained that it is somewhat moot whether they should be considered separate individuals. They exemplify organisms so integrated with each other and with the environment that they themselves construct that it is difficult to consider the organism as only what is within each unit (e.g., Turner 2000). Some forms branch in a simple treelike way as they grow, with branches occurring in a more or less random pattern but affected by external conditions. Coral polyps grow on acellular (corallite) stalks that occasionally divide into two similar units, each forming a new stalked polyp, overall resulting in a treelike structure that approximates a fractal pattern.

More Complex Animals: Basic Characteristics

Much of our knowledge about the remaining forms of animal life is generalized from the intense study of only a few model organisms (e.g., mice, fruit flies, nematodes, sea urchins) and a few special traits (e.g., eyes, early embryonic stages, limbs), usually studied for historical reasons (because so much is already known) or convenience (transparent embryos, short lifespan, easy to manipulate or raise in the laboratory). But the broad picture is likely to be representative of the general array of “strategies” used.

These “higher” forms of animal life essentially are committed to multicellular life derived from a single starting cell. Subsequently, a highly orchestrated four-dimensional dance of development in space and time takes place. Differentiated organ development includes a few simple processes: make a ball; change its shape by differential, asymmetric growth; fill space by local division of similar cells; pinch it; make a local bulge outward or indentation inward; close or open a hole in the ball forming tubes; and split growing areas into two or more branches. Cells then differentiate to produce specific substances such as extracellular materials that provide structural rigidity or other properties, perhaps comprising dead cells or protein debris (e.g., hair, lenses), or may become mineralized and quite hard, as in shell, bone, or cartilage. Some cells secrete particular functionally important products such as hormones or digestive enzymes.

Despite sharing these basic characteristics, sometimes in a highly programmed way within a given species, there is great variation in most aspects of development among species, sometimes even among closely related species. A few rounds of rather unspecialized cell divisions generically referred to as *cleavage* turn the single fertilized egg into a simple primary shape, generally a hollow ball (*blastula*). But there are various basic patterns, including very regular and symmetric division in echinoderms, many asymmetric patterns, and the initial formation of a syncytium, a large single cell that contains many nuclei not surrounded by nuclear membranes or cell walls (in many insects) (Gilbert 2003).

Commitment of cells quickly occurs in two basic ways that have been defined by manipulation of embryos, or more recently, of gene expression. *Polarity* is also established early and involves basic axes of symmetry mentioned above. In insects like *Drosophila*, polarity is established through gradients of maternal gene products in the syncytium. Vertebrates first separate a yolk or nutrient section from a section that will generate the embryo itself. Then, the AP and DV axes of the embryo form, instructed by signals produced by centers known as *organizers* that affect nearby cells that proliferate as growth extends away from the organizer region. For example, Spemann's organizer is a region of cells capable of establishing the early AP and DV axes in a vertebrate (the homologous structure has different names in different species), as is shown by its removal or surgical relocation in an embryo and by mutations that alter or prevent it from developing.

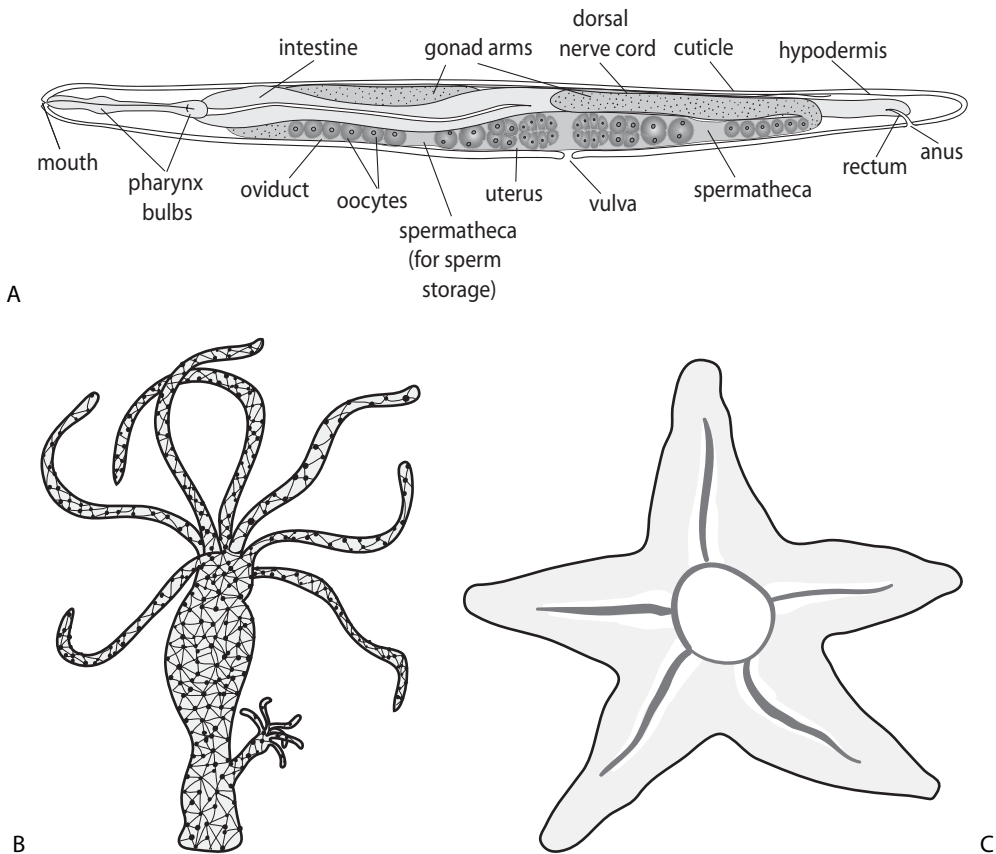


Figure 8-2. A bestiary of basic anatomical body plans: (A) Nematode, experimental model system, *C. elegans* with tissues identified (for use below); (B) hydra; (C) starfish illustrating radial symmetry; (D) fossil fern from Carboniferous age around 350 million years ago (Mya); (E) fossil trilobite from the Cambrian (Burgess shale) about 500 Mya; (F) Richard Owen's famous archetype of a standard, shared vertebrate body plan. Also see the frog, fly, and plant body plan figures below. Sources: Fern, (White 1899); trilobite, (Walcott 1918); (figures D–F reproduced from Moore 1987; 1989, with permission).

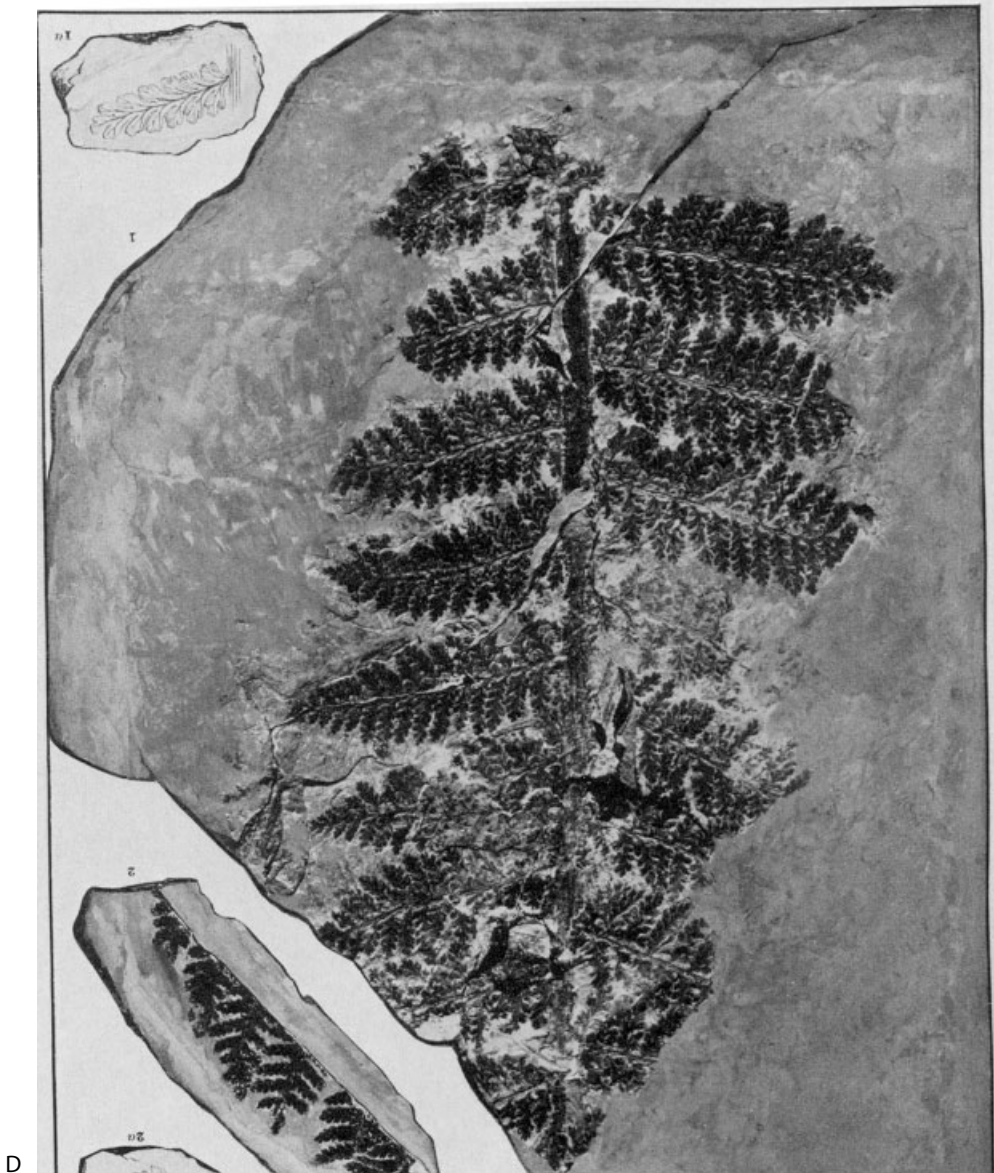
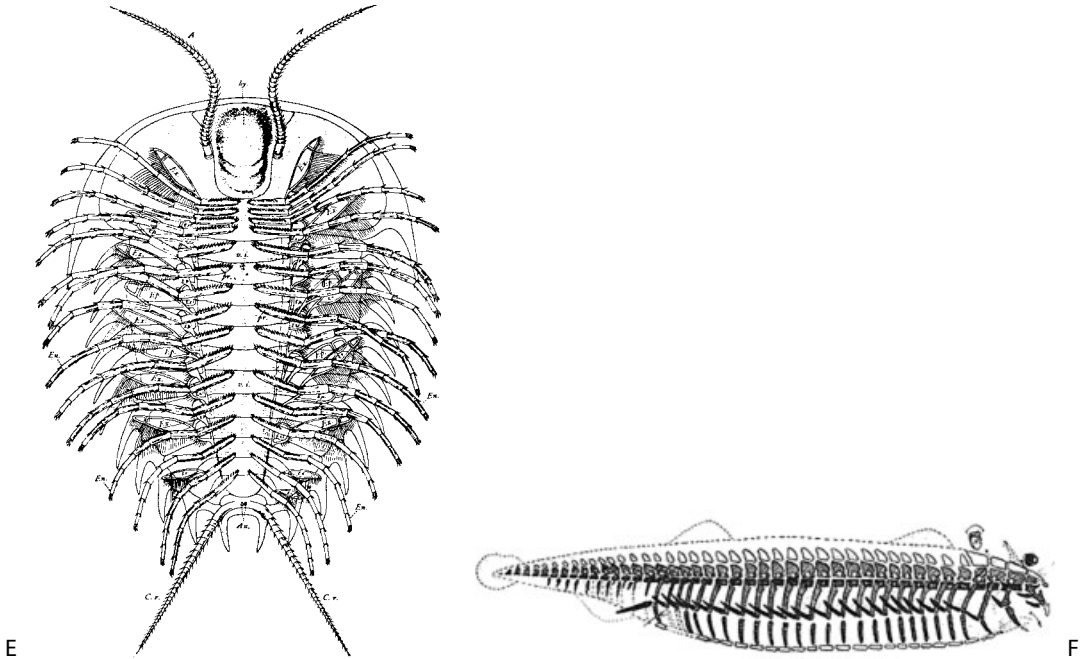


Figure 8-2. *Continued*

The notion of an organizer is somewhat imprecise, but in a general sense there are organizers from head (see Chapter 15) to tail (Agathon et al. 2003) and many places in between, and in subsequent chapters we will see how some of them work.

The notion of an organizer, like others in biology that use a term borrowed from everyday life, should be understood with circumspection. The term is used to identify signals in a practical sense and does not mean that these cells somehow come from the outside, as the word “organizer” often implies in other uses. Developmental organizers result from prior cells in a continuum of a developmental process.

Figure 8-2. *Continued*

Pentamerous animals such as echinoderms (e.g., starfish) have an additional kind of axis, comprising the five “arms” arrayed around the central body location; they have a central radial axis and a single, repeated PD axis going out along each arm. The genetic and developmental homologies with linearly patterned animals are not yet well understood (Popodi and Raff 2001).

Cells in different regions of the blastula become committed to three basic lineages that are the precursors of different sets of tissues in the final organism. Broadly, *ectodermal* cells will generate the outer covering and nervous system, cells that make up the *mesoderm* will become muscle and connective tissue, and *endodermal* cells will become the gut and its associated organs. Organisms formed from these three basic tissues are referred to as *triploblastic*. These first steps in the hierarchy of histological differentiation are highly varied in detail, even if the result is somewhat similar. Figure 8-3 shows a selection of developmental tissue “fates,” in different classes of animals, that form in a hierarchy of commitment (e.g., see Carlson 1999; Gilbert 2003; Wake 1979). By “fate” is meant that descendants of the tissue in question differentiate into a unique hierarchy of cell or tissue types. This is a manifestation of the hierarchical, histological sequestration that is the major outcome of development, which is why the pattern can be drawn cladistically (in a treelike diagram of descent and differentiation). Commitments are made at different stages of development; these are not yet well understood, but some late-developing organ systems may already be *preprogrammed* in cells at very early developmental stages (the cells await a subsequent proper environment to begin elaborating their program).

The formation of these layers involves a process known as *gastrulation* by which the committed cells take their relative positions. Gastrulation in most animals

involves the movement of cells from the outer cell layer to the inner, a process dependent on cell division and migration. Basically, through inpouching, the formation of holes in the outer layers or simply the expansion of dividing cells into the space inside the embryo, a layer forms that will develop into an inner pouch or tube that will become the gut—essentially a tube running through the animal and open to the outside world. Mesodermal cells are caught between the outer ectodermal covering and this inner tube. The higher animals are classified as protostomes (basically, mollusks, worms, insects), and deuterostomes (chordates), depending on how the gut is formed.

Among the most important invertebrate model species is the nematode *Caenorhabditis elegans*, which develops from a single cell into an adult with a stable 959 cells; lineages of all 959 have been entirely traced through a stereotypical developmental journey. Each organ or structure develops from a specific, very small number of cells, and gastrulation and other basic processes are correspondingly simple.

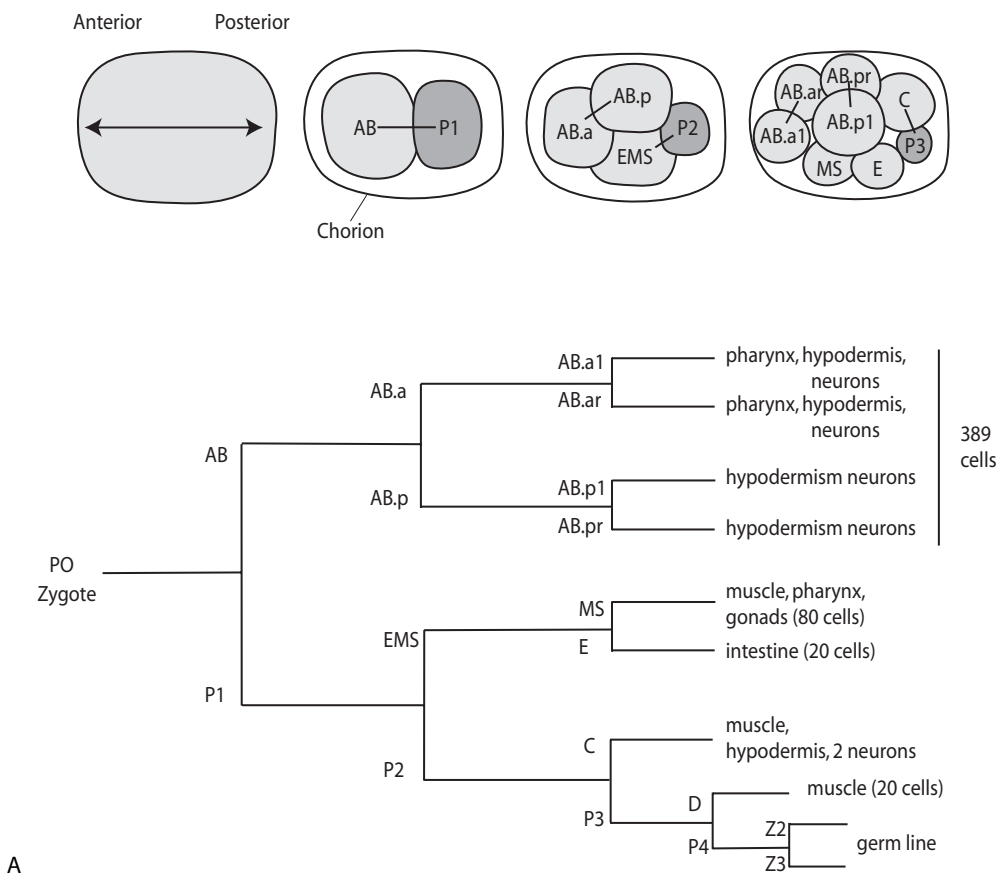
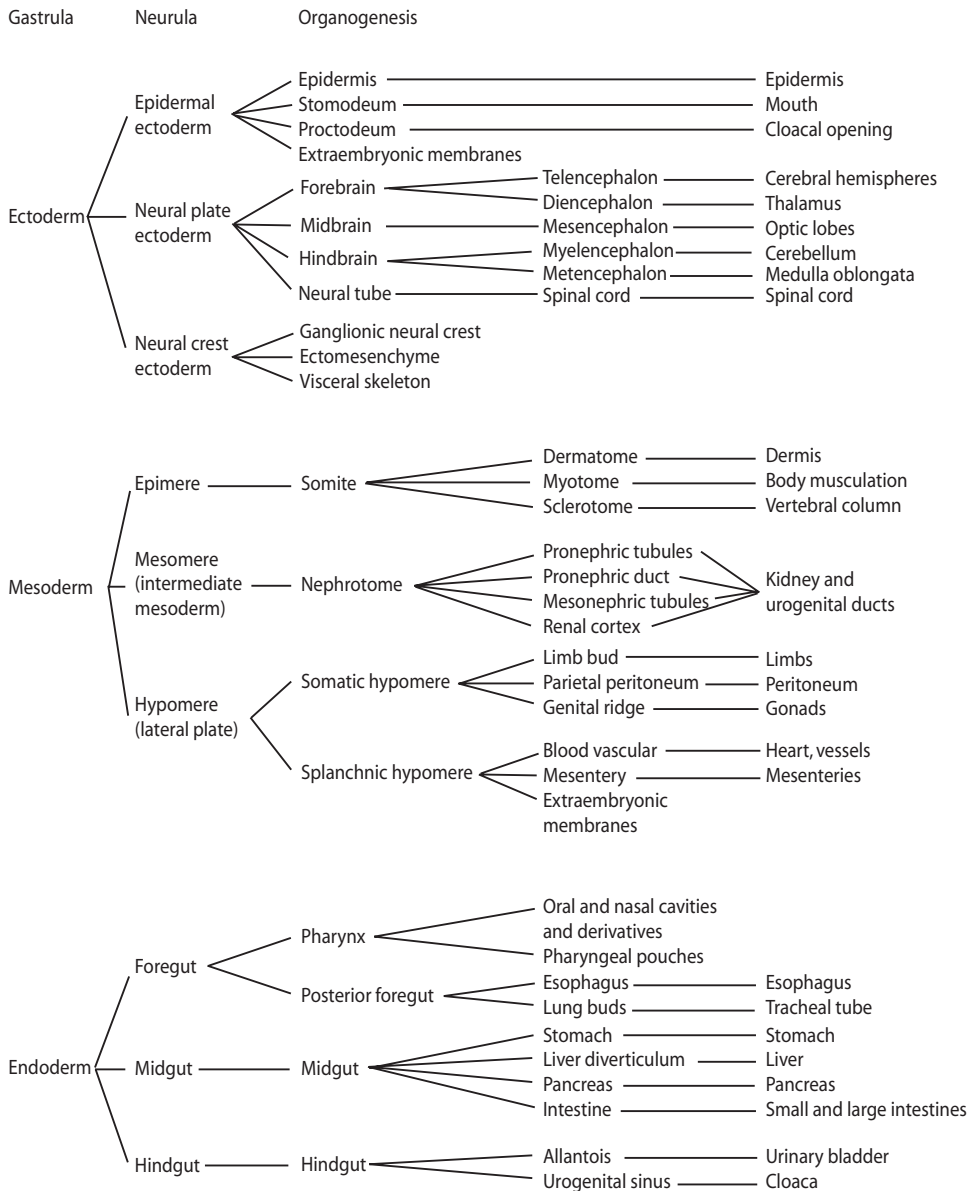


Figure 8-3. Embryological fate maps. (A) The development of the nematode *Caenorhabditis elegans* from a single cell to 558 differentiated cells (these will further proliferate to a final total of 959 cells, but without further change in cell type); (B) the tissues descended by hierarchical differentiation of the three germ layers in vertebrates (example is from humans): endoderm, mesoderm, ectoderm; neural crest interactions not specifically identified.



B

Figure 8-3. Continued

Regarding arthropods, more is probably known about *Drosophila* than any other (Lawrence 1992). Cell walls form around the nuclei in the syncytium to form a hollow structure along which there quickly appears a series of segmental structures that differentiate along the AP axis as the embryo grows. Local organ primordia called *imaginal disks* develop at specific locations along the axis. The imaginal disks themselves serve as organizers for cellular subsets that are precursors of the various parts of the antenna, mouth and tailparts, legs, wings, sensory bristles, respiratory spiracles, and the like. At appropriate times, ingrowth or outgrowth starts the

process of organogenesis from the imaginal disks, and these structures themselves develop AP and DV axes, and then develop PD axes as they grow in relation to the body of the embryo.

In vertebrates, developmental processes are more complicated, although the basic principles are similar (Gilbert 2003). Figure 8-5 shows one pattern, as seen in *Xenopus*, the laboratory frog. The basic cell types (e.g., gut, muscle, photoreceptor, neural) are similar to those in invertebrates like *Drosophila*, but vertebrates have to deal with some other complications. They are large, which is made possible by having an internal rather than external supporting skeleton (though in their early evolution they were covered, protected, and supported by an external scale-like skeleton), they have more complex organ systems, and they tend to live longer and have more renewable tissues (especially organs lined by epithelia, tissues with ever-dividing *stem cells* that renew locally differentiated special cells like those of the gut and skin), and they have more elaborate immune systems. Large internalized bodies entail thermal, mechanical, and respiratory demands and the circulatory and neural apparatus that goes with them.

Vertebrates, like arthropods, are organized with an external wrapping around a hollow digestive tube, having organ outpockets such as the pancreas. Also like arthropods, which are only distantly related, they have a longitudinal hollow nervous

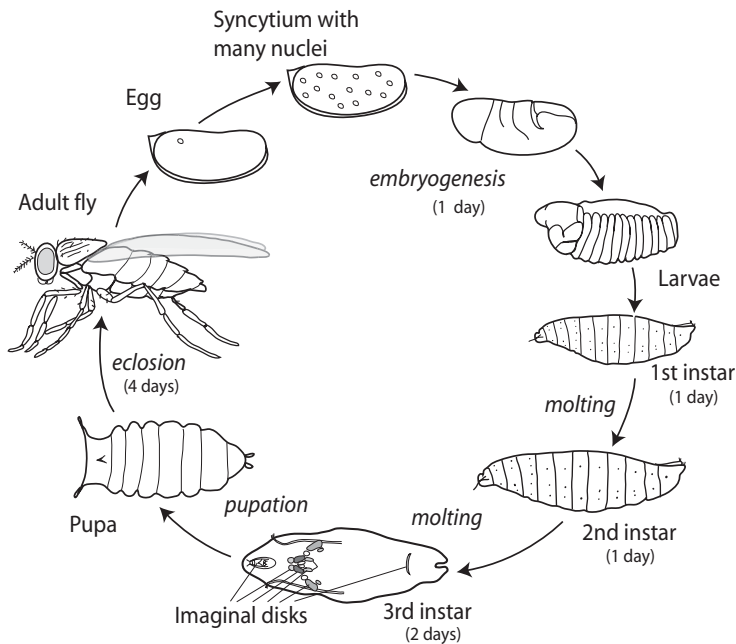


Figure 8-4. Basic stages in insect development; *Drosophila*, from fertilized egg to embryogenesis, larval stages, pupation and eclosion, or metamorphosis, to adult fly. In the important transition from the syncytium to the cellular embryo, the nuclei in the former locate around the periphery of the egg, and membranes form to separate them into separate cells. The embryo forms stripes of gene expression (not shown) that are precursors of the axial segmentation that is important from the early embryo stage through to the highly segmented adult.

system, with the brain at one end and differentiated branching along the AP axis, referred to as an “inverted” body plan. The vertebrate circulatory, gut, and nervous systems, however, develop with more complex turning and folding, by differential growth at various places along the axis, versus the corresponding system in arthropods, which is more or less straight along its major axis. In both groups, local signaling centers or organizers guide the development of structures like the distal tips of developing limbs and the midbrain and hind brain regions of the central nervous system, teeth and feathers, and the like. The neural axis is dorsal in chordates but ventral in arthropods, and the digestive systems have reverse relative locations. There is healthy debate about what homologies if any might be involved, as will be discussed in Chapter 9.

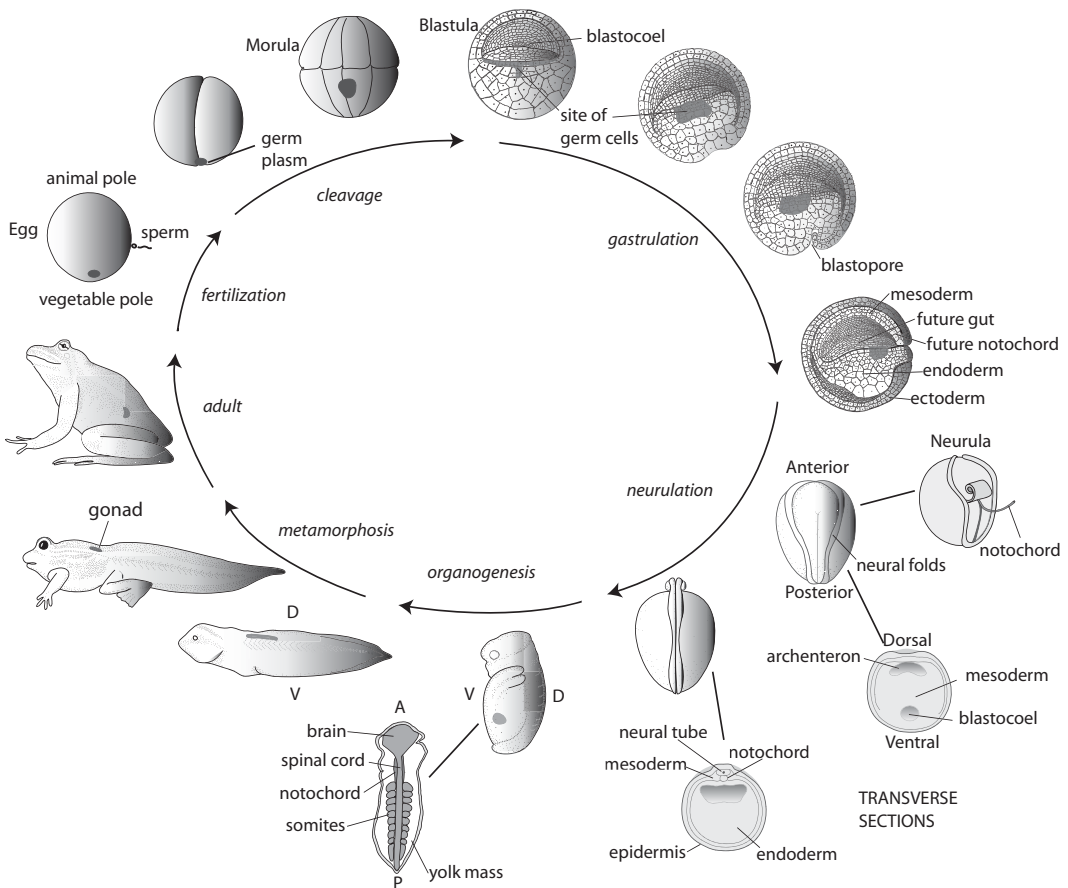


Figure 8-5. Basic stages of *Xenopus* development, from fertilization to cleavage (cell divisions that produce the blastula, a spherical layer of cells that surround a fluid-filled cavity), gastrulation (invagination of cells from the blastula to form the gastrula, a two-layered sphere consisting of ectoderm and endoderm surrounding an archenteron that communicates with the exterior via the blastopore that will form the gut), neurulation (embryonic stage at which the neural tube develops), organogenesis and metamorphosis to the adult frog. Shaded spots indicate site of development of germ cells and gonads at each stage.

Among the defining characteristics of vertebrates is a cell lineage known as the *neural crest* (NC), which derives from the early dorsal neural ectoderm. The NC cells migrate from positions along the dorsal midline, intercalating through mesodermal tissue to reach various locations where, in contact with overlying ectoderm, inductive tissue interactions trigger the initiation of various organs, like hair, feathers, mammary gland, teeth, and others (NC cells migrating from the head region are sometimes specially referred to as Cranial Neural Crest, or CNC). These inductive *epithelial-mesenchymal interactions* (EMI) are location and context specific and are responsible for so many important characteristics of vertebrates that some researchers give the NC a special place as a fourth primary germ layer (Hall 1999; Hall 2000), making them “quadroblastic.” Among other structures, the segmental patterning of parts of the head have been correlated with EMI and NC migration (but the role of EMI is not entirely clear; see Chapter 9).

At least one reason EMI is important in vertebrate development is that many organs can develop only after the embryo has reached a certain stage of complexity. At that stage, however, it may be difficult to initiate a dispersed structure because the tissue bed is too large for effective signaling. For example, teeth cannot develop until there is a pharynx or jaws to develop in, taste buds cannot develop until there is a tongue, and so on. The use of NC cells and EMI allows tissues to be preprogrammed for certain developmental fates, but then not be activated until an appropriate organizational level. Neural crest cells can then come into contact with the inductive tissue located where the structure will develop. Other organ systems form in vertebrates by inductive interactions that involve mesoderm. Other forms of differentiation occur in cells programmed in some home location but that then circulate freely, as in the blood and lymphatic systems. Consistent with the notion that early program is important in later development, invertebrates establish basic regional differentiation very early in their development.

In addition to the widespread presence of epithelia, vertebrate organs typically contain both primary “functional” tissue, called the *parenchyma*, often in the form of repeated units, like pancreatic islets, nephrons in the kidney, osteons in bone, hairs, taste buds, or ovarian follicles. The functional tissue in each unit secretes hormones or enzymes, filters blood, transmits neural signals, and so on. There is usually also supporting tissue, called the *stroma*, such as Schwann or glial cells to protect neurons, or a cellular matrix to hold and structure the organ. Invertebrates are also organized in a similar patterned way, for example, with discrete eye subunits including sensory elements of various kinds that will be described later (not to mention the joints for which arthropods were named).

Many if not most organ systems have periodic, modular, or segmented, organization, with multiple regularly spaced and perhaps regionally differentiated subunits. Some systems do not segment linearly but ramify—literally—by physical branching, as in the nervous, vascular, and respiratory systems. Some organ systems, like limbs in tetrapods, are both segmented and branching.

It is easy to think of development as having to do with morphology, but much if not most of what goes on in life is “virtual,” that is, is physiological and has to do with chemical interactions rather than physical structure. Some of these take place within the cell, but some take place in the internal circulating fluid, like lipid transport and respiratory gas exchange—in some ways comparable to those of the external environment, to the extent that some believe that the environment should

not be considered as separate from the organism itself (Turner 2000). Metabolic differentiation at the gene level involves many of the same characteristics found in genes responsible for physical systems: the use of multiple subunits often involving related gene products, which interact with each other. The processes of differentiation of physiological systems are often hierarchical in time or cell lineages and demonstrate homeostatic interaction with the environment, in response to signaling such as of hunger, hormones, pheromones, exogenous infections, and the like.

PLANTS

Plant life histories differ from those of most animals in several basic ways; in part, their system is different because the rigid cell walls prevent the kind of migration so important in animal development. Many plants have life cycles that include diploid and haploid stages, perhaps in alternating generations. Although not reproduction in the sense we have already described, because everything stays connected, some plants nonetheless effectively propagate via *rhizomes*, part of their root systems, that push up through the soil to start a new above-ground plant. Plants that reproduce via single fertilized cells do not sequester a separate germ line, and gametes are independently produced in many places (different flowers) on the same plant. Plants also continue to undergo differentiating development and morphogenesis throughout their lives, as stem and root meristems retain the ability to develop new structures at their ends because they maintain a core of pluripotent stem cells, resulting in plants being able to respond to their environments in ways that are less stereotypical or fixed than most animals.

Plant embryos also generally form three basic tissues, known as *dermal*, *ground*, and *vascular* tissues, but this is not established by gastrulation. Plants have an active vascular system for the upward and downward transport of nutrients, but it forms differently from these tissues (Figure 8-6). Like animals, plants use a limited repertoire of basic processes, including growing, elongating along the stem and root axes to form branches, and differentiating some stem tips into flowers, leaves, or other specialized structures. Thus plants have a vertical axis (separately behaved in stems and roots), as well as corresponding axes along stem and root branches.

The First Few Cell Divisions

Not surprisingly, early plant development varies among the many plant species that have been studied (Weigel and Jurgens 2002). Under normal conditions, the fertilized zygote undergoes a few rounds of development to form initial root-tip and leaf structures, thus providing the means of survival through its early days. The new plant quickly develops a few primary cell types or layers, which lead to the apical root and apical shoot meristems. Development occurs radially around the meristems. Several primary tissue layers form distinct parts of the plant: dermal tissue forms the outer protective layer, vascular tissue forms the hollow fluid transport tubes, and ground tissue grows in between these layers.

The Basic Plant Body Plan

The two meristems grow to become roots and stems, respectively. Cells behind the growing meristem divide to allow elongation of the shoot or root. At intervals determined by growing conditions and the genotype of the plant, buds of meristem tissue

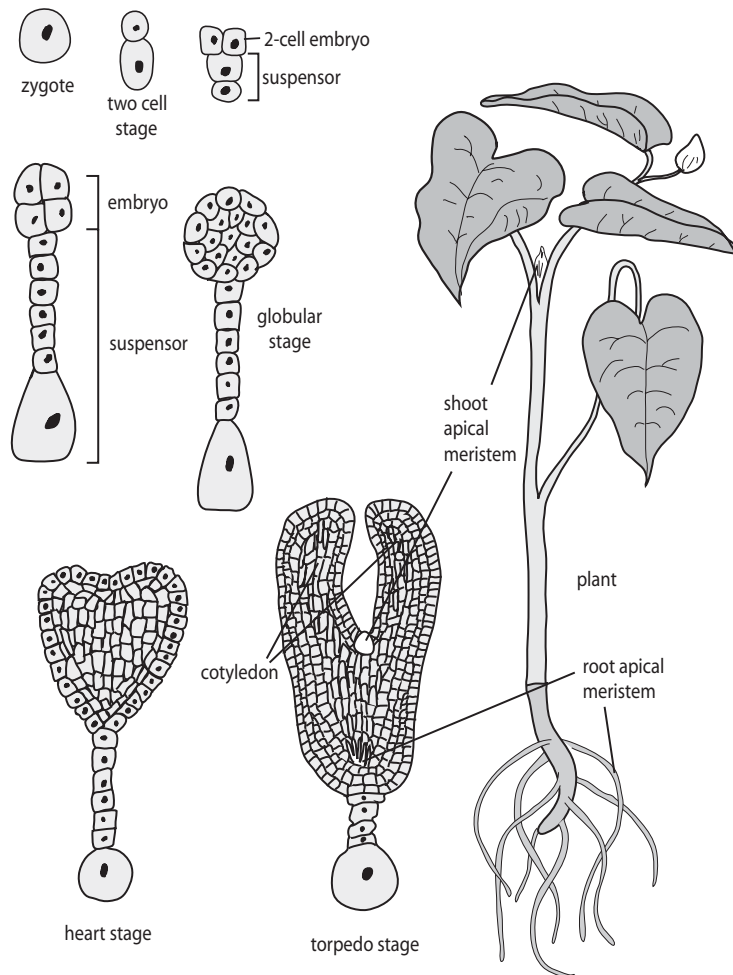


Figure 8-6. Schematic of plant development. Figure depicts the stages of a dicot; monocot development is basically the same.

separate laterally, to initiate branches. In the shoot, each such branching normally develops into a leaf and a small meristem, which itself can generate a new branch or shoot (depending on species, conditions, and the like). Signaling from the primary shoot meristem may control the timing of development of other shoot meristems, maintaining “apical dominance” of the primary shoot. An interesting feature of plant development is that it achieves a kind of spiral symmetry as new shoot meristems are formed at relatively species-specific angles of rotation around the developing apical meristem. Furthermore, plants have hierarchical branching symmetry, with each new shoot or root having radial and PD axes of their own. At points determined by internal and external conditions, a shoot meristem will differentiate into a specialized structure, a flower. This can be male, female, or both. Plants vary in whether they have only one, both, or “bisexual” flower types. Root meristems also send off lateral branches at intervals.

The notion of “stem” cells in animals and plants is similar, and there are at least some genetic similarities shared between stem cells in the two kingdoms (Benfey 1999; Laux 2003; Weigel and Jurgens 2002). This suggests that in the common ancestral period, multicellularity may have been a widespread sometime trait among otherwise single-celled organisms, perhaps due to the effects of gene(s) (today with homologs *Zwille* in plants, *Piwi* in animals, of those currently identified) related to basic cell division or adhesion. In any case, the properties of stem cells are similar in the two kingdoms, but plant stem cells are essentially totipotent in adults, able under appropriate signaling to generate all the plants stem, or root tissues, unlike animal cells which are “stem” for a tissue but not all tissues (Laux 2003; Weigel and Jurgens 2002). It is this that makes each stem and root rather independent, unlike the case in animals where each tissue can further differentiate, but generally only down its respective fate-map pathways. A comparable notion in animals would be the cells that give rise to each hair or limb. But in a more substantial way, each branch of stem or root is like a separate organism (see below). Plant cells have less rigid developmental fates in that differentiated cells can more easily dedifferentiate to become stem cells.

Plants are simpler in their developmental patterns than animals because plants do not develop as many organs or as much interstructure communication as animals may. Plants do not have to deal with mobility and hence have simpler communication systems. Plants can respond to predation, but not in ways that require the degree of movement flexibility and hence muscular or nervous system complexity of animals. This does not mean that plants are all alike or are simple or that they have little sense of their environments. Plants do differentiate many different cell types and tissue structures, they have to maintain rigidity often at great body sizes, and they have to maintain mutational integrity over very long lifespans. They must be able to find nutrients, light, and water (and mates) and must be able to protect against infection or predation while remaining immobile (or, for floating plants, without real control over their mobility). Plants have distribution systems (although they are not closed as are the circulatory systems of vertebrates) and internal as well as interindividual communication. Although the systems differ in detail, there are many fundamental similarities in the molecular mechanisms by which they are achieved. Plants sense many aspects of their environments and integrate that information to “make decisions” relevant to their kind of life. To achieve this, plants also have evolved a variety of diverse ways of branching (Sussex and Kerk 2001), and developmental fate maps, even if they are not implemented in as unitary way as animals’ are (Jurgens 1994).

SOME ADDITIONAL GENERAL PRINCIPLES

The foregoing attempted to give a general, if superficial, picture of the basic patterns of animal and plant development and form, stressing the processes or phenomena that are repeatedly observed among species or within individuals during their development. So brief a catalog gives a poor reflection of the diversity of mechanisms, and we have to be careful about how we interpret the data, perhaps especially in regard to what appears to be widely conserved developmental characteristics because they often are quite different when examined in detail. This is clear when multiple species (sometimes even closely related species) are compared.

NOT EVERYTHING WE SEE IS “NECESSARY”

Simpler organisms have no clear-cut body plan, some can develop a whole new organism from any cell or at least from cells located throughout their body, and some have no clear-cut “embryo” stage. Plants have an organizational “plan” and have essentially totipotent cells throughout their bodies. In the case of animals with classic body plans, the hierarchy of development and phylogenetic similarities led biologists to form a general idea of how development works, and this then became intimately related to the study of evolution (Richards 1992). Going back to the leading embryologist Karl Ernst von Baer in the early 1800s, it had been shown that there is typically less phenotypic divergence early and more divergence later in development, consistent with a hierarchical dependency of later structures upon earlier foundations. Before the discovery of evolution, the idea was that similar animals developed their adult forms by modification of a largely identical early form. Around Darwin’s time, the changes among species were widely considered to have come about by sequential building of new structures upon *existing* adult stages over the generations, a process called “terminal addition.” This explained how complexity arose or evolved. Essentially, natural selection was treated as if it worked only on adults.

The famous and classic example has to do with the stage of vertebrate embryos known as the *pharyngula*, at which gill arches, somites (prevertebral segments), dorsal nerve cord, and early limb buds were present. Ernst Haeckel (see Figure 8-7), Darwin’s energetic and prolific defendant and science-popularizer in Germany, developed his well-known recapitulation theory of morphological evolution and called it the Biogenetic Law, that Ontogeny (development) Recapitulates Phylogeny. This theory held that in development an organism passes through the adult stages of its ancestral species, adding on later stages leading to its current more advanced form.

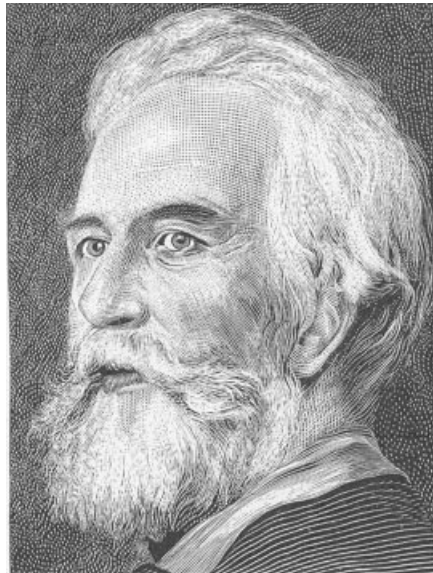


Figure 8-7. Ernst Haeckel. 1899 drawing by F. von Leubach, in Haeckel 1906.

In Haeckel's famous figure, diverse vertebrates are shown passing through pharyngula stages, which are essentially the same, presumably representing their ancestral state as a kind of primitive gilled fish (Figure 8-8B). An "hourglass" model developed over the years in which the different forms of vertebrate cleavage had been canalized to produce a stereotypical pharyngula common to vertebrates, which then diversified later in development to the varying adult morphologies. We know now (and in truth it was known to Haeckel) that the pharyngula stages are not as identical as depicted; even more importantly, however, the earliest stages from cleavage onward are in fact quite diverse among vertebrates, and indeed the supposed stereotypical traits vary in their timing of appearance (Bininda-Emonds et al. 2003; Raff 1996; Richardson et al. 1997; Richardson and Keuck 2001; 2002). Nonetheless, there are similarities, even if they were exaggerated, and they provide important evidence for shared ancestry.

As a digression, historians and philosophers of science have many times remarked on the ubiquitous, perhaps even necessary, element of public advocacy (or even propaganda) that is necessary for a controversial view to gain acceptance. One can go further and note that one of the problems in science is that there is, and always has been, a whole lot of "fudging" going on: theory is routinely defended by selective use, presentation, collection, and manipulation of the evidence, ignoring some facts and stressing others, redefining terms, and retrospectively and conveniently reinterpreting data. This cannot be justified formally, but it in many classic cases was necessary, in the face of imperfect data or measurement (or understanding), in order to reach a general understanding of things. Gregor Mendel's supposed cheating in his analysis of pea plants is a famous example (Weiss 2002b), and so it was with the classical Copernican revolution in astronomy that occurred despite many totally wrong inferences and facts incompatible with the theory as proposed at the time. Many theories are wrong, but even correct theories typically require commitment to a framework even in the face of such evidence. Darwin's wrong ideas about inheritance and the problem of the age of the Earth did not prevent recognition of the overall truth of common ancestry.

Even in related animals that end up with similar adult morphology, developmental stages *subsequent* to the pharyngula can vary greatly (Raff 1996). There are several well-studied examples of this: sea urchins, anurans (frogs), and ascidians (primitive chordates including tunicates, or sea squirts). From genetic phylogenies of contemporary species, among closely related species with similar adult forms, some are found to pass through an intermediate juvenile or larval stage, whereas others do not. Furthermore, the similarity of larval stages across the phylogeny suggests that it probably was the ancestral state (at least, it is an ancient state). However, a larval stage has been dispensed with in several subsets of the descendant lineages. Figure 8-9 shows the "pluteus" larval stage in sea urchins. Several frog lineages have subsets of species that pass through an immature, tadpole stage. The converse situation also appears to be true: among sea urchins, there are genetic differences in how *similar* developmental stages are achieved (Kissinger and Raff 1998; Nielsen et al. 2003; Raff 1996; Raff 1999). Comparable statements appear to apply to the development of flowers among angiosperm species (Kramer and Irish 1999; Ma and dePamphilis 2000).

It is clear that there has not been an absolute, or clear-cut, advantage to going through one or the other form of development—a ladder of developmental sophis-

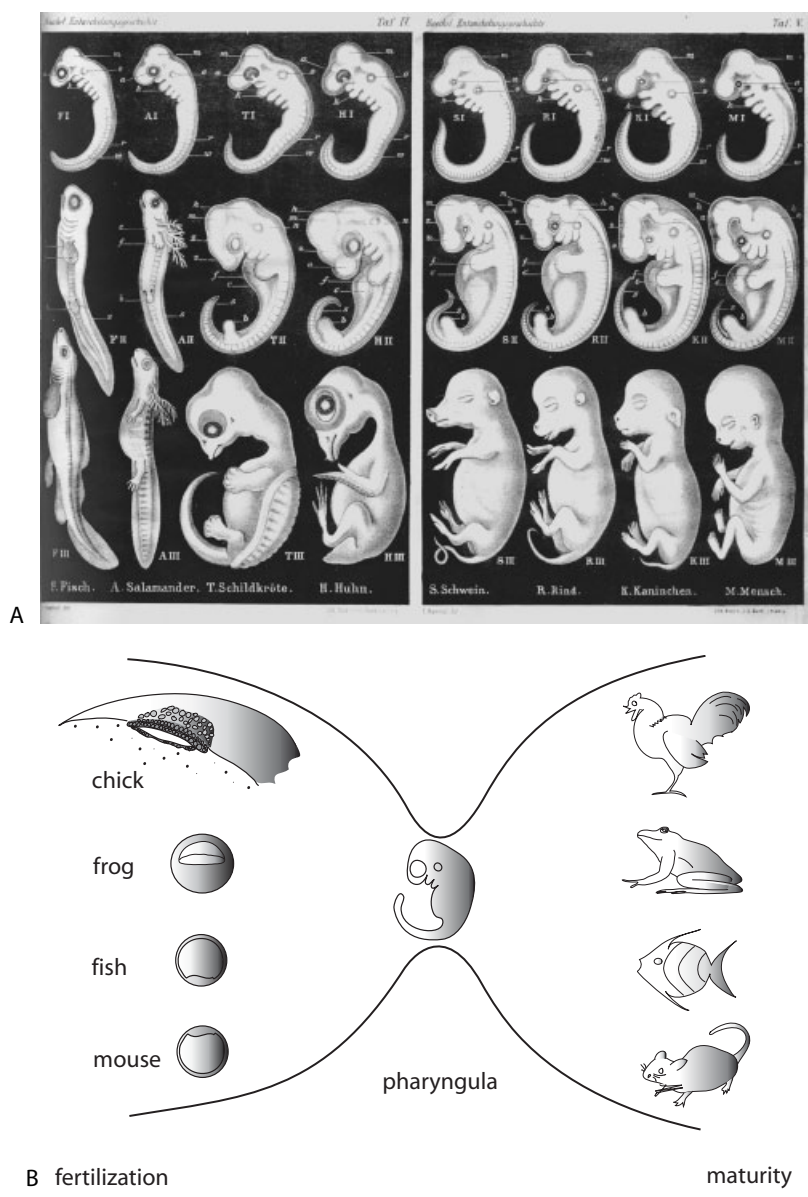


Figure 8-8. (A) Ernst Haeckel's famous figure showing divergent adult stages from similar pharyngula stages; (B) developmental constraint notion, that divergent blastula stages among vertebrates all pass through a common pharyngula stage and then diverge again to the adult form. (A) photograph courtesy Michael Richardson; (B) modified after Richardson, Hanken et al. 1997 with permission.

tication or improvement is not evident. Direct and indirect developers with very similar morphology are doing well today. A larval stage is not a necessary aspect of a frog's solving the "problem" of becoming an adult; some do, some don't, and axolotls stop at the larval stage. If the idea of evolution by natural selection is correct, then it must be that under some local environmental conditions, when the

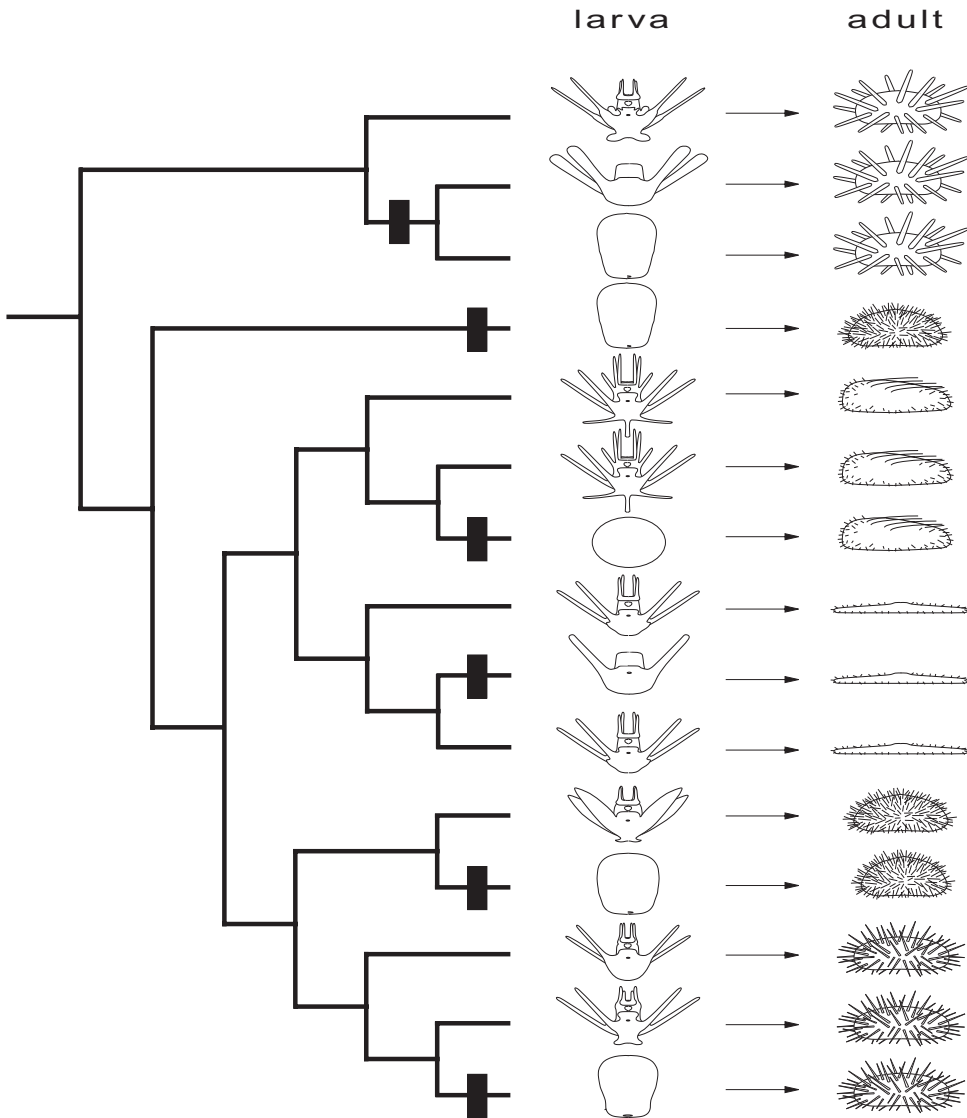


Figure 8-9. Variation in sea urchin development showing the repeated evolution (or repeated loss) of the pluteus larval form in different branches. Modified from (Wray et al. 2003), reprinted with permission.

appropriate mutations were present, one route of development provided an advantage of some sort.

Repeated evolution of a structure is plausible if the mechanism is not too intricate. A potential alternative compatible with modern theory is that the structure was present in the common ancestor, and was repeatedly lost. A variation on this theme would be that the changes occurred by phenotypic drift: that both developmental strategies existed in a population because of some tractably simple genetic mechanism, and one variant replaced the other by chance (or, was not needed and its loss was tolerated by the screen of natural selection). When we see something

spotted throughout a phylogeny, such as wings in some insect clades, tadpoles and optic cups in some amphibians, pluteal stages in some sea urchins, and some aspects of olfaction or color vision we face these explanatory possibilities.

GROWTH, HETEROCHRONY, AND ALLOMETRY

One of the persistent questions in the evolution of morphology or development is how what appear to be complex changes can be brought about in a practicable way in the time available and by the simple process of natural selection. It has long seemed—correctly, it appears—that it can't be that each variation on a theme is due to the arrival of a new gene specific to the purpose. For example, it would seem to be impractical for every stripe on a tiger, or vertebra, or cusp on a tooth to come about by the action of a new gene.

Discussion of this topic goes far back in the history of biological thought (Schlichting and Pigliucci 1998), but the best-known systematic attempt to explain the phenomenon of changes in shape in modern scientific terms belongs to D'Arcy Thompson (Thompson 1917), who stressed concepts of *allometry* (size differences in the same structure that may or may not be linear in all dimensions), *heterochrony* (changes in timing of the same structure), and *heterotopy* (changes in the placement of a structure in an organism) during development. His idea was to explain the structures in organisms in terms of the universal laws or properties of physics, rather than invoking biology-specific explanations. Whatever the proximate mechanism (which we would attribute to genes, cell characteristics, structure proteins, and the like), biological structures achieve their variation due to physical constraints and temporal aspects of their growth. A number of authors have recently dealt with these general topics (e.g., Calder 1984; Gould 1977; Hall 1999; Raff 1996; Schlichting and Pigliucci 1998; Wilkins 2002; Wolpert, Beddington et al. 1998). The close study of developmental timing has shown this to be a source of variation in vertebrate body plan, in contrast to the notion of a conserved pharyngula stage in vertebrates mentioned earlier. Timing seems to be simple to change, but can have what appear to be complex effects: a powerful tool for evolution.

In modern genetic terms, we seek to identify molecular mechanisms that will bring about these ends and to understand their relative importance. The appeal of an allometric or heterochronic approach is that in principle it can provide a means by which rather simple molecular processes can account for a wide and otherwise perplexingly complex diversity of shape or other higher-level complexity. Shape can be substantially changed simply by changes in the speed of or the differential timing of developmental processes like tissue growth, the onset of segment-formation, and so forth. This can be achieved by mutations in the dynamics of gene expression or the interaction of products of the same genes, without a need for new genes or mechanisms to arise. This may be a major means of morphological evolution (e.g., Carroll et al. 2001).

In fact, there are many examples of substantial differences in the timing and pattern of growth and development among related species, showing how variable this can be or how weakly correlated it is with phylogeny (e.g., Chipman et al. 2000). However, neither heterochrony nor allometry explains all morphological variation. For many changes, even among closely related species, more profound genetic mechanisms seem to be responsible (e.g., Raff 1996).

A famous example of the work of such mechanisms in evolution is the idea that humans are an example of *neoteny*; that is, we retain juvenile morphological proportions, particularly in brain size, as adults; this enables us to have relatively large heads compared with our closely related primate ancestors or contemporaries. Of course, we aren't really just grown-up babies (even if we act that way), but the differential shape and growth history resembles that of an arrested maturity.

Many aspects of development within a generation are responsive to environmental conditions, and some may be somewhat heritable genetically (e.g., mating type in yeast) or epigenetically (a well-nourished mother may produce a larger egg), a point that cannot be stressed enough (Lewontin 2000; Schlichting and Pigliucci 1998). If the environment persists, the effect is effectively heritable, even though not in genetic terms (Chapter 3). Many such factors, including effects on chromosome packaging (akin to genetic imprinting, e.g.), may remain to be discovered. Removal of the relevant environment may lead to a reversion of the organisms to their former state, but it may be that successful survival or mating will come to depend on the "environmental" trait. If so then if mutations arise and are favored by selection, genetic assimilation can occur (Hall 1999; Waddington 1953; 1956; 1957; Wagner 2000; Wagner and Misof 1993). Then the trait becomes inherited in the genetic sense. For example, by being bigger, an animal might be able to utilize a different food source, which could lead to selective favoring of other traits related to that change: animal gets bigger because the egg was larger because climate was favorable for its parents; because of larger size, animal can eat larger seeds than rivals from less favorable parental environment; genes for eating larger seeds hitchhike to high frequency as a result.

FROM CELLS, TO ORGANISMS, TO COMMUNITIES, TO . . . "GAIA"?

There are numerous ways in which cells aggregate to form organisms, or organisms aggregate for common purposes, or ways that cells are connected physically, logically, or temporally or by social aggregation. This suggests that an appropriate notion of an "organism" should be broader than is usually considered. As human organisms that sequester a germ line, we tend to think that the "individual" in our usual sense is the essential unit of evolution, the very definition of a living being. But this is rather arbitrary (e.g., see Buss 1987).

The nature of association among cells carrying genotypes within a "species," defined as those that can potentially reproduce together, is variable. Cells within our bodies aggregate necessarily, and only the germ line connects us cellularly to other humans. But trees and sponges do not sequester a germ line, and the reproductive cells of species like some annelids change during the individual's life. Is there a major distinction in regard to the logic of evolution between the cells in our various organs and the ants in a hive? Bees can be viewed as having a restricted, sequestered, specialized germ line, in the form of the reproductive cells of the queen. This differentiation is induced, for example, by royal jelly, but is that conceptually different from the induction of our germ line by diffusible hormones? Our limbs develop more or less autonomously relative to other parts of our body, but is this so different from the "limbs" of a hive—its individual bees—relative to each other?

Sexually reproducing organisms are not completely separate units of evolution (except hermaphrodites). But how completely separate are the different humans in a population? We are cellularly connected internally by the tree of our developmental life histories and externally by our reproductive history. We communicate among disconnected cells in our own bodies by use of circulating free cells or signaling molecules and to others by the air using pheromones and their receptors and through transmitted vibrations (sound) received by a different kind of receptor, and so on. As we will see, the mechanisms are very much the same even at the fundamental level of genes. We have cellular dependence in our bodies, but the bodies (and hence cell aggregates) of our offspring depend on others in their population, certainly until they reach maturity. It is possible to take a more seamless view of biological connectivity, all through the common phenomenon of cell division. This is much the notion Bateson was trying to express long ago (Bateson 1894; 1913) in the sense of different individuals being extensions of a single organism.

Some of the subtleties, and perhaps the reasonableness of our suggestion that the notion of the individual organism is not so clear as is usually thought, can be seen in plants (Halle 1999). Because of the relative independence of the different shoot meristems, plants can shed gametes from different locations rather than having a single sequestered genome. Somatic mutation can lead these different reproductive units to become genetically variable over the many stems and long life of a tree. Further, the individual units compete with each other during the life of the plant. The individual stems draw sustenance from a common root system (which, however, has its own independent repetitive units), but they act in other ways as individuals.

We can extend the same ideas even more broadly to include the self-reproducing, homeostatic communities we call ecosystems. They have “organic” behavior, and even the very diverse species that make up an ecosystem have common ancestry, if very ancient. There is also complex communication among the organisms (e.g., via sounds, smells, and in many other ways). They eat each other, but how different is that from apoptosis or other aspects of degrading and recycling of its constituents that routinely occurs in complex organisms?

Notions of these sorts have been taken to their global extreme by a few investigators (Lovelock 1979; 1988; Lovelock and Margulis 1974). They suggest that it is productive to consider the entire global biosphere as a single homeostatic, self-evolving superorganism. This idea has been called *Gaia*, after the Greek goddess who drew the living world forth from Chaos. The Gaia idea has sometimes been used to argue that the Earth is an almost self-aware system that tries to maintain harmonious balance, resisting the pressure toward disorder from the forces of entropy (e.g., discussed by Turner 2000). Actually, within the confines of modern empirical science, roots of the Gaia hypothesis can be found in the work of a founder of modern geology, James Hutton (Hutton 1788), who viewed the world as a closed physical system subject to the physical laws of nature.

The basic idea is simply that, through various feedback mechanisms, the diverse forms of life maintain on Earth a stable physicochemical environment that would not be possible were the planet inert and life-free. Unlike passive physical structures, living forms are homeostatic, and their interactions depend on their evolutionary connectedness (for example, species consume each other in part because the prey is made of constituents similar to the predator’s and hence reflect what the latter needs). In evolutionary terms, an excess of some resource can stimulate the evolution of organisms to fill it, and extinction can be viewed in some senses as a

way to remove imbalance. At least, any multicomponent system with inputs, outputs, and interactions can achieve regularities of structure or various states of homeostasis or equilibrium (or can spin out of control, but that has not yet happened to the biosphere). No mystic or conscious component need be invoked in considering the biosphere as a single interacting system.

Indeed, because evidence suggests that all life is cellularly connected through evolution, and currently interconnected in numerous ways, the question is whether we gain scientific insight by thinking of the biosphere as a unit. Certainly, we do this in reconstructing phylogenetic trees from DNA sequences or in using the biochemical pathways found in common to try to infer the origins of life. And many ecologists have thought about the overall energetic aspects of life, such as its relationship to the conservation of energy and thermodynamic efficiency, as a criterion for selection and for the evolution or balance of ecosystems, food chains, and the like. It is valuable, at least, to remove conceptual constraints in trying to understand how organisms manage to get through life.

CONCLUSION: MANY WAYS OF “LIVING THE DREAM”

The great biologist Francois Jacob supposedly said that the “dream” of every cell is to become two cells (so said his friend Jacques Monod) (Monod 1971)) (if DNA can dream, would it be of the ecstasy of base pairing?). In his classic book written shortly after the DNA coding system had been worked out, in which he had a major role, Monod considered that all of life is about the single “project” of reproduction. He refers to this as *teleonomy* to try, perhaps only partly convincingly, to avoid invoking teleology. The driving dream of reproduction is fulfilled in many ways; but rather than the project of life being *to* reproduce, it may be more accurate to say that a core *characteristic* of life is *that* it reproduces.

What is done to be big or complex are but different aspects of the same processes of basic cell biology and replication and of differentiation that started when they were small and simpler. The three to four billion-year-old unbroken membrane and its contents continually ooze off buds that, when they stick more or less together we call the “development” of an organism, when they separate we call the “reproduction” of a new organism, and when they no longer join cells during their life history we call “speciation.” But these are all forms of variation on one, long, connected largely branching process. This is a remarkable fact, and it makes the evolution of life as we see it much more believable.

The single phenomenon “make more of life” has been a conserved trait of all the lineages of life forms that have survived to the present. But *how* making more is done varies tremendously. There are many ways to become long, large, mobile, complex, or to make progeny. We presume that the growth, development, and reproductive phenomena we have discussed at so many levels came about through the agency of random genetic change and natural selection. In modern biology, we are not satisfied that we understand a phenomenon until we understand it at the genetic level. This is but one way, and perhaps not even the best way to understand life and its development (Keller 2002). But our major purpose in this book is to understand the role of genes in the nature and evolution of life, and we now turn to what is known about the genetic basis of these phenomena, as they relate to development. We will see that genetic mechanisms, like the phenotypes to which they are related, may be conserved for long time periods, but are also quite fluid.

Chapter 9

Scaling Up: How Cells Build an Organism

A single cell can develop, more or less on its own, into a full adult organism. This basic phenomenon was noted nearly as long ago in the history of Western thought as we can go back, by Aristotle. In *On the Soul*, he wrote, “It is a fact of observation that plants and certain insects go on living when divided into segments; this means that each of the segments has a soul in it identical in species, though not numerically identical in the different segments, for both of the segments for a time possess the power of sensation and local movement.” William Bateson (Bateson 1913) likened organismal growth to the basic process of cell division, and Bonner (Bonner 2000) suggests that the beginning of complex organisms was the failure of cell division products to separate, presumably in the early ocean.

A single cell somehow contains within it all the requisite “information” to go beyond its first division. Here, we will describe how a repertoire of a relatively few basic and logically simple processes is widely used to produce organismal diversity. Much of this chapter may seem like gene name-dropping. Listing genes is not very informative, and we are most likely to have found only what we know how to find, not every gene involved in the trait or process of interest, but the results do make sense. Not only are the same processes used and reused, but very similar genes or genetic mechanisms are involved over and over again to bring those processes about. These repeatedly used processes include what we would expect: signaling factors and receptors, transcription factors, activating and inhibiting effects. By mixing the order, frequency, and timing of their use—that is, the repeated contextually varying expression of subsets of the same set of genes—an organism can bring about its own transformation from a simple beginning to an adult.

HOW CELLS MAKE ORGANISMS: BUILDING MORE WITH LESS

In a way, cell division was a fundamental trick in making more of life. The mechanisms by which the constituents of a cell replicate and a cell divides are related to

the way cells adhere, move, or cooperate (e.g., Bonner 1998; 2000). Thus materials in spindle fibers are similar to those used in flagella or cilia, and various speculations can be made about how both cell division, with its rules for the distribution of materials into daughter cells, and other multicellular functions evolved. Much of what happened in the rest of evolution is a use and reuse of the many things that were needed for basic cell biology and division.

One ultimate question that we touched on earlier is *why* cell division has led to the aggregation of cells, and their dependence on each other, to the point that only some of them are even able to copy themselves into new colonies (individuals). We can imagine biological challenges or potential opportunities that single cells could not answer very well. Autonomy can be liberating in many ways, but we can imagine (and observe) circumstances in which large aggregates of cells can better withstand environmental changes or acquire resources. Bacterial aggregates appear to have this effect: bacteria in biofilms are much more difficult to destroy with antibiotics than planktonic bacteria, for example, due to a number of antibiotic resistance mechanisms that have recently begun to be elucidated (e.g., Stewart 2002; Stewart and Costerton 2001). Directed mobility enables organisms to pursue potential food if there is a local shortage (or avoid becoming somebody else's food).

ON BEING BIG

It has been said that there is usually an open ecological niche at the top of the size range (e.g., Bonner 1998). Several times in evolutionary history a new branch of organisms has evolved a subset of lineages of ever-larger species, a premise sometimes known as Cope's law (see, e.g., Bonner 1988). Thus the first fish, land creatures, dinosaurs, birds, etc. may have been small, but larger species ensued. Yet, to reiterate a point we have made before, small is beautiful, too.

Large species have experienced several major die-offs, so it may be precarious to sit atop the size pyramid even if it gains you an empty niche when you first move in, but there are still a lot of large creatures around. Perhaps a kind of arms race develops among aggregates of cells (that eventually are reproductively isolated enough to become different species). Multicellularity has arisen many times in many different ways. Even when "bigger" was evolving, small members of the same lineages also typically did very well. Tiny reptiles scurried through the giant legs of dinosaurs. Single-celled and other simple organisms are thriving still and may be what is left standing at biological Armageddon.

Being large and complex implies a division of labor, which in turn requires specialized cells or tissue structures that can communicate with each other, but must be kept chemically sequestered so they can *be* specialized. This means carrying messages and nutrients to them and porting waste and messages from them, which usually means formalizing communication. It means protection against micro- and macro-attack and processing information from the outer world as well as among parts of the inner world. As a result, we see specialization for respiration, digestion, neural control, immune resistance, sensation, and the like (Bonner 1988). A large organism must also be sufficiently supported physically. Water can do this (internally or externally), and large organisms like seaweed or octopuses have very little skeletal structure. But most large creatures, land and sea, plant and animal, have skeletal structures of some kind (and so do most small creatures). Plants have rigid

cell walls, and animals have external (arthropods) or internal (vertebrates, some gastropods) skeletons.

We cannot answer the general question of why cells originally began to aggregate, but we can address the question of how that aggregation led to something more than the sum of its parts: a differentiated organism. In this chapter we will first look at some of the general differentiation “strategies” and then at the repertoire of their use in the development of complex organisms. Much of the discussion will relate to morphology, but physiological systems can be similarly characterized in terms of their biochemical pathways.

DIFFERENTIATION IS HIERARCHICAL

Differentiation is a contingent process, in that what a cell does next depends on its current state. Because of the sequestration of gene expression patterns within cells, and the mitotic inheritance of those patterns, an organism develops a hierarchy of different parts. Cells and their descendants in this sense are partially autonomous, in a nested way, over time. However, they are not completely sequestered, and they may be directed to change their state by circulating signals from headquarters, or from the environment. Hierarchical differentiation means that if cells of a given type are needed in a particular context, the local precursor cells must provide them by division and changing gene expression, and if signaling is to be a part of this process, the cells must be prepared to be able to detect the appropriate signal.

It is a kind of developmental dogma that differentiated cells rarely revert to undifferentiated cells. In vertebrate organ systems, there are beds of stem cells that are committed to a particular organ but have not yet undergone terminal differentiation. When the time is right they differentiate into a stem cell and a descendant cell that undergoes this final state change process. These decisions are made during development but also throughout life as part of homeostasis or response to circumstances. In some cell lineages, like those of most vertebrate neurons, the final differentiation is thought generally to occur only once (mature neurons cannot undergo further mitosis). Some stem cells give rise to the panoply of different types of descendants, as in the generation of the diverse blood cell types. The epithelia that line most vertebrate organs, a layer of stem cells, undergo terminal differentiation repeatedly as needed. For example, cells in crypts between the villi that line the gut continually divide to replace the villus cells as they slough off into the lumen. Figure 9-1 shows an example of hierarchical differentiation of cells that form the central nervous system, a cellular rather than tissue fate-map.

The idea in animals generally is that differentiated cells within an organism have made too many gene expression commitments—often meaning modification of their DNA or its packaging—to go backward. This is analogous to Dollo’s “law” for species: evolution does not reverse itself (Marshall et al. 1994). But unlike species, the cells in an animal retain the genome and under some circumstances can reverse their differentiation, whereas once mutations have occurred in a species it can’t really regenerate its prior state. The recent engineering of cloned individuals derived from chromosomes obtained from differentiated somatic cells demonstrates that this can be done under artificial conditions (such as somatic chromosomes being placed in enucleated host undifferentiated cells to be stripped of the changes that differentiation had already made in it). Of course, like species, somatically derived chromosomes may bear some somatic mutations, a minor caveat to their

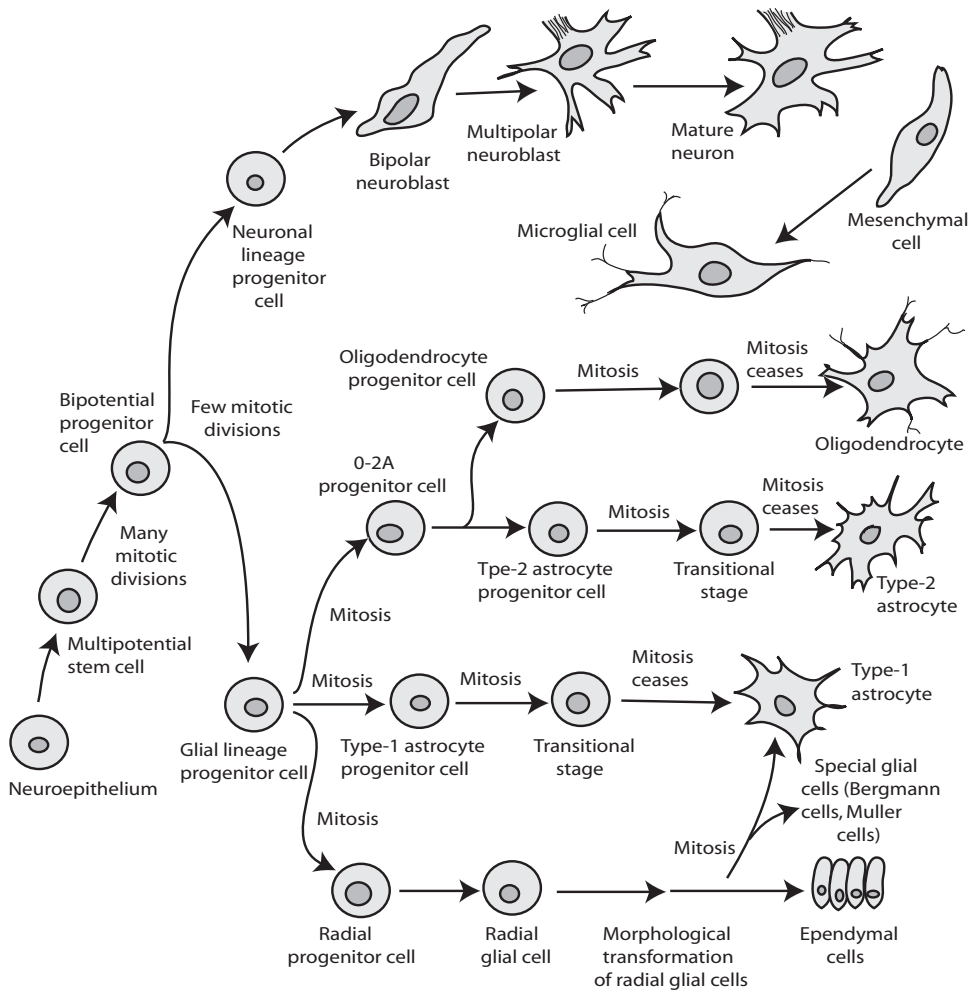


Figure 9-1. Cell lineages in the developing central nervous system, showing hierarchical cytological differentiation from stem cells. Modified after (Carlson 1999).

differentiation capabilities for most genes (though not in all, as will be seen in future chapters).

Cloning is one example of a growing realization that animal cells may not be as wholly committed as had been thought, but the generalization about normal context-dependent differentiation, to whatever state, seems generally to hold. And as we also noted in Chapter 8, plants are much less rigidly committed to their cell fates. Nonetheless, the use of hierarchical differentiation to produce sequestered, differentiated cells is a fundamental aspect of the development of complex plants and animals.

ORGANISMS ARE BUILT BY DUPLICATION AND REPETITION

William Bateson (Bateson 1894; 1913) wrote extensively and prominently on the problem of reproduction and replication, in particular the *meristic*, or repetitive,

nature so common to biological traits (Carroll et al. 2001; Carroll 2001). As he said (Bateson 1886), “greater or less repetition of various structures is one of the chief factors in the composition of animal forms.” We have cited Bateson many times in this book because of his interest in repetitive traits. He coined the term “genetics” and was an early advocate of Mendel’s work on heredity, but he did not think repetitive traits could be accounted for in mendelian terms, one of the reasons that he did not think natural selection could explain evolution. We now know that genes can indeed be responsible for such traits, and we are finding the genes, as this chapter will illustrate.

In fact, the living world surrounds us with duplicate structures. Vertebrae and fingers are examples of this, but the same applies to leaves and petals, starfish arms, insect antennae and legs, worm segments, feathers, hair, and scales; branches in nerves and arteries; pancreatic islets, intestinal villi, and taste buds. Examples of meristic structures (that have repeated segmental elements) can be seen in Figure 9-2. Duplication of structures is clearly a fundamental aspect of complex organisms, and we have already stressed the repetitive nature of the underlying genetic and biochemical systems themselves.

MORPHOGENESIS IS STIMULATED BY INDUCTIVE SIGNALING

Because an organism is not just a simple linear array of different structures, but a complex three-dimensional structure assembled from a simple beginning, each component can only begin developing when the proper context exists. Organs are typically built when cells capable of the appropriate differentiation cascade are *induced* to begin differentiating by some form of context-specific signaling. We introduced examples of this in Chapter 8.

The Spemann organizer is an area on an early vertebrate embryo that patterns axial polarity, and an area between the future mid- and hindbrain is an organizer for brain regionalization. Epithelial-mesenchymal interaction (EMI) involving overlying epithelium and migrating NC cells was mentioned in Chapter 8 as an important characteristic of vertebrates. EMI is responsible for inducing sites where teeth will form along vertebrate jaws; subsequently, local spots of signaling factor (SF) release called *enamel knots* serve as organizers for cusp development within teeth. These are examples in which the Fgf signal spreads from its source to induce cell growth in surrounding cells.

All the cells in a tissue field may be capable of differentiating in some particular ways, such as to initiate tooth development, but only those that receive the SF signal will be induced to respond. Surrounding areas may then be inhibited from forming the structure. In arthropods, corresponding to vertebrate organizers, the imaginal disk precursors of wings and other structures are organizers for those structures. In plants, cell differentiation of pluripotent meristem cells is induced by cell-cell interactions and hormones in response to external conditions such as day length, temperature, water and so forth. Shoot and root meristems in plants can be viewed as containing organizers, and some of the signaling and indicator genes that maintain the source of stem cells will be described below (e.g., Benfey 1999; Laux 2003; Weigel and Jurgens 2002).

Inductive signals may come from adjacent to distant cells, as communication from one organism to another via pheromones or behavioral cues such as displays, or simply the presence of conspecifics, or from environmental information (olfaction, vision, immunity). The use of terminology for interorganismal communication that

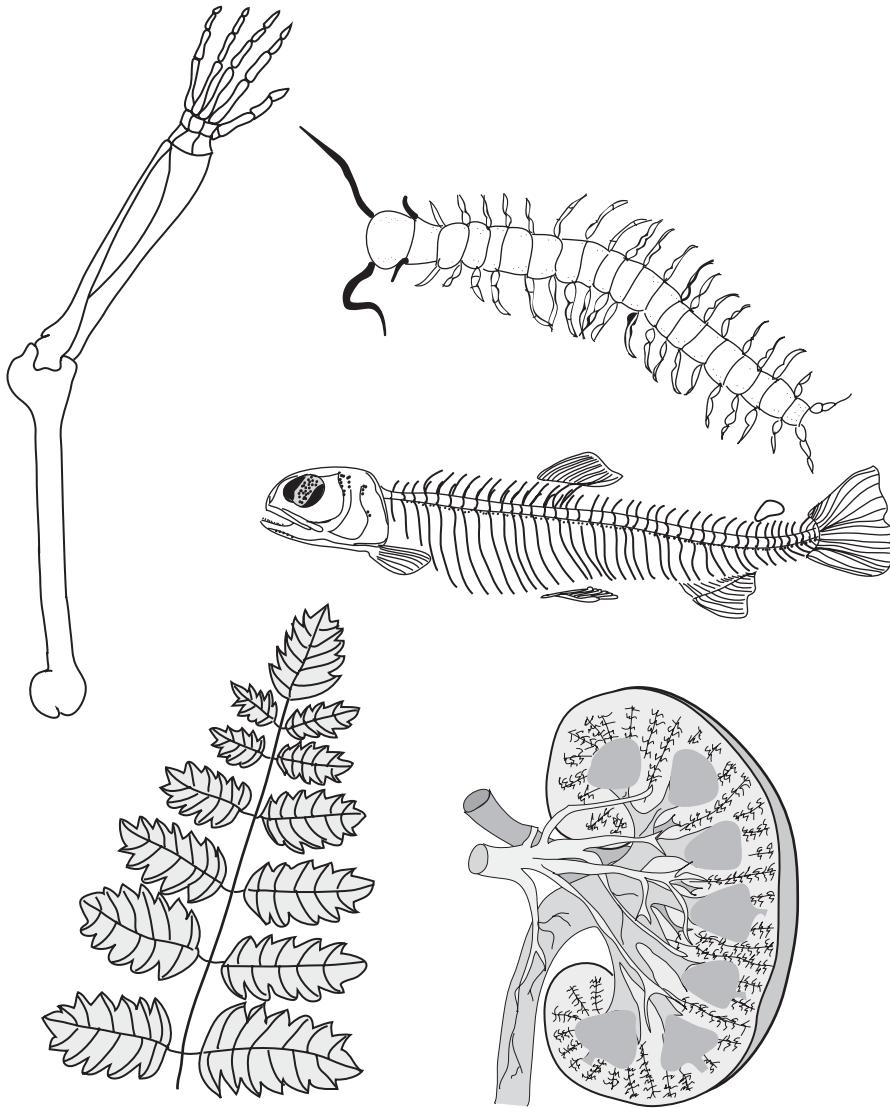


Figure 9-2. Duplication of structures: whole-body serial homology in centipede and fish; nested repetition of structures in plants; segmental nature of tetrapod limb; structural repetition within an individual organ (primate kidney).

is similar to that for cell-to-cell communication is justified because similar genes or at least information transfer mechanisms are involved.

POSITION IN A REGION CAN BE SPECIFIED BY SIGNALING GRADIENTS

Local differentiation can be specified by concentration gradients of extracellular SFs (in this context known as *morphogens*). Morphogens can be protein products, such as those from the *Hedgehog* gene family, or other small, diffusible molecules like retinoic acid. The cells across an otherwise undifferentiated tissue field inter-

pret their position by sampling the local concentration of the morphogen (see, e.g., Tabata 2001; Teleman et al. 2001; Wolpert 1969; 1981; Wolpert et al. 1998), using appropriate receptor mechanisms. Gradients of concentration can occur if a substance is produced in a localized source and physically diffuses away from it along the tissue.

A concentration gradient can form either simply as a passive ink-in-water process or by the binding of diffusing signal molecules to extracellular receptors removing them from the flow, by being diminished by active degradation by a factor produced at another source, and probably in other ways. Signals may move in various ways, including by passive diffusion or even by being relayed cell to cell (see, e.g., Figure 9-3). Gradients can induce gene expression changes if the signal concentration exceeds a threshold (Figure 9-3A), for example, by binding a large enough number of cell-surface receptors or by leading to the binding of enough REs flanking a target gene to induce its expression.

A classic example of gradient signaling involves establishing anterior-posterior (AP) polarity by the diffusion of transcription factors (TFs) within the early *Drosophila* egg (Figure 9-3B). Maternally deposited *Bicoid* mRNA diffuses from the anterior, and *Nanos* mRNA from the posterior ends (the early stage of the embryo is essentially a single cell with many nuclei on its inner side). Initially, mRNA from the genes *Caudal* and *Hunchback* are distributed rather uniformly throughout the egg and are translated in the cytoplasm. At the anterior end *Bicoid* message is translated, and Bicoid protein diffuses toward the posterior end. As described in Chapter 7, Bicoid protein binds to and inhibits the translation of *Caudal* mRNA, generating a gradient of increasing Caudal protein posteriorly establishing the first gradient. A corresponding Hunchback protein gradient is established because Bicoid protein binds to REs upstream of Hunchback, inducing expression of that gene. At the same time posteriorly, *Hunchback* mRNA is bound by Nanos protein, which is concentrated there.

As cell walls develop in the syncytium forming the cellular blastoderm, the intracellular environments become more sequestered, autonomous, and precisely controllable. Hunchback, Bicoid, and Caudal proteins are involved in expressing the “gap” genes that are the first zygotic genes to be expressed in an axially segmenting pattern, as the larva develops (of which more below).

A second example of a gradient patterning mechanism controls dorsoventral (DV) patterning. There, *Bmp*-class SFs (Dpp and Screw) are produced along with the proteins Tolloid in the fly blastoderm (Srinivasan, Rashka et al. 2002) (Figure 9-3C) (terminology and nomenclature can be complicated; *Dpp* is a fly relative of the *Bmp* class of *TGFβ* SF genes, Bmp being the name coined for vertebrate genes). Sog protein diffuses dorsally from a ventrolateral source in the embryo which at this stage consists of a layer of cells surrounding an inner extracellular space. Tolloid (Tol) and Tolkin (Tok) proteins bind and degrade the Sog protein as it diffuses dorsally, and another protein, Dynamin (Dyn) retrieves undegraded Sog molecules.

Together, these extracellular interactions establish a Sog concentration gradient, decreasing from the lateral source to the dorsal sink for this protein. Since Sog protein binds Bmps so they cannot serve as receptor ligands, and there is more active Sog laterally than dorsally, there is an inverse DV Bmp gradient. Its source is dorsal, and Sog serves as a sink. The resulting Bmp gradient affects gene expression differences that compartmentalize the dorsal embryo, as we will see below in a broader context.

A third example of morphogenic gradients patterns the vertebrate limb. At specified sites along the side of the embryo, diffusible Fgf8 protein induces (or at least is associated with) the initial outgrowth of limb buds. Anterior-posterior (AP) limb polarity is initially established by the induction of the SF Shh (a vertebrate *Hedgehog* family gene product) in a posterior (caudal, or tailward) part of the future limb known as the *zone of proliferating activity* (ZPA). ZPA-derived Fgf4 diffuses to induce the asymmetries associated with the posterior side of the limb. The organization of the distal part of the limb begins in the Fgf4-negative part of the limb bud. As the limb develops, regional variation in signaling molecules helps establish combinations of *Hox* gene expression, in roughly summed-sequential ways, along different limb axes, as will be discussed below.

These are tidy stories, but similar phenomena need not be due to the same mechanism, even within the same animal, even if a similar logical means is used. For example, gradients of *Dpp* activity are also used in AP patterning of the *Drosophila* wing imaginal disk. But in this case, unlike DV patterning in the early embryo, the gradients are established not by interaction with degrading proteins, but because a separate protein, called Thick-Veins (Tkv) inhibits the rate of Bmp diffusion away from its source (Srinivasan, Rashka et al. 2002) (Figure 9-3D). It does this because Bmp molecules are tightly binding ligands to Tkv and this stops their further

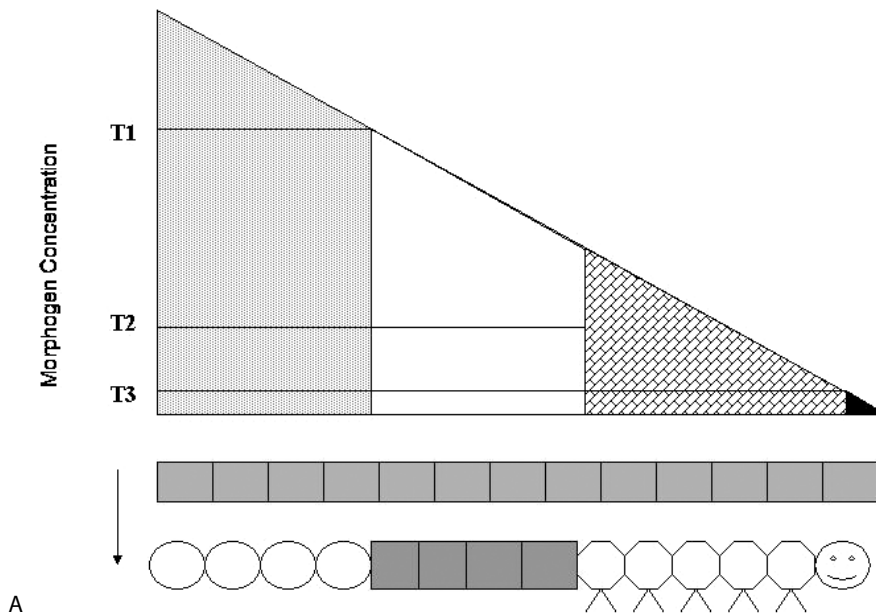
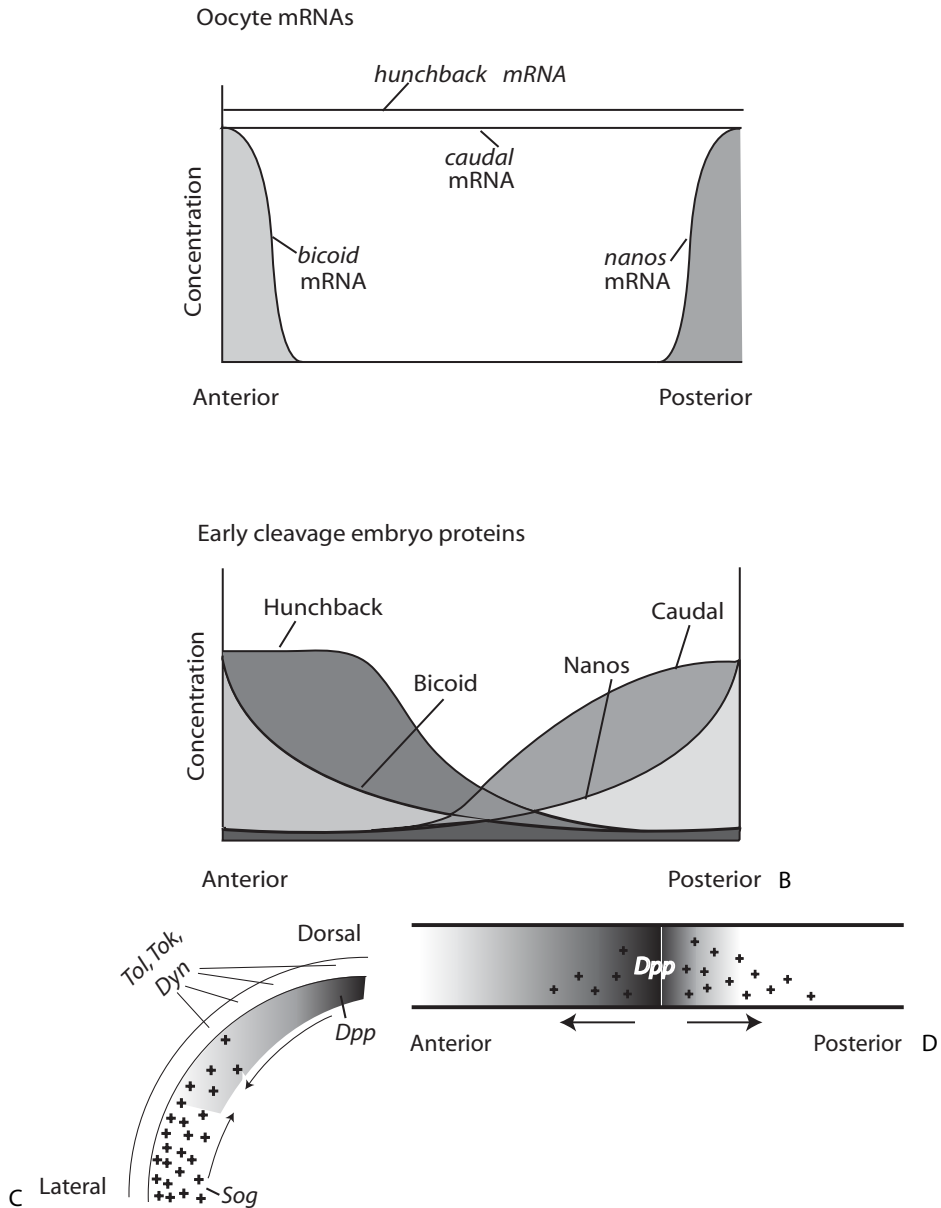


Figure 9-3. Gradient patterning mechanisms. (A) Multiple concentration thresholds (T1, T2, T3) established across a tissue field affect local cell gene expression, inducing regional differentiation (bottom) of previously identical cells (middle, induction indicated by arrow); (B) AP patterning in the *Drosophila* syncytium; (C) DV patterning in the fly cellular blastoderm: product of the *Sog* gene (+) diffuses dorsally intercepting Dpp (gray) diffusing laterally from its dorsal source. *Tol*, *Tok*, and *Dyn* gene products inactivate or intercept *Sog* molecules dorsally; (D) AP patterning in the *Drosophila* wing imaginal disk: Dpp (gray) gradient spreads asymmetrically from a source because the concentration of its ligand Tkv (+) is greater posteriorly; cell-to-cell movement of Dpp is aided by *Dyn*, not shown. For C & D see (Srinivasan et al. 2002).

Figure 9-3. *Continued*

diffusion, and in the mainly cellular tissue of the wing disk, Dynamin here acts to move Dpp from cell to cell to help it diffuse from its source.

Figure 9-3C described the workings of a simple concentration gradient mechanism, for DV patterning in the fly. This establishes the dorsal dominance of Dpp signal that, among other things, inhibits neural development, which takes place more ventrally. This is all part of a broader DV gradient patterning mechanism, that results in regionalized DV areas that develop into the major tissue fates in the fly body plan. This process is worth describing briefly to show the nested or layered

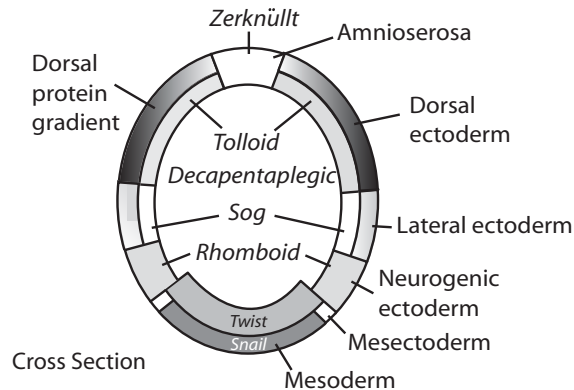
Dorsal inhibits *Tolloid*, *Dpp*, and *Zerknüllt*Dorsal activates *Twist*, *Snail*, *Rhomboid*

Figure 9-4. Gradient mechanism for dorsoventral body plan patterning in *Drosophila*. The center figure shows a cross section of the early embryo, identifying locations of gene expression as well as (right side) the tissue fates of the various regions. Above the figure is shown schematically that Dorsal protein inhibits genes that must be expressed for dorsal tissue fates; below the figure is a schema of gene induction by Dorsal, of *Twist* and *Snail* expression to induce mesodermal ventral fates, and inhibiting *Rhomboid* whose expression (ventrolaterally) is associated with neural fates. For details, see text and (Gilbert 2003).

nature of patterning and the numerous ways it can be brought about (Figure 9-4, and for details see Gerhart and Kirschner 1997; Gilbert 2003).

Early DV polarity involves a cascade of effects beginning with a protein, called Gurken (that is, coded by the *Gurken* gene) whose gradient is provided by the oocyte nucleus, which by the process of oocyte formation is located dorsally within an ovarian follicle that also contains within it 15 other cells, called *nurse cells* that supply mRNA and other materials to the oocyte (for details see Gerhart and Kirschner 1997; Gilbert 2003; Lawrence 1992). Gurken protein inhibits ventralizing signal from forming in the dorsal site of its (Gurken's) production. Another protein, Dorsal (i.e., a TF encoded by the *Dorsal* gene), in part provided by the follicle to the oocyte, is initially distributed throughout the egg. However, Dorsal's effect is to inhibit dorsal structures, so that it must form higher concentrations in the ventral side (the gene is so named because in its absence the entire embryo takes on dorsal fates). A series of events involving 11 known genes causes the translocation of Dorsal protein to the ventral nuclei in the syncytium.

As indicated in Figure 9-4, dorsal tissues form at the lowest concentration of Dorsal. Dorsal is complexed with a gene product, Cactus (not shown in the figure), and is activated only when released, which occurs via the Nudel protein whose concentration is highest in the ventral part of the egg, because Gurken protein, which is dorsally concentrated, represses *Nudel* transcription. Thus, a DV gradient of active Dorsal TF is established; via a series of threshold-level effects such as those shown in Figure 9-3A (where the cartoon at the bottom suggests AP patterning), Dorsal

acts as a selector for differential gene expression initiating the major embryonic tissues. Ventrally, where the Dorsal concentration is greatest, it activates genes including *Twist* and *Snail* that form mesoderm and undergo gastrulation when external tissue moves inside the embryo to form the gut and other internal structures. Just lateral to that, the lower Dorsal concentration enables the gene *Rhomboid* to be expressed, and so on. On the dorsal side of the embryo, where Dorsal concentration is least, dorsalizing genes including *Tolloid*, *Zerknüllt*, and the familiar *Dpp* are activated. We saw in Figure 9-3C how Sog and other proteins, whose expression is induced as a result of the Dorsal gradient, then interact to establish the Dpp gradient, that in turn inhibits neural development, that occurs in cells in the region shown in Figure 9-4.

The Dorsal gradient mechanism is interesting in that it is a TF not a signaling factor, showing that it is the gradient, and not the type of gene that establishes it, that is important in the logic of patterning. In this case also, depending on the circumstances, the Dorsal protein appears to activate some genes and inhibit others. The same TF can have both kinds of effects, depending on the REs to which it binds near a target gene—for example, what else must bind there, or must be prevented from binding there.

DEVELOPMENT IS PLASTIC . . . AND OTHER CAVEATS

The patterning results just discussed are among the many phenomena that have been carefully studied experimentally. Many if not most of the genes involved were first found in mutant flies, and even named for those mutant effects, and their expression was then documented and manipulated to identify their effects. But traits are often studied first in terms of expression patterns of known genes, and it is important to be careful about interpreting expression patterns because they can be misleading: expression of a gene in a tissue or developmental stage does not automatically mean critical function at that stage. For example, the importance of EMI in pharyngeal development in vertebrates seemed somewhat perplexing because pharyngeal region patterning seemed to have arisen in evolution before neural crest. Experimental ablation of NC cells in the chick to prevent their migration did not disrupt early pharyngeal arch patterning (Veitch et al. 1999). Thus, although NC may be involved, normal development does not entirely depend on it. In a similar way, it is a common experience that experimental inactivation of developmental genes in mice does not affect a structure in which the gene is expressed and/or in the way the expression pattern might suggest. Unfortunately, it is not practicable to do experimental manipulations of all gene expression patterns, especially in large or long-lived species.

The *Bicoid* and *Dorsal* stories are classics in the discovery of morphogenic gradients. However, organisms are robust and the actual effects are subtle. Bicoid gradients can be manipulated experimentally in *Drosophila*, and the result shows that to a considerable extent the embryo somehow compensates for experimentally induced aberrant dose levels and develops more or less normally (Driever and Nusslein-Volhard 1988; Namba et al. 1997). This is a point to stress something we have noted earlier, the role of chance even in the biochemical concentrations of cellular constituents. Organisms must be resistant to such changes. Bicoid concentrations have been found to vary considerably among embryos (Houchmandzadeh et al. 2002), probably buffered by scale-information built into the Hunchback concentration.

This variation may be a reflection not just of chance variation in transcription rates but of variability even within species in the structure and sequence details of enhancers more generally (Wray et al. 2003). We referred in earlier chapters to the fluidity and variability in enhancer binding location, number, and sequences.

But there is more. This early patterning might seem to be quite a fundamental aspect of development, one hard to change, but a fruit fly does not all insects make. Fruit flies are “long-germband” insects, in which the blastoderm (the part from which the embryo develops) takes up most of the egg and the body segments are specified simultaneously by processes such as those we have described. By contrast, “short-germband” insects develop more structures outside of the egg cell itself, sequentially during development. Recent work on the short-germband beetle *Tribolium* (another popular model arthropod species) shows that two other genes, *Orthodenticle* (*Otd*) and *Hunchback*, play the *Bicoid* role in establishing AP patterning (Schroder 2003). Whatever the common ancestor, there has been a substitution of mechanism for a conserved basic polarity-establishing phenomenon, a kind of phenogenetic equivalence and perhaps a manifestation of phenogenetic drift.

Genes from all four *Hox* clusters (denoted *HoxA*, *HoxB*, *HoxC*, and *HoxD*) are expressed in mouse embryonic development and are thought to be required for pattern formation. Many examples of effect, including homeotic shift (of identity of segments) have been observed in natural or experimental mutation of these genes. But the picture is not always so straightforward. As one of several known examples, mice experimentally lacking their *HoxC* cluster genes establish normal patterning (they have subsequent other problems, so the genes, which are used in many tissues, do have important function) (Suemori and Noguchi 2000).

SOME BASIC DEVELOPMENTAL PROCESSES AND SOME GENES THEY USE

We have identified a few generic principles by which spatial and temporal differentiation takes place. We can now describe how these same principles are employed in a repertoire of basic developmental processes that characterize much of development in animals and even plants. The same sets of genes appear again and again in different developmental contexts, even in the same organism. Many or even most regulatory pathways have homologs in vertebrates as well as invertebrates, or even in animals and plants—the kind of deep conservation referred to in Chapter 3 and elsewhere. These are often used in logically similar developmental patterning across diverse species, and in fact this pathway homology has helped establish homologies between traits that had been thought for more than 150 years (since Darwin) only to be analogous, that is, similar in function but independently evolved. Indeed, deep conservation and multiple use of genetic mechanisms have seriously challenged the concepts of analogy and homology themselves.

PRIMARY AXIS SPECIFICATION

An embryo quickly establishes structured asymmetry (polarity) along at least one and usually two or three anatomic axes. Subsequently, many new axes or asymmetries are established, often in a nested way. Position along a temporal or physical axis is correlated with the expression of different genes or gradients of the same

genes. Primary axis specification is fundamental and, once laid down by evolution, has stubbornly retained many of its essential features.

We have described how concentration gradients establish aspects of axes of the fly embryo. Homologous genes are used in vertebrates in an interesting way that relates to long-standing questions about the evolutionary relationships between the two major groups of animals—and showing what was, when discovered, surprising evidence of our common ancestry. As shown in Figure 9-5, the relative DV positions of the central nerve axis and gut are oriented similarly relative to early expressed genes, even though these signals are “inverted” between vertebrates and invertebrates relative to the DV axis. Proteins coded by the homologs *Chordin* in vertebrates and *Sog* in invertebrates bind the homologous proteins from *Bmp4* and *Dpp*, respectively, preventing the Bmp4/Dpp signal from being received by local cell surface receptors. Product encoded by the homologs *Xolloid/Tolloid* inactivates Chordin/Sog, allowing Bmp4/Dpp signal to be received by the receptors inducing neurogenic ectoderm. In a sense, the default cell fate of ectoderm is neural, so this antagonistic relationship prevents ectoderm in nonneural regions from becoming neural. Thus the *Chordin/Sog* genes “protect” a region from becoming neurogenic ectoderm. These are quantitative inductive interactions and have been shown experimentally to work across the vertebrate-invertebrate divide (e.g., vertebrate genes having their same effect in invertebrate embryos). Sometimes a conserved series of pathway interactions is found in both groups though performing different functions, but sometimes there is functional homology that reawakens notions of the unity of animal form that had gone out of acceptability a century or more ago.

It was hypothesized in the early 19th century by Geoffroy St Hilaire that invertebrates were “inverted” vertebrates (or vice versa). Figure 9-5A shows a forced example to position an invertebrate—a cuttlefish, and a vertebrate—a bird, suggested by two little-known authors, trying to show that the two types of animal were of essentially the same design; this figure was drawn by investigators who themselves had been drawn into a famous heated debate in 1830 between Geoffroy and his former colleague Georges Cuvier, about the extent to which the similarities were true (Appel 1987). Although there are clear similarities between vertebrates and at least some invertebrates (mouth and head at one end, anus and reproductive organs at the other, limbs branching out along the side, and so forth), at that time the homology of this overall organization, much less the skeletal, neural, digestive, and other systems was not clearly established. The uncertainty was probably made worse by the fact that this was before Darwin and hence before common ancestry was a serious explanation of body-plan similarity (Richards 1992).

Even after Darwin, and despite these overall differences, the known ancestral forms and fossils were such that it still seemed fanciful to suggest that a vertebrate was an inverted insect. However, the kind of genetic patterning data recounted above breathed surprising new life at least into the general idea. It seems clear that today’s deuterostomes (chordates) and protostomes (worms, arthropods) have dorsoventrally inverted body plans relative to each other, both in terms of morphology and the expression of genes associated with corresponding structures (Figure 9-5B). In both groups, *TGF β* genes (*Bmp/Dpp*) repress, but antagonists (*Chordin/Sog*) locally counteract that signal to permit, neural commitment. How this works in flies was described above; in vertebrates the combined action of the developmental signal β -catenin and *Nodal*-related proteins activates *Chordin* in

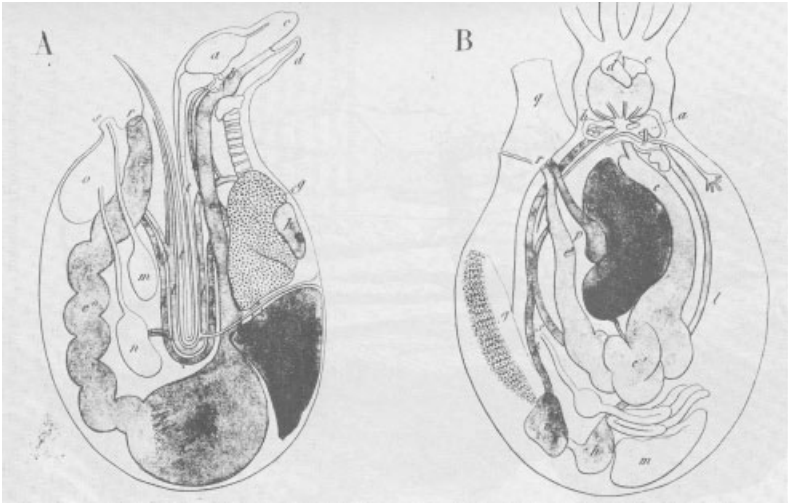
the midline, repressing *Bmp* expression and allowing neural development (De Robertis et al. 2000; Gerhart 2000; Nielsen 1999).

Does this indicate that St Hilaire was right after all (Holley et al. 1995)? Clearly this seems to be so in the sense that the conserved but inverted anatomic relationships are matched by conserved but inverted gene induction patterns. However, an inversion would require one of the two forms to be ancestral, and rather than an inversion per se the common ancestor might have had a less dorsoventrally specific body plan, or have been intermediate in other ways between what we consider today to be two distinct body plans; the means of expression of these homologous genes in the two groups is different and some primitive animals have less distinct DV orientation than either vertebrates or arthropods (Gerhart 2000). Modifications over evolutionary time of the pattern of migration of cells that will take neural fate, relative to the locations of the mouth at the blastopore stage, could provide an explanation without requiring a real and more complicated inversion event (Fitch and Sudhaus 2002; van den Biggelaar et al. 2002).

Since the discovery of this conserved aspect of polarity, which made a wonderful story of the resuscitation of a prescient kind of guess that had been long ridiculed, many more kinds of homology have been observed between vertebrates and invertebrates. It is probably fair to say that major elements of most corresponding systems, and many if not most basic cell types, share at least some homologous gene expression. Table 9-1 provides a number of these, and Wilkins (Wilkins 2002) presents a thorough discussion of the prominent examples known to date. Of course, the concept of homology relative to specific structures is challenged by the fact that many of the shared genes, such as the *Hox* and many other TF and SF genes, have numerous sites of expression, making it likely that some similarities might occur even by chance, an implication of the powerful “strategy” of evolution to use and reuse a limited toolkit.

Darwin would be pleased, because this confirms his ideas about common ancestry, but with entirely new types of data. However, it is tempting to overstate the homologies. Geoffroy’s idea of “inversion” included the exoskeleton of invertebrates corresponding to the internal skeleton of vertebrates (e.g., a vertebra corresponding to the exoskeletal elements of an arthropod limb). We know that *Hox* and other genes like *Bmps* and *Distalless* TFs are used in both, but they really are not very similar. Even if homologous pathways are used in similar ways, they are embedded in different overall organizational contexts. Thus the fly wing and vertebrate forelimb are both limbs, and their overall polarity is established by some strikingly

Figure 9-5. Comparison of stereotypical body plans for vertebrates and invertebrates to show the idea that they are inversions of the same overall plan. (A) Alleged similarity between vertebrate (bird) and invertebrate (cuttlefish) sent to St Hilaire in the 1830s by otherwise unknown authors Meyranx and Laurencet, and published by his opponent Georges Cuvier (Cuvier 1830); (B) vertebrates and invertebrates have a common body plan in many ways, but with a 180 degree dorsoventral rotation. Top row is invertebrate, bottom vertebrate. Left are correspondingly oriented copies of an image of Geoffroy’s inversion hypothesis by Wilder (Wilder 1909), right are transverse sections of the body plans of each animal type showing the major basic structures; to the right of that are genes involved in early dorsoventral patterning showing their conserved usage. For more homologies see Table 9-1, and see text.



A

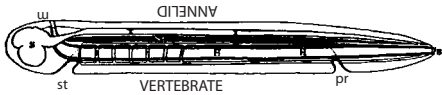
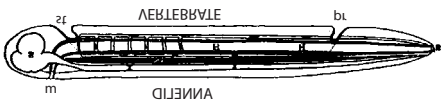
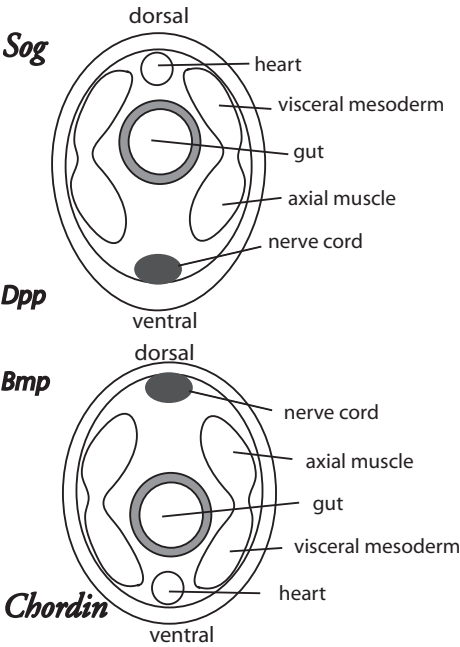


FIG. 140. Reversible diagram illustrating the Annelid theory. Reversible designations, applying to both forms; S, brain; X, nerve cord; H, alimentary canal. Designations applying to Annelid only; m, mouth; a, anus. Designation applying to Vertebrate only; st, stomatodæum; pr, proctodæum; nt, notochord.

ANNELID/ARTHROPOD



CHORDATE (VERTEBRATE)

B

TABLE 9-1.

Some shared gene expression patterns suggests genes that may have been responsible for patterning and development in the common ancestor of bilateran animals (vertebrates and invertebrates).

AXIS OR TRAIT	SHARED GENE EXPRESSION
Anterior-Posterior	Summed combinations of <i>Hox</i> gene expression specifies region along the axis; Segmentation behind CNS: <i>Hairy</i> , <i>Engrailed</i>
Dorsoventral	<i>Sog/chordin</i> dorsal, <i>dpp/TGFβ</i> ventral
Gut	Para-hox genes: <u>Anterior</u> , <i>Lab/Gsx</i> ; <u>central</u> , <i>Bcd/Xlox</i> , and perhaps an Antp/Ubx-like gene; <u>posterior</u> , <i>Cdx/Abd-B</i> .
CNS	<i>Otd</i> , <i>ems</i> , <i>Hox</i>
Photoreception	<i>Pax6</i> , <i>opsin</i>
Chemosensation	<i>Odorant receptors</i>
Body axis outgrowth	<i>Dll/Dlx</i> , <i>Fgf</i>
Heart	<i>Tinman/NK2.5</i>
Mesodermal structures	<i>Snail</i> , <i>Twist</i> , <i>Slug</i>

See (Arendt and Nubler-Jung 1994; Campbell 2002; Carroll, Grenier et al. 2001; De Robertis and Sasaki 1996; Gerhart 2000; Scott 1994; Wilkins 2002).

similar gene hierarchies and inductions; but they are very different in their details: your legs and a fly's are not the same.

REPEATED STRUCTURES PRODUCED BY PERIODIC PATTERNING PROCESSES

Many traits—plant and animal—have periodic (repeated) and/or hierarchically nested patterns. What might be responsible at the gene level? Modular structures can develop Roman candle fashion by being generated from a source region and growing away from that region; the continued replacement of reptile teeth is an example. Combinatorial expression of TFs like the *Hox* genes can be responsible for initiating different cascades in the repetitive elements along an axis. However, neither combinatorial enhancer binding nor signal concentration gradients can explain all periodic patterning. For traits like hair, scales, feathers, intestinal villi, leaves, or fly sensory bristles or ommatidia (individual units of the compound eye), it is not plausible that each individual unit would be programmed by its own unique gene combination. There are too many units, and they are also often too similar to each other. Some other patterning process must be responsible.

In one such process the individual elements develop from initiation sites periodically spaced along an axis or in a tissue field. Once an initiation site becomes committed for the appropriate differentiation cascade, the unit can proceed autonomously without communication among the individual units; for example, the structure may develop when isolated experimentally.

Bateson developed an “undulatory” theory of life (Bateson 1894; 1913; Hutchinson and Rachootin 1979; Webster et al. 1992; Webster 1992; Weiss 2003a). He likened repetitive systems to interference patterns seen in wavelike phenomena being studied intensely according to field theories of late nineteenth century physics (Bateson had some notions derived from his time, that are not far from a kind of vitalistic inherency of built-in patterning, but we need not accept all of someone’s views to see where perceptiveness may lie). Diffusion usually establishes simple gradients, as described earlier, but in 1952 the famous computer scientist Alan Turing (Turing 1952) suggested a *reaction-diffusion* process in which the interaction between two chemicals with different diffusion characteristics could establish spatially heterogeneous and/or repetitive patterns. Essentially, interference waves are established. In the archetypal simple model, the pattern can be described by two differential equations, one for the dynamics of change of concentration in space and time of each factor.

In a basic activator-inhibitor system of that type (Figure 9-6A) the interacting elements are SFs. One, the activator, catalyzes its own production in cells that detect it and also induces the production of a second signal that inhibits the activator. Both diffuse from sources of production across a cellular field. In this basic model, pattern formation requires that the inhibitor diffuse faster than the activator. This single *process* can in principle generate a series of spaced initiation sites in which an organ element will develop (that is, sites where the activator exceeds some expression threshold), surrounded by inhibition zones where no structure develops. The nature and stability of the resulting pattern depends on the production and turnover rates of these key substances (or on how many of them there are, and so on) (e.g., Bar-Yam 1997; Meinhardt 1996; 2003; Murray 1993).

That such processes were involved in biological patterning was suggested by the striking similarity between mathematical models and computer simulations and a host of observed traits including natural pattern traits such as the locations of hair, feather, teeth, butterfly spots, fish coloration, seashell and mammalian fur patterning, and others (see, e.g., Figure 9-6B and Kondo 1995; Asai 1999; Meinhardt 1996; Meinhardt 2003; Meinhardt 2000; Murray 1993; Nijhout 1991; Salazar-Ciudad 2002; Jung 1998). Very different patterns like butterfly wing patterns, fish stripes, and seashell coloration can be generated by similar processes worked in different ways. At the same time, small changes in the characteristics of a single process can generate very different patterns—just what one would want of a system that can evolve, can generate complex outcomes, but is itself not so complicated.

As one example, the mantle edge of a seashell is also a kind of linear *progress zone* in relation to shell color patterns. The shell grows out from the mantle forming a hollow spiral (hollow except for the organism itself). The mantle constitutes a line of cells *along* which periodic waves of different color generation are produced by a reaction-diffusion-like process. Wavelike color differences form along this line but then grow out and away from the mantle to become the new part of the shell. As the linear process “unrolls” from the mantle like a window shade, a two-dimensional color pattern results (Meinhardt 1996).

Simulations by various authors attempting to model different living patterns have used variations on the basic equations, and the process may never be exactly as specified by simple differential equations. But the nature, or *logic* of the patterning processes is probably similar to what is modeled. In fact, processes of this general

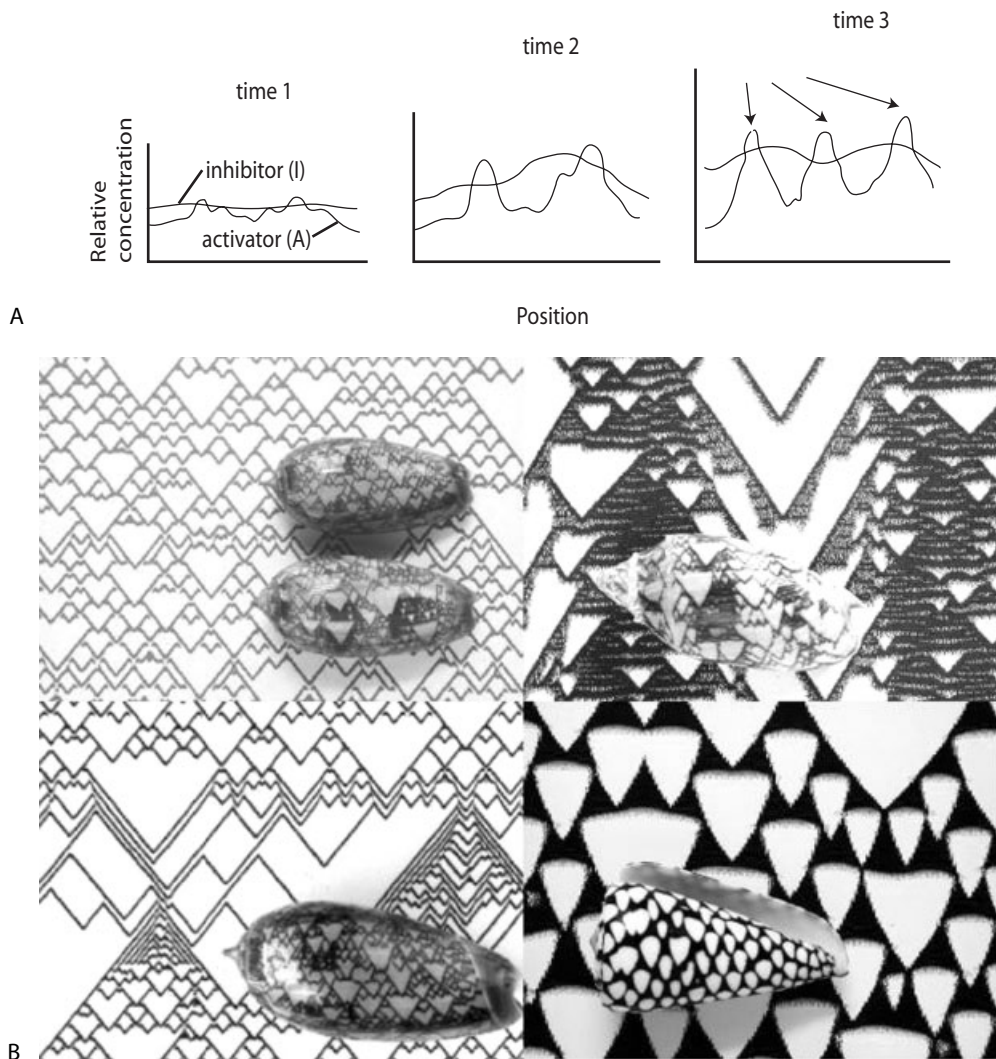


Figure 9-6. Reaction-diffusion systems can generate repetitive patterns. (A) The principles of a basic model shown at sequential developmental times; the horizontal axis represents position along a layer of otherwise similar cells. Cells produce an activator, **A**, that catalyzes its own continued production. **A** diffuses across the tissue, where cells detecting it are stimulated to make **A** as well as a rapidly diffusing inhibitor, **I**. Cells whose receptors detect **I** inhibit their production of **A**. Where the concentration of **A** relative to **I** exceeds a threshold level, gene switching occurs leading to activity peaks (arrows) where structural units, like feathers, teeth, or tooth cusps form. (B) Seashells real and simulated. Courtesy H. Meinhardt. For model see (Meinhardt 1996; 2000; 2001; 2003).

type seem likely to be responsible for a large fraction of the organization of complex organisms. There has been direct molecular demonstration of reaction-diffusion-like patterning in the use of *Fgf* and *Bmp* signaling as activators and inhibitors, respectively (Hogan 1999; Jernvall and Jung 2000; Jernvall and Thesleff 2000; Jung et al. 1998; Jung et al. 1999). Here we add the qualifying terminology, “reaction-diffusion-

like” to indicate that these repetitive patterning systems vary in most of their details, the nature and number of interacting agents, their diffusion mechanisms, and the like.

In this kind of *dynamic patterning*, the frequency and location of the waves, peaks, or zones of expression are determined not by the specific ingredients of the system but by its dynamic *parameters*: timing and intensity of expression, number of cells, diffusion rates, receptor/enhancer binding efficiencies, and the like. Combinatorial expression is also involved, because it is the combination of activator(s) and inhibitor(s) and the cascade of gene expression they trigger that specifies the zones of growth and inhibition, and while there may only be quantitative differences between the two, combinatorial differences consequent to the signaling process account for the structural differences.

While nested patterns may involve several patterning signals, there is not a separate gene or even gene combination for each hair, feather, or tooth cusp. Instead, each unit is produced by a reinvocation of the *same* developmental cascade. The dynamic interaction of SFs determines when, where, and how many such instances will occur. There are genes for the process, but not “for” each individual component. The units *and* their intervening regions are part of a single trait.

For repetitive organ systems, an initial process called *prepatternning* occurs, which involves the priming of genetic switches that are “remembered” in the cells (or their descendants); the “enabled” genes are activated only at some subsequent time and tissue context. That is how they “know” to express *receptors* for the SFs they will have to be able to detect in order to be induced. Traits like dentition or taste buds, mammalian coloration patterns, and perhaps feathers in birds or respiratory spiracles along the sides of insects are examples. EMI may be the triggering phenomenon, when migrating NC cells in the jaws encounter inductive signals from specific locations in overlying prepatterned or prepared epithelium.

Feathers and hair are arrayed along what first appear as linear axes and then become more spatially distributed. Extensive studies have been done on the patterning of chick feathers (Figure 9-7) (Jung, Francis-West et al. 1998). Lines of comparably pluripotent cells expand in the epithelium, ultimately forming stripes, detectable experimentally by the presence of SFs including Shh, Fgf4, and Ptc (from the *Patched* gene). The stripe then organizes into local “condensations” of cells, or slightly thickened *placodes* that are the initial indication that a structure will develop. Primordial placodes serve as autonomous local signaling centers or organizers, which interact inductively with underlying NC-derived mesenchyme to differentiate into feathers. Zones of activation that then generate surrounding zones of inhibition are typical of periodic patterning processes. SFs Bmp2 and Bmp4 are produced and diffuse away from the feather placode, where Bmp receptors on surrounding cells receive these ligands, and are inhibited from forming placodes, thus producing inhibition zones around each placode. Bmp antagonists like Noggin or Follistatin in the placode prevent self-inhibition. Much the same pattern is responsible for hair patterning (Jung, Francis-West et al. 1998; Oro and Scott 1998) and as noted earlier teeth as well (Jernvall and Jung 2000; Weiss et al. 1998).

This general and rather simple process is nested in various ways. As the inhibiting signal diminishes with distance from an initiation site, new placodes appear. The original stripe differentiates into periodic feather buds along a line, but the diffusion process then produces new feathers laterally, to produce the two-dimensional array. Different types of feathers arise even among adjacent locations, and in dif-

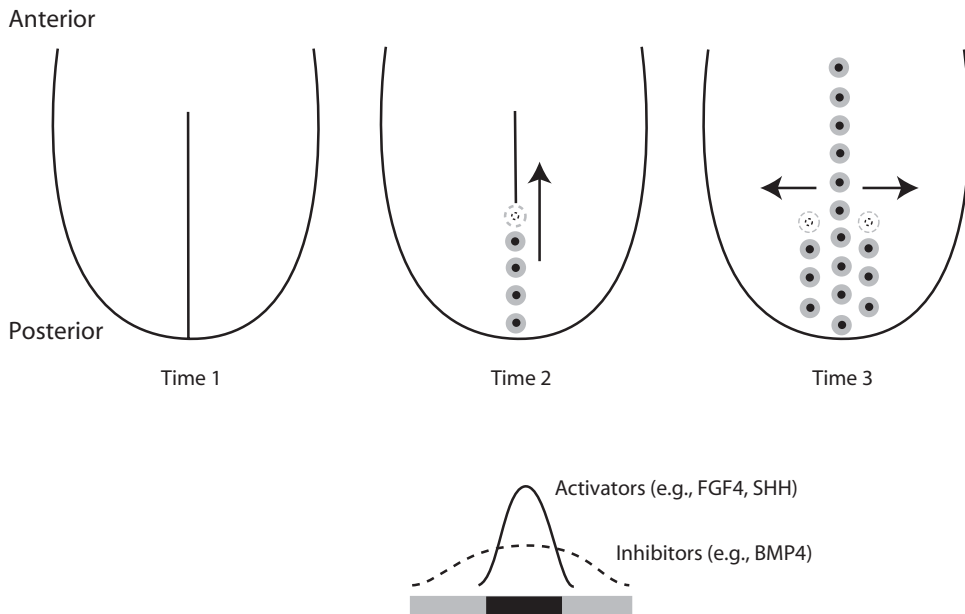


Figure 9-7. Feather patterning reaction-diffusion schematic at standard times during chick development. Original signal develops along a line. The signal then develops activation spots where initial feather buds arise. As the process continues, diffusing activator and inhibitor signal leads to the formation of additional rows lateral to the original one. Bottom figure shows the activator-inhibitor gene system with gray-shade that corresponds to the top figure. Redrawn after (Jung, Francis-West et al. 1998).

ferent parts of the bird (see Figure 9-12), much as different color patterns are generated in different parts of a mammal's fur or different types of vertebrae or segments appear along the AP axis of vertebrates and invertebrates, respectively. Hair, fish scales, mouse facial sensory whiskers, and mammary glands are patterned structures that develop similarly.

Although elegant experiments have shown the reaction-diffusion-like nature of this process, knockout and other experimental manipulation of these signals also show that they are not all required for successful patterning. This suggests that other factors are involved (e.g., Chuong et al. 2000). Of course they must be: as we have noted in many places, at the very least the cells must be primed to express the SFs, receptors, and signal processing machinery and to make the appropriate parts of chromosomes available to respond to the signal.

In vertebrates, the oral epithelium expresses the SF *Fgf8*, but this signal becomes inhibited by zones of *Bmp4* expression, and local *Fgf8* foci remain that will become tooth-forming placodes along the upper and lower jaws (and/or elsewhere in the oral cavity, depending on species). Other genes, including the SF *Shh* and the TF *Leff1*, are also expressed in the initiation sites. The first known indicators are organizer-like zones of SF production that are similar to those just described for feathers in birds (indeed, hair and teeth can be made to grow in similar areas in some experimental manipulations in mice) (Jernvall and Jung 2000; Jernvall and Thesleff 2000; Thesleff et al. 2001).

Specific genes in the underlying mesenchyme (e.g., TFs *Msx1*, *Msx2*, *Pax9*) whose presence allows the NC-derived cells to respond to these epithelial inductive signals and others (e.g., *Bmp2*) appear to inhibit laterally adjacent mesenchyme from participating, thus focusing the mesenchymal response into a local responding center under the epithelial initiation site, in which a tooth will form. Subsequently, the local signaling centers within each tooth, the enamel knots (EKs) referred to earlier, appear and secrete similar (though not entirely identical) combinations of SFs. Interestingly, the EKs themselves do not express receptors for the factors they secrete, but the surrounding cells do (Kettunen et al. 2000); this stimulates cell division and downgrowth around the EK, which eventually undergoes apoptosis, but not before secondary EKs repeat this process to generate other cusps. The dentition has several axes: along the jaw, the breadth and width of individual teeth, and the crown-to-root vertical axis. Each is produced by signaling systems, sometimes involving the same genes. In the end, the pattern can be modeled closely by reaction-diffusion-like models for the array of teeth along jaws (Kulesa et al. 1996), and for periodic cuspal variation within teeth (Salazar-Ciudad and Jernvall 2002).

Feathers and teeth are among the model systems that illustrate what are much more widespread phenomena (there are many similar examples in insects and other animals and the equivalent in plants). In the case of teeth, experimental evidence suggests that the epithelial tissue layer has been prepatterned long before the initiation sites appear (Weiss, Zhao et al. 1998). Taste buds have comparable spatial patterning on the tongue, and direct experimental evidence suggests that these locations are prepatterned as early as gastrulation (Barlow and Northcutt 1998; Northcutt and Barlow 1998).

There are good reasons why this early preparation might be expected, which also illustrate why differentiation is hierarchical: a tissue like hair or feathers cannot form until there is skin to form on. Upper and lower teeth develop in anatomically separate regions of the jaws that suggest that prepatterning may have occurred very early there as well (Weiss, Zhao et al. 1998) and we referred above to evidence that the limb may also be prepatterned before it becomes a physically distinct structure. Another indicator that prepatterning occurs is that closely related animals can achieve similar results even though they are of very different size (such as the stripes on a Bengal tiger and a small tabby house cat, or teeth in an elephant and a mouse). One might say that this solves the allometric problem: the same basic mechanism is likely to be involved across the species size range, and the initial patterning event is likely to occur at a developmental stage when the embryos of all the species are more or less the same size (e.g., Murray 1993). The early stage can be much earlier than the first manifestation of the actual trait in the embryo. Then later, growth differences can generate structures of size appropriate to the species and these can be allometrically stretched or distorted by the growth factor (a subset of SF class gene products) patterns.

In a regionally patterned organ system, it is typical that the simplest and most global aspects of structure arise first and then more complex structures are produced. Axial region-specific expression of *Hox* genes is induced by gene- and region-specific enhancers (e.g., Shashikant et al. 1998). Once a region is specified, differentiation of hindbrain segments, mouthparts, vertebrae, and limbs occurs (Figure 9-8). Thus the *Hox* system is involved in the regionalization of major AP zones like thorax and abdomen, but these have multiple nested structures within

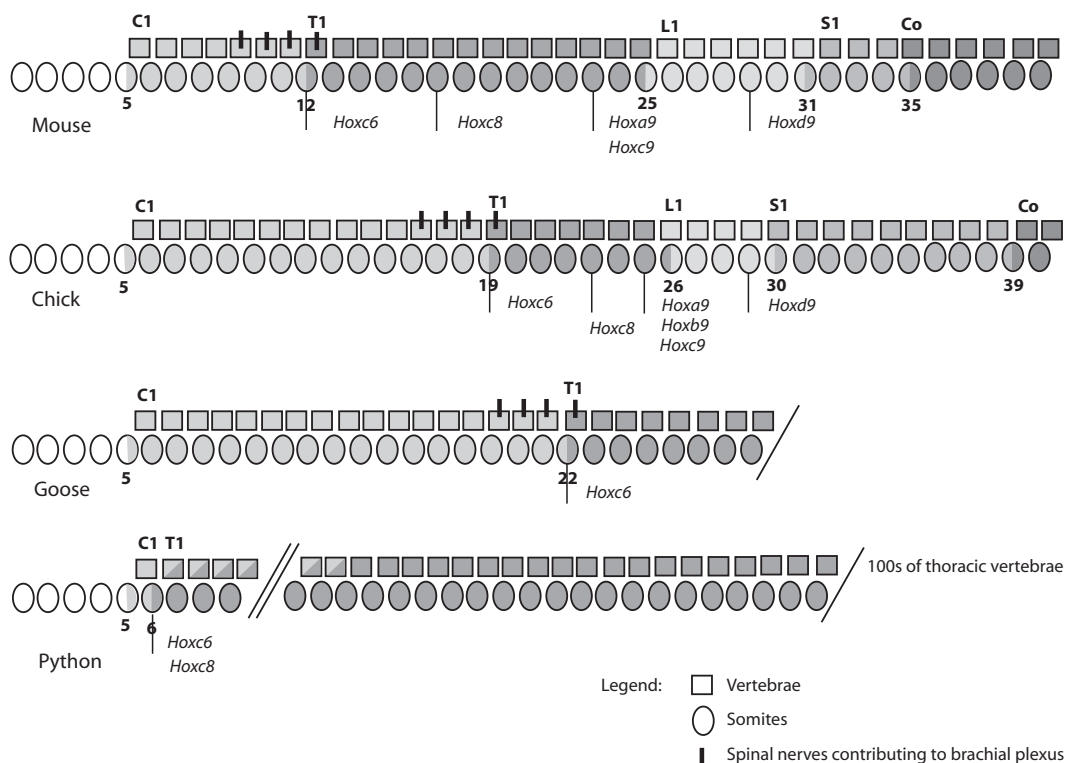


Figure 9-8. Functional conservation of axial *Hox* expression boundaries in vertebrates. Despite differences in detail, such as the number of vertebrae, gene expression patterns along AP axis are conserved relative to major homologous boundaries in the organism, such as the origin of nerves to the brachial plexus (that services the forelimbs). For corresponding body regions in other animal taxa, see (Carroll, Grenier et al. 2001). Redrawn from (Burke et al. 1995).

them (vertebrae and ribs, for example, that themselves vary and/or have patterned subunits). The homologous boundary of *Hox* gene expression switches shown in the figure indicates that an earlier broader patterning process is in place as the *Hox* system engages. Such layering of what may be rather similar kinds of patterning process may be involved in the structure of most meristic traits (and that means a high fraction of all traits in complex organisms).

Each of the roughly 800 ommatidia in the compound eyes of fruit flies has a basically fixed number of cells that are programmed, or patterned, by a reaction-diffusion-like activation-inhibition system. Each has about 20 cells of seven different types, including eight photoreceptor cells, seven of which are arranged hexagon-like around the other (known as R8); six surrounding accessory cells; and four lens-generating cone cells. The eye develops from an imaginal disk of about 20,000 cells. The cells in the eye imaginal disk are apparently equivalent and equipotent (e.g., all expressing the *Eyeless/Pax6* TF). These cells are induced to form ommatidia by a sweep of activation-inhibition activity induced by an indentation called the *morphogenetic furrow* that moves wavelike from posterior to anterior across the disk.

As the furrow passes it induces cells to express *Hh* (a *Hedgehog* gene family homolog of the vertebrate *Shh*), which then induces *Dpp* (the *Bmp* homolog) in the furrow to drive it anteriorly. In the R8 cell, the SF Boss (*Bride of sevenless*) is released and induces the *Sevenless* gene in the neighboring R7 cell in the little prephotoreceptor cluster. This signal to R7 initiates a cascade of differentiation of the other photoreceptors and surrounding supporting cells. The Wingless SF is also active in the disk as an inhibitor expressed ahead of the morphogenetic furrow, which helps establish anatomic polarity in the eye. Again the result is a somewhat elaborate pattern of structures generated by a single and relatively simple quantitative, nested signaling process.

GROWTH AND INHIBITION IN PATTERNING

By themselves, pattern-generating processes might produce continually changing patterns, but the system can be stopped in various ways including apoptosis and mineralization (e.g., calcification of teeth and vertebrae, the moving away of shell material from the active mantle cells). One mechanism that establishes inhibition zones surrounding initiation sites is the *Delta-Notch* system (Table 7-2). This is used in the establishment of sensory bristles and ommatidia in insects and in feathers, teeth, and other structures in vertebrates. In the case of bristles (whose structure and nature will be described in Chapter 12), activation zones arise in a set of comparably prepared cells; cells expressing high Delta levels inhibit surrounding cells, and a few of those cells become the precursor of bristles. The successful cell (if being a bristle is more “successful” than being the surrounding tissue!) arises by chance.

The similarity of biological patterns to patterns simulated by computer (noted above) justified the inference that activation-inhibition processes were responsible for such traits in the real world. This was confirmed with the examples we have discussed. However, as so often happens, evolution frustrates what experience excites. The early *Drosophila* egg manifests stripes of expression of various genes, including *Even-skipped* (*Eve*) (the gene was named for the effect of its mutants on the stripes of early fly embryos). This seemed a clear example of a reaction-diffusion system, yet experimental work has shown that there are separate stripe-specific *Eve* expression mechanisms (Carroll, Grenier et al. 2001; Ludwig et al. 2000; Ludwig and Kreitman 1995; Ludwig et al. 1998). There is no rule without an exception—or is it that there is no exception without a rule?

IMPORTANT ASPECTS OF QUANTITATIVE PATTERNING

Turing’s and other similar processes can be modeled mathematically in terms of the relative properties of interacting factors. For many years this was thought to be fanciful but rather meaningless (or even mystical and nonscientific) biological theorizing, because it could not be operationalized (proven in experiments, for example). As we have seen, that is certainly now done successfully. In a sense these processes can be viewed as “mathematical” in that it is not necessary to know what the factors are. If they have the specified properties, the effects are seen automatically. This can be said whether the process is by diffusion signaling, mechanical interactions, or temporal or physiological interactions among molecules rather than among physical cells. The result will be pattern; it can be ever-changing, divergent, or steady state depending on the parameters of the interactions.

This is an example of what we mean by the *logic* of gene use and mechanism in biology. We referred earlier to the functional arbitrariness but logical necessity of mechanism, meaning that it did not matter for the specific structural attributes of a trait what patterning process brought it about. If we think of processes in terms of their logic, it is the interaction of entities that is the process and, in a sense, not the entities themselves. Unfortunately, mathematical verisimilitude does not imply identity. There may be multiple mathematical ways to generate a similar outcome, which can be expressed one in terms of the other. In terms of the logic of patterning, it may not matter which equations are used; but for discovering the actual mechanism in any given case it matters considerably how many and which genetic factors are involved—and this becomes especially important in trying to understand homologies in species or processes we can't study directly from ones we have. Phenogenetic drift shows this as well.

But not all repetitive patterns can be simulated by a single set of equations, and in fact, meristic structures raise all sorts of subtle questions about homology (e.g., Hall 1999; Wagner 1996). The usual description of meristic traits is that they represent *serial homology*. The notion there is that each unit is a separate use of the same process, not among species but among regions of the same embryo.

Reaction-diffusion-like processes can generate repeated structures as described, and it is tempting to explain any such pattern by these elegant processes. When true, each iterate of the structure (e.g., each hair or scale) is part of a single *continuous* process. However, by a different route, each unit might more literally be a repeated but *independent* use of the same process, something more akin to the idea of homology between structures. As with reaction-diffusion processes, the same genes would be used in each unit. Mutations in those genes would therefore affect all units, demonstrating the unity of the overall system.

Another way to generate repeat units is for the same genes to be used, but to be invoked by separate means in each part of the embryo. In such case, unlike a standard reaction-diffusion patterning system, the separate domains could be sequestered from each other. For example, the regulatory region of a gene that is involved in the cascade that develops each unit can have different enhancer cassettes, each responding to different TFs and in a specific compartment. This is the case, for example, with the *Eve* gene in *Drosophila*, where enhancers for different TFs control expression in different stripes in the egg and this helps establish subsequent AP segmentation. Here, mutation within the regulatory region will affect only those units that use the specific enhancer that was mutated.

There are probably instances of all three of these phenomena, and the latter two require that in some way a prepatterning is laid down so that the repeated invocation of the same expression cascade takes place. Local reaction-diffusion like processes may be involved.

The patterning of butterfly wing spots seems to be a good example of these phenomena (e.g., Carroll, Grenier et al. 2001; McMillan et al. 2002; Nijhout 1991; Nijhout 1994). Wings are divided into cells (compartments, separated by veins), within which quantitative activation-inhibition patterning involving the development of expression sites ("organizers") of the genes *Dll* and *En* and others, leads to a ray of *Dll* that grows from a stripe of expression along the distal wing margin to the middle of each wing segment. There, it becomes a focal location of expression for a color spot. Other genes, involving the veins separating the segments, participate, and the

genes include some of our familiar early polarity-specifying genes (*Dpp*, *Wg*, *Rhomboid*, and others) (Keys et al. 1999; McMillan, Monteiro et al. 2002).

Artificial selection experiments in which butterflies were bred to have larger or more, or fewer or smaller spots, have found that the effects are correlated among spots in the same individual (Beldade and Brakefield 2003; Monteiro et al. 1994). The effect of selection was to favor whatever combinations of genes might affect spotting activity. The result shows that in a sense the wing spots together constitute a single trait. However, other experiments have shown that mutations can affect (e.g., delete) individual spots within the row of spots (each in a different wing compartment), without affecting the rest of the spots (Monteiro et al. 2003). This reveals a strong modular element of control than normally achievable by a single reaction-diffusion-like process, and suggests that a complex enhancer was mutated. Some prepatterning process must, however, still be present, even if moved earlier in development.

Here we have an example of multiple components generating a serially homologous trait. The interpretation of the various results is not yet unambiguous, but begins to show how a complex meristic patterning process can evolve and vary (and scales are developmentally related to sensory bristles, and the focal spot of *Dll* is produced by the coopting of an evolutionary prior use of *Hh* signaling to separate anterior and posterior wing compartments) (Carroll, Grenier et al. 2001; Keys, Lewis et al. 1999). It may not be a simple, single patterning process. But it is a series of familiar processes, occurring apparently separately, within the cells of the wing, whose nature and origin are as we would expect.

OUTPOUCHING

Most animals are hollow tubes relative to their overall AP, DV, and left-right symmetries, but they have structures like limbs and antennae that grow outward from placodes or imaginal disks along the main axis, as we described above. A variety of new patterning signals are then expressed in spatiotemporal order, to establish the new main proximodistal (PD; sometimes called mesiodistal) axis as well as DV and AP axes within the outgrowing structure.

The same genes can be involved in multiple axes even within the same structure. A good and well-studied example is the formation of tetrapod limbs (Carroll, Grenier et al. 2001; Davidson 2001; Gilbert 2003). Genes from all four *Hox* clusters are used first to establish the overall AP axis. Then, locally, *HoxD9* and *10* are expressed as the most proximal region, or *stylopod* (e.g., humerus), develops in the outgrowing limb bud in a summed-sequential way, roughly from anterior to posterior. Along the PD axis, *HoxA9–13* are expressed. In the formation of the next segment, or *zeugopod* (e.g., radius, ulna), *HoxD10–13* are expressed in an AP summed-sequential combinatorial manner. Finally, as the *autopod* (hand) forms, *HoxA13* as well as *HoxD10–13* are expressed again in an AP manner as the digits form (in an axis separate from the PD axis, or in a modified, curved single axis that wraps around the end of the limb to induce the digits).

In each case, *Hox* expression is associated with the formation of cartilage models that later ossify as the bony elements. *HoxC* genes (along with other known genes) are expressed differently between forelimb and hindlimb. In this way, the same genes expressed at different times control different aspects of the hierarchical, regionalized, multiple-axis morphogenesis (see Figure 9-9).

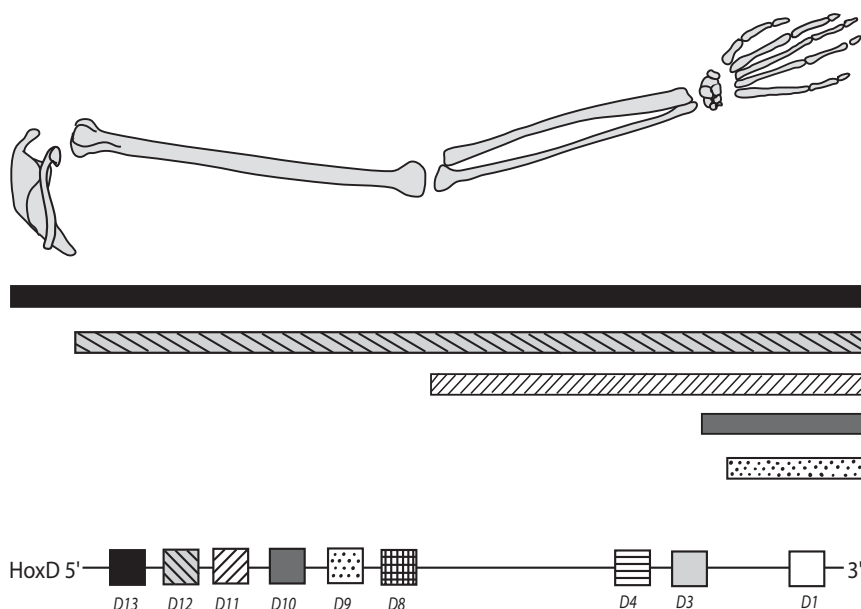


Figure 9-9. Combinatorial *HoxD* gene expression during development and its correspondence to sections of the adult tetrapod limb. Redrawn with permission from (O'Day 2003).

An interesting aspect of the vertebrate limb has to do with how its PD axis is formed and the *Hox* expression patterns are initiated. A structure called the *apical ectodermal ridge* (AER) forms as a kind of line that caps the distal edge of the limb bud. The AER has been thought to be a dynamic generative structure, or *progress zone*, so that as cells divide under the AER, the longer their precursors have been in that region the more they are prepared to make the segments that they do (stylopod, zeugopod, autopod). This seemed to be similar to the model of seashell patterning along a linear progress zone described earlier. Unfortunately, like so many stories of this kind, continued research shows that the story is not so tidy. The AER had been thought to be perhaps a stripe of reaction-diffusion-like process too slow to repeat before the limb becomes mineralized or stops growing for other reasons, but recent evidence suggests that the AER is already prepatterned in miniature, perhaps like the dental arch or tastebud patterning of the tongue, the three major limb segments ready to be “revealed” in the process of growth (Chiang et al. 2001; Duboule 2002; Dudley et al. 2002; Sun et al. 2002).

Wings grow out from the main body axis in flies and butterflies, representing a different kind of limb. AP coordinates are specified by sequential activation of genes called *Engrailed* (a homeobox TF), and the SFs *Hedgehog* and *Dpp*. Only the posterior part of this region expresses *Engrailed*, which induces *Hh*. Anterior cells express the *Ptc* receptor for Hh signal; receipt of this signal induces a line of *Dpp* expression. DV compartments are separated by dorsal expression of the *Apterous* homeobox TF, which induces expression of *Serrate* and *Fringe* components of the *Notch* signaling system; *Notch* receptors are expressed across the whole region but only receive *Serrate/Fringe* in the posterior part, and that induces in turn a stripe of the *Wg* (*Wingless*) SF expression.

Wg, like Dpp, diffuses in both directions from this line, inducing different subsequent expression leading to compartment-specific differentiation. Some of the SFs work across only a few cell diameters, such as Hh, but induce others that can travel more (e.g., up to about 20 for Dpp), in a kind of relay system. Two TFs, *Vg* and *Sd*, are expressed in all wing cells and enable other wing-specific differentiation to occur. When these two genes, along with compartment-specific TFs, are activated in the same cell they induce further compartment-specific gene expression, all resulting in compartment-specific combinations of expressed genes.

BRANCHING

Budding occurs when a cell or cluster of cells in a tissue begins to proliferate locally and grow outward or inward from the initial tissue layer. Branching occurs if secondary buds form from a primary, and if this occurs repeatedly over time a treelike structure results. The processes are not unique; for example the hand can be viewed as a branched structure coming of the original budlike structure of the limb. We think first of branches and roots in plants, naturally, but similar patterns occur in animals. Branching can occur from an initiation site outward, as in plants, or inward as in the branching lobes and bronchi of lungs. Other examples include the branching of vascular systems, and the development of nephrons in the kidney, the ductal structure in mammary glands, and nerve networks. The branching of the major nerves of the human nervous system is shown in Figure 9-10A, and the circulatory system in Figure 9-10B.

Some branchlike organs are produced by a process known as *clefting*. Ingrowth of cells from the periphery of a bud can form a wall that divides the bud in two. Along with the formation of buds and branches, these processes can generate many of the structures we see in complex organisms (Hogan 1999). One can view the hand as a clefted as well as a branched structure. A pad forms first, in which the digital ray cartilages form, but then apoptotic processes cleft the spaces between the digits (but not in webbed species).

Branching systems arise in various ways. Some begin developmentally as a single initial indentation or sac (rather than a placode), which divides into a nested hierarchy of internally descendant branches. The first or major divisions of branching systems are often highly stable among individuals, as in the left and right lobes of the mammalian lung or the major arteries, veins, and nerves (which is why anatomy students have to memorize them). Eventually, these systems divide into more numerous and/or highly variable branches like capillaries (which is why anatomy students *don't* have to memorize them).

Some systems, like the circulation, are closed (branches must connect so that blood can circulate), whereas others, like the lung, are open-ended (branches can diverge ever farther from the original trunk). The mammalian circulatory system forms as an initial plexus of connected vessels and then proliferates by sprouting new vessels and collapsing or destroying others to form the large, deeply branched, but closed system in the adult. Vessels can form “on demand,” induced by the release of signals from tissues starved of oxygen—including tumors—so that a branching pattern can often best be characterized by the *process* that generates it rather than by a map of the final structures.

Some deeply branched systems produce ever more numerous and smaller branches that resemble a *fractal* phenomenon. This means that the branching *pattern*

is essentially the same no matter at what point you observe it. The developmental process continues to generate new copies of that pattern, though in many cases of ever-smaller absolute size. Mathematical models can mimic such structures, and give a compelling explanation of the type of continual process that generates the pattern, but some patterns that look as if they “must” be fractal really are not literally so (Metzger and Krasnow 1999). Instead, what is most useful is to say that, like the periodic patterning discussed above, a process is at work rather than a specific program for each element in the system. Indeed, because new branches can be induced at least in some systems (blood vessels, plants), this must be so for them.

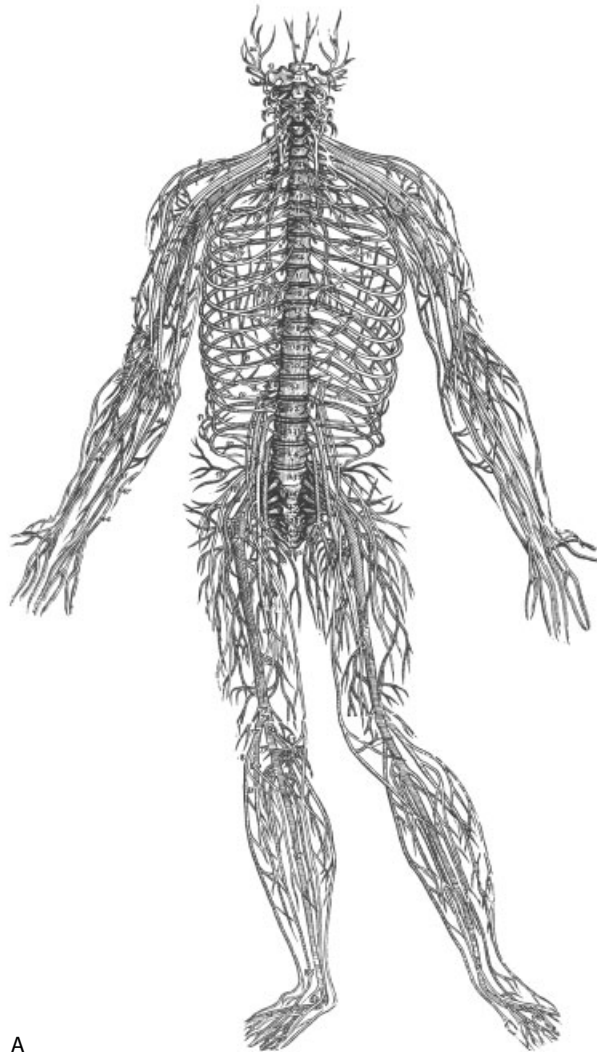


Figure 9-10. Branching patterns are important and widespread in life. (A) Branching of the major nerves in humans; (B) branching of the venous drainage system in humans. There are corresponding branching structures in the arteries and lymph ducts. From Andreas Vesalius’ classical drawings of 1543, that helped introduced the modern era in anatomic studies (Vesalius 1543).

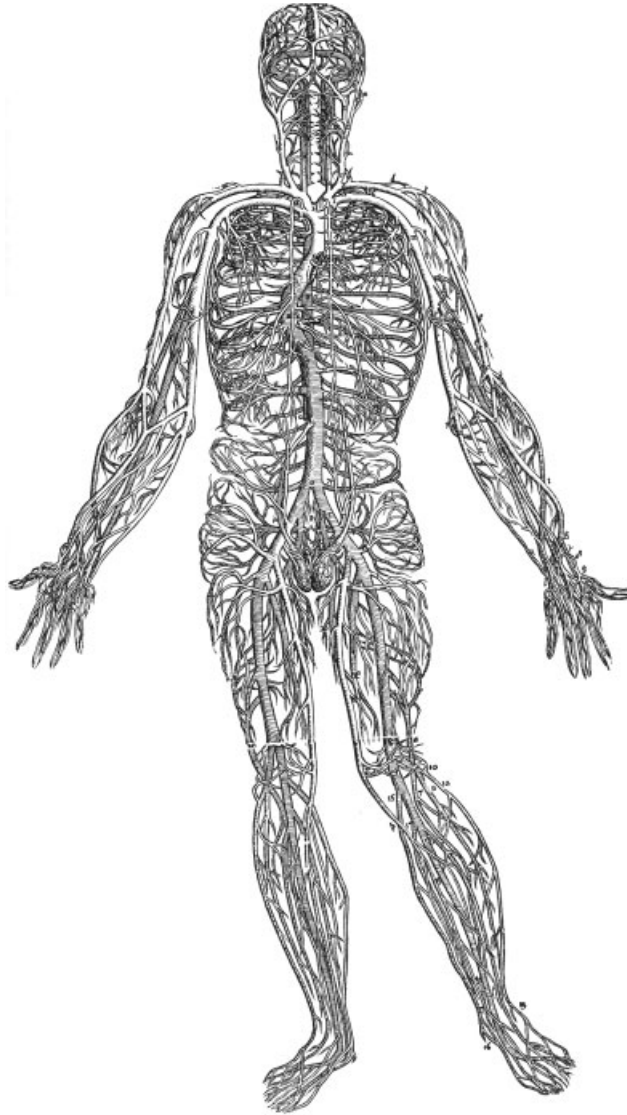


Figure 9-10. *Continued*

That the result has a fractal dimension (a numerical way to characterize the richness of the branching pattern) would be an incidental finding rather than something built into the process: there seems to be nothing about fractal geometry that is physiologically necessary per se.

Along an insect body, tracheal sacs form and invaginate to begin a branched tracheal system that carries oxygen to the internal tissues. The openings are known as *spiracles*. Once located along the embryo, each sac generates a structure of six primary, about 25 secondary, and hundreds of tertiary branches. There is considerable variation among species, but in the “archetypal” condition (i.e., in species like *Drosophila* where this has been specifically studied) the entire tracheal structure is formed by movement and morphological changes in an initial population of cells

rather than by the cell growth by which mammalian bronchial trees develop. It is in this sense an autonomous developmental unit. It is adaptable, however, in that eventually the smaller branches grow toward signals produced by oxygen-deprived cells and connect with branches from other trees.

In *Drosophila*, spiracles initiate in 20 sites of expression of the *Trachealess* TF gene along the midgestation fly embryo, involving about 80 cells each (what locates the spots is not known) (Davidson 2001; Metzger and Krasnow 1999). *Trachealess* and Tango protein (bHLH class TFs) expression presumably regulates downstream genes necessary for tracheal sac development. A cascade of expression of the same genes occurs in each sac, which develops autonomously, by growing inward, forming secondary and tertiary branches, each of which appears to be a module involving similar genes.

The secreted *Branchless* (an *Fgf* homolog) is activated in five additional locations surrounding the initial *Trachealess* expression zone; expression appears to be delimited by determinants of the overall AP and DV axes relative to the sac (Metzger and Krasnow 1999; Samakovlis et al. 1996; Sutherland et al. 1996). *Breathless Fgf Receptor* homologs on nearby cells receive the Branchless signal, triggering a cyto-dynamic cascade that guides the migration of cells to their budding locations in each of the five primary branch sites. Meanwhile, a second secreted factor represses *Fgf* genes to create an inhibition zone surrounding the cells forming the primary branches. *Branchless* turns off in the cells as they move along the branch but then switches on again in the distal cells as the next round of branching takes place. Cells moving toward the Branchless signal express branch-related genes not found earlier, as well as a second inhibitor that confines the location of the secondary branching. Terminal branching again involves Branchless signaling but also oxygen-sensitive signaling; together, they generate the more variable subsequent branching that is needed to respond to local tissue oxygen requirements. Tens of downstream SF and TF genes that affect the different steps in this process have been identified by mutation analysis (that is, mutations lead to aberrant development).

The process of tracheal ramification is, however, not simply a repeated invocation of the same branching signal (Metzger and Krasnow 1999). At different stages along the tree different genes are expressed, and mutational analysis suggests that different branches within a tree are controlled by specific genes (although a common set of *Fgf* pathway genes are also expressed). In the latter sense, as with so many similar systems, subsequent branching may be more like a single process analogous to reaction-diffusion processes, a type of mechanism that can generate roughly fractal branching. At least, although there may be some differences along the developing tracheal tree, hundreds of branch-specific gene signals are not required. In having formed from a single primordial set of cells rather than by mitotic growth, the tracheolar system can be viewed as the unfolding of a program somehow latent or prepatterned in those cells.

Homologous genes are expressed in mammalian tracheal and bronchial branching, which goes through 6 to 20 or so generations depending on the size of the animal. The major initial stages (trachea to two primary bronchi, etc.) are essentially the same among individuals and species. *Fgf10* and its receptor are involved in triggering branching in the end of a growing tracheal bud, with inhibitory signal (perhaps of *Shh*) expressed in the center, so that the outer sides grow but the center does not (Figure 9-11). Experiments have indicated that mesenchyme is important in providing patterning cues, but the epithelial layer is also needed. As with flies,

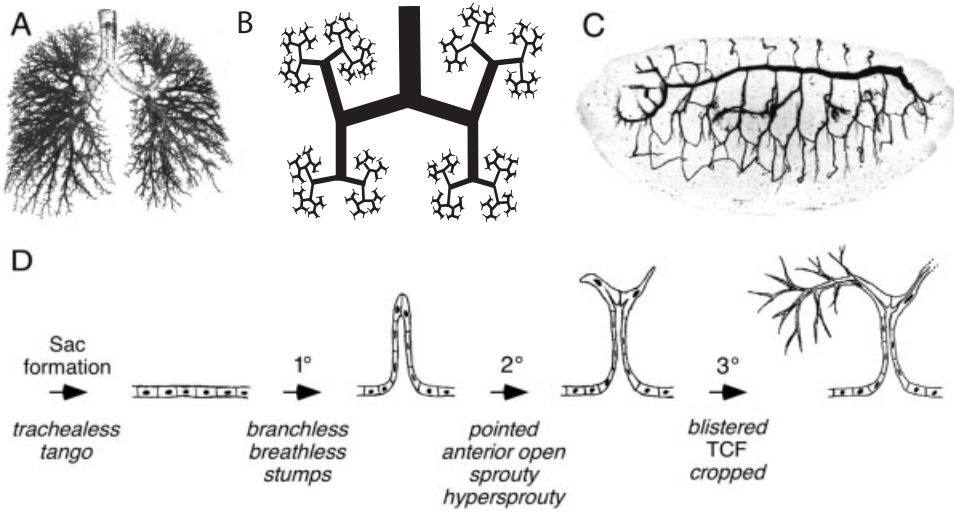


Figure 9-11. Animal budding and branching patterns. (A) Mammalian lung; (B) a fractal-like pattern generated by computer resembles natural budding and branching of this sort; (C) similar branching in the tracheole system of insects; (D) the steps in forming the tracheolar branching as in C are shown along with gene expression specific to each. Modified from (Metzger and Krasnow 1999), reprinted with permission.

the lung budding process is basically autonomous (can be achieved in isolated cultures of bud primordium), and the same signaling is reused in subsequent branches. Receptors are expressed more diversely than the cells secreting Fgf, which is expressed in a temporally dynamic pattern in the mesenchyme. Buds grow toward the nearby source region at any given time. As elsewhere in the embryo, Bmps are antagonists of Fgf, and in the lung they appear to inhibit growth and limit branch formation. Mesenchymal differentiation produces supporting structures that are more complex than in insects and involves further gene cascades.

A simple computer-generated fractal pattern is shown in the center of Figure 9-11. This has been said by many authors (but not those of the paper from which the figure was derived) to represent the kind of branching divergence found in traits like lungs where, for example, such a process generates the maximum surface area packed into a given volume. The figure shows only the first few branching iterations of what would be a space completely packed with ever more nested versions of exactly the same branching (what the term “fractal” refers to). The idea is a good metaphor, perhaps representing the “objective” of a branched lung to pack as much surface into a given volume as possible, but the lung is not exactly fractal nor symmetric. The left and right human lungs have two and three lobes respectively and are more different when seen in cross section (though not easily seen in the format of Figure 9-11), and this is a standard pattern not a chance difference.

NESTED COMPLEXITY BY BUDDING AND BRANCHING

Branching and budding can go together, and feathers provide an interesting and well-studied example (Yu et al. 2002). Feathers have three levels of branching: rachis or main axis to barb, barb to barbule, and barbule to cilia or hooklike structures

Jung, Francis-West et al. 1998; Metzger and Krasnow 1999; Patel et al. 1999; Yu, Wu et al. 2002).

Shared gene use is no surprise, but some aspects of the presumed gene action are interesting. Although Bmp4 inhibits tooth initiation and may be involved in inter-feather inhibition zones, it appears at the initiation sites within feathers. Bmp2 is a Bmp4 antagonist in teeth but has a generating effect in barbule generation. Shh is here not a growth stimulator but instead appears to be involved in apoptosis that removes tissue to make spaces between barbs. We noted above that Shh may have inhibitory effects in bronchial branching. There may be signaling gradient effects in feathers as well as switchlike effects. Overall in these and other systems, a nested, repeated use of the same genes occurs in different and sometimes at least partly opposite roles.

“EXPLORATORY” BRANCHING

Branching development may be “random” or fractal or stereotypical, but it can also be “exploratory” (e.g., Gerhart and Kirschner 1997). NC cells migrate through mesoderm to peripheral locations where, if they get there, they are stimulated to proliferate, aggregate, or differentiate by SFs emitted by overlying ectoderm of the appropriate type. Angiogenic factors secreted by tissues (including tumors) induce differentiation of angioblasts to form new blood vessels. Vascular branching at the local level seems largely random. Endothelial (vessel lining) cells present Fgf and other surface receptors, and chemotactic growth occurs as the cells proliferate to follow concentrations of growth factor ligands produced by oxygen-deprived cells. Tracheogenesis in insects may “seek” local oxygen-deprived cells, but in vertebrates oxygen is provided by the circulation rather than directly by the lungs; this means that vertebrates must supply lung branches with ample vessels.

In plant branching, which has some similar exploratory characteristics, the meristem acts as an organizer, as mentioned above. Central meristem cells express the homeobox TF *Wus* that maintains meristem status, but induces expression of a secreted protein, *Clv3* in adjacent cells. These cells are maintained in undifferentiated state by the TF *Stm* (*Shoot meristemless*), that represses a cascade of expression of differentiation genes (including TFs called *As1* and *Knat1*) (e.g., Laux 2003; Weigel and Jurgens 2002). As the meristem grows upward these cells are displaced peripherally, away from this repressive signal, lose their *Clv3* expression, and become capable of differentiating into primordial cells for a new branch, flower or leaf (as part of the repetitive induction of competent states, the central cells in a flower regain *Clv3* expression). There are new genes here, but also some homologs. *Zwille/Pinhead* in plants are homologous to *Piwi* and its relatives in animals, and both help maintain stem cell state (Benfey 1999). This brief description is mainly from the *Drosophila* of the plant world, the mustard relative *Arabidopsis*, but similar and/or homologous mechanisms are found in other plants, differing, of course, as the plants themselves differ. Even where the mechanisms are not homologous, however, there is nothing new here in that the logic of the processes is similar to those in animals.

Leaves are also internally branched. They provide rich venation for the cells through diverse patterns among species from reticular patterns in broadleaf plants to parallel veins in many grasses. As with lung or other branching, there can be a hierarchy of branching order and branch (vein) size. Reaction-diffusion-like mech-

anisms seem plausible, perhaps involving transport of signaling factors like auxin, or other interaction gene products (Dengler and Kang 2001). That a quantitative kind of process is involved in leaf shape is suggested by experiments showing that changes just in the timing of expression of a single gene, *Knox1*, can modify a leaf from a simple to a complex shape (Bharathan et al. 2002).

These various patterns are exploratory in the sense that growth takes place without a predetermined autonomous plan, and without a prepatterning process, and can occur in response to local factors. Indeed, there can be great variation in a trait among individuals with the same genotype (e.g., inbred identical plants or animals), even if the environmental conditions are essentially the same. The genetic program is the mechanism for the branching, not the branch.

FLOWER SEGMENTATION

Plant apical meristem tissue corresponds in some senses to the apical tissue in animal organ buds, but the analogous *process* does not involve homologous genes. Plant differentiation decisions are controlled more by external than internal conditions, including temperature, light, and humidity. Time since last branching also has determining power. Plants grow and bud in response to signaling molecules (generally small nonprotein molecules). Diffusible substances, including auxins and cytokinins, act as hormones to trigger these differentiation processes (see Chapter 10). Basic branching patterns in plants are shown in Figure 9-13. As with other examples in life, the apparent complexity of variation is probably brought about by relatively simple modifications of a basic process (Sussex and Kerk 2001). One of these appears simply to be in the differential timing of the development of axillary (side) branches off the main (“dominant”) apical meristem; this appears to be controlled by hormones as will be seen in Chapter 10. Gene mapping studies (Chapter 5) have been done and some QTLs (candidate chromosome regions) have been found, that is, genes that quantitatively affect branch pattern and proliferation. One gene, a TF called *Teosinte-branched 1* (*Tf1*), related to tissue proliferation, appears to affect branching architecture in maize by suppressing lateral branching (Sussex and Kerk 2001).

In some cases, particularly of flower differentiation, a cascade of transcription factor expression follows (Ng and Yanofsky 2000; Weigel and Meyerowitz 1994). The stereotype of this cascade is the “ABC” system (Figure 9-14), in which three different sets of genes in classes denoted A, B, and C are expressed in different positions around the meristem (looking at it end-on). The genes in the model species in which the system has been studied are mainly plant members of the *Mads* TF family, and the system is very similar in spirit to the combinatorial use of members of the *Hox* family in axial patterning in animals.

Essentially, a meristem is induced to produce flowers by meristem identity genes including *Leafy*, *Unusual Floral Organs* (UFO), and *Apetala1*, 2 (Ng and Yanofsky 2000). The meristem is arranged in roughly concentric whorls in which different A, B, and C genes are expressed. Cells expressing the A genes *Apetala1* and *Apetala2* alone lead to differentiation of sepals (green leaflike parts) around the outside of the inflorescence. Expression of the C gene *Agamous* leads to carpel (female part) formation in the inside. A + B gene expression generates flowers, whereas B + C generates stamens (male parts). Mutually antagonistic interactions among these and

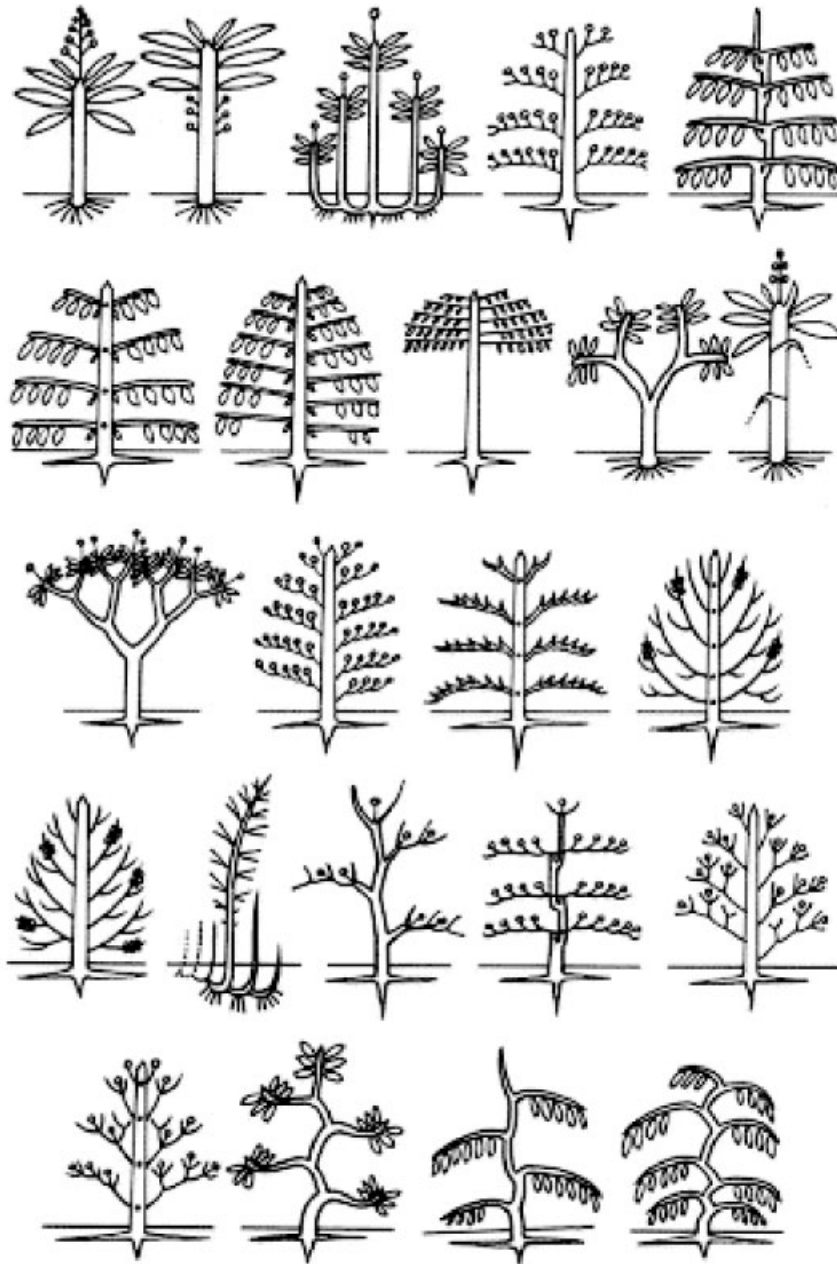


Figure 9-13. Twenty-three basic branching patterns in plants. Reprinted from (Halle 1999), with permission from Elsevier.

other genes generate the different expression zones early in the developmental cascade, which allows differentiation to occur. Experimental manipulation of these genes has produced homeotic mutations, which shows the combinatorial nature of this system of identity determination. Plants like maize with separate male and

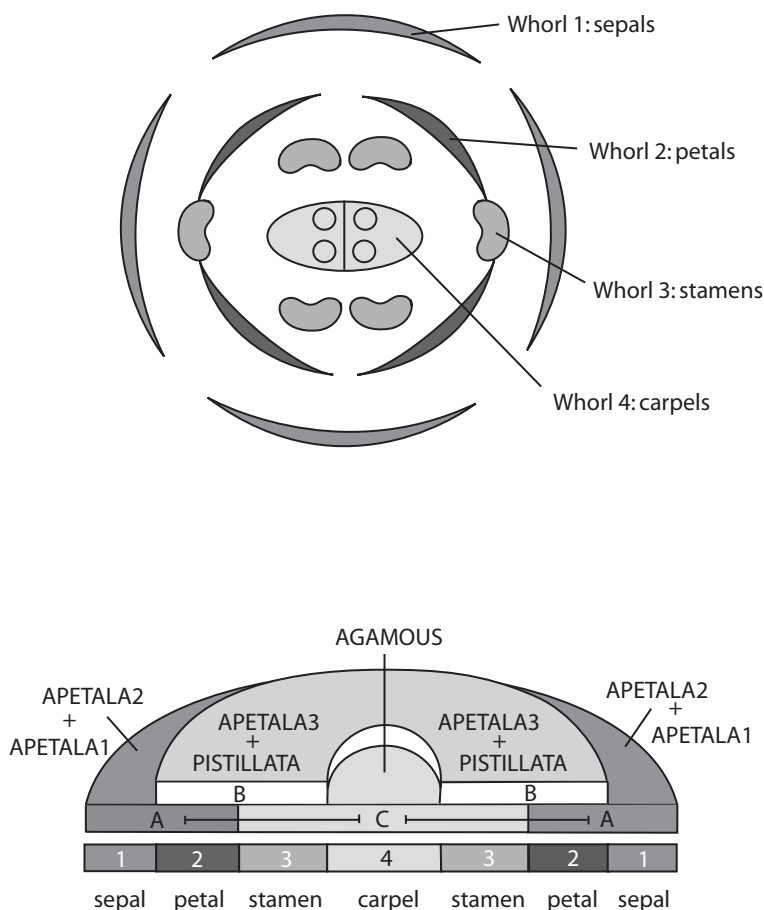


Figure 9-14. The basic ABC model of combinatorial gene expression and the development of flower organs. Three classes of homeotic genes specify the identity of the floral whorl components: sepals, petals, stamens, and carpel organs. Class A genes (dark grey) specify sepals, classes A and B (white) specify petals, classes B and C (light grey) specify stamens, and class C alone specifies carpels. For details see (Ng and Yanofsky 2000).

female flowers use homeotic gene differences to produce single-sex flowers (e.g., Ma and dePamphilis 2000), and similar changes can be brought about experimentally with the *Silky1* B-class gene homolog.

A general combinatorial system involving many of the same ABC system *Mads* and other TF genes is conserved among angiosperms (flowering plants), but the details, specific genes, and functions of specific genes vary (Kramer and Irish 1999; Ma and dePamphilis 2000). As with other examples noted earlier, such as tadpoles or pluteal larvae, phenogenetic drift seems to have been at work during the 130 million or so years of angiosperm evolution: either some flower structures have evolved repeatedly, each time recruiting new genes, or an original angiosperm flower has lost structures in different lineages while phenogenetic drift has modified genes or gene use in what is otherwise a generally conserved ABC system. As in the other systems we have discussed, gene duplication with divergence and overlap in function has also played a role.

SCULPTING BY CELL DEATH

Organs are sometimes said to be “sculpted” during development—but the metaphor is somewhat inapt, because sculptors work from the outside. Michelangelo may have “seen” *David* within a solid block of newly cut Carrara marble, but David did not make himself from within that stone. Perhaps we should view even the simplest embryo as much more a master sculptor than Michelangelo. One trick is that the embryo first makes a rather crude form of a structure and then forms the elegant final version by selectively removing unwanted material. This is done by apoptosis, using various pathways mentioned in earlier chapters.

Even some life cycles of bacteria involve autolytic cell death. Cells can die if they are starved of some nutrient or growth factor. Neural connections in the brain are formed through “exploratory” growth of some cells, leading others to follow them; the guide cells die once connections among the latter are made. Neurons can die if they fail to reach cells in an appropriate target tissue, a process triggered by the failure of receptors on the neuron to be bound by ligands secreted by cells in their destination. As mentioned earlier, vertebrate limbs form as paddlelike structures, but once the cartilage segments that will be the bones in the digits form, the tissue between the future fingers or toes dies away (when this doesn’t happen in humans, anomalous webbing can result). In a somewhat different way, deciduous plants shed their own leaves by sealing them off from nutrient supplies when faced with stress such as drought or the cold of winter.

Apoptosis occurs in two basic ways. The death sentence can come from the outside. Cells presenting apoptosis-related receptors on their surface undergo a cascade of self-destruction when a ligand (Tissue Necrosis Factor, Fas, or others) is received. This type of externally triggered mechanism is involved in the targeted destruction of infected cells by the immune system. *Bmp* signaling can also lead to the degradation of cells during development, as occurs in some reaction-diffusion patterning. Apoptotic genes include the oncogenes *p21* and *p53* (named for their protein size), which antagonize genes that promote cell growth, preventing cells from overgrowing their normal tissue structure. In one common pathway, signal reception activates a member of the *Caspase* (cysteine-aspartic acid protease) family whose proteolytic activity digests the materials in the cell by anomalous phosphorylation or dephosphorylation of cellular proteins affecting their function. For example, the phosphorylation regulator *Pten* modifies second-messenger molecules. Apoptosis can be achieved by disrupting the normal cell cycle, so that the cell can no longer divide and eventually withers.

Apoptosis can also be managed by internally derived pathways. Some of these also involve *Caspase* genes. In one example, an *apoptosome* containing various proteins forms when internal cellular damage releases Apaf1 from an Apaf1/Bcl2 complex bound to the mitochondrial surface membrane. Damaged mitochondria also release cytochrome *c*, which complexes with Apaf1 and Caspase9, to activate a cascade of Caspase-based proteolysis that cleaves proteins in the cytoplasm. Apoptosis can also destroy chromosomal DNA, another effective way to stop a cell in its tracks. Essentially, the logic of apoptosis is to use external signaling or internal cues to produce compounds that effectively mutate (interrupt) existing pathways in the cell, and as expected there are many ways to achieve the net result of cell death.

TIMING AND ASYMMETRIC GROWTH

We referred to spatial and temporal asymmetry in Chapter 8 as another set of ways of making a complex organism by simple mechanisms that affect scale and shape (Calder 1984; Carroll, Grenier et al. 2001; Hall 1999; Raff 1996; Thompson 1917; Wilkins 2002). The timing, level, and persistence of growth factors are responsible for these phenomena. Simple beginnings can be amplified hugely in terms of general growth, leading to considerable differences even among closely related species. We referred earlier to the likelihood that some pattern is laid down very early in the embryo and that this can explain how very similar traits like teeth or color stripe patterns can be generated across a large size range of species so closely related that they clearly must use essentially the same patterning mechanism (e.g., similar stripes in very small and very large cats; teeth in voles and elephants). No new duplicate genes or new mechanisms are required, only differences in their expression.

Timing is a relevant phenomenon—when an activity starts, how long it goes on, and its intensity during that period. Hormone- or induction-receptive periods are established in which a signal can be interpreted properly. This is true even at a higher level of trait complexity, such as language learning ability, songbird song learning, rat olfactory pathways, puberty, developmental patterning, and blossoming/leaf abscission. What determines the sensitive times? This is unknown at present for most examples, but some genes are known that relate to periodic events in development.

Chronobiology is a rapidly growing field in genetics. Various genes, including *Clock* and *Chairy*, are known to be involved in expression timing, the latter having 90-minute activity cycles in some systems. Melanopsin is involved in calibrating light-related cycles.

CONCLUSION

Diverse biological processes in diverse species use a limited number of genetic regulatory factors and their interactions, raising questions about how local specificity is established or how the requisite complex combinatorial patterns are controlled in a stable way. These processes include ligand-receptor binding for signal transduction that alters gene expression, gene regulation by the binding of *cis* enhancers by TFs, message transduction mechanisms such as the protein kinase networks, and combinatorial use of a limited set of factors to establish numerous unique contexts.

The fingerprints of gene duplication are all over these generalizations, so that in a real sense the diversity of mechanisms is derived from only a few ancestral instances. The widespread presence of the same mechanisms across the biosphere justifies referring to them—and in particular, their “logic”—as basic principles of life and shows that not much is around that is really new, in the regulatory sense, compared with the diversity of final function. Relatively simple mechanisms are used to make relatively complex and variable structures. However, the same network is not always preserved to have the same function across taxa. So it is not the specific regulatory network that counts so much as the product of that network—the final phenotype.

None of these phenomena is specifically predictable as a fundamental of life from the formal theory of evolution or its supporting theory of population genetics, which in this sense do not provide a very useful theory for development. *Fgf4* and *Shh* are expressed in limbs and teeth and many other structures, *Distal-less* in insect wing

spots and vertebrate teeth. There is nothing about these genes that is physically tooth- or limblike, or even mouselike (because the same genes do similar things in many other species). As we noted in earlier chapters, regulatory genes constitute a toolkit whose own identity is purely arbitrary relative to what the tools are used to make. A *Hox* or *Fgf* gene can be involved in the regulation of any gene that has the appropriate flanking enhancer sequences. In a meaningful sense, a trait is not really due to the genes that bring it about, but to their interactions.

Phenomena like periodic patterning affect a diversity of traits, and even simple coloration has widespread uses including camouflage, mate attraction, age grade or sex marking, and the like. Similar developmental phenomena have varied functional effects, such as the functions of different types of feathers or hair (display, flight, thermal protection). Jointed locomotor structures, suckers in octapi, tentacles (pentameric symmetry), protective scales, worm segments, cactus spines, and leaves and their internal structures all reflect similar processes, some but not all due to the action of homologous genes.

Development was one of the key factors in Darwin's thinking, for example, in explaining embryonic similarities as due to common ancestry and using development as a kind of recapitulation of prior evolution. Indeed, the word "evolution" (which literally means "unfolding") had previously applied to what we call development. However, for many decades in the twentieth century geneticists considered developmental biology to be but a crude and empirical business of little generalizability or rigor (Gilbert et al. 1996) that could not be accounted for by the one-by-one evolution of genes via selection of allelic variation that was taken to be the real theory of evolution. Fortunately, this is no longer true, but we still need to make progress in reconciling the evolution of the more organismal phenomena of development and the more particulate perspective of evolutionary genetics. It is not enough simply that they are not inconsistent with each other.

Chapter 10

Communicating Between Cells

The specialization in cells and tissues that multicellularity allows depends on mechanisms for regulating and integrating the function of the different cells and organs. One cell must be able to influence the physiology of another, on the basis of its context or conditions. In a sense, this is what makes what we term an “organism” as a coordinated biological entity possible. The organism coordinates many functions that involve communication between neighboring cells or among more distant cells or even communication from a single location to everywhere else in the organism. The functions involved are often programmed, specific, or stereotypical rather than responses to open-ended situations. In this chapter, we will describe ways by which complex organisms communicate internally to achieve these ends.

Intercellular communication or at least its basic mechanisms probably arose before multicellularity, because, as we have seen in earlier chapters, single-celled organisms are capable of communicating with each other. These mechanisms include cell surface receptors, G proteins, protein kinases, and the like, many homologous mechanisms used by the cells of more complex organisms.

A number of cell interactions play a role in the timing of gene expression and changes in the biochemistry of the cell to regulate growth, development and homeostasis, and the like. These interactions can be triggered by environmental cues that can induce cellular functions such as photosynthesis or the running of sap in plants, hibernation in animals, or diapause (biological dormancy) in insects or by intercellular signaling via molecules like nutrients such as sugars or ions or via hormones, organic substances that in minute concentrations have defined effects on cellular activity.

Two basic regulatory systems control cell-modulator interactions in animals. One is a nervous system, and in more complex organisms a central nervous system (CNS), which sends electrochemical signals to peripheral systems and receives electrochemical messages from the periphery. This occurs in a relatively specific way to connect the two endpoints of the communication. The other, in which communication is more diffuse, is the endocrine system and its analog in plants. The endocrine system consists of a number of glands throughout the body that synthesize organic

substances, polypeptides or steroids, which are released into the bloodstream or extracellular space and circulate to target organs whose cells present specific receptors, stimulating a variety of functions, one of the most important of which is regulation of gene expression. Given the genetic message mechanism, the sending cells must be prepared to use the sending mechanism under appropriate circumstances, and cells that need to respond must be prepared to receive the signal.

NEURO-IMMUNO-ENDOCRINOLOGY

The vertebrate CNS and endocrine systems interact with each other, and current knowledge suggests that the two systems are not as functionally, mechanistically, or evolutionarily distinct as had long been believed. Until recently the nervous system was viewed as a regulator of brief, rapid, localized responses, whereas the endocrine system slowly initiated more prolonged and generalized responses. However, understanding of the overlapping functions of components of these distant regulating systems has led to a more synthetic view, completely consistent with a more unified view of life and the common origin of systems and genes, as well as of species. No new phenomena are involved, but familiar ones are.

Nerve cells are secretory, in that they release chemical *neurotransmitters* at synapses to trigger fleeting neuronal responses; inhibitory cells ensure that the triggering signal can be received only briefly. However, some nerve cells are specialized for translating neural signals into chemical rather than electrical stimuli. These *neurosecretory* cells release *neurohormones* into the bloodstream at *neurohemal organs*, the synapse between the axon of the neurosecretory cell and a blood capillary, or so close to their target that they need never enter the bloodstream at all. Additionally, the endocrine system regulates the function of the nervous system, and the nervous system controls the blood flow and secretory activity of endocrine glands, because those glands are innervated and their cells receive neural signals. Neurohormones can travel to distant target cells.

The signaling molecules of the nervous, endocrine, and immune systems are often the same. Immune function is affected by hormones, and the endocrine system is affected by cytokines, mediators of the immune response (Chapter 11). Immunologically reactive cells have recently been found to secrete hormones that had previously been thought to be synthesized only in the brain or pituitary gland (Ojeda and Griffin 2000), and excretions from glands long thought to have exclusively endocrine function, such as norepinephrine from the adrenal medulla, are now known to be neurotransmitters as well. So, although we will concentrate in this chapter on the action of hormones, the isolation of the endocrine system in this way is an artificial classification that ignores the essential interactions between the different systems, interactions being discovered and characterized very actively today.

An important point about cell signaling is *that* it happens, and that this, rather than *how* it happens, is what natural selection has acted upon. Understanding the distinction can help explain many things in biology, such as why the G protein-coupled receptor signaling cascade is involved in so many signaling pathways in tetrapods and arthropods but so few in plants and the fact that kinases are important to signaling in all organisms but the particulars of the amino acid that gets phosphorylated or the shape of the transcription factor that results are irrelevant to the general process and what natural selection “sees.” Cellular communication mechanisms can vary even though the regulated trait has remained similar.

Hormones can be divided into two general classes. *Regulatory* hormones are involved in the control of metabolic and homeostatic processes, whereas hormones that control the onset of irreversible physiological switches are referred to as *developmental* hormones. The latter basically trigger events such as molting cycles or reproduction. However, the difference is in function, not mechanism of action. The general modes or logic of action of plant and animal hormones are similar and involve processes we visit many times in this book.

But what is a hormone? The classic definition is an organic substance produced in an endocrine gland and released into the bloodstream or the sap of a plant in very small amounts for transport to distant cells or organs prepared to receive it and where it affects physiological activity via both genomic and nongenomic pathways. However, in fact, hormones need not travel long distances, nor need they be released into the bloodstream or hemolymph or extracellular space for transport to distant cells. As described in Chapter 6, they may act at various distances, for example, paracrine function being local and autocrine or intracrine function affecting the hormone-producing cell itself.

Hormones might not even be synthesized in a gland traditionally seen as part of the endocrine system—in mammals adrenocorticotropin (ACTH), β -endorphin, prolactin, and luteinizing hormone-releasing hormone (LHRH) are all produced by immunologically reactive cells, and hormones are released by cells lining the stomach and small intestine, among many other examples. Some growth factors work in a nondiffusible way, when the membrane-anchored growth factor on one cell binds to its receptor on the membrane of an adjacent cell (as, for example, the *Delta/Notch* signaling system).

The more we know the more life defies categorization, but if we need a more inclusive definition of a “hormone,” it might be simply an endogenously synthesized organic substance that, when bound by a receptor or binding protein, triggers a specific cellular response. Yet even this is not quite right, because there are other endogenously produced substances—for example, SFs, that induce changes in gene expression to alter cell physiology—that are not classed as hormones. The connotation of the term “hormone” is bound up, like so much else, in history and the traditional notions that predate modern molecular biology. Ultimately, given the blind nature of evolution by phenotype and the striking connectedness of life reflected in the way genes and genetic mechanisms evolved by divergence from a common ancestor, it is likely that any kind of grouping that we might conceive is an overlay of our own construction for our own convenience, rather than a discretely programmed part of nature.

PLANT SIGNALING

In plants, as animals, growth and development are controlled by responses to environmental or inter- or extracellular triggers. Environmental triggers of plant responses include features of light signals such as quality (red light activates different genes than blue light), duration, direction, and quantity, changes in temperature that can alter the fluidity of cell membranes, chemical signals (air pollutants, phytotoxins, or elicitors from other organisms), changes in water availability and thus cell turgor, insect or pathogenic attack, which can trigger the immune response, gravity, available nutrients, proximity of other plants, wind, and other variables.

Intracellular signals, generally activated by receptors that bind a variety of extracellular signals, include second messengers such as inositol triphosphate (IP₃), which binds to and activates a calcium channel in the endoplasmic reticulum, and Ca²⁺, an important mediator of the activity of many proteins inside the cell, and metabolites such as sugars and glutamate (an amino acid). Cellular activity is also driven by intercellular communication. Electrical signals, small molecules (<800 Da), and some mRNAs can pass between cells for a short distance via the plasmodesmata. Finally, plant hormones, which are released from a cell into the *apoplast* (the extracellular space) and transported to other cells, are important in altering gene expression and physiology.

Like animal hormones, plant hormones are compounds synthesized in the plant that at low concentrations elicit physiological responses either in the cell in which they are synthesized or in other cells when translocated. Plant hormones are transported from cell to cell through intercellular spaces or through vascular bundles, the xylem and phloem. Plant hormones have synergistic and antagonistic effects; they inhibit and stimulate other hormones. These signals thus “circulate” in the organism, but this is not the rapid, closed circulation that we are familiar with in vertebrates, because the upward and downward flows are essentially independent and there is nothing corresponding to a capillary bed to connect them.

A number of classes of phytohormones have been isolated (see Table 10-1). *Auxin*, *cytokinin*, *gibberellin*, and *brassinosteroids* are, by and large, growth stimulators, and *abscisic acid* and *ethylene* can generally be called growth inhibitors, whereas *salicylic acid*, *jasmonic acid*, *systemin*, and *oligosaccharins* are synthesized in response to stress and initiate plant defenses. Another class of hormone, the *phytoecdysteroids*, is noxious to nonadapted insects and herbivores and probably evolved as a plant defense. Generally, the growth stimulators and inhibitors affect cell division, elongation, and differentiation, but their effects are modulated by the type of target cell, the life cycle stage, interaction with other hormones, and hormone concentration, so that hormones that are usually categorized as growth stimulators can, depending on the cellular context, sometimes inhibit growth, and vice versa.

At the level of the cell, hormones primarily induce, and sometimes inhibit, gene expression and enzyme action, although some hormones act via nongenetic pathways, such as by being cofactors in reactions among existing constituents of the receiving cell. The mode of action of phytohormones is most easily characterized through the study of mutant plants, whose phenotype seems to be the result of over- or underproduction of a hormone. The biosynthetic pathways of a number of plant hormones are not yet understood, primarily those of hormones for which mutants have not been available for study.

These core signaling pathways are widely shared among plant taxa and used over and over in many different plant developmental pathways and in the maintenance of homeostasis, and most have a long evolutionary history, of gene duplication and diversification, as they are found in plants as old as the primitive mosses (McCarty and Chory 2000; Schumaker and Dietrich 1998).

Auxins are involved in a large diversity of developmental processes, in response either to environmental cues or to intracellular stimuli. Synthesized in shoot apical meristems (differentiating tips of branches), young leaves, and embryos, they induce cell division, promote elongation, and inhibit growth of lateral buds (a phenomenon known as apical dominance), and early in development they transcriptionally activate a set of early genes that mediate processes such as determination of the

TABLE 10-1.
Plant Hormones.

Hormone	Place of Synthesis	Action
Growth regulators		
Auxins	Shoot apical meristems, young leaves, embryos,	Cell division, promotion of elongation, inhibition of flowers, fruits, and pollen, growth of lateral buds, axial growth, induction of tropism, vascular patterning, and differentiation, lateral organ outgrowth in root and shoot
Gibberellins	In immature seeds, the root and shoot apical meristems and in young leaves	Stem elongation, germination, growth of some fruit, development of male sex organs in some flowers, control of juvenility in some plants
Cytokinins	Most in roots, in angiosperms of flowering plants	Cell division, shoot and root formation, apical dominance, xylem formation, leaf senescence, solute mobilization, cotyledon expansion
Brassinosteroids		Regulate gene expression, stimulate stem growth, inhibit root growth, promote xylem differentiation, retard leaf abscission
Absciscic acid	Fruits, root caps, mature leaves	Inhibits cell growth, promotes seed dormancy, involved in opening/closing of stomata when leaves wilt
Ethylene	Anywhere, but primarily apical buds, stem nodes, senescing flowers and ripening fruits, wounded tissues	Promotes ripening
Defense Mechanisms		
Salicylic acid		Disease resistance, control of heat production, essential to systemic resistance response
Jasmonic acid/ Methyl jasmonate		Help regulate plant growth, defend against fungi, involved in leaf senescence
Systemin		Disease resistance
Oligosaccharins		Disease resistance, induction of SRR

bilateral axis of the developing embryo (Christensen et al. 2000; Ouellet et al. 2001). Later in development, auxin is involved in vascular patterning and differentiation and lateral organ outgrowth in the root and shoot (Christensen, Dagenais et al. 2000), cell division, phototropism, gravitropism, and gene induction.

High levels of auxins stimulate the production of ethylene, which essentially counteracts auxin activity. Leaves remain attached to the stem as long as there is auxin moving from the leaf blade down the petiole (leafstalk). When auxin is interrupted, as the leaf begins to senesce, the cell walls of the cells at the base of the petiole (the *abscission zone*) undergo dissolution and the leaf falls off (Cleland 1999). This is called abscission, and it also happens in fruit when the seeds cease transporting auxin through the fruit pedicle. However, the route by which auxin reaches the abscission zone is important in determining the speed with which senescence occurs. Auxin reaching the abscission zone from the tip promotes abscission, whereas auxin reaching the structure from the opposite end inhibits it. (Abscission is a major example of a plant process that depends on interaction between different hormones to catalyze the reactions that break down the cell walls. Once bound by their intracellular receptors, ethylene stimulates the synthesis of the enzymes involved in destruction of cell walls and abscisic acid speeds the required senescence. Gibberellin inhibits abscission by promoting growth rather than senescence.)

Three *Auxin* gene families have been characterized; the *Aux/Iaa* gene family, the *Gh3* gene family, and the *Saur* gene family (Ouellet, Overvoorde et al. 2001). The *IAA*, or indole-3-acetic acid, gene family encodes the predominant group of auxins in higher plants. *Arabidopsis*, for example, has at least 25 *Aux/Iaa* genes (Rouse et al. 1998). *IAA* is synthesized mainly in the apical meristem shoot tips, young leaves, and developing fruit. A number of pathways seem to be involved, and *IAA* can be catalyzed by endogenous plant enzymes as well as by enzymes produced by plasmids of plant pathogens. The endogenous production is not yet well understood (Zhao et al. 2001), although it seems to be dependent on the amino acid tryptophan or the breakdown of glycosides, carbohydrates.

Cytokinins (CKs) promote cell division and are involved in shoot formation, root formation, apical dominance, xylem formation, leaf senescence, solute mobilization, root growth, and cotyledon ("seed leaf") expansion. They are produced in growing areas such as meristems and are derivatives of adenine. CKs interact with auxins in mitosis and the initiation of shoot and root primordia. The antagonistic action of auxin versus cytokinin is involved in apical dominance (predominance of one meristem relative to others in the plant) and xylem development, that is, the process depends on the relative amounts of these hormones. A similar antagonistic interaction determines the relative growth of roots, shoots, and callus (the undifferentiated tissue that grows at the edge of a wound) (Zhao, Christensen et al. 2001). A high *IAA/CK* ratio promotes root growth, and a lower ratio promotes growth of the callus and shoots.

The CK signaling pathway is not yet well understood, although it has been proposed that it is a two-component signaling system. Two-component systems control signal transduction pathways in many microorganisms but have only recently been recognized to be important in plant signaling (Hwang and Sheen 2001; Inoue et al. 2001; Moller and Chua 1999). The two components include a histidine protein kinase that detects the signal and transmits it to the second component, the response regulator, which mediates the plant response. The signal is transmitted by phosphorelay—that is, the protein kinase autophosphorylates at a histidine residue, taking the

phosphate from a donor ATP molecule, and then transfers the phosphate group to an aspartate group on the response regulator. This activates the regulator, generally a transcription factor, inducing it to repress or stimulate expression of a target gene. A cytokinin receptor, a histidine kinase, has been identified in *Arabidopsis*, *Cre1* (*Cytokinin response 1*), but the regulator is not yet known. Presumably it will be a transcription factor (Hwang and Sheen 2001; Yamada et al. 2001).

Gibberellins are a group of more than 100 related compounds, all of which contain a chemical structure known as a “gibbane ring” (Cleland 1999). No plant synthesizes all of the gibberellins, but all plants seem to have more than a few. These hormones are produced in immature seeds, in the root and shoot apical meristems, and in young leaves. Gibberellins promote stem elongation, induce enzymes involved in germination of grass seed, and promote the growth of some fruit, the germination of some seeds, the development of male sex organs in some flowers, and the control of juvenility in some plants. Their effects are enhanced in the presence of auxins.

Less is known about the gibberellic acid (GA) signaling pathways than those of other phytohormones. GA synthesis is catalyzed by the enzyme copalyl-diphosphate synthase (CPS), which is encoded by *Gai1* (*Ga-insensitive*). The *Gai* genes were first identified in a plant with reduced GA sensitivity. *Gai1* mutant plants do not synthesize GA and are extreme dwarf, have reduced male fertility, and germinate poorly (Moller and Chua 1999). Other GA mutants lack GA repressors; the *Spy* (*Spindly*) mutant has a phenotype similar to that of wild-type plants that are overexposed to Ga. *Rga* (*Repressor of Ga*) represses the GA signaling pathway in *Arabidopsis*. It encodes a protein of the *Gras* family, which includes *Gai*. The repressor activity of *Gai* and *Rga* can be inactivated by GA and is triggered by the upstream gene, *Spy*.

Brassinosteroids are key regulators of plant responses to light. They are steroid-like compounds, related to cholesterol as are animal and insect steroids, that elicit growth responses by regulating the expression of genes associated with development. They stimulate stem growth, inhibit root growth, promote xylem differentiation, and retard leaf abscission. Their effect on stem cell elongation and xylem differentiation are auxin-mediated processes. This class of hormones works by binding to a brassinosteroid receptor on the cell surface, which induces a serine/threonine receptor kinase signaling pathway. Brassinosteroids have recently been found to be involved in inducing resistance to disease in higher plants as well (Nakashita et al. 2003).

One brassinosteroid receptor gene has been identified in *Arabidopsis*: *Bri1* (Wang et al. 2001). Homologs to *Bri1* have also been found in *Arabidopsis*, but their role in brassinosteroid signal transduction, if any, is not yet known. *Bri1* encodes a leucine-rich repeat (Lrr) transmembrane receptor kinase. Structurally, the repeat region contains 25 Lrr's, interrupted by a 70-amino acid island, a transmembrane domain, and a cytoplasmic serine/threonine kinase domain. Because plants do not have nuclear steroid receptors, any genomic response a plant steroid hormone initiates must begin at the cell surface. The *Bri1* receptor ligand-binding domain, therefore, is extracellular and ligand binding initiates signal transduction via the cytoplasmic kinase domain of the receptor, which autophosphorylates serine and threonine residues. Other components of this signaling cascade have not yet been identified (Li et al. 2001; Mussig and Altmann 2001). Even the nature of the ligand is still unknown (Wang, Seto et al. 2001).

Ethylene, derived from the amino acid methionine, is a gas that promotes ripening. It can be produced anywhere in a plant but primarily is produced in apical buds, stem nodes, senescing flowers, and ripening fruit as well as wounded tissues. As a gas, ethylene diffuses readily to other cells in the same plant and even to nearby plants, so this hormone need not be produced endogenously to trigger ripening, although the extent to which plants receive or respond to signal from other plants is debated.

Ripening is a series of events leading to senescence and, ultimately, rotting. Ethylene initiates the cascade by inducing the breakdown of cell walls, changes in pigments, and formation of flavor compounds. It is a factor in leaf and fruit abscission and facilitates the withering and death of petals of flowers, among other effects. Many of the processes now known to be due to ethylene were once attributed to auxin, because ethylene is often produced as a response to high concentrations of the former.

Ethylene synthesis is controlled by the ACC (1-aminocyclopropane-1-carboxylate) synthase family of enzymes. ACC is the precursor molecule to ethylene. The genes that convert ACC to ethylene comprise multigene families in different plant species. Expression of ACC synthase is induced by stimuli that lead to increased ethylene production—wounding, senescence, fruit ripening, etc. This suggests that the rate of ethylene production depends on ACC synthase activity. Oxidases also seem to be involved in regulation of ethylene production (Johnson and Ecker 1998).

Like the cytokinin signaling pathway, the ethylene pathway also shares homology with bacterial two-component systems. Here, the receptor is a histidine kinase that autophosphorylates an internal histidine residue of target proteins, and the response regulator includes a conserved aspartate residue that is the recipient of the phosphate group from the histidine kinase (Johnson and Ecker 1998; Stepanova and Ecker 2000). Unbound ethylene receptors activate a negative regulator, Ctr, a protein kinase in the *Raf* family, which induces a cascade that ultimately leads to the repression of the *Ein2* (*Ethylene insensitive*) gene. But when ethylene is bound to its receptors, Ctr is inactivated, releasing *Ein2* repression and leading to activation of the ethylene response gene (Moller and Chua 1999; Ouaked et al. 2003).

Abscissic acid (ABA) inhibits cell growth but, like so many biomolecules, has multiple functions. ABA levels, for example, rise during times of environmental stress; thus ABA is sometimes called a stress hormone. ABA promotes seed dormancy and is involved in the opening and closing of stomata when leaves wilt. A 15-carbon acid, ABA is made in fruits, root caps, and mature leaves in response to environmental signals. Water stress, or freezing temperatures, for example, result in massive and rapid ABA synthesis. Movement of ABA is through phloem and xylem and by diffusion between cells. ABA seems to be antagonistic to auxins, cytokinins, and gibberellins.

Five *Aba-insensitive* genes, *Abi1* through 5, have been cloned in *Arabidopsis* and shown to have homologs in at least two other plant species. Some components of the *Aba* pathway are even found in animals (Gampala et al. 2001). *Abi1* and *Abi2* encode serine/threonine kinases with various downregulatory functions in genetic responses to cold or drought and regulation of ion channels, and *Abi3*, 4, and 5 encode genes for various transcription factors (Gampala, Finkelstein et al. 2001).

Apical dominance, the control of development by the main shoot meristem, characterizes the branching and differentiation of many plants, but not all. In general, branching seems to depend on relative amounts of CKs, ABA, and IAA, with

relatively high CK:IAA ratios affecting and/or reflecting the location and relative growth rates of nodes that develop in axial locations on shoots, and the pattern is then modified by ABA (Sussex and Kerk 2001). We noted earlier that CK:IAA ratios seem to be important in root development as well. The distribution of relative levels of these hormones may function as a kind of quantitative patterning process, possibly reaction-diffusion-like, in which simple alteration of the parameters (the quantitative levels of the hormones), are responsible for the diversity of branching patterns shown in Figure 9-13. As discussed in Chapter 9, the genetic mechanisms responsible for these quantities are not yet identified.

Another group of plant hormones appears primarily in response to wounding and pathogen attack. *Salicylic acid* is involved in disease resistance and control of heat production in some species and is essential to the systemic resistance response. *Jasmonic acid* (JA) and *methyl jasmonate* (MeJA) help regulate plant growth, defend against fungi, and are involved in leaf senescence. MeJA has been found to be synergistic with ethylene and activates enzymes involved in defense against wounding. JA is probably confined to the cell in which it is produced in most cases. MeJA can act between cells, as well as between plants by signaling between infected plants and unaffected plants. Although controversial, it has been suggested that volatile MeJA, released into the air by a plant under pathogen attack, can induce increased resistance in nearby unaffected plants (Dolch and Tschardt 2000; Karban et al. 2000; Preston et al. 2001). If true, this is another instance of altruistic cooperation that many biologists would argue requires specific evolutionary explanation. MeJA also promotes tuber formation and storage protein formation.

Systemin is involved in disease resistance, although how ubiquitous it is as a hormone is not known. It acts by inducing proteinase inhibitors throughout the plant. Action may be via the synthesis of JA, which acts intracellularly as a second messenger in the induction of proteinase inhibitors. Plant cell walls are composed of a mixture of complex carbohydrates. When these break down, small pieces are released, some of which have biological activity. These are *oligosaccharins*, and they are produced during pathogen attack and signal other cells to prepare to defend against attack.

As will be discussed in Chapter 12, one way that plant defenses are triggered is via gene-for-gene resistance pathways, the rapid defense response a plant has when its *R* genes are induced by a pathogen's *Avr* genes. This hypersensitive response (HR) involves apoptosis of plant cells in contact with the pathogen. The precise mechanism of induction of this response is not well understood, but it is hypothesized that *R* proteins "guard" plant proteins that are the targets of attack by *Avr* proteins ("guardees") and that the HR is triggered by *Avr*-guardee interactions (Glazebrook 2001).

The HR triggers the systemic resistance response (SAR), which protects the entire plant and is long-standing. The SAR involves the signal molecule salicylic acid and the expression of a characteristic set of defense genes. Some, but not all, response pathways use the signal molecules JA or ethylene, and cross-talk between the pathways is required by some responses. JA and ethylene mutants, either in the production or the perception of these hormones, are more susceptible to disease than their wild-type counterparts. The mechanism by which they are involved in defense responses, however, is not clear; it may be via a synergistic interaction with genes in various defense pathways (Ellis and Turner 2001). Many of these genes can be induced by the application of JA in the absence of wounding or attack (Reymond

et al. 2000), showing the importance of this hormone in the transduction of signal and the initiation of defense responses.

ARTHROPOD HORMONES

INSECTS

Insect hormones control a broad range of processes in the insect life cycle, both developmental and physiological. These include the metabolism of carbohydrates and lipids, the maintenance of water balance, including excretion of water after a blood meal, stimulation and inhibition of the circulatory system and the firing of muscles, growth, including molting and metamorphosis, diapause (a period of suspended growth or development), apoptosis, reproduction, caste determination (which occurs in social insects, such as bees and ants, whose fate as queen or worker is determined not by genes but by differential feeding and pheromone exposure at the larval stage), and aspects of behavior during molting and migration, reproductive and other social behaviors, and response to pheromones.

Insects produce two general classes of hormones, as do vertebrates, lipid or steroid (lipophilic) hormones and polar (hydrophilic) peptides, and their mode of action depends on how the hormone interacts with the cell membrane (peptide hormones are coded directly by genes; the other forms are the product of enzymatic reactions for which the enzymes, but not the final hormone product, are genetically coded). Being hydrophobic, the lipid hormones pass readily through the cell wall to bind with receptors in the cytoplasm or nucleus of the cell. The hormone-receptor complex in turn binds with specific DNA sequences to initiate, enhance, or inhibit gene expression. The peptide hormones, however, do not easily pass through the cell membrane but instead bind with receptors on the cell surface, which transduce the signal to second messengers inside the cell, often via a G protein cascade. Target cells are not continuously receptive to hormone stimulation. There are “hormone-sensitive periods,” and although hormones circulate throughout the hemolymph, exposing all tissues to the same levels of hormone at the same time, the tissues are not all equally receptive. What controls the timing of the sensitivity of a tissue is not known, but clearly both the signaling and receiving cells need to be prepared, implying a prior element of differentiation in these two, often distant, locations in the body.

Insects synthesize hormones in two distinct organ systems: endocrine tissues and neurosecretory cells. The glandular endocrine tissues are specialized for the synthesis and release of hormones. The most important endocrine glands in insects are the *prothoracic glands*, which secrete ecdysteroids during development, and the *corpora allata*, where the juvenile hormones are produced and secreted. The ovaries and testes of many adult insects also produce ecdysteroids.

Endocrine Tissues

The gross morphology of the prothoracic glands varies widely throughout insect phylogenies, both in size and in location in the body, but the cellular structure is quite uniform (Nijhout 1994). As in vertebrates, probably because of shared ancestry, the distinction between the nervous and endocrine systems is not clear-cut; homologous nerves of the CNS in all species generally innervate the prothoracic glands, and some of these are neurosecretory and are involved in regulating secre-

TABLE 10-2.
Insect Hormones.

Hormone	Site of Synthesis	Function
Prothoracicotropic (PTTH)	Neurosecretory cells in the brain	Stimulate secretion hormone of ecdysone
Ecdysteroids		Promote growth, control molting, embryonic development
Ecdysone (an ecdysteroid)	Thoracic glands	Induce apoptosis, cell division, degradation of old cuticle, production of new
Juvenile hormones	Corpora allata	Development, metabolism, behavior
Bursicon	Abdominal ganglia	Control of hardening of new cuticle

tions from the prothoracic glands (Nijhout 1994). These glands secrete, but do not store, *ecdysone*, a steroid that promotes growth and controls molting. In most insects, the prothoracic glands undergo apoptosis during the metamorphosis of the larva to adult stage and ecdysone is then no longer synthesized and released.

Molting is essential if an insect is to change size or shape because the insect's hard outer shell, or cuticle, cannot accommodate growth by expanding (in contrast to the endoskeleton of vertebrates, which grows along with the individual). The process is complex, involving the formation of a new cuticle *within* the old one while the old one is digested and the proteins reused. Therefore, the new cuticle must be protected against degradation by the digestive enzymes and at the same time remain pliable enough to expand when the old cuticle is shed (Nijhout 1994). Chemical assault to interfere with normal molting is one insecticide strategy.

Molting cycles are triggered by the secretion of the *prothoracicotropic hormone* (PTTH), which is produced by neurosecretory cells in the brain. The only known function of PTTH is to stimulate the secretion of the molting hormone ecdysone by the two prothoracic glands in the thorax. These two hormones trigger every molt, whether larva to larva or pupa to adult. Levels of a different set of hormones, the *juvenile hormones* (JHs), control metamorphosis.

The corpora allata are a pair of small glands found along the main vessel in the neck of most insects and are attached to the brain by a nerve that passes through the *corpora cardiaca* (additional small glands just anterior to the corpora allata). The corpora allata produce JHs. Innervation of the corpora allata is by nerves that conduct impulses, as well as by neurosecretory neurons. In some insects the corpora allata are also neurohemal organs for some neural secretions from the brain. The brain and these associated neurohemal glands form the *brain-retrocerebral neuroendocrine complex*, a control, synthesis, and excretion system that is the most important neuroendocrine organ in the insect endocrine system.

JHs serve both as regulatory and developmental hormones and have a role in every aspect of insect life, from development to metabolism to some behaviors. Three major forms of JH are known; some insects secrete only one, others two or

all three. After secretion from the corpora allata, JHs bind to the *juvenile hormone binding proteins*, which increase the solubility of the hormones in the hemolymph and protect them from degradation. JHs are lipid hormones, but their mode of entry into the cell, and subsequently into the nucleus, is not yet definitively known (Davey 2000). It may be that juvenile hormone binding proteins chaperone JHs into the cell and regulate their binding to JH-specific receptors once there.

As in plants, insect hormones have an antagonistic or complementary relationship with each other and highly tissue-localized, specifically timed gene expression. The presence of JHs prevents a juvenile insect from becoming an adult (thus the nomenclature) by suppressing secretion by the brain of hormones involved in molting and metamorphosis. The process is complicated and depends on JH-sensitive periods during the molting cycle. Basically, when the insect is JH sensitive and JH is present, the insect does not molt. If JH is absent, the developmental stage changes. This generally depends on the secretion of ecdysone having already initiated the next molt. Different sections of the epidermis have different JH-sensitive periods. The onset of JH-sensitive periods is independent of presence or absence of JH, and it is during this period that a cell can be committed to a specific developmental fate. Usually, it requires the action of *ecdysteroids*, however. JHs are pleiotropic and play different roles at different stages in the life cycle (see below).

Ecdysteroids are a family of steroids that includes ecdysone and its analogs and metabolites. These compounds are found in insects and crustaceans and in some plants as phytoecdysone. Nearly 100 of these compounds have been described in insects and other arthropods, and more than 200 have been isolated from plants (Sadikov et al. 2000). These hormones promote growth and control molting and play a role in embryonic development. Ecdysteroids are lipid hormones and act by enhancing or inhibiting gene transcription. The hormone-receptor complex usually binds a G protein inside the cell to initiate the signaling cascade.

Ecdysone is produced by the thoracic glands and is a relatively inactive *prohormone* that becomes active when converted by the fat body and epidermal cells into *20-hydroxyecdysone*, the most important molting hormone in insects (Nijhout 1994). Ecdysteroid action is the same at all stages of insect life; these hormones act on epidermal cells to induce apolysis (the first step in the synthesis of a new exoskeleton), cell division, degradation of the old cuticle, and production of the new. As with other hormones, ecdysteroid concentrations are affected by the concentration of other hormones. In this case, ecdysteroid secretion depends on the pattern of PTTH secretion in the brain.

When the new-stage insect emerges from the old cuticle of the previous stage, initiating a new *instar* (the phase between molts) in a process called *ecdysis*, the new cuticle must harden or *tan*. This is controlled by a neurosecretory hormone called *bursicon*, except in the higher Diptera in which it is controlled by a set of neurosecretory hormones called *pupariation factors*. The principal source of bursicon is the abdominal ganglia, and the hormone is released into the hemolymph from the abdominal previsceral organs, although it is synthesized throughout the nervous system, including the brain.

The ecdysone signal is transduced by the *ecdysone receptor*, a heterodimer formed by the joining of two nuclear receptor proteins, EcR (Ecdysone receptor) and Usp (*Drosophila* Ultraspiracle receptor), a homolog of the vertebrate nuclear receptor RXR (Retinoid X receptor) family of ligand-dependent transcription factors (TFs) (Arbeitman and Hogness 2000; Ghbeish et al. 2001; Mouillet et al. 2001). The RXR receptors have two signature domains, the DNA binding and ligand

binding domains. The ecdysone receptor requires not only heterodimerization for DNA binding, as do the RXR family of receptors, but also for ligand binding, a characteristic it does not share with the RXR receptors. Through alternative splicing and two promoters, the *EcR* gene encodes three protein isoforms, EcRA, EcRB1, and EcRB2, each with different quantitative control over transcription (Mouillet, Henrich et al. 2001). Each of the three isoforms is able to dimerize with the Usp receptor, and this may explain how it is that ecdysone can initiate the large variety of responses, at various stages in the life cycle, that it does (Mouillet, Henrich et al. 2001).

The ecdysone signal activates a hierarchical response when bound to the receptor, turning on a small set of early genes that in turn activate a larger set of genes downstream. In vertebrates, activation of the homodimeric steroid receptors depends on the presence of a molecular chaperone-containing heterocomplex (MCH), which interacts with the receptors, probably to facilitate protein folding and DNA binding by their ligands. The presence of an MCH seems to be required to activate the ecdysone pathway as well (Arbeitman and Hogness 2000).

Molting is a good example to illustrate the importance under some circumstances of centrally coordinated signaling. We mentioned in Chapter 9 that some structures like limbs or teeth cannot develop until a place has been prepared for that to happen. In this case, prepatterning of cells to respond to later signals is likely to be a mechanism that enables this delayed, context-specific response to occur. An entire insect must be ready before molting can occur. At the appropriate time, a central signal is released that affects the whole body.

Neurosecretory Cells

The second insect organ system that secretes hormones consists of the neurosecretory cells of the CNS, neurons that produce small polypeptides, or neurohormones. These cells tend to be localized in the brain, although they are found in all the ganglia of the CNS; they have axons that end in *neurohemal organs* or *areas*, where the secretions are released directly into the hemolymph.

Adult insects do not undergo molting or development, so in adults the same ecdysteroids, JHs, and various neuroendocrine hormones that controlled these processes earlier in life control adult processes like diapause, migratory behavior, and reproduction—the production of yolk proteins, maturation of the ovaries, and synthesis of eggs (Hartfelder 2000; Nijhout 1994). As a generalization about hormone action in insects, a given hormone can have different effects in different target tissues and different effects on the same tissue at a different time in the life cycle. Once again we see the use and reuse of a mechanism, showing the importance of context specificity in the inducing mechanisms (which themselves may have multiple uses). As we have mentioned in Chapter 9 in context with the apparently inconsistent inductive/inhibitory effects of Dorsal protein, the specific effect of a signaling substance ultimately depends on the nature of the regulatory regions associated with a gene, and the combination of factors that must bind jointly there to activate or inhibit a gene.

OTHER ARTHROPODS

Crustaceans have a number of endocrine mechanisms in common with insects, including production of *methyl farnesoate* (MF), a hormone that is the precursor of insect JH. MF seems to play a role in reproduction in crustaceans. Interestingly,

although many of these substances are found in insects and crustaceans, they do not necessarily play the same role in both.

The eyestalk neurosecretory complex of decapod crustaceans is the location of the sinus gland, the source of a number of neuropeptide hormones, including those involved in pigment migration (*red pigment-concentrating hormone* and *pigment-dispersing hormone*), regulation of carbohydrate metabolism (*crustacean hyperglycemic hormone*, CHH), molting (*molt-inhibiting hormone*, MIH), and gonadal growth (*gonad-inhibiting hormone*, GIH) (Webster 1998). These neuropeptides have been found to play an inhibitory role, acting on endocrine tissues that produce the hormones that initiate reproduction and molting. DNA sequencing has shown that the genes for these hormones are structurally related. Again, they are also pleiotropic; CHH, for example, has a role in molting and in reproduction as well as in carbohydrate metabolism. Other neuroendocrine centers are located throughout the CNS, including, as well as the eyestalk, the brain and the subesophageal ganglion.

VERTEBRATE ENDOCRINE SYSTEM

As in other organisms, because hormones are a fundamental way that multicellular organisms “solve” the problem of intercellular communication and the regulation of local gene expression to serve the whole, vertebrates have evolved a substantial array of hormone-producing glands, genes, and systems that affect all aspects of life: reproduction, growth and development, maintenance of a stable internal environment, mental functions, physical activity, food seeking and satiety, many behaviors, and regulation of energy balance (Ojeda and Griffin 2000). Rudiments of these systems are found in chordate “relics” like amphioxus. As noted earlier—and as we have come to expect—at least some of these are evolutionarily related to substances found in invertebrates.

ENDOCRINE GLANDS AND HORMONES

The vertebrate endocrine system classically consists of a number of ductless glands that produce hormones, with different functions, for release directly into the bloodstream (Table 10-3). However, a number of other glands and cells secrete hormones, and in fact some cells produce hormone for use within the cell and do not secrete it at all. These “nonclassical” hormone-secreting organs and their products are listed in Table 10-4 (Kacsoh 2000; Ojeda and Griffin 2000).

Hormones can have several effects on a target tissue, and several hormones may have the same effect. Also, as in plants and insects, hormone concentrations are controlled by feedback mechanisms, that is, they are up- or downregulated by the concentration of hormones or other compounds in the blood. Increasing levels of glucose in the blood, for example, stimulate the release of insulin, a polypeptide hormone, from the pancreas. Increased amounts of insulin in the blood in turn stimulate glucose uptake by the liver and its conversion to glycogen, and subsequent lower blood glucose levels lead to decreased secretion of insulin and slower glucose uptake.

The pancreas illustrates an interesting point about the evolution of organ systems. It is an evolutionary and developmental outcropping of the gut and also secretes digestive enzyme from a different set of cells; this indicates the general

TABLE 10-3.
Classical Endocrine Glands and their Hormone Products.

Endocrine Gland	Unit	Main secretory products
Pituitary gland (hypophysis)	Anterior lobe	Growth hormone (GH); prolactin (PRL), adrenocorticotropin (ACTH); gonadotropins (follicle-stimulating hormone (FSH); luteinizing hormone (LH)); thyroid-stimulating hormone (TSH); β -lipotropin, β -endorphin
	Intermediate lobe	Melanocyte-stimulating hormone (MSH), β -endorphin
	Posterior lobe	Oxytocin (OT); arginine vasopressin (AVP or antidiuretic hormone (ADH))
Adrenal gland	Cortex	Aldosterone; cortisone (F); androstenedione; dehydroepiandrosterone (DHEA); DHEA-sulfate (DHEAS)
	Medulla	Adrenaline (epinephrine); noradrenaline (norepinephrine)
Testis	Leydig cells Sertoli cells	Testosterone (T); estradiol (E ₂); inhibin; Müllerian inhibitory hormone. (MIH or anti-Müllerian hormone (AMH))
Ovary	Hilar cells	Testosterone
	Follicles	Estradiol (E ₂); androstenedione; testosterone; inhibin
	Corpus luteum	Progesterone, estradiol (E ₂); inhibin
Thyroid gland	Follicles	Thyroxine (T ₄); triiodothyronine (T ₃)
	Parafollicular cells	Calcitonin (CT)
Parathyroid gland	—	Parathyroid hormone (PTH)
Pancreatic islets	—	Insulin; glucagon; somatostatin (SRIF); pancreatic polypeptide
Placenta		Human chorionic gonadotropin (hCG), human placental lactogen (hPL), progesterone, estrogen
Pineal gland (epiphysis)	—	Melatonin, biogenic amines, several peptides

TABLE 10-4.

Nonclassical Endocrine Glands and their Hormone Products.^a

Organ	Unit	Main secretory products
Brain	Especially hypothalamus	Corticotropin-releasing hormone (CRH), thyrotropin-releasing hormone (TRH), luteinizing hormone-releasing hormone (LHRH), growth hormone-releasing hormone (GHRH), somatostatin, growth factors (fibroblast growth factors), transforming growth factor- α (TGF- α), transforming growth factor- β (TGF- β), insulin-like growth factor I (IGF-I)
Heart		Atrial natriuretic peptides
Kidney		Erythropoietin, renin, 1,25-dihydroxyvitamin D fibroblasts
Liver, other organs,		IGF-I
Adipose tissue		Leptin
Gastrointestinal tract		Cholecystokinin (CCK), gastrin, secretin, vasoactive intestinal peptide (VIP), enteroglucagon, gastrin-releasing peptide
Platelets		Platelet-derived growth factor (PDGF), TGF- β
Macrophages, lymphocytes		Cytokines, TGF- β , proopiomelanocortin (POMC)-derived platelets
Various		Epidermal growth factors (EGF), TGF- α , neuregulins, neurotrophins

^aFrom (Ojeda and Griffin 2000).

nature of endocrine organs as related to other developmental systems. *Amphioxus*, a primitive chordate thought to resemble the common ancestor of all chordates, has been found to express pancreatic hormones in the region of its gut, but without any morphologically distinct pancreatic organ.

HORMONE FAMILIES

Like insects, vertebrates have directly coded hormones and others produced by enzymatic reactions. Vertebrate hormones are classified by their chemical nature into four major groups: amines, which are modified versions of the amino acid tyrosine; peptide hormones, which are short chains of amino acids; polypeptide hormones, strings of up to 200 or so amino acids; and steroids, which are lipids synthesized from cholesterol, the only class of hormones that are not direct gene products. These molecules differ in size, affinity for water, and locus of action, but, like hormones in plants and insects, and in fact all signaling molecules, they all bind to receptors on or in their target cell to effect action.

Expectedly, many hormones can be grouped into families of homologous molecules. Members of the same family have high DNA sequence homology, but they also tend to cross-react with the receptors for other members of the family, and they generally use similar mechanism or secondary messenger system to transmit their signal. The affinity with which a hormone binds to its receptor can determine the efficiency with which it transactivates a gene and thus the level at which that gene is expressed. For a given concentration of hormone, low-affinity binding will lead to the synthesis of less protein than high-affinity binding.

The amines include thyroid hormones, melatonin, and catecholamines, all derived from the amino acid tyrosine. These hormones are synthesized in the thyroid gland in the neck, are hydrophilic, and can cross cell membranes by diffusion; in fact, however, thyroid hormones tend to exit the cell by transport, packed into granules and released from the cell by exocytosis. This is the same mechanism cells use for the secretion of a number of other compounds.

The protein families of peptide and polypeptide hormones include the *insulin*, *glycoprotein*, *growth hormone*, and *secretin* families. The insulin family includes *insulin* and *relaxin*; the glycoproteins are *luteinizing hormone*, *follicle-stimulating hormone*, *thyroid-stimulating hormone*, and *chorionic gonadotrophin*; the growth hormone family includes *growth hormone*, *prolactin*, and *chorionic somatomamotropin*; and the secretin family includes *secretin*, *glucagons*, and *gastrointestinal polypeptide*.

These hormones are synthesized in the rough endoplasmic reticulum (ER) of their secreting cells. The protein as first translated is longer than the mature hormone will be and at this stage is called a *prohormone* or a *preprohormone*. The transient leading amino acid sequence, or *leader*, is hydrophobic and allows the hormone to be moved into the ER in an inactive state. The leader is cleaved when the hormone has crossed the membrane of the ER to be transported to the Golgi apparatus, where it assumes its mature conformation and then is stored in a granule that fuses with the cell membrane to be released into the perivascular extracellular space, again through exocytosis.

There are five major classes of hormone in the steroid family including the estrogens (estradiol, estriol, estrone), progesterone, androgens (DHEA, testosterone, androstenedione), glucocorticoids (cortisol, cortisone), and mineralcorticoids (aldosterone). Cholesterol is the precursor molecule, and it is transported in the bloodstream to the testes, the ovaries, and the adrenal glands, endocrine organs that convert it into steroid hormones under the direction, in turn, of hormone signals from the brain. Because they are by and large hydrophobic, steroids can readily pass through cell membranes, both to exit the cell of synthesis as well as to enter the target cell.

Inside the cell, steroids trigger gene expression by binding to nuclear receptors, which bind to DNA to transactivate gene expression. They may also have a non-genomic effect; receptors for steroid hormones also are found on cell surfaces, and receptor-hormone complex on the cell surface initiates not gene transcription but changes in intracellular concentrations of various ions by activating a second-messenger cascade.

Although these hormones are chemically related and activate gene transcription in a similar way, they have different roles in cellular physiology. The glucocorticoids and mineralcorticoids are involved in the regulation of cellular homeostasis and metabolism, whereas the estrogens, progesterones, and androgens are sex hormones,

involved in the development of secondary sex characteristics as well as reproductive function.

HORMONE TRANSPORT

Once in the bloodstream, hydrophilic hormones can circulate freely but hydrophobic hormones, the steroids, are generally bound to a protein carrier, which is either specific, such as *sex hormone binding protein*, or general, like *serum albumin*. As with most generalizations about the endocrine system, there are exceptions to these rules. Some hydrophilic hormones, such as IGF1, circulate bound to proteins, and this greatly extends their half-life, whereas some hydrophobic hormones, such as aldosterone, circulate unbound. Control loops and feedback mechanisms regulate the concentration of hormones in response to physiological processes and needs, and the mode of circulation determines the rate at which hormones are cleared from the bloodstream.

Hydrophilic hormones mix freely in aqueous solution that is chemically charged but cannot normally pass through the uncharged lipid cell membrane. As a result, information from hydrophilic hormones must be transduced by ligation with specific cell-surface receptors on appropriate cells that are receptive to the signal. Receptors for hydrophilic hormones are located on cell surfaces, whereas receptors to the hydrophobic hormones, the steroids, are intracellular, usually located in the nucleus. There are exceptions: Some lipophilic hormones, melatonin and the eicosanoids, like steroids, do bind to cell surface receptors to initiate cellular response.

Hormone signal transfer via extracellular ligand-receptor binding works via second-messenger cascades, mostly via single-transmembrane receptors. The basic means of information transfer was described in Chapter 7. The intracellular receptors for the hydrophobic hormones that pass through the cell membrane generally also function as TFs; the estrogen and glucocorticoid receptors are examples. When the receptor is bound with hormone, a conformational change in the receptor allows it to bind regulatory DNA sequences, which then activate gene expression. Some receptors bind to regulatory sequence in the absence of hormone but act as transcription *repressors* until bound with hormone; the thyroid hormone receptor is an example. When bound to thyroid hormone, however, the receptor-ligand configuration stimulates transcription of thyroid hormone-inducible genes (Koenig 1998).

The location of a protein within a cell is generally well regulated, and this determines the other molecules with which it interacts and thus the signaling cascade in which a signaling factor participates at any given time. In Chapter 6 and elsewhere we have referred to the sequestration of chemical cascades, themselves hierarchical and contingent based on the availability of substrates at each stage and the like. Sequestration within the cell controls the level of cross-reaction between systems, especially when they might use some of the same constituents, like amino acids or second-messenger components. The cell is also able to control the timing of the entry of transcription factors into the nucleus and thus the timing of gene expression (Downward 2001).

HORMONE EVOLUTION

Hormones cannot bind to DNA on their own but must first be bound to a receptor. How this linkage evolved is not clear. Did every mutation in a receptor have to

weather the blows of natural selection until a comparable mutation arose in its ligand? This seems unlikely, but complex organisms depend extensively on such coevolution, the mechanisms of which in general are not well understood. This is another element of the “problems” of specificity, cross-reactivity, and coevolution that face systems that have multiple components evolving by modularity and duplication. However, some aspects of hormones and their receptors may help elucidate this.

A number of “orphan” nuclear receptors have been identified (by their shared amino acid sequence similarity with receptors for which ligands are known) for which a ligand has not yet been found and may not exist. Indeed, some receptors (i.e., TFs encoded by genes in a receptor gene family and having the prototypical receptor structure) seem to be able to function as TFs independently of ligand. Two possible evolutionary scenarios have been proposed: either the ancestral nuclear receptors were orphan and bound to DNA as homodimers, some of them later gaining ligand-binding capability independently (Laudet 1997), or the ancestors were liganded and some of the receptors independently lost their ligand-binding capability (Escriva et al. 2000). This supposes that the ligands—hormones—existed before ligand-receptor binding became part of the working cellular repertoire, and in fact, as we have seen, plants make steroid hormones, as do fungi, so the ancestral molecule was clearly ancient.

The affinity with which a ligand binds to its receptor varies, and indeed, some hormones bind to more than one receptor and some receptors bind more than one ligand. This kind of cross-reactivity might in principle “confuse” the organism, and there is certainly variation in nature that could reflect this (e.g., deviations from stereotypy in morphological or behavioral sexual dimorphism among individuals), but probably also provides a measure of protection via redundancy.

Finally, all of the nuclear receptors that have been identified in cnidarians are unliganded, suggesting that this is the ancestral state (Grasso et al. 2001). The most parsimonious explanation of the evolution of hormones and receptors is perhaps that an ancestral receptor used in some other way evolved characteristics that also would bind a ligand and, after a gene duplication event and subsequent mutation, each receptor acquired the capability of binding to different ligands. This could have involved a period of cross-reactivity (which if too harmful would have been removed by selection), followed by divergence of function. We are largely handwaving here, however, because this does not explain why receptors in different subfamilies now bind ligands with no structural homology (that is, they are not themselves related to each other), how photoreceptors came to respond to light rather than chemical ligands, whether protohormones were TFs that alone could bind DNA, and why they may have lost that ability, among other provocative questions.

Perhaps the tightly paired ligands and receptors we observe and catalog today are those for which appropriate cross-reaction or coevolution happen to have occurred, whereas other signals evolved independently of ligand (e.g., some of the orphan receptors, which in fact are the majority of nuclear receptors) or ligands evolved independently of receptors (nitric oxide is an example) or have disappeared without them. The exchangeability of regulatory mechanisms means that in principle any one of many possible receptor-ligand pairs could carry out a given regulatory function and what we see today may just be those that happened to have been used. At the same time, similar systems and the use of gene family members in different species for the same job, or within species for related jobs, clearly suggests that coevolution has been important.

In this context it is interesting to ask why a truly unliganded receptor would maintain a viable ligand binding domain (that is, why would mutation not have erased the organized nature of that domain, given no selective pressure maintaining its ligand binding structure?). This could, of course, simply reflect limits on our current knowledge; ligands continue to be found for “orphan” receptors (Gustafsson 1999), thus causing the receptor to be termed “adopted” (Chawla et al. 2001). It is now recognized that some adopted receptors play a role in lipid sensing, forming heterodimers with retinoid X receptors, with low-affinity dietary lipids as ligands (Chawla, Repa et al. 2001). Also, as research goes, a gene may be long studied in the context in which it was originally found, and only later discovered to have entirely different functions. Such discoveries often tip off distant homologies among the functions, but this need not be the case.

However, orphan receptors commonly bind DNA as monomers rather than the heterodimers that steroids bind as, suggesting that this class of receptor acts in a way that is different from other receptors, whether by being truly unliganded or something else not yet understood. If any of these receptors truly have no necessary ligand, the fact that they still share the structure of liganded receptors may represent their occasional ability to cross-react in ways that have enough importance to be supported by selection, such as some undocumented specialization. On close inspection, history suggests that genes in the same family may not be as completely redundant as some experiments initially suggest.

STEROIDS

Steroid hormones seem to have been highly conserved until the divergence of jawed from jawless vertebrates some 450 mya. Figure 10-1 shows the structure of a number of steroid hormones. The structural diversity is produced by variation in the synthetic pathways and hence the genes that encode the synthesis of these molecules. Genes for nuclear hormone receptors form a large gene superfamily.

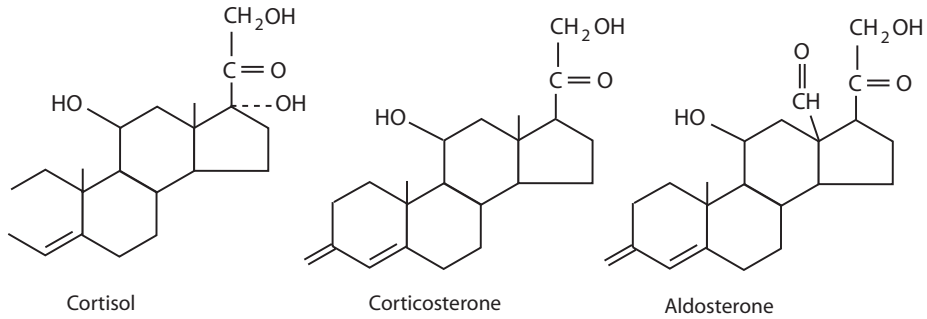
PEPTIDES

Peptide hormones are typically very short strings of amino acids. Comparative sequencing suggests that after the divergence of jawed vertebrates peptide hormone genes underwent periods of sustained and rapid bursts of change, although there seems to be significant variation in the basal evolutionary rates of these genes (Liu et al. 2001; Wallis 2000). As has been suggested by studies of other genes in the vertebrate lineage, genes for protein hormones show some evidence of two whole-genome or at least large chromosomal duplication events early in the evolution of vertebrates; for example, when there is a single gene for a given hormone in *Drosophila*, four paralogous genes are found in vertebrates, creating large gene families for many hormones (Ohno 2001), and some teleost (bony) fish are tetraploid. Nonprimate mammals and the bush baby, for example, have a single-copy growth hormone gene, whereas humans and the rhesus monkey have five (Liu, Makova et al. 2001).

COEVOLUTION AT HIGHER LEVELS

Genes related to life history stages, such as different types of metamorphosis in various animal groups, must experience various forms of coevolution with other

A. Adrenal Steroid Hormones



B. Gonadal Steroid Hormones

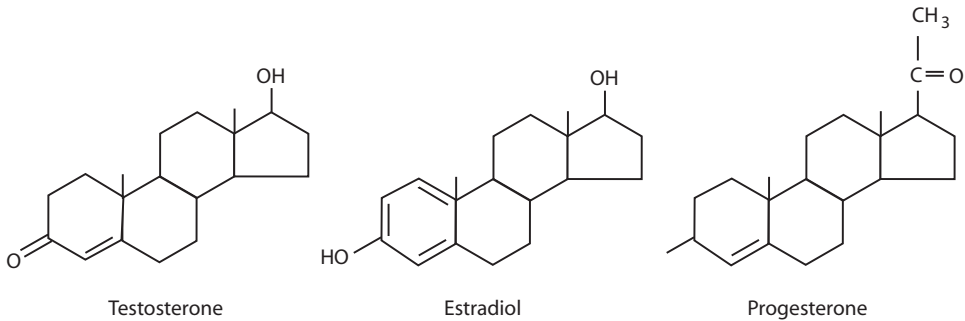


Figure 10-1. The chemical structure of (A) adrenal and (B) gonadal steroid hormones.

functions. These would seem to be complex, but they can evolve relatively rapidly. Thus some insects have complete metamorphosis (holometabolism) whereas others have no larval or pupal stages. Similarly, in groups of sea urchins, tunicates, or amphibians, we find that *within* some branches of the phylogeny some species experience more and others less complex development (Raff 1996). Tadpoles are found in various branches of frogs, the stage apparently having evolved independently.

All insects make ecdysone, and ecdysone controls insect metamorphosis, but metamorphosis requires many physiological changes and thus many other proteins and functions. Genes for apoptosis, for example, genes for the synthesis of new cuticle, and genes for the hormones that control other stages of metamorphosis must have all coevolved, more or less tightly, yielding the array of stages of metamorphosis that we see today.

The signaling coordination responsible for these developmental traits may involve many genes and their responses but clearly is something easy and relatively simple to evolve. Some amphibians, such as the Mexican axolotl, do not undergo metamorphosis in the normal life course, but metamorphosis is inducible with exposure to thyroid hormone. Therefore, the axolotl is still genetically capable of losing its neotenuous state, but this can only happen given the provocative physiological

state. The key probably lies in the quantitative or qualitative control of a small number of critical developmental, signaling, hormonal, or growth factors.

Some invertebrate hormones have marked homology to vertebrate counterparts and thus are likely to have common ancestors, which these days is no surprise. However, homologous hormones do not necessarily perform homologous functions across species, just as homologous genes are not necessarily used in homologous pathways, and other hormones seem to be found only in invertebrates, suggesting that they arose after the vertebrate-invertebrate divergence.

Like the plant story described earlier, animal hormone function (e.g., reproductive cycling and behavior in vertebrates) may involve timed, quantitative differences among a series of hormones. Not surprisingly, in some instances at least the hormones are either functionally diverged members of a gene family, or modifications of a common base molecule (steroids). Once again, we see modular evolution by duplication with divergence leading to partial sequestration of related function.

RECEPTORS

We introduced nuclear receptors in Chapter 7 (Table 7-3). They bind ligands that make their way directly into the cell, rather than intercepting them in the extracellular space. The first nuclear receptor is likely to have arisen some 1000–800 mya, among the first metazoans. Nuclear receptors have not been found in plants. Virtually all vertebrates have the same six steroid receptors (estrogen receptors α and β , progesterone receptor, androgen receptor, glucocorticoid receptor, and mineralocorticoid receptor). Three steroid receptors have been found in agnathostomes rather than six, including an estrogen receptor, a progesterone receptor, and a corticoid receptor but no androgen receptor (Thornton 2001; Thornton and Kelley 1998).

Despite the lack of structural homology between steroid hormones, thyroid hormone, and vitamin D₃ molecules themselves, their nuclear receptors are homologous enough to be considered a steroid receptor superfamily. Nuclear receptors as a class are modular, consisting of four or five domains that function autonomously, and can be interchanged between related receptors without loss of function. This modularity presumably reflects a history of exon shuffling early in the evolution of the genes that code for these molecules (e.g., Patthy 1999).

Phylogenetic analysis of nuclear receptors led Laudet to propose six subfamilies of receptors, with members grouped by how they dimerize and bind DNA (Laudet 1997). Most of the six subfamilies are ancient and have receptors in both arthropods and vertebrates. The diversification of the superfamily may be the result of two waves of gene duplication (Escriva, Delaunay et al. 2000), which occurred before the divergence of lamprey and jawed vertebrates (Baker 1997; Thornton 2001; Thornton and Kelley 1998; Wallis 2000). There is no link between the ligand a receptor binds and the subfamily in which it is classed. That is, receptors that are closely related phylogenetically bind ligands with totally different biosynthetic pathways, suggesting that if nuclear receptors coevolved with their ligand, it was not a straightforward gain or loss of function during a gene duplication event, for example.

Phylogenetic analysis suggests that the ancestral steroid receptor was an estrogen receptor of some kind, with the androgen receptor emerging sometime after the vertebrate divergence. In lamprey, it is estrogen that regulates the development to sexual maturity in males as well as females, not androgen, suggesting that hormonal control over sexual dimorphism is relatively recent (Thornton 2001). A min-

eralocorticoid receptor has not been found in fishes or other lower vertebrates, indicating that it is the most recent of the steroid receptors. Agnaths apparently make most steroids, including testosterone, for which they have no receptor (personal communication, Thornton).

G PROTEIN-COUPLED RECEPTORS

G protein-coupled receptors (GPCRs) were described in Chapter 7. As noted briefly in Table 7-4, there are several classes. The genes can be grouped into families in a number of different ways: by the molecular weight of their ligands, by the structure of their α -subunits, or by conserved amino acid structure. Five or six classes are conventionally agreed upon, and these include family I, the largest group of receptors, related to rhodopsin receptors; family II, including calcitonin-, PTH-, glucagon-receptors, and so forth; family III, containing metabotropic glutamate receptors and others, including a subgroup of vomeronasal receptors; family IV, comprised of STE2 yeast pheromone receptors; family V, yeast STE3 hormone receptors; and family VI, receptors related to slime mold cAMP receptors (Josefsson 1999). (These classes are sometimes designated by letter, A–F).

Phylogenetic analysis based on DNA sequence yields three large clades and several minor ones, all of which arose more than 800 mya (Josefsson 1999; Wess 1998). In this analysis, families I, II, V, and VI plus some unclustered receptors form one large group, family III forms a smaller clade essentially as already classified, and family IV forms a final cluster. GPCRs in these families generally bind classes of ligands with similar functions, but with much diversity between the classes (Josefsson 1999). Receptors from these families were present in the acoelomate flatworms of the Precambrian, showing that intercellular signaling is an ancient characteristic of multicellular organisms and probably of single-celled organisms.

Phenogenetic drift seems to have played a role in GPCR evolution. These genes comprise a large fraction of the genome (about three percent in mammals) and are used in a huge diversity of pathways. There is much structural diversity in the receptors as well as their ligands, even though they all operate similarly at the molecular level, so over evolutionary time there have been substitutions in the details but conservation of the mechanism and its basic logic.

GPCRs have been found in plants, but they do not seem to be as important in intercellular signaling in plants as they are in metazoans. A single gene for the α -subunit of the heterotrimeric G protein and one for the β -subunit are known in *Arabidopsis*, and homologous genes have been found in other plant species. In plants, calcium is the most important transducer of intracellular signals, as well as being involved in control of turgor pressure, cell growth, cell division, and other processes, with downstream effects including ion channel activation, gene expression, and vesicle fusion (Malhó et al. 1998). It is also an important mediator of environmental and intercellular signals. How cells interpret calcium-transduced signal is still unclear, but calcium-binding proteins are involved. One set, the *Calmodulin* genes, bind to and regulate a wide variety of signaling proteins such as kinases, which phosphorylate transcription factors, cAMP phosphodiesterase, which degrades cAMP, as well as other enzymes, ion channels and pumps, various cytoskeletal components, and the like. Calmodulin is found in all eukaryotic cells. Each Calmodulin molecule binds four Ca^{2+} ions and changes conformation on binding, which allows the complex to bind to and activate molecules downstream.

Beyond its role in inducing conformational change in binding proteins, calcium action in a cell is dependent on spatiotemporal factors, such as the timing and location of transient membrane ion channels. Distribution of Ca^{2+} is not uniform throughout the cell but, on signaling, spikes at specific intracellular locations. At rest, cytosolic calcium concentration is maintained at between 10 and 100 nM, but it can peak to between 1 and 5 μM . These transient spikes, or waves, are complex and not well understood, but it is known that the wave is initiated at a defined site by a defined signal or by passing through the plasma membrane at a single location. The wave is not transmitted by diffusion but by continuous release of calcium from sub-cellular stores, the ER, mitochondria, or other vesicles, followed by reuptake by cell compartments as the wave moves along. The wave can be constrained to specific locations in the cell by the factors involved in its production, and thus different parts of the cell can be separately regulated.

In Chapters 6 and 7 we described ion channels and the families of genes whose members complex to form these pores in cell membranes. These, too, have an evolutionary history. For example, there are 1 or 2 Na^+ channel genes in invertebrates but 10 in mammals. Mammalian channel genes are found on four chromosomes, suggesting multiple duplications of an ancestral chromosome, which happened before divergence of teleosts and tetrapods. Sodium channel genes are linked to *Hox* gene clusters (Lopreato et al. 2001).

CONCLUSION

We have looked at some of the specific ways that cells communicate with each other within and even between organisms. The purpose of such communication, if “purpose” is an appropriate word, is for one cell to effect changes in other cells, and this generally means changes in gene expression. Because this is so central to multicellular, organized organisms it is no surprise that many, elaborate mechanisms, with their own checks and balances, have evolved in plants and animals.

Communication of this sort is inherently four-dimensional, involving space and time (and perhaps one could suggest additional dimensions, because time and space mean different things within and between cells, and perhaps there is an additional “concentration” dimension). From immediate changes to gradual life span changes that can take decades (e.g., puberty, tooth eruption, and perhaps even some aspects of aging), or be seasonal (flowering, dormancy), cells communicate with their diversity of molecular mechanisms. It is this that produces life cycles and enables organisms to undergo development and to maintain homeostasis. In this regard, a high level of predictability is often vital. For example, stages of early development follow quickly on each other and developmental stuttering can be fatal. Similarly, mating requires that both sexes be prepared at the same time. As we mention in several contexts, the same mechanism is used *between* organisms, as in mating-related signaling, adding subtlety regarding the nature of what we call an “organism” or “being” if the reproductive behavior of one depends intrinsically on the behavior of another.

These statements apply within an organism, but we have discussed aspects of the compatible change between receptors and ligands that must occur over evolutionary time. As gene families proliferate, selection and drift must maintain at least a sufficient set of compatible ligand-receptor sets for existing function and to evolve new or modified function. In fact, additional elements of coordination are required,

because the proper receptor, among its frequently numerous gene family alternatives, must be expressed in cells if they are to respond and signal-sending cells must express the appropriate ligand gene. Selection by phenotype keeps the internal chaos within bounds, but how this happens is not yet understood in detail.

But if this control is coordinated and predictable among cells, there is also a role for chance. With some exceptions such as in early development, two individuals rarely undergo changes at precisely the same time or age, and this applies to genetically near-identical animals like clones or inbred strains of mice, or plants. The similarities may be great, and we do not yet know the relative roles of somatic mutation, environment, and chance, but these factors certainly do play a role and perhaps provide opportunities for selection to act. Generally, the earlier in development, the fewer cells in an organism are committed to a particular fate and the greater the effect of a chance interruption in the nominal flow of events.

Similarly, the ability to respond to signals from the outside in itself provides organisms with a tolerance for and an ability to respond to environmental change or stress, but only within limits. Plants, for example, produce hormones to protect against drought or temperature extremes, but beyond predictable thresholds—extended drought or freezing or extreme heat, for example—they cannot survive. It is important to note, too, that the limits vary by species: tundra plants tolerate colder climates than jungle plants, flamingos tolerate warmer temperatures than penguins. Variation in response to change in environment is sometimes called the *reaction norm* (Schlichting and Pigliucci 1998). Many factors are involved, most of them unknown, but they surely will be shown to be intimately connected to cell signaling mechanisms.

Many aspects of homeostasis and life cycles are basically hardwired into the organism's makeup, with stereotyped responses sometimes rather precisely determined by genetic mechanisms. Temperature sensitivity, mating seasons, and the like may be examples. These are rather specifically predictable. Yet many vital aspects of the environment cannot be predicted specifically, and perhaps the ability of organisms to respond to things they cannot anticipate in advance is the most interesting aspect of the nature of complex organisms—even plants. Much of the remainder of this book is about what we refer to as “open-ended” sensory systems—senses such as vision and hearing, smell and taste. The organism is enabled to detect various *kinds* of externally derived input, but cannot in advance know what its details will be—yet has to be able to recognize and respond. The organism can recognize pathogens, neurons store memories of unique experiences, the heart can slow or speed its beat in response to the amount of work the organism is doing, animals can perceive a huge number of smells, we can see light within specific frequencies, and so forth. Its response involves internal communication and coordination of the sorts described here, among others.

Yet, if these abilities are crucial for the life-ways of the organisms that possess them, in fact they and their responses are not entirely open-ended or unrestricted. They typically fail above or below predictable thresholds. Primates cannot see UV light, our hearts cannot beat any faster after a certain point, we cannot defend against all types of invading pathogens, etc. The fact that a system is not perfect and has its failure limits, however, is less curious evolutionarily than is often thought: if the system is here, it has been good enough to get here. That is the only interest of evolution. It is to these systems that we now turn.

Chapter 11

Detecting and Destroying Internal Invaders

A substantial fraction of organisms that have ever lived have probably died as a result of predation, and much of evolution and of the structures of organisms can be viewed in the context of the battle to eat or be eaten. And as life would have it, many organisms alive today survive only by consuming other forms of life. Predation, broadly defined as obtaining nutrition from other living organisms, is a primary way in which organisms secure their energy needs. As a consequence, most animals and plants are meals-on-a-platter for some other organism(s). *Not* being consumed by somebody else before reproducing is a primary requisite for darwinian fitness.

Avoidance of predation by large organisms has clearly been important, and the living world has evolved many mechanisms of detection, defense, and escape. However, probably the first, and certainly still among the most dangerous, forms of predation is attack by microorganisms such as bacteria, viruses, and other parasites. These organisms are too small to be perceived by the cognitive or “organismal” sensory systems by which larger predators are perceived. Micropredators enter the body, where they can reproduce and grow and attack their prey from within. Typically they can reproduce rapidly, becoming in a sense an internal megapredator. There thus ensues a race against the attack.

Not only are microorganisms small, but they frequently attack through molecular rather than physical means, by recognizing and binding to various cell surface molecules within the prey and entering the cells, or even insinuating themselves into the host’s DNA. But if their small size makes their presence difficult to detect cognitively by the host, their size means that they can be defended against by molecular means. This is convenient for complex organisms whose life is itself based on means to interact by molecular signaling and recognition. Defense against microorganisms can do double duty as defense against nonreplicating but potentially threatening exogenous microscopic or molecular entrants into the body—toxins, pollen, foreign protein, and the like. Even if not alive, such agents can be dangerous if exposure is extensive or extended over time.

All multicellular organisms have evolved mechanisms for protecting against molecular attack. Given that the number of pathogens and their ability to evolve rapidly greatly outstrips that of most of their prey, self-defense is no small task. It would seem, on the face of it, an almost impossible battle for big, long-living (and sometimes stationary) organisms to outcompete the swift and nimble microparasite attack. Yet this obviously happens.

The challenge in molecular self-defense is not just a matter of recognizing alien organisms within. In some way or another, an organism fighting molecules with molecules must obey the dictum “above all, know thyself.” That is, organisms must be able to distinguish their own cells from pathogens, and to preferentially target the pathogens for destruction, a daunting requirement. Yet complex organisms are themselves molecularly very diverse and, due to somatic mutation and cellular differentiation, changeable during their lifetimes. Some organisms have very complex mechanisms to deal with these challenges, but others get along very well with simple mechanisms.

Some aspects of immune defenses are shared between extremely divergent organisms. In this chapter, we will discuss the immune systems of insects, plants, and animals and what is currently thought about how they evolved. Because it includes most of the components found in the immune systems of other organisms, we will begin with the most complex system, that found in vertebrates. In addition, although at first blush the necessity for an immune system at all seems to confirm the idea that life is primarily a battle to the death between competing organisms, in fact, many if not the most successful pathogens do not kill their hosts but instead have evolved a coexistence; most organisms peacefully serve as hosts to many microorganisms, and indeed cannot do without them. The challenge to live a completely aseptic life might be—or at least has proven to be—beyond the capacity or the interest of evolution.

VERTEBRATE IMMUNE RESPONSES

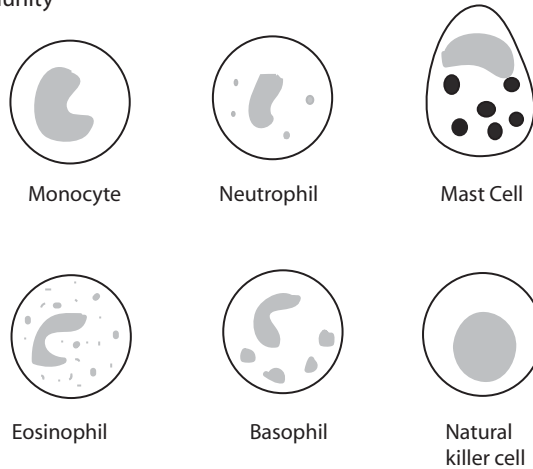
All vertebrates have what is traditionally termed an *innate* immune response, a rather generic, nonspecific system. Gnathostomes (jawed vertebrates) also have an *adaptive* response, that “learns” via antigen-antibody reactions to combat specific pathogens. Innate immunity does not improve with subsequent exposures to a pathogen. Adaptive immunity is specific and has *memory* in that the adaptive response to a pathogen improves with each subsequent exposure during an individual’s life, but this knowledge is not transmitted to the next generation. This does not mean that jawed vertebrates have a more effective immune system in general; agnathic vertebrates are still around, and their rarity relative to gnathostomes is generally attributable to their having been outcompeted by jawed vertebrates, not pathogen attack. However, it certainly raises the question as to what forces led to the evolution of the complex adaptive system (see below).

INNATE VERTEBRATE IMMUNE RESPONSES

The overwhelming majority of multicellular organisms face the world with innate immunity alone. These systems may work against particular types of pathogen, but do not depend on recognizing specific organisms. Innate immunity begins with physical barriers, such as the skin or bark, mucous membranes, the cough response, chem-

ical barriers such as pH of the stomach, or temperature, which produce conditions in which microorganisms cannot live. If a microorganism breaches these first lines of defense, the innate immune system detects this and launches an immediate response. Innate immunity in vertebrates is mediated by cells of a general class known as *phagocytes*, of which specific types called *macrophages* and *neutrophils* use surface receptors for common bacterial components to trap, engulf, and destroy the pathogen. Cells of the immune system are shown in Figure 11-1. The *complement system*, described below, is also activated, to *opsonize* the bacteria, that is coat

Innate Immunity



Acquired Immunity

Antigen
Presenting
Cells

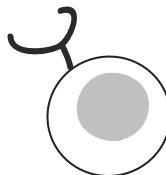


Dendritic Cell



Macrophage

Lymphocytes



T-Cell



B-Cell

Figure 11-1. Basic cell types of vertebrate immune systems.

them for recognition and destruction by phagocytes, or to destroy it outright. These receptors are encoded in the germline, but unlike the components of the adaptive response (see below), they are inflexible. These cell lineages are part of a generalized hemopoietic differentiation process that generates the circulating blood cells.

To establish an infection, a pathogen must either avoid detection by the innate immune system or defeat or overwhelm it. To evade recognition by phagocytes, many extracellular bacteria (that is, those that do not enter the host's cells but circulate in the blood or live in mucosal tissues) develop a thick polysaccharide coating or capsule. Some pathogens have developed a way to grow inside the phagosome (inside a cell that has engulfed the pathogen).

If enough bacteria enter the body they can overwhelm the innate response, and the adaptive immune system (in jawed vertebrates) must then respond if the organism is to survive the attack. If a pathogen evades or overwhelms the innate response, the early induced responses, humoral and cell-mediated effector mechanisms to be described below, are activated. This second wave of responses may contain the infection until the adaptive responses are ready to mount a defense, and they also influence the kind of adaptive response that is mounted. Some of the same mechanisms of the innate immune response may later be enlisted to eliminate the pathogen.

On activation, phagocytes release *cytokines* into the circulation. When received by cytokine receptors, these molecules alter the behavior or induce the proliferation of the recipient cell types in the immune system. *Cytokine* genes include a family of genes called *Interleukins*, which induce the liver to produce proteins that activate complement and the opsonization of microbes. Cytokines can also trigger a fever response that can be helpful, because many bacteria do not function well at elevated body temperature.

The innate immune system fairly quickly elicits a localized *inflammatory response* at detected sites of infection. This involves the recruitment of more phagocytes and effector cells (lymphocytes that are immediately able to mediate the destruction of a pathogen without having to undergo the further differentiation that other classes of cells in the adaptive immune system require) to the infection and the release of cytokines that have local effects like inducing increased blood flow to the area, helping, for example, by improving access for more effector cells. There is also an increase in vascular permeability, which allows the local accumulation of fluid (with concomitant pain and swelling of infection) and an increase in the number of immunoglobulins and complement, among other effector cells, in the area (Janeway 2001).

Phagocytes release many other molecules as well, such as nitric oxide, toxic oxygen radicals, mediators of inflammation, and the like, that both fight the pathogen and facilitate further adaptive response. One class of molecules called *defensins* is a family of short antibacterial peptides that help to permeabilize bacterial membranes so that they can be destroyed. Similar defense mechanisms are found in plants. Viral infection induces the production and secretion of proteins known as *interferons*. Interferons, true to their name, interfere with the replication of viruses by binding to interferon receptors, initiating a signal cascade that ultimately activates the transcription of genes that degrade viral RNA or otherwise inhibit viral replication inside a cell, preventing the spread of infection to neighboring cells.

These diverse measures, each involving families of gene products that activate other cell types and thus trigger the production of many attack molecules, besiege

an infected site with generic defenses. The same process induces changes in the endothelial cells (the lining of blood vessels) during inflammation to induce these cells to trigger blood clotting in small vessels near the site of infection. This reduces the ability of surviving pathogens to enter the bloodstream and travel to other sites in the body. Meanwhile, phagocytes that have engulfed pathogen are carried by the fluid that leaked into the area at the early stages of infection to nearby lymph nodes, where they trigger an adaptive immune response in which T cell receptors (TCRs) or antibodies (see below) recognize the foreign peptide on the surface of the cells and are prompted to proliferate. This leads to the destruction of the infected phagocytes (Janeway 2001).

THE COMPLEMENT SYSTEM

This system enhances the ability of the adaptive immune system to destroy bacteria, thus the term *complement*. It is particularly important in the destruction of bacteria. It seems to be a part of the innate immune system that was partly coopted in the evolution of adaptive response. Comprised of some 20 serum proteins, the system works as a cascade of cleavage reactions, one reaction activating the next component of the system. The effector mechanisms of complement include opsonization of the surface of pathogens so that phagocytes can recognize and engulf them, direct killing of microorganisms by creating holes in their surface membrane, chemotactic attraction of leukocytes to sites of infection, and activation of leukocytes.

The system is activated via three pathways. The first is called the *classical pathway*, activated by antigen-antibody complexes and active in both innate and adaptive immunity. The second is the *mannan-binding lectin pathway* (MBLectin pathway), which responds to the binding of a serum protein called mannan-binding lectin to carbohydrates on bacteria or viruses that contain mannose, a type of sugar molecule. Finally, the *alternative pathway* is activated when the surface of a pathogen is bound by a previously activated complement component (Janeway 2001).

These “innate” systems are invoked generically, without regard to the specific pathogen involved. In that sense they are preprogrammed for a kind of blind defense. A diversity of rather crude mechanisms is used, like destroying cell walls, that are possible because innate defenses recognize generic components of broad classes of pathogen, or exploit common constituents of these organisms (such as mannose) which the pathogens cannot shed because they are so intrinsic to their survival. In that sense the innate system evolved as a response by complex organisms to constituents of bacteria in place possibly since their origin. An assault on cells that is too generic can also damage the host as well, but if the response occurs early enough and is kept localized, the damaged area can be regenerated or sloughed harmlessly, and the infection is overcome.

THE ADAPTIVE IMMUNE RESPONSE SYSTEM

Bacteria and viruses can mutate and evolve much more rapidly than animals that reproduce slowly like vertebrates, and we might expect the microbe always to be able to outevolve any specific adaptation by the host—a race that must always go to the swift. Although many slowly reproducing species nonetheless manage,

immune systems in higher vertebrates include a component that is *adaptive* on a scale to match that of microbial parasites.

In vertebrates both the innate and adaptive immune responses are mediated cellularly by leukocytes. Two phagocytic classes, macrophages and neutrophils, were mentioned earlier; along with *monocytes* they are primarily involved in the innate immune response system. Phagocytes mount a first-line defense in the immune response, which is immediate but generic.

The adaptive immune response is specific but can take up to seven days (in humans) to prepare to defend against a pathogen while the innate response attempts to rid the body of it. A class of leukocytes known as *lymphocytes* are the workhorses of the adaptive response. These cells recognize specific intra- or extra-cellular pathogens. The two predominant types of lymphocytes are known as *B* and *T* cells, named for the location in which they have been thought mainly to develop. In mammals, B cells differentiate in the liver in the fetus, and in bone marrow post-natally whereas T cells develop in the thymus. B and T cells acquire specificity to pathogens via receptors they produce, known as *antibodies*, that recognize *antigens*, molecules on the surface of the target pathogen or a toxin that it produces.

As currently understood, the major differences between the receptors on B and T cells are that the B cell receptor has two identical antigen recognition sites and can be secreted into the circulation whereas the T cell receptor always remains anchored to the cell surface and has only one recognition site. Antibodies circulating freely in the blood or lymph are called *immunoglobulins* (*Ig*). Other antibodies are anchored on the surface of B or T lymphocytes.

However, although the antibody-antigen ligation is specific at the molecular level, antibodies are not programmed in advance by specific antibody genes or alleles. Instead, lymphocytes with an essentially random assortment of antibodies, called *naïve lymphocytes* because they have not yet come into contact with antigen, circulate from the blood into the lymphoid tissues. There, an adaptive selection process unfolds in which the specific microbial antigens themselves are used to select antibodies to which they can be bound.

B Lymphocytes

A given B cell produces only a single antibody type (see below). B cells present this antibody on their surface (*surface immunoglobulin*), but once the antibody recognizes (binds) an antigen, two signals induce the lymphocyte to multiply and differentiate into immunoglobulin-secreting plasma cells. The first signal is induced by the ligated receptor and the second by a costimulatory signal in the form of a B7.1 or B7.2 molecule, coded by a member of the *B7* gene family. The first signal initiates the synthesis of one subunit of the *API*, or antigen-presenting, TF and the second signal directs the synthesis of the other subunit. The API TF recognizes enhancers such as those for interleukin gene expression. Once the B cell differentiates it can proliferate and/or produce large quantities of immunoglobulin—its particular antibody—that bind to other circulating copies of the antigen that originally activated the cell (Medzhitov and Janeway 1998). B cells can also be induced to differentiate in the spleen into memory cells that will quickly produce antibody if the organism is subsequently rechallenged by the same pathogen.

In Chapter 7 we described a variety of reactions that involve ligand binding by cell surface receptors that triggers response via the receptors' intracellular domains. Immunoglobulins work in a logically similar way. They have two major functions,

each carried out by different parts of the molecule. One region binds to antigen, and the other mediates *effector* functions, that is, the destruction of pathogens specifically based on type of infection and the pathogen's life cycle stage. There are a number of effector systems, and the same systems are used by the innate and adaptive immune systems. The simplest is neutralization, when the antibody simply binds to the pathogen and neutralizes its activity. Another is phagocytosis, where antibody activates phagocytic cells to engulf the pathogenic cell and degrade it in one of several ways. Some antibodies, as well as TCRs on T cells (see next section), when they recognize antigen, induce cytotoxic reactions that kill the pathogen outright, either by perforating the membrane of the target cell or by inducing apoptosis.

The two functions of immunoglobulins are structurally separate on the molecule. Antibodies are formed by the pairing of heavy and light polypeptide chains (see Figure 11-2). The antigen binding region, the arms of the **Y**, that include the light chains, varies enormously in several open-ended ways and thus is called the *variable (V) region*. The rest of the molecule, the leg of the **Y** composed of the heavy chains that are responsible for the effector functions, is not nearly so variable and so is called the *constant (C) region*.

There are five Ig isoforms (alternative polypeptides), each coded by a different C-region gene, and these isoforms define the functional classes of immunoglobulins in higher mammals: IgG, IgM, IgA, IgD, and IgE. IgG is the predominant immunoglobulin in humans, comprising 70–80% of the total. It is found circulating in the blood as well as interstitially (in between cells). The structure of IgGs is remarkably variable among mammals, and they cross the placenta in some but not others. Maternal IgG confers immunity to newborns for several months, until the infant is able to make its own.

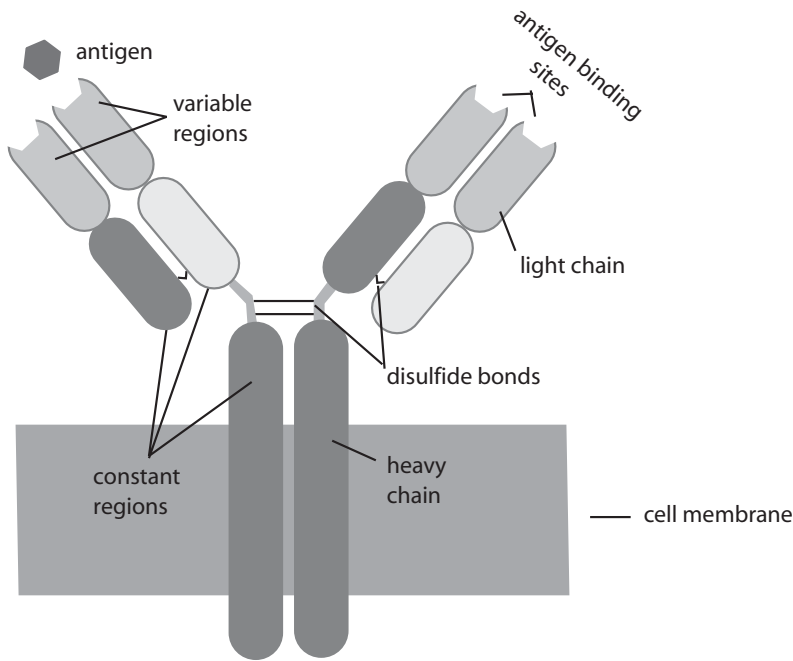
IgM comprises about 10% of the body's immunoglobulins. It is generally found circulating in the blood and is the first antigen-induced response to infection. IgA accounts for 10–15% of human immunoglobulins. Secretory IgA is the primary immunoglobulin in saliva, colostrum, milk, and other secretions. IgD accounts for <1% of total immunoglobulin but exists at high levels on the surface of B cells. Its function is not known, but it may have to do with the development of the fetal immune system or with lymphocyte differentiation. IgE is usually scarce in human serum, but probably is involved with immune response to parasites as well as in allergic responses (Roitt et al. 1998).

T Lymphocytes

T cells play a variety of roles in the adaptive immune system. As currently understood, some are involved in controlling the development of B cells and their antibody production, *helper T cells* help phagocytes destroy the pathogens they have internalized, and *cytotoxic T cells* recognize and kill virally infected cells.

T cells recognize antigens that are presented on the surface of a cell by molecules of the *Major Histocompatibility Complex* (MHC) (see Figure 11-3). The TCR is specific for antigen-MHC complexes. T cells function either by releasing cytokines, which signal between cells during an immune response, indirectly by activating macrophages to destroy organisms they have engulfed, or directly by interacting with infected cells. Like B cells, once a T cell has recognized a pathogen, it multiplies to vastly increase the amount of TCR available, and because the TCR is specific this generates *clones* of the T cells appropriate to the pathogen.

A. Antibody Molecule



B. T-Cell Receptor

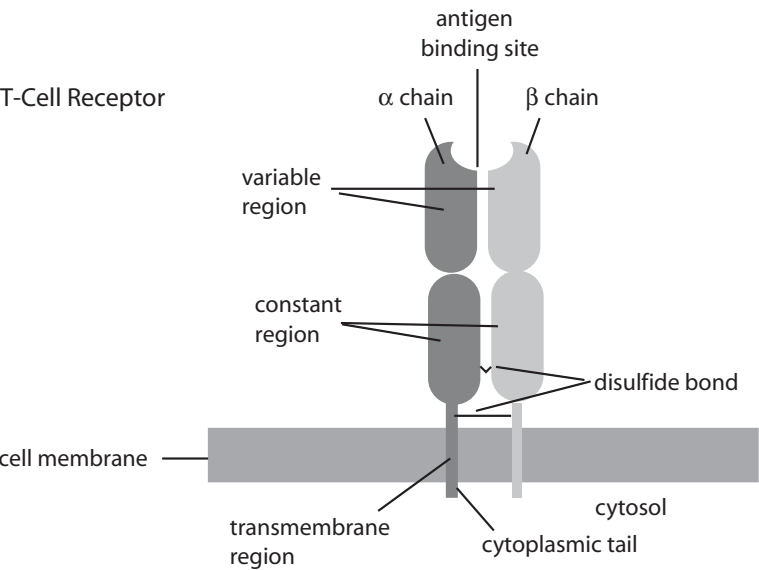


Figure 11-2. Structure of immunoglobulin proteins, showing (A) the joined light and heavy chains, constant and variable regions; (B) T cell receptors, showing their similarity to immunoglobulins.

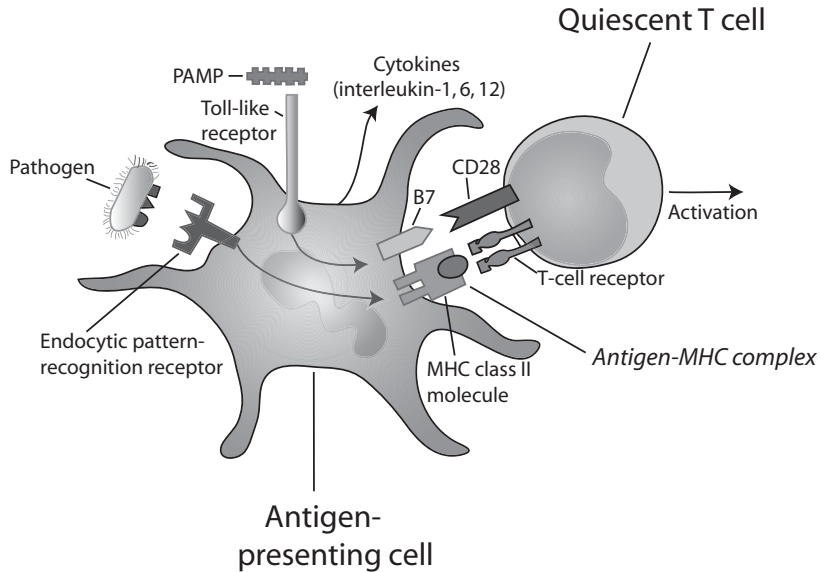


Figure 11-3. Antigen presenting cell with signals, T cell and receptors. Redrawn from Medzhitov and Janeway with permission; see that reference for details. Original figure copyright 2000, Massachusetts Medical Society, all rights reserved.

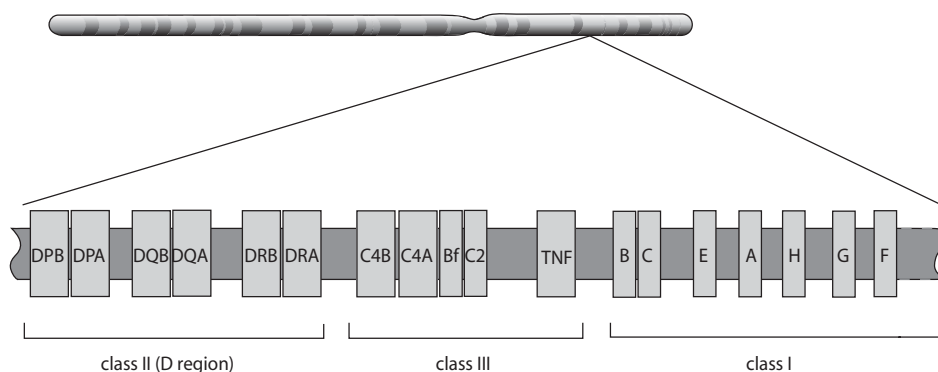
There are approximately 30,000 TCRs on the surface of every T cell. There are two types of these receptors, both formed of heterodimers composed of two polypeptide chains joined via a disulfide bond (Janeway 2001). The chain of the predominant type of receptor is made of α - and β -subunits, whereas the second, and less common, type of TCR, is a dimer of γ - and δ -chains. These receptors recognize antigen differently, and in fact, the role of the TCR is not yet fully understood. These heterodimers are physically associated on the T cell surface with five polypeptides called the CD3 TCR complex, which is required for the expression of the TCR at the cell surface, although it is not yet clear how. Ninety to ninety-five percent of T cells are $\alpha\beta$ T cells and the rest are $\gamma\delta$ T cells (Roitt, Brostoff et al. 1998).

Leukocytes express many different molecules on their surfaces, which are used to classify cell types by a *cluster designation* (CD) type. $\alpha\beta$ T cells are classified as CD4+ (“helper”) or CD8+ (“killer”) cells. Helper T cells activate other cells to destroy pathogen, whereas CD8+ T cells kill directly. Although $\gamma\delta$ T cells comprise a small fraction of circulating T cells they are frequent in mucosal epithelia. It is thought that these T cells may be important in protecting mucosal surfaces from some kinds of viral and bacterial infections.

The Major Histocompatibility Complex

The family of genes involved in tissue rejection comprises the MHC. This was first described in the mouse; the human analog is called the *human leukocyte antigen* (HLA) system (the organization of both systems is shown in Figure 11-4). *Histocompatibility antigens* are encoded by *histocompatibility genes*. The HLA complex is on human chromosome 6, and includes more than 200 genes, 40 of which encode leukocyte antigens. These are among the most polymorphic of all mammalian genes

Human HLA complex, chromosome 6



Mouse H-2 complex, chromosome 17

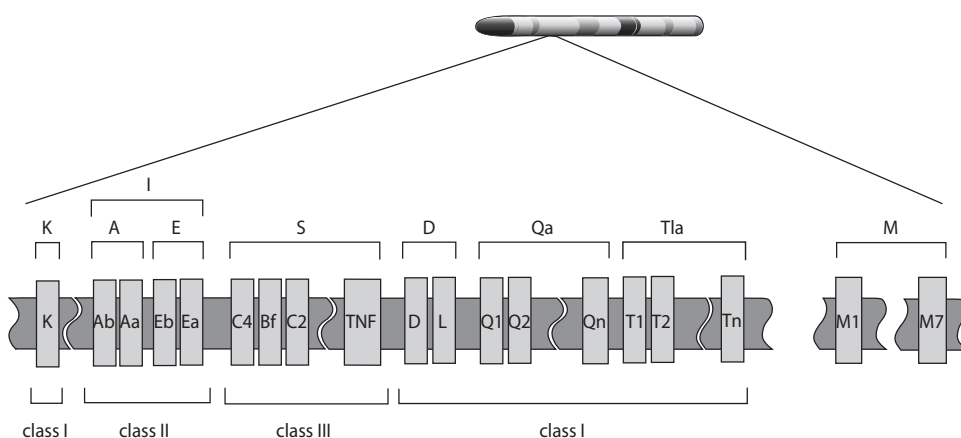


Figure 11-4. Organization of the human and homologous mouse major histocompatibility (MHC) complexes and position of major genes. The regions span three to four megabases.

(Lechler and Warrens 1999). They bear coding domains homologous to the immunoglobulin and TCR genes, all part of the large cell adhesion molecule (CAM) superfamily mentioned in Chapter 7. Other genes within the large HLA chromosomal complex encode complement, cytokines, and other molecules with a role in the immune system.

The MHC is complex incompletely understood, and varies among species. Essentially, there are two major MHC classes, encoded by *class I* and *class II* genes, although the “other” genes in the complex, physically linked but structurally unrelated to the class I and II genes, are sometimes referred to as MHC *class III* genes. The details of the structure and function of the two major classes are distinct, but the basic role of each is to present pathogen-derived peptides to the surface of cells for detection by the TCR of T cells; this happens as part of the process of cellular

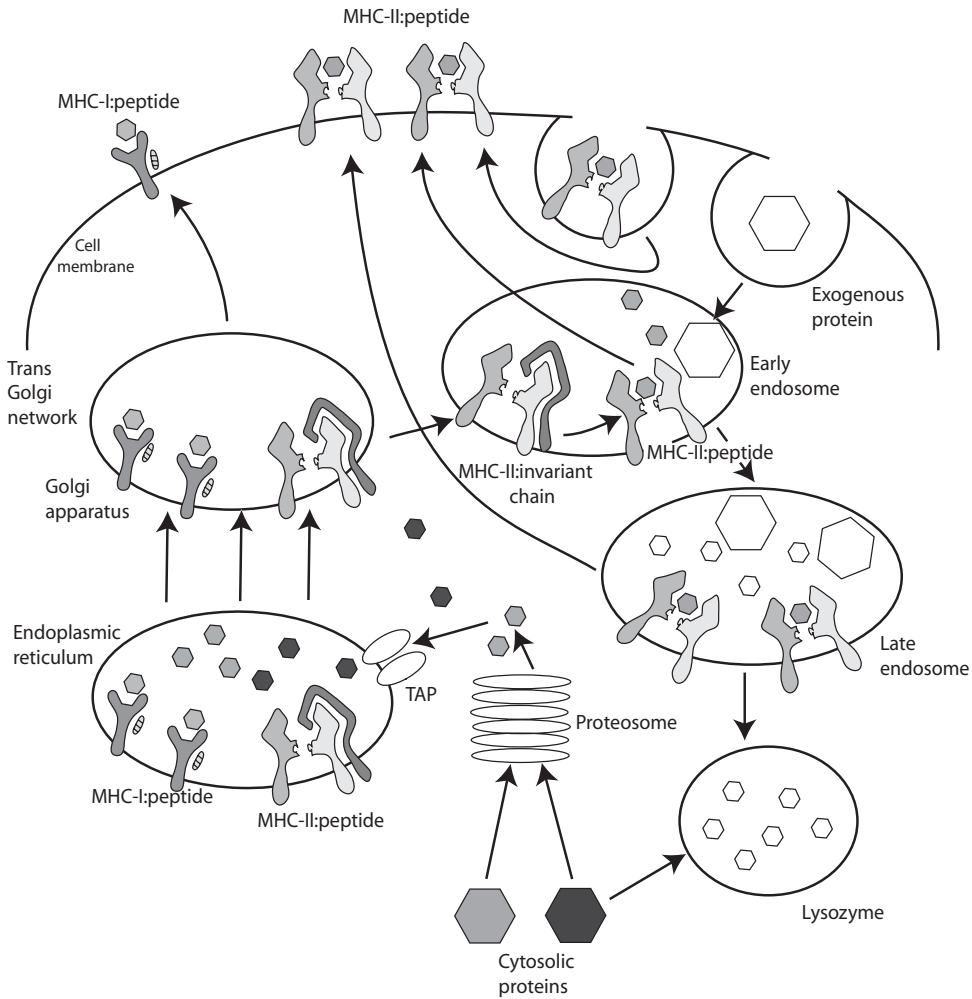


Figure 11-5. Stages in antigen processing and presentation to the surface of the cell by the MHC class I and II gene products. Redrawn after (Paul 1999) with permission. Original figure copyright Lippincott, Williams and Wilkins, 1999.

waste disposal. Class I genes are expressed on the surface of most somatic cells, whereas class II genes are expressed only on some types of immune cells, including B cells, activated T cells, macrophages, dendritic cells, and thymic epithelial cells (Klein and Sato 2000). In particular because most cells present an individual's class I molecules, the MHC has been viewed generically as a form of self/nonself recognition signaling.

Generally speaking, class I molecules carry endogenously derived peptides (intracellular, from viruses, for example) whereas class II molecules transport exogenously derived peptides (from extracellular self or foreign proteins such as bacteria or viruses that have not yet entered a cell), although this is not always true. Class I molecules plus antigen are recognized by killer T cells, whereas class II molecules and antigen are recognized by helper T cells (Klein and Sato 2000; Roitt, Brostoff

et al. 1998). Thus an antigen presented by a class I molecule elicits a T cell killer response, whereas if the antigen is presented by a class II molecule, a helper T cell response follows.

Worn-out or misfolded proteins inside a cell are marked for degradation by a molecule called *ubiquitin*. The marked proteins are unfolded with the help of chaperone molecules and fed to proteasomes, which fragment them. The resulting peptides are either degraded into amino acids, which float in the cytosol of the cell and are reused, or transferred into the endoplasmic reticulum (ER) for transfer to the cell surface. This is done by *transporters associated with antigen processing*, or *TAPS*. In the ER, class I molecules pick up these peptides for transport to the surface of the cell, where they are displayed.

Extracellular proteins are treated differently. They are surrounded by invaginations from the plasma membrane of a cell, which pinch off as endocytic vesicles and fuse with lysosomes to form endosomes. Class II molecules transport these peptides to the cell surface; they do not bind to peptides in the ER, but instead the class II molecules are enclosed in membranous vesicles that fuse with endosomes to form the MHC class II compartment where exogenous proteins are degraded and subsequently transported to the cell surface.

Degradation of endogenously derived proteins, including self proteins, happens all the time in most cells, and MHC class I molecules are constantly transporting peptides to the surface of cells. The action of class II molecules on exogenously derived proteins is less ubiquitous, however, generally restricted to cells that are specialized in phagocytosis or endocytosis. In any case, most cells are studded with MHC-peptide complexes all the time, hundreds of thousands of them, and the vast majority represent self peptides (Klein and Sato 2000).

Natural Killer Cells

A third kind of lymphocyte is the *natural killer cell (NK)*. These cells originate in bone marrow. They do not express antigen receptors but have the ability to destroy some tumors by lysing tumor cells, and they play a role in innate immunity by killing cells infected with viruses.

PRODUCTION OF IMMUNOGLOBULIN AND T CELL RECEPTOR DIVERSITY

So much for the mechanics. To restate the evolutionary challenge presented to macroorganisms, pathogens have a much shorter life span and can rapidly out-evolve the ability of large organisms to defend against them. If recognition is the challenge, then immune systems are faced with ever- and unpredictably changing diversity. One strategy for keeping up is the generation of huge amounts of diversity in the adaptive immune systems of higher vertebrates. Although somewhat different in structure and function, the fundamental tools of this system, the TCR and immunoglobulins, are both produced by a very interesting *somatic* DNA recombination mechanism that is unusual because during the development of each lymphocyte lineage, the inherited genome in the antibody coding complexes of that cell is permanently altered. This mechanism has been traced to the emergence of the ancestral jawed vertebrate about 450 million years ago.

The immunoglobulin and TCR gene complexes contain sections of coding regions known as V (variable), D (diversity), and J (joining) elements, in which there are many duplicate coding elements (especially in the V region). Active recombination-

TABLE 11-1.

Number of Functional Gene Segments in V Regions of Heavy and Light Chains, Human.

Segment	Light chains		Heavy chain
	κ	λ	H
Variable (V)	40	30	65
Diversity (D)	0	0	27
Joining (J)	5	4	6

Source: (Janeway 1999).

generating mechanisms randomly select one each of the V, D, and J regions during the somatic development of lymphocytes. This is what is transcribed, while the remaining alternative segments in these regions are discarded from the genome and from the cell. The repertoire from which this random selection process chooses provides a huge variety of possible VDJ combinations, and hence different lineages. Table 11-1 shows the number of functional gene segments in human V regions.

The total number of possible different receptors is enormous, and the total number of different lymphocytes circulating in an individual at any time is on the order of 10^8 (Janeway 1999). The V and C (constant) regions are connected by the short J region and, in the heavy chain, also a D region. It is the C region that determines the class of antibody and how it destroys the pathogen once it is identified and bound, whereas the variable region determines which antigen the antibody or TCR binds. Sequence variation is greater in the V region, and there are many related genes in tandem in the V region, whereas the C region has more standard genelike function. Because the former determine antigen-binding specificity of the coded molecule, this means there is a comparable diversity in the molecules.

This *somatic recombination* is triggered by the products of the recombination-activating genes *Rag1* and *Rag2*. The V, D, and J gene segments are flanked by recombination signal sequences (RSSs) that consist of conserved heptamer and nonamer sequences separated by 12- or 23-base-pair spacer sequences. The *Rag* genes encode a recombinase that catalyzes site-specific cleavage of DNA between the RSSs; two coding ends of the excised pieces are then joined in an imprecise way, with additions of random nucleotides and/or short deletions, yielding a unique new combination of V(D)J segments (Roitt, Brostoff et al. 1998) (see Figure 11-6).

Diversity in immunoglobulins can result from any one or a combination of five different processes; the different possible combinations of Ig heavy and light chains, gene conversion, recombination of V, D, and J genes (V and J for the light chains), variability in joins, or somatic hypermutation introducing mutations into the V regions of activated B cells in a way that is not yet completely understood but that involves regions of numerous point mutations, mutational hot spots, and a mechanism in which DNA breaks initiate error-prone DNA synthesis by faulty DNA polymerases in V(D)J regions (Diaz and Casali 2002). The triggering mechanisms are not yet known, but generally are well within the familiar repertoire of evolution (mutation, duplication), gene expression (context-dependent transcription, splicing), and development (combinatorial signaling).

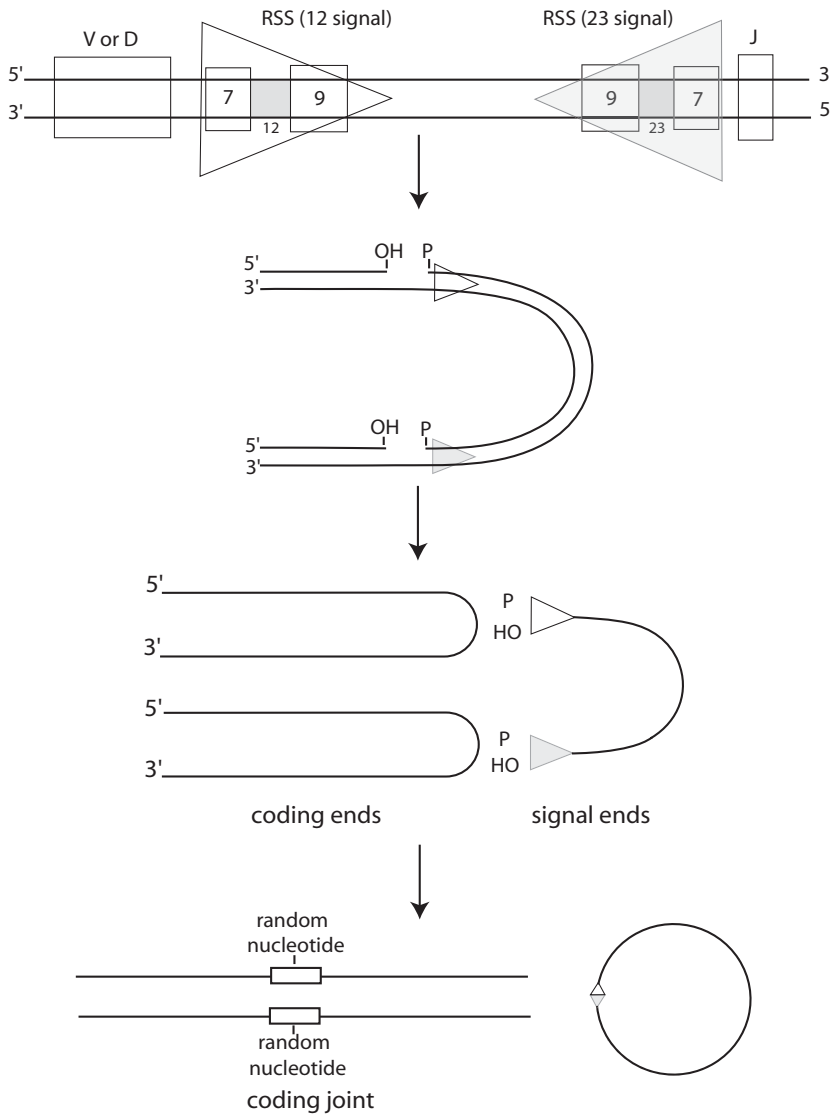


Figure 11-6. *Rag1* and *Rag2* mediated V(D)J formation (see text). Redrawn from (Ohmori and Hikida 1998) with permission.

This somatic recombination happens in *cis*, that is, along the TCR and heavy and light chain Ig gene regions. At the same time, there is a *trans* phenomenon of *allelic exclusion*, by which the same region of the homologous chromosome is inactivated, so that the cell expresses only one combination of its genes (for the TCR or its Ig, depending on the cell type) derived from only one of its chromosomes. Each of these events happens independently in each of the cell lineages. At least one mechanism of allelic exclusion seems to involve asynchronous replication in mitosis (Singh et al. 2003). The earlier-replicating chromosome can in principle produce

some substances or conditions that prevent recombination from occurring on the homologous chromosome. Of course, in any evolving system, there will be diversity among species. A variety of ways in which antibody diversity is generated in vertebrates is depicted in Figure 11-7. The system in sharks depends on a large number of individual complete genes, much like that of olfactory reception (Chapter 13) and the *R*-gene system in plants (see below). We see again our common finding: in their own way, each works well enough to have evolved.

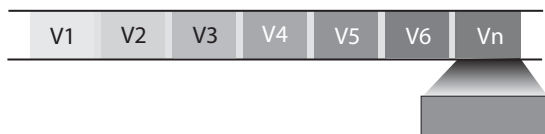
The first stages of B cell development occur in the bone marrow, where the process of gene rearrangement takes place. This stage generates an immature B cell that carries an antigen receptor. This antigen receptor is in the form of cell surface IgM, capable of interacting with antigen. ("IgM" means that in addition to its chosen V, D, and J region, the cell is using its "μ-" or M constant chain to form the immunoglobulin). Immature B cells at this stage that do not encounter antigen either die or are inactivated (become *anergic*), thereby eliminating many potentially self-reactive B cells from the repertoire of antibody-producing cells, because B cells that crosslink their own IgM are programmed to die (Janeway 2001; Roitt, Brostoff et al. 1998). Cells that survive this process are subsequently free to leave the bone marrow and enter the lymphoid system, where they may encounter foreign antigen and be activated. Activated mature B cells proliferate clonally, to become either antibody-producing plasma cells or memory cells that can respond quickly to reinfection with the same pathogen. If activated in later stages of infection, the constant chain switches from M to G, etc., depending on the context. The stages of B cell development are shown schematically in Figure 11-8.

T cells are bone marrow-derived stem cells, called *thymocytes* when they are immature lymphocytes. They differentiate and mature in the thymus, undergoing recombinational events like those that produce the differentiated antibodies of B cells. They also undergo the same kind of selection process as B cells: 98% of thymocytes die, by apoptosis, before they mature, although, unlike B cells, T cells undergo positive as well as negative selection. The process is complicated and not well understood, but it is based on the recognition by T cells of self peptide-self MHC complexes and eliminates those that are highly self-reactive, while selecting for those that have a moderate affinity for self. This negative selection thus ensures that these cells become self-tolerant. Experimental work has also shown that T cells are unable to recognize and bind to foreign peptide-foreign MHC complexes. That is, positive selection of T cells, based on moderate affinity to self MHC, ensures that T cells recognize foreign peptide-self MHC complexes (Janeway 2001; Roitt, Brostoff et al. 1998).

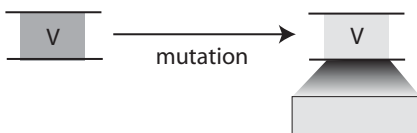
Several aspects of the system are worth noting. The recombinational possibilities come about because of the number of selectable regions that can be used (thanks to a history of duplication), because there are two chromosomes for each gene complex to choose from, and because recombination is an error-prone system (especially in the D and J regions). Since each activated B or T cell expresses only a single binding variant, the proliferation of just that variant represents a kind of natural selection within the organism that favors what works without proliferating a lot of antibodies that do not work against a particular pathogen (but could generate anti-self reaction). This makes the system modular in an interesting way and sequesters its specificities in controlled ways—but that do not require the organisms as a whole to "know" which of some fixed prior repertoire of antibodies to activate and which not to.

A.

1. Multiple V genes, each encoding a different V region



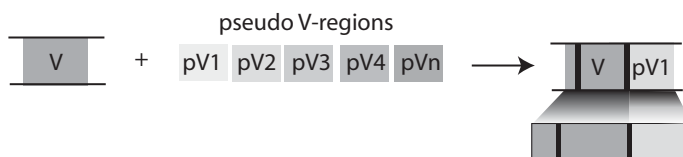
2. Somatic mutation during B cell ontogeny, producing different B cell clones



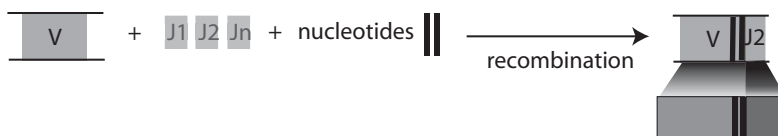
3. Somatic recombination during B cell ontogeny



4. Gene conversion



5. Nucleotide addition, when V and J regions are cut, before joining



A

Figure 11-7. Generation of antibody diversity in vertebrates (A) Various vertebrate antibody recombination mechanisms including multiple V-region genes in the germ line, somatic mutation, somatic recombination between elements forming a V-region gene, gene conversion, and nucleotide addition; (B) antibody diversity in the mouse is due to somatic recombination; in sharks to numerous antibody genes rather than somatic recombination; chickens have few antibody genes but a high level of gene conversion. (A) Redrawn from (Roitt, Brostoff et al. 1998) with permission from Elsevier, original figures copyright Elsevier 1998; (B) redrawn from (Gerhart and Kirschner 1997) with permission.

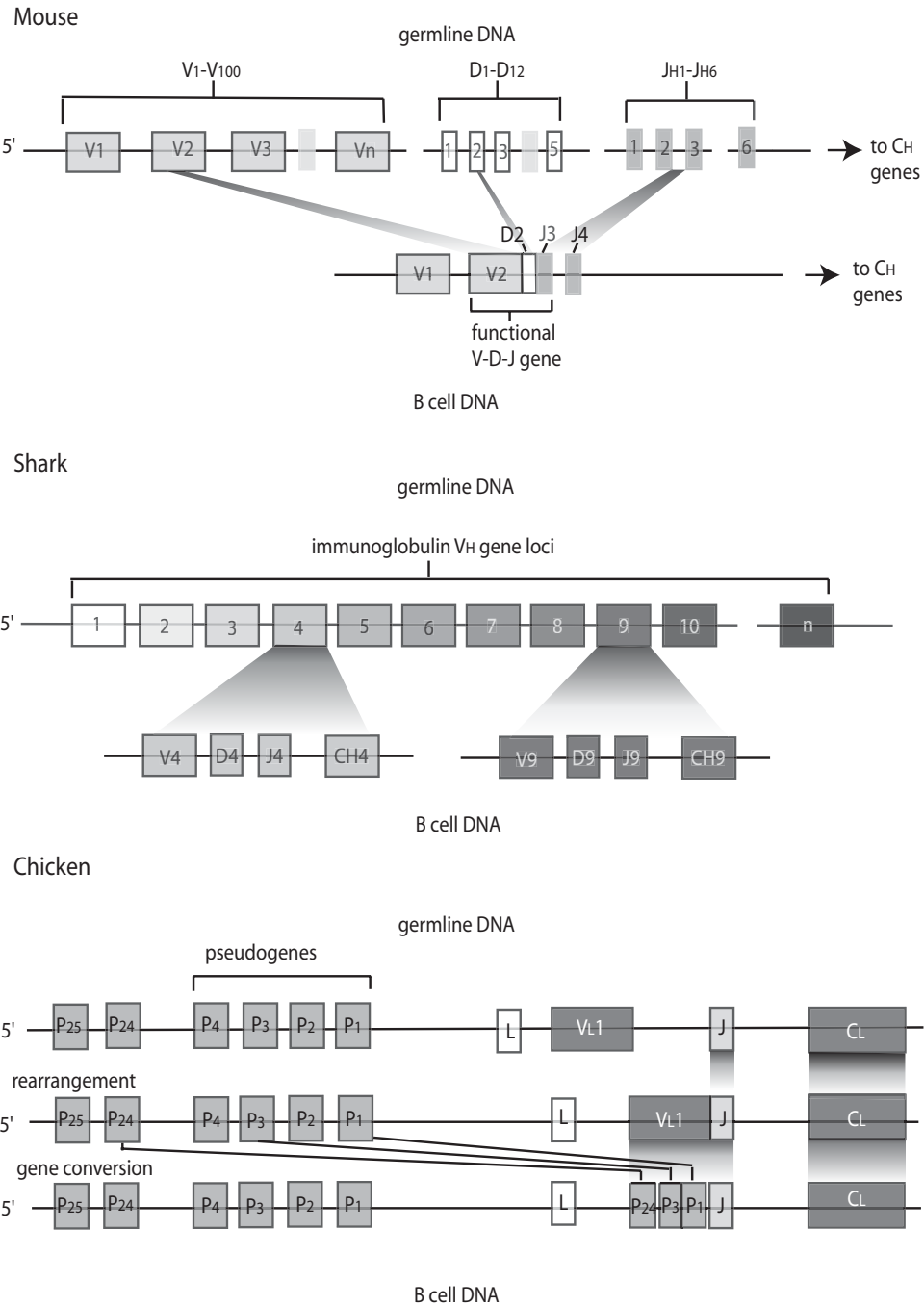


Figure 11-7. Continued

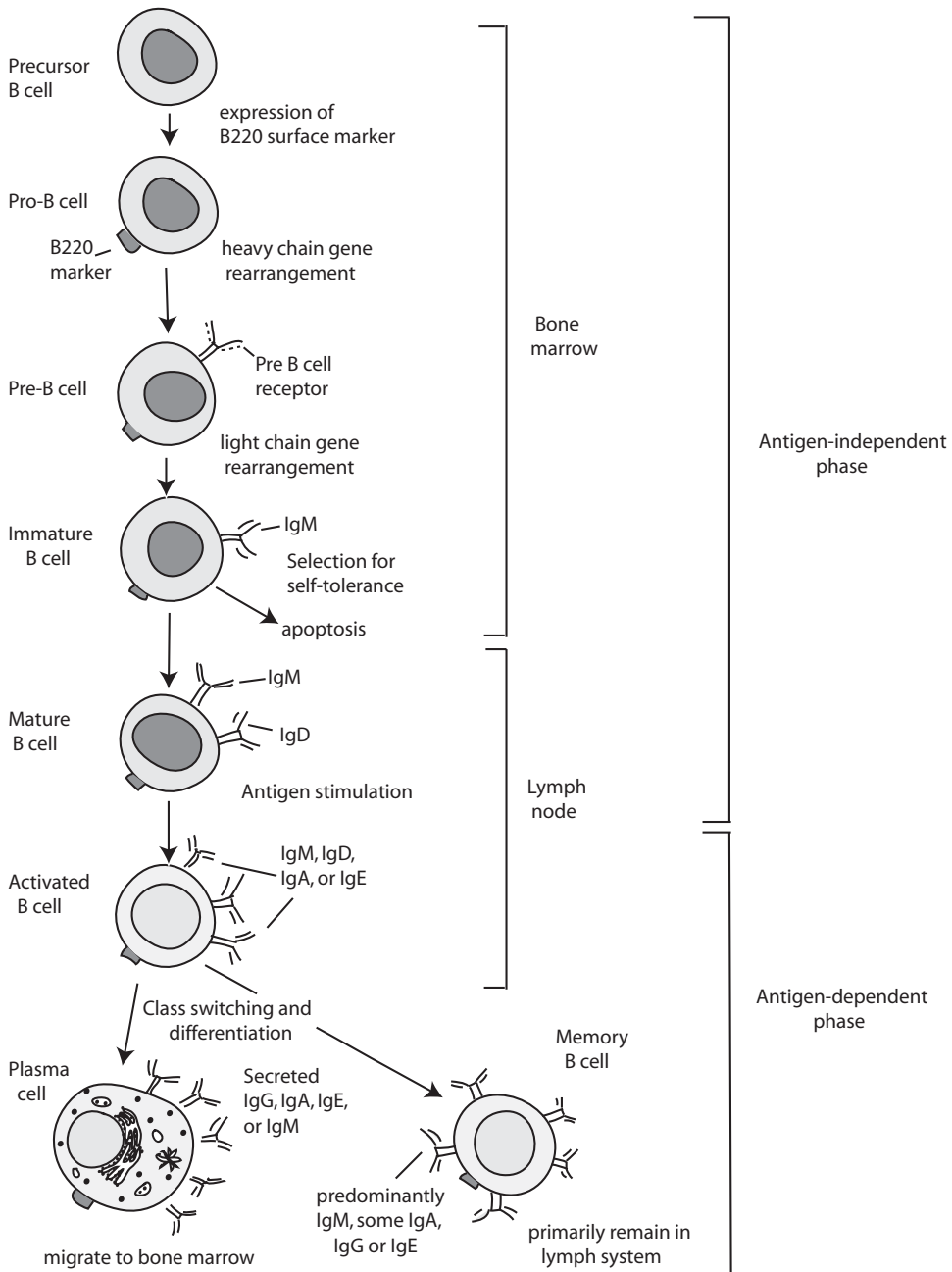


Figure 11-8. Stages of B cell development from precursor to mature cell.

Many more combinations are possible than are circulating in the blood at any one time, and these cells are being recycled throughout life. The system generates variation in a combinatorial way, but it need not be exhaustive: tiny voles and huge elephants have essentially the same immune diversity, with orders of magnitude

differences in the number of lymphocytes they can circulate. There are no major differences in adaptive immunity associated with life span or body size in higher vertebrates (and fish can live very long lives). For most pathogens, the millions of antibodies present at any time appear to be sufficient to ensure that enough cells recognize the specific pathogen that the body can mount an effective defense. Thus the system is not limited by these factors (*or* it is not as important for survival as we tend to think).

Note that the immune memory that an individual develops during its lifetime is not genetically inherited (or at most is inherited in the limited extent of temporary effects of transplacental resistance from mammalian mothers to children). What is inherited is the (randomly mutated) germline repertoire and the sloppy recombination process from which any new attack is mounted. As currently understood, each new individual must generate its own set of specific antibody molecules. It starts fresh, but the mechanism is such that this is not a particular handicap relative to its benefits; and to a considerable extent, the pathogenic environment is always evolving anyway.

The system is powerful but neither necessary nor perfect; most animals do not have such a system, so it is clearly not a prerequisite for survival or “adapting” to a world filled with tiny, hungry pathogens. Also, as noted earlier, we live in commensalism with many microparasites (indeed, we could not survive without them, as illustrated by the need in vertebrates for bacterial colonization of the gut to aid in various stages of digestion; usually, these stay out of the body, where they are less vulnerable to immune attack) and, despite effective immune defense, more animals of many species may be killed by micropredation than by any other single cause.

SOME GENETIC ASPECTS OF LYMPHOCYTE SIGNAL TRANSDUCTION, DIFFERENTIATION, AND RESPONSE

When cell surface receptors on lymphocytes bind to lipopolysaccharide (LPS), a component of Gram-negative bacterial cell walls, peptidoglycan (PGN) from the cell wall of Gram-positive bacteria, or interleukin1 (IL1), this initiates a signal-transduction pathway, triggered by Toll-like receptors, (TLRs), which releases a TF, NFκB, from its inhibitor IκB (see Figure 11-9). This allows NFκB to enter the nucleus and bind to various promoters, activating genes involved in adaptive immunity as well as proinflammatory molecules, cytokines, that signal between cells during an immune response. Tlr4, with additional factors, is required to sense LPS, whereas the *Tlr2* gene product is required to sense PGN (Takeuchi and Akira 2001; Takeuchi et al. 1999). TLRs, a family of leucine-rich proteins, are homologous to *Drosophila* Toll protein, a membrane receptor that is also thought to function in insect microbe recognition, as well as being used in the development of dorsoventral patterning in flies as we saw in Chapter 9, among other functions.

THE INNATE RESPONSE INDUCES ADAPTIVE IMMUNITY AND SELF/NONSELF RECOGNITION

Traditional discussions of vertebrate immunity treat innate and adaptive responses as separate systems, with innate immunity being derived from the ancient response to general characteristics of pathogens and homologous to the immune systems of lower vertebrates or invertebrates. Its role was viewed as, at best, keeping pathogens

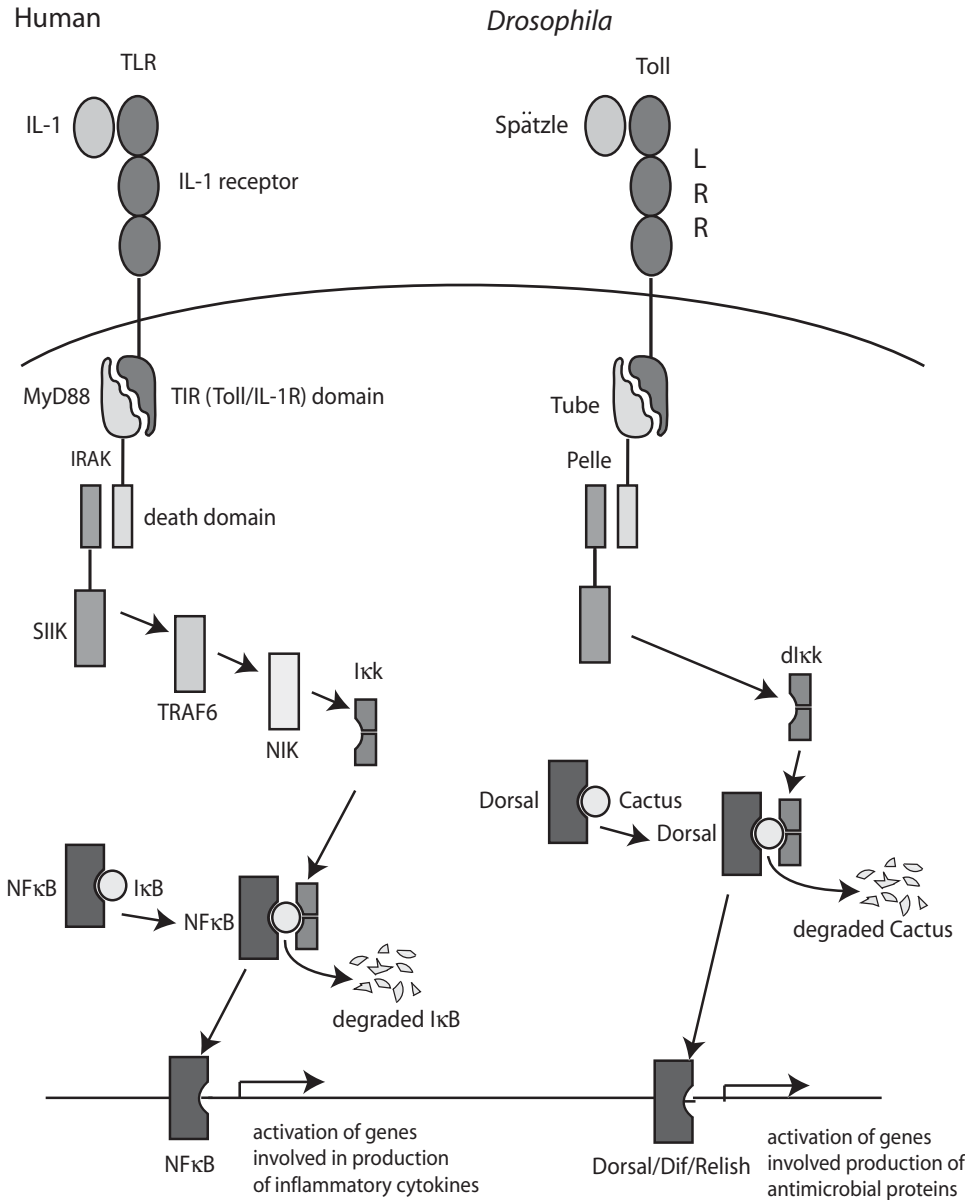


Figure 11-9. *Toll* and Toll-like signaling pathways in humans (left) and *Drosophila* (right). In human lymphocytes, the IL1 receptor phosphorylates IkB via the IRAK kinase cascade, the homolog of the *Drosophila* Pelle cascade. Once phosphorylated, the IkB protein (the *Drosophila* homolog is Cactus) is degraded, which allows the human transcription factor coded by *NFκB* (Dorsal in the fly), to enter the nucleus where it regulates the transcription of lymphocyte specific genes involved in the immune response. In flies, the same pathway effects dorsoventral patterning during development and the transcription of antimicrobial proteins in adults.

at bay while the adaptive immune system geared up to properly defend the organism. This was always a bit of a forced argument since most species keep pathogens “at bay” indefinitely by their innate system (since they have no adaptive system). The adaptive system was tailored so that it responded to characteristics of invading cells that were not found in the host’s own cells (bacteria, for example, have different components of their cell walls compared with animal cells).

According to this view, the adaptive response is more recent, more highly evolved, and (to scientists) much more interesting. More recently, however, there has been discussion of the role of the innate immune system in eliciting and regulating the adaptive response (Hoffmann et al. 1999; Janeway 2001; Medzhitov and Janeway 1997; 1998). Again, we should not be surprised if a newer, but related, system is connected developmentally with its still-present predecessor. Janeway and his colleagues have proposed that the adaptive immune system is activated by a signal from the innate response. The innate immune system induces the expression of costimulator molecules on antigen-presenting cells, and without this costimulatory signal there can be no recognition of antigen by T or B cells, and thus no clonal expansion of the specific antigen-recognizing cell.

These costimulatory signals are based on the innate immune system’s ability to use what Medzhitov and Janeway (Medzhitov and Janeway 2000) have called PRRs (*pattern recognition receptors*) to detect PAMPs (*pathogen-associated molecular patterns*) on invading cells, of which there are many known examples (e.g., Bauer et al. 2001; Moriwaki et al. 2001). PRRs are encoded by host genes and are specific to common molecular signals on pathogens because, these authors suggest, PRRs have coevolved with pathogens whose PAMPs are molecular structures so crucial to the pathogen that they cannot mutate to avoid detection without fatal consequences to themselves (Medzhitov and Janeway 1997). PRRs are found on the surfaces of host cells likely to be the first to encounter pathogens, such as mucosal membranes or surface epithelia as well as other types of cells of the innate immune system. We will see relevant comparisons in plant immunity below and later in Chapter 13 in relation to olfaction versus pheromone reception.

Because the V(D)J system can produce an effectively unlimited variety of antibodies, it can be expected also to produce antigens against the cells of the host’s own proteins. However, if the innate induction theory of adaptive immunity is correct, then lymphocytes carrying anti-self antibodies would not destroy self proteins because a second costimulatory signal is required for lymphocyte stimulation and self proteins do not have a recognizable costimulatory molecule capable of sending this signal.

The existence of autoimmune diseases, however, perhaps weakens this idea to some extent. Human examples of autoimmunity include many life-threatening diseases: type 1 (juvenile) diabetes mellitus in which the host destroys its own pancreas, the nerve demyelination disease multiple sclerosis, systemic lupus erythematosus, which attacks a variety of organs, and many others. However, these may not simply reflect aberrant anti-self attack, because there is some evidence that they occur with various epidemic patterns in populations, including after the host has experienced an infection. An idea has been that some infections can closely mimic self antigens and the similarity triggers an autoimmune reaction as a consequence of an attack on the infectious agent. In any case, the mechanism by which purging of self-recognition molecules is controlled, once thought to be known, seems now not to be so well understood.

To some immunobiologists, the self/nonself dichotomy, however explained, does nothing to further our understanding of how the immune system works, or of its evolutionary origins. Such categorization is a human concept, not necessarily a function of the immune system as it actually evolved. The host itself is always changing depending on its developmental stage, response to environment, somatic mutation, and the like. What is “self”? How would a host be programmed to detect somatic mutation as not “self”? Should it be programmed?

In addition, the host may be large enough to withstand attack on itself so long as the latter is triggered by a genuine exogenous invasion that is cleared before too much host is self-destroyed. That the system occasionally goes awry, sometimes with lethal consequences, is not unique to immunity (and many of the known autoimmune diseases generally kill postreproductively). We do not really know why an adaptive system was favored, but clearly it was not “needed” in any ultimate sense. A system that worked well enough to combat infection better than its predecessor or competitor (in some environment that vertebrates may have had to suffer through) would tolerate the evolution of an adaptive system that simply was not too aggressive against self. In fact, given the rapidity of evolution of pathogens, the diversity of vertebrate cell types, and the often rather loose screening by selection, it may not be surprising that the system is only rather loosely discriminatory between self and nonself. If you live long enough to reproduce, evolution does not care, or even see, whether you destroy yourself subsequently or not.

Another view is that the adaptive system may have evolved to maintain the somatic integrity of the individual and only incidentally reacts against exogenous pathogens. The immune system could thus be viewed as a messy “dialogue” between the body and the cells of the immune complex (Cohen 2000). According to this view, autoimmune T cells in fact contribute to body maintenance, and play a role in maintaining the central nervous system (CNS) (Moalem et al. 1999): autoimmune T cells, reacting to myelin antigens, protect the CNS against posttrauma secondary degeneration by reducing nerve conductivity, which allows the nerve to heal. The same cells can also attack the CNS, causing disease, but the point is that autoimmune T cells are found in healthy individuals, as well as those with autoimmune disease. This, Cohen suggests, is because the business of the immune system is to maintain the body, not to discriminate between self and nonself. That such a system would, once around, come to be useful against exogenous molecular variants is not a surprising side benefit. Such discrimination may be a dispensable luxury (and was, for most of the history of evolution), if the immune reaction is timely and tempered and sufficiently localized and slows down after disposing a particular pathogen. Many vertebrate organ systems use continually renewed epithelia, and this could make the organism vulnerable to ill effects of somatic mutation, but whether this is relevant or not is wholly speculative at present.

The discussion of why an organism’s immune system does not destroy the host would be a fairly arcane side branch of biology if the understanding did not have such important ramifications for how we go about developing therapies against autoimmune diseases and protecting against rejection of transplanted organs and other such medical problems. Perhaps the most important evolutionary question is why only one tiny branch of the biosphere has developed a complex adaptive system because perhaps 98–99 percent of all multicellular animals do perfectly well with innate response alone. Trees, squid, some insects, and so on are large potential pathogen dinners that often manage to live a very long time.

DETOXIFICATION MECHANISMS

Other defense mechanisms also exist. One source of exogenous molecular attack is from the nonbiological world of chemicals. Many chemicals to which an organism can be exposed through the air, food, or other means may not amplify by growing within the host, but may react in damaging ways with the organism's own biochemistry. There are a number of metabolic systems that serve to inactivate or detoxify exogenous molecules that have certain properties. As an example, the cytochrome P450 (*CYP*) genes serve this function (the name comes from the history of the discovery of these enzymes).

Cytochrome P450s comprise a phylogenetically old class of heme-containing monooxygenases largely expressed in the liver that function to detoxify xenobiotic (exogenous) hydrocarbon molecules. The class of *CYP* genes appears to have evolved from related steroidogenic ancestral genes at the time of the separation of animals and plants (the latter are still found in prokaryotes and are expressed on mitochondrial and other intracellular membranes, where they help synthesize steroids and other substances used to maintain cell wall integrity).

CYP response works by catalyzing the oxidation of xenobiotic substances, which, along with other reactions, makes them more water soluble and excretable as well as inactivating them. The diversity of *CYP* genes in animals suggests that the system may mainly have evolved to combat substances derived from plants, in an animal-plant predator-prey competition.

CYP genes are mainly active in the liver but, like many large, old gene families, have other sites of activity and probably other functions. The genes are important in human responses to the environment but, ironically, can turn harmless human-made substances into toxic ones and can inactivate medicines or make them toxic. So the system has many facets when it comes to human affairs.

INVERTEBRATE IMMUNE RESPONSES

Immunity has, for obvious reasons of practical application, been studied most intensely in vertebrates. The vast bulk of animals must rely on their own innate, inducible, broad-spectrum, nonspecific immunity, and a few model invertebrate systems have been studied in detail, with some information available from others as well. Generally, although many invertebrates have short lives and high reproductivity (*r* strategies) as a possible defense mechanism—for the species, not the individual—even these organisms can be destroyed by pathogens and they have evolved protective defenses to combat them. In invertebrates the system relies on phagocytosis or encapsulation of the pathogen.

INSECTS

Most insects are *r* strategists that reproduce rapidly and have short generation times. They are also mobile. From this point of view, insect species might in principle be able to survive pathogens by reproducing fast enough to tolerate heavy loss of individuals or might be mobile enough to readily escape predation. However, some insects are relatively immobile, or large, or long-lived, so we cannot invoke generic evolutionary strategy arguments with regard to their immune capabilities.

Insect immune mechanisms begin with their hard exoskeleton, which serves as a first line of defense against pathogens or injury. The peritrophic membrane that sep-

arates the gut epithelium from the lumen (the passageway of the gut) also serves to inhibit infection by separating the contents of the lumen from the body interior, as does the lining of the trachea (Khush and Lemaitre 2000). Insects maintain a low pH and accumulate digestive enzymes and antibacterial lysozymes in their midguts as well, creating an unfavorable environment for most pathogens. However, we should note that in mammals the stomach is an acidic environment not usually hospitable to pathogens but to which a group (helicobacters) have adapted.

If an insect is wounded, or a pathogen evades the structural defenses, insects are able to mount a fast, strong response, both humoral and cellular. The system is not adaptive, nor does it have memory. It is likely that the antimicrobial response is based on the binding of receptors on circulating antibacterial proteins to general microbial epitopes (sites on these molecules that antibacterial proteins recognize and bind). The involvement of several different pathways has been described (Lee et al. 2001), but neither the receptor genes nor the mechanism that initiates insect response to wounding are as yet well understood.

Two immediate responses to wounding, infection, or the presence of other foreign objects in insects are the release of phenoloxidase and clotting of the hemolymph. Phenoloxidase induces the formation of melanin, which surrounds the wound and encapsulates parasites. Within a few hours of infection, a number of antibacterial proteins and peptides begin to build up in the hemolymph. More than 150 of these have been identified (Chung and Ourth 2000). The most significant is a family of peptides called *cecropins*, which quickly lyse bacteria. Another group of bactericidals is the *defensins* (with vertebrate homologs), which act more slowly but also permeabilize microbial or fungal cell membranes, and a third group is the *attacin-like* molecules, which disrupt dividing cells of *Escherichia coli* and other bacteria, causing the breakdown of their outer membrane. Genes coding a number of these antibacterials are known (Lee, Cho et al. 2001).

The regulatory regions for these peptides contain binding sites for transcription factors containing the Rel domain, and some of these Rel proteins are homologs of transcription factors with similar roles in immunity in mammals, the *NFκB* family. The insect gene *Toll* does not function in microbe recognition but is activated on an immune challenge. A number of TLRs, in addition to *Toll*, have been identified in *Drosophila*, and these do interact directly with microbes (Khush and Lemaitre 2000).

As in vertebrates, the immune response in insects is activated by recognition of common proteins on microbes by PRRs. Peptidoglycan recognition protein (PGRP), for example, has been identified in the moth *Trichoplusia ni* (Khush and Lemaitre 2000). It binds to Gram-positive bacteria, as does peptidoglycan recognition protein in vertebrates. Gram-negative binding proteins (GNBP) have also been isolated from insects, again with homologies to proteins in vertebrates that recognize Gram-negative bacteria and initiate an immune response (Ochiai and Ashida 1999).

Inside the body invading microbes are attacked by hemocytes, that is, phagocytic cells in the hemolymph. Antibacterials and antifungals are produced in the *fat body*, the functional equivalent of the mammalian liver. The genes that encode these peptides, including cecropins, attacins, diptericin, defensin, drosocin, drosomycin, andropin, and diptiricin-like protein, are induced by intracellular signaling cascades with homologies to the activation of *NFκB* in mammalian immunity (Lagueux et al. 2000). Other antimicrobial responses include the production of nitric oxide, which

is toxic to some parasites, and perhaps sequestration of iron to limit bacterial infection (Khush and Lemaitre 2000).

Activation of the immune response in an insect, either because of microbial infection or wounding, also causes systemic and behavioral changes including reduced feeding, and developmental delay, and there is even a possible febrile response, marked by the insect's pursuit of warmer locations in its environment (Hultmark 1993). The type of immune response an insect mounts depends on the type of microbe against which it must defend. Fungi and bacteria elicit different responses in insects, as they do in vertebrates.

MOLLUSKS AND OTHER INVERTEBRATES

Studies in mollusks show that at least some of the mechanisms found in insects have earlier origins. Two members of the insect defensin family of antibacterial peptides are found in mollusks (Charlet et al. 1996). PRPs specific to molecules on the surface of pathogens initiate the immune system in these animals, as in the vertebrate and invertebrate systems. Among other responses, PRPs initiate the *prophenoloxidase* (ProPO) cascade, the same melanin release system found in insects. The ProPO system recognizes nonself proteins and targets them with antimicrobial peptides and other toxic metabolites.

Some of the immune response of these animals relies on clotting, and the clotting protein has been cloned and found to be a member of the vitellogenin family. Echinoderms have a complement system, somewhat analogous to that of vertebrates.

IMMUNITY IN PLANTS

PLANTS HAVE SPECIAL VULNERABILITIES

Because they have the basic characteristics of life, Aristotle hypothesized that plants have a property that we might call awareness, and we are learning that in many ways this is indeed the case. Rooted plants are sessile and many remain in the same place for a very long time, especially relative to the life spans (and mutation cycles) of microparasites. Plants can be very large and are attractive hosts for macro- as well as microparasites. One would expect that plants simply could not persist as a form of life if they were passive meals for pathogens. Although the mechanisms are different from those in animals, plants do indeed sense attack and respond defensively. But, interestingly, so far as we know, they have not evolved adaptive immune responses.

Plants have developed a number of mechanisms to defend against a wide range of bacteria, fungi, viruses, nematodes, and insects. When plants detect a pathogen they quickly launch one or more of these molecular defenses. In his seminal work on plant pathology, specifically the resistance of flax to flax rust, H. H. Flor in 1956 first suggested a *gene-for-gene* pathogen-specific response by plants to pathogens (Flor 1956), and molecular genetics has subsequently confirmed and expanded knowledge of this relationship. The gene-for-gene response involves the interaction of the products of *resistance* (*R*) *genes* in the plant and *avirulence* (*Avr*) *genes* in the pathogen. *R* genes code for receptors to products of *Avr* genes: the pathogen releases the product of the *Avr* gene into the plant at the site of invasion, and it

interacts with the product of the *R* genes, whose expression is induced by the presence of Avr protein.

The *R-Avr* gene interaction activates a cascade of defensive gene expression changes. Details are just becoming known in a few model plants and pathogens. However, the signal is transduced by the *R* gene, which as well as having a ligand-binding (LRR) region to serve as a receptor, also has nucleotide-binding (NBS) regions to serve as a transcription regulator. The resulting defensive reactions include the *oxidative burst*, which can trigger rapid localized cell death at the infected site, an apoptotic reaction known as the *hypersensitive response (HR)*. This involves homologs of apoptosis-related genes in animals.

Plants have tougher cell walls than animals, but these can be penetrated by pathogens. Plants can respond by crosslinking (chemically toughening) their cell walls against further invasion, expressing at least eight classes of antimicrobial compounds including defensins, thionins and snakins, toxic secondary metabolites, and hydrolytic enzymes (Almeida et al. 2000; Garcia-Olmedo et al. 1998; Klessig et al. 2000; Segura et al. 1999).

These same responses can also be triggered by a nonspecific host response to attack. The oxidative burst, in which levels of reactive oxygen species and hydrogen peroxide rapidly increase, is the most common plant defense and is often triggered by nonspecific defense reactions. This activates the hypersensitive response, which involves an apoptotic reaction that destroys the plant's own cells in the area under attack and thus deprives the pathogen of nutrients, confines the pathogen to the site of necrosis by creating a physical barrier composed of dead plant cells, and induces the production of antimicrobial proteins (see, e.g., Mittler et al. 1999). The HR is triggered by the production of nitric oxide, which induces the expression of genes that synthesize antimicrobial compounds (Delledonne M et al. 1998).

These are localized responses that occur at detected sites of infection and serve to seal off the infected tissue and the pathogen, depriving it of the opportunity to grow or spread. However, plants have also evolved a mechanism to resist attack systemically, known as *systemic acquired resistance (SAR)*, triggered by infection with a necrotizing pathogen. SAR establishes a state of heightened resistance throughout the plant to protect against subsequent attack, not only by the pathogen involved in the primary attack but also by a broad range of pathogens. This response is characterized by an accumulation of salicylic acid systemwide and the production of a set of *pathogenesis-related (PR)* proteins that confer increased nonspecific resistance to the plant, probably through antimicrobial activity (Feys and Parker 2000). SAR, once triggered, has been found to last as long as an entire season.

Another form of systemic resistance has recently been described: *induced systemic resistance (ISR)* (Pieterse et al. 1998). Independent of salicylic acid and PR proteins, plants respond to nonpathogenic colonizing bacteria, often found in the roots, by enhanced protection against pathogenic attack. ISR is triggered by signaling pathways regulated through *Npr1* that lead to the release of plant growth hormones, jasmonic acid, and ethylene. Jasmonic acid and its derivatives (known collectively as jasmonate) trigger the release of genes encoding defense-related proteins including thionins and proteinase inhibitors. Ethylene induces the expression of members of the *PR* gene family, and jasmonate and ethylene together synergistically stimulate *PR* gene expression (Xu et al. 1994) and activate the expression of genes encoding plant defensins (Penninckx et al. 1996).

The gene-for-gene response mechanisms seem to be relatively rare elicitors of plant defenses (Klessig, Durner et al. 2000) compared with the nonspecific host response. However, the gene-for-gene mechanism is phylogenetically diverse and widespread among plants, suggesting that it arose early in plant evolution. Plant resistance genes can be divided into at least five classes of related sequences, defined by the structural characteristics of the proteins for which they code (Ellis et al. 2000). They can be distributed across the genome, and there are at least 200 of these genes in *Arabidopsis* alone, comprising about 1% of its entire genome (Ellis, Dodds et al. 2000), comparable in scope to the mammalian olfactory genes (Glusman et al. 2001). Gene-for-gene immune mechanisms were thought to apply to vertebrates as well, until somatic recombination was discovered.

The majority of plant resistance genes isolated to date share NBS-LRR domains (Feys and Parker 2000; Young 2000). A major subclass also has a "TIR" region, with homology to *Toll* of *Drosophila* and interleukin receptor-like proteins in mammals, which are involved in nonspecific cellular immunity (e.g., Girardin et al. 2002). Toll-like receptors in vertebrates and insects mediate an inflammatory response that is similar to the plant defense responses: in both, the response includes production of antimicrobial active oxygen species or free radicals with bactericidal and microbicidal properties and, ultimately, death of the infected cell (Dixon et al. 2000). Some *R* genes have protein kinase intracellular domains and work through the classic message-transduction means, but others lack such an internal domain (Ellis, Dodds et al. 2000).

In the gene-for-gene model, plant and pathogen genes are under different evolutionary pressures; unlike animals, plants cannot flee their predators, and many plants have long life spans. One might expect that micropathogens that can divide (and hence mutate) in hours would easily out-evolve any defense on the part of plants. Of course, plants maintain pluripotent cells throughout their lives, they reproduce from many different places, and they reproduce by producing huge numbers of seeds (with the recombinational and mutational variation that can generate). Thus the attack of a pathogen on one branch will not necessarily destroy the reproductive capacity of other branches. However, because plants can live long lives the pathogen need not be in any hurry. Pathogens infecting seeds would cause a particular problem.

Plants must detect *Avr* proteins to trigger their disease resistance responses, but the pathogen benefits from evading detection. It is not likely that the role of *Avr* genes in triggering plant defenses that thwart their replication explains why these genes have been maintained by selection. Indeed, in the absence of the corresponding *R* gene, *Avr* genes play a role in the virulence of a pathogen. Virulence is recessive; if a pathogen is to become virulent in the presence of its specific receptor in the plant, its *Avr* gene must undergo a loss-of-function mutation so that the pathogen can evade detection by the plant's *R* gene (Richter and Ronald 2000).

R genes are subject to diversifying selection. To resist a new pathogen, the plant must gain a new resistance function. *R* genes are highly polymorphic, with high recombination rates and high evolvability (Ellis, Dodds et al. 2000; Richter and Ronald 2000). They typically are in clusters of many tandem duplicate genes along with a number of pseudogenes (probably the detritus of frequent recombination events), and several carry transposons, deletions, or frame shifts (Ellis and Jones 1998; Richter and Ronald 2000) on the same and on different chromosomes.

This reservoir of variation may be essential for the generation of new *R* genes in the face of evolving pathogens. However, it may not suffice to produce the amount of pathogen-specific resistance that has been found. Several instances of disease resistance conferred by the interaction of more than one *R* gene have been documented (Botella et al. 1998), which may give a combinatorial defensive capability comparable to some nonrecombining systems in animals (e.g., olfaction, discussed in Chapter 13). Additionally, some *R* genes can only elicit the hypersensitive response if the pathogen expresses an *Avr* gene and *Hrp* (*hypersensitive response and pathogenicity*) genes as well.

Of course, pathogens have evolved mechanisms to evade host detection. Pathogens lose avirulence (become virulent) much more quickly than plants gain resistance genes. Also, some pathogens are known to take advantage of programmed cell death to avoid the activation of host defenses. For example, necrotrophic fungi *B. cinerea* and *S. sclerotiorum* are pathogens that trigger the hypersensitive response and apoptosis to exploit necrosis to colonize the plant, in this case, *Arabidopsis* (Govrin and Levine 2000).

The genes involved in *R* gene signal transduction vary by plant and pathogen. The pathways involved have begun to be understood by the study of plant mutants; when specific genes are mutated, expression of downstream genes in the pathway may be blocked and resistance compromised or absent. Aarts (Aarts et al. 1998) has suggested that there are at least two pathways initiated by *R* genes, and the structure of the *R* protein in part determines which pathway it invokes. Some *R* genes belong to the leucine zipper subclass of *NBS-LRR* genes, and others belong to the class that encodes a protein with amino-terminal similarity to the cytoplasmic domains of the *Drosophila* Toll and mammalian Interleukin1 transmembrane receptors, the TIR-NBS-LRR (Aarts, Metz et al. 1998; Glazebrook 2001). Genes in the first pathway are marked by downstream activation of resistance genes that requires *Eds1* (*Enhanced Disease Susceptibility 1*), and in the other, *Ndr1* (*Non-Race Specific Disease Resistance 1*) is required (Aarts, Metz et al. 1998). There are *R* genes that require neither *Eds1* nor *Ndr1*, although these genes seem to play a role in their defense signaling, so it seems that there is at least a third pathway, the genetics of which are not yet known (Glazebrook 2001).

Some further specifics about the *Ndr1* and *Eds1* pathways are now known. For example, *R* genes that require *Eds1* also require *Pad4* (*Phytoalexin Deficient 4*). Genes that require *Ndr1* also require *Pbs2* (*avrPphB Susceptible 1*) (Glazebrook 2001) but are independent of *Eds1* (Aarts, Metz et al. 1998). A number of *R* loci are known to require *Eds1* (the *Resistance To Peronospora parasitica* genes, *Rpp2*, *Rpp4*, *Rpp5*, *Rpp21* and *Rps4*, for example). Mutant *Eds1* plants no longer have resistance to *P. parasitica* conferred by these *Rpp* genes (Aarts, Metz et al. 1998).

Plants also produce a class of steroid hormones, *phytoecdysteroids*, which are homologous to insect *ecdysteroids*. These compounds are noxious to herbivores and insects, both in flavor and effect, and can have physiological or even “xenohormonal” effects on other species, such as inducing abnormal molting or infertility in insects and clover disease in sheep, which affects the fertility of ewes by inducing histopathological changes in the uterus (Oberdörster et al. 2001). These hormones act through nongenomic pathways as well as the better-understood receptor-mediated initiation of gene expression and are often antagonistic to the predators’ endogenous hormones. This shows the mixed effects of some of these plant “hormones.” We would not normally use the term for substances that are toxic to preda-

tors, but as so often happens, functions even of the same substance are not neatly confined. As stated in Chapter 10, neurological, immunological, and endocrinological systems are in many ways related.

OTHER MOLECULAR ANTIPREDATOR MECHANISMS

The role of microdefenses (gene products) against microscale internal invasion is the subject of this chapter, but the term “immune” resistance can be extended a bit to allow us to mention that microscale mechanisms exist against macro- and external predators. This is the world of toxic or bad-tasting compounds plants produce to keep predators away (it works at least to some extent, but often some predator becomes resistant and eats the plant). The same is true in some molecular ways for animals (octopus ink, skunk odor, etc.)

THE MOST PRIMITIVE IMMUNE SYSTEMS

Even simple organisms like bacteria are food for viruses and other parasites (sometimes including other bacteria). Not unexpectedly, bacteria have immune responses of their own. One response is the ability to resist natural antibiotics that can exist in their environment. Naturally occurring antibiotic resistance genes from a number of bacteria have been cloned and are routinely used in laboratory microbiology and molecular genetics research. Often, these genes are found on plasmids, simple circular DNA molecules that can be exchanged between bacteria (this exchange along with a rapid life cycle explains the ability of bacteria under antibiotic siege to develop resistance rapidly). Resistance genes on plasmids work in conjunction with other chromosomal genes in bacteria.

Bacteria can be invaded by viruses, sometimes known as *bacteriophages*. At least one major way in which they defend themselves is by the use of *restriction endonucleases* that they encode in their genome. These are a large class of genes that bind DNA at short recognition sequences and cleave it (the sequences are often palindromic in nature, taking into account the reverse complementarity of DNA, such as **GAATTC**, so that the restriction enzyme can bind equally well on both strands in opposite directions). This will destroy incoming viral DNA. The bacteria protect themselves by protecting their own DNA from cleavage, as with sequence-comparable restriction methylases that modify their DNA so the restriction enzymes will not cut it.

Another primitive defense system involves RNA interference (Carthew 2001; Sharp 1999), described in Chapter 7. Double-stranded RNA (dsRNA) complementary to a section of the mRNA from a gene can inactivate that RNA and hence prevent translation. This is a system used by various invertebrates in defense against viruses.

THE EVOLUTION OF IMMUNITY

PHYLOGENY OF INNATE RESISTANCE SYSTEMS

The innate responses are varied and much older than the adaptive responses. Many if not most of the systems are found in vertebrates and invertebrates, and indeed often in plants as well.

For example, four *Defensin* families are found in eukaryotes; *Alpha-Defensins* and *Beta-Defensins* in mammals, insect *Defensins*, and plant *Defensins* (Hoffmann, Kafatos et al. 1999). This general mechanism arose long ago and clearly has remained useful up to the present. The production of nitric oxide by plants for use by the hypersensitive response, in inducing apoptosis and the expression of genes that induce the production of antimicrobials, is enzymatically similar to nitric oxide production in animals (Clarke et al. 2000), which is also used as an antibacterial toxin and against tumor cells (an appropriate type of autoimmunity) (Roitt, Brostoff et al. 1998). The associated cell death that occurs in the HR reaction in plants is functionally analogous between plants and mammals; it is even found in bacteria. Cytotoxic T cells can signal a targeted cell to undergo apoptosis (Roitt, Brostoff et al. 1998). We have seen many ways, including apoptosis, by which normal development is based on internally derived signals, screening, stimulus, and assault.

AN EVOLUTIONARY INTERFACE BETWEEN INNATE AND ADAPTIVE IMMUNITY?

There are similar levels of parallel in signaling pathways using *Toll/Cactus/Dorsal* in *Drosophila* and *IL1 receptor (IL1R/NFκB)* in mammals and *R* genes with their *NBS-LRR* domains in plants (Medzhitov and Janeway 1998). Not surprisingly, elements of the system have some correspondence with genes in bacteria, such as the NBS with that of animal apoptosis genes (Young 2000). Again, these are relatively simple or generic defenses that have stood the evolutionary test of time.

However, the *R* gene resistance mechanism seems not to be completely straightforward. Some *R* gene clusters show evidence of diversifying selection, which may be related to a general fusillade response, whereas others clearly show that classic darwinian allelic selection matching specific *R* alleles to pathogen *Avr* genes has been important. That is why the gene-for-gene system is “innate” rather than “adaptive” in the sense these terms are used here.

Although both involve a history of repeated tandem gene duplication and recombination, the *R* gene system is fundamentally different in tactic from the vertebrate adaptive immune system, which requires somatic rearrangement and mutability, rather than inherited allelic “memory,” to mount a defense against a specific pathogen. However, the *R* gene system does involve combinatorial mechanisms and may be viewed as a kind of intermediate state between wholly generic chemical innate resistance and highly adaptive immunity. We do not imply that *R* genes represent an evolutionary missing link, because the common ancestor of plants and vertebrates did not have a multigene system from which *R* genes and immunoglobulins have descended. However, the vertebrate system *did* have an *R*-gene-like antecedent (see below).

Although deep homology of individual genes or gene domains shows that the basic genetic mechanisms are in some sense conserved and used for new purposes in immune defense, phenogenetic drift is always possible. In plant disease resistance responses, HR can be elicited by *R-Avr* gene interaction but also by non-*R-Avr* interaction. Some *R* genes pick up *Avr* signal cytoplasmically, some extracellularly, some are transmembrane transducers, some *R-Avr* interactions elicit an HR that involves apoptosis, and some do not. Some *R* genes do not transduce signal without another *R* gene involved; others do not need interaction.

PHYLOGENY OF THE ADAPTIVE IMMUNE RESPONSE

We tend to think of the adaptive response system as that which evolves during life, particularly the antibody generation mechanisms that we see in gnathostomes. But this system is innate in agnaths, who rely on combinational response rather than recombinational or gene-for-gene responses (“combinatorial” means that it uses multiple inherited genes, in the “hope” that at least one will work, whereas “recombinational” means using rearrangements within the organism’s lifetime that are not inherited). However, it is possible in principle that natural selection in the form of a major pathogen could favor the proliferation of one of the many antibody genes possessed by an agnathic animal, as it does in the case of *R* genes in plants.

In principle, the same could still occur in gnathostomes, but this goes against the prevailing view that the adaptive system involves so much mutation and recombinational imprecision that it became more efficient and effective to generate resistance by recombination than by combinatorial attacks with inherited alleles. The gnathostome system is so mutable that classic allelic (darwinian) selection might not even work. A specific V gene allele favored in one individual would be unlikely to be sufficiently prominent among the array of somatically generated antibodies that, in other individuals, would happen to be a major aspect of the latter’s defense against the pathogen. Selection would not be able to, and would not need to, single out a particular allele in a particular V region of an immunoglobulin gene cluster. Still, selection by epidemic can be severe and rapid, and it may be that more classic kinds of selection do in fact pertain to the components of antibody genes.

Some of the basic pathways of resistance are ancient and serve (and perhaps initially arose to serve) other purposes. Signaling pathways used in the vertebrate adaptive immune response, with invertebrate—and even plant—homologs, function to activate innate immunity in the absence of an adaptive response. Earlier we noted the homology of the amino terminus of other closely *NBS-LRR*-type genes to the *Drosophila* Toll or human interleukin receptor-like (TIR) region. Several human homologs of the Toll (h-Toll) protein have been isolated and shown to activate the *NFκB* pathway, which mediates lipopolysaccharide-induced cellular signaling and initiates the vertebrate adaptive response by production of inflammatory cytokines. The NBS domain of plant *R* genes is characterized by several sequence motifs found in the recently described *Ced4* and *Apaf1* animal genes. The latter genes regulate the activity of proteases that can initiate apoptotic cell death. Should we add development to the common neuro-immuno-endocrine system? Probably it is simpler to repeat that the definition of a “system” is a human artifice laid upon an evolutionary reality.

It is difficult to account in a convincing way for the evolution of the somatic recombinational adaptive immune system because so many millions of species do well in similar environments without such a system. What selective force would have given animals an advantage from producing and fine-tuning the adaptive immune system? Perhaps one major adaptive advantage of the recombinatorial response is its role in resistance to reinfection, provided by immunological memory (Parish and O’Neill 1997). This may be a plausible argument given that the adaptive recombinational system depends on initial antigen recognition by the noncombinatorial innate system. Or, perhaps as noted earlier, the adaptive immune system evolved primarily for “body maintenance” (wound healing, tissue repair, angiogenesis, cell regeneration, and the degradation of old or abnormal cells (Cohen 2000)) and defense against pathogens is a by-product.

The immunoglobulin family with its many V, D, and J regions has had a long evolutionary history. The cell adhesion family that includes these genes (and most of the MHC genes) serves a wide variety of purposes, all mediated through similar domains (intracellular, transmembrane, and extracellular). The array of immunological *Cam* genes found in different vertebrates shows their history. In terms of immunological (and/or self-recognition or other related) functions, the MHC/T cell receptor system can be traced phylogenetically as far back as elasmobranchs (cartilaginous fishes, including sharks) and teleosts (bony fishes, including most of the more common fishes) (Nakanishi et al. 1999; Roitt, Brostoff et al. 1998). These genes generate high clonal diversity among white cells, but the mechanisms vary and have changed during the hundreds of millions of years the system has had these functions.

The repertoire of antibody diversity in vertebrates varies greatly. The genes are highly polymorphic in fish, as they are in mammals. In light of the results of allograft and other experiments, it is thought likely that fish MHC genes have an immune function similar to that in mammals (Nakanishi, Aoyagi et al. 1999; Roitt, Brostoff et al. 1998). However, potential diversity in elasmobranchs is not nearly as extensive as in higher vertebrates. This is because the heavy chain of shark Igs is constructed by the stringing together of V, D, and J segments as multiply repeated units and this eliminates the possibility of recombination between the separate gene segments (Roitt, Brostoff et al. 1998).

Chicken antibodies show considerable diversity, but this is largely due to gene conversion rather than Ig gene recombination (Roitt, Brostoff et al. 1998). In the light chain system, there is only one V, one J and one C segment gene, and in the heavy chain system, only single V and J segments. There are 16 D segments, but they do not exhibit much sequence diversity and thus do not contribute appreciably to a diverse repertoire.

Classic molecular types of response that we would describe as being an immune system are necessarily much older, because they are complex and hence had to evolve over some time period. On the basis of DNA sequence analysis, the origins of the adaptive immune system can be traced phylogenetically to the precursor of the recombinase activating genes, *Rag1* and *Rag2*, which arose approximately 450 million years ago (Agrawal et al. 1998). It is in part because this system seems to have arisen fairly well intact in jawed vertebrates, that the somatic recombinational machinery may have been transferred horizontally rather than evolving from endogenous precursors (Marchalonis and Schluter 1998; Marchalonis et al. 1998; Marchalonis et al. 1998), although this idea is not universally accepted (Hughes 1999) (horizontal transfer is thought to be difficult once specialized eukaryote cells evolved, as we described earlier). In any case, the use of the same set of genes for immunelike function probably antedated the recombinational mechanism for making increased clonal diversity.

AN INTERESTING VARIATION ON THE THEME

The molecular battle between micropredator and the prey that is the object of its attention has led to recombinational as well as combinatorial systems. An interesting phenomenon that is conceptually related is the genetic evolution of venom proteins in a group of around 500 species of predatory *cone snails*. These diverse species have several gene families related to venom protein production. Although the

sequences of their signal peptides, responsible for processing the proteins after they are coded, are relatively conserved, the venom peptide domains are hypervariable (Olivera et al. 1999). Different exons in these genes may have very different mutation rates, with the effect of generating toxin peptide diversity. These venom proteins attack ion channels involved in their victims' neurotransmission and hence act as neurotoxins. The hypervariability in these genes is a combinational strategy, like that of the *R* genes in plants and that we will revisit in Chapter 13 when we discuss olfaction. A possible explanation is that this reflects the evolution between their venom characteristics and evolution in the target molecules of their prey.

A BIT OF A MISNOMER

The distinction between innate and adaptive immune systems actually distinguishes between generic and somatically evolving active responses to infection. There is, however, a more direct sense in which organisms “adapt” to infection, and that is in the classic evolutionary sense, and sometimes specific to the attacker.

Mutations in genes related to the innate system or the mechanisms underlying the adaptive system can generate generalized immune deficiencies. Presumably, those aspects of immunity are maintained by darwinian selection. We also know of examples of passive resistance to, rather than active attack against, pathogens that evolve in the usual sense (alleles selected relative to specific pathogens). Selection can directly favor such host alleles, raising them to higher frequency.

The classic examples are the malaria-resistant hemoglobinopathies like sickle cell anemia. What these do is block some aspect of the pathogen's life cycle or biological needs, rather than actively attacking its cell surface antigens, etc. A stage of malarial parasites, for example, must take place within red blood cells, which are filled with hemoglobin. Mutant hemoglobin that prevents parasite entry into the red blood cells effectively resists infection. The example of sickle cell anemia is instructive also in that there is a disadvantage to having sickled hemoglobin. This leads to a balancing selection and a stable allele frequency polymorphism between “normal” hemoglobin that is a good oxygen transport molecule and “abnormal” sickle cell hemoglobin that is resistant to malaria but not a good oxygen transporter.

Many pediatric diseases in humans are recessive in the sense that a mutant allele by itself does not lead to disease. Yet the frequency of alleles associated with the disease sometimes seems too high, given the seriousness of the affected genotypes, which are often nearly lethal. Phenylketonuria, cystic fibrosis, and Tay–Sachs disease are among many examples. It is thought that the “pathogenic” alleles may have experienced a history of some as yet unknown balancing selection in these diseases, even if less dramatically than is the case with sickle cell anemia. If such explanations are made specific and prove true, it will extend the array of mechanisms against infection that are not, but probably should be, classified as part of “immune” systems. Perhaps the reason we don't now classify them as such is in a sense evolutionary: these “new” mechanisms arrived via different evolutionary pathways than the complement/antibody/MHC systems.

CONCLUSION

All organisms are under attack. If they respond well enough to go on to reproduce, they win. There is a great variety of ways that organisms respond, from simply repro-

ducing soon and often, to having very complex, staged molecular machinery to identify, isolate, and destroy pathogens at the molecular level and, in some organisms, to remember what happened earlier so that in the event of subsequent exposure the invader can be destroyed more quickly. The organism might also just tolerate or become commensal with the pathogen or may out-reproduce or simply endure it.

It is too easy to reconstruct “had to” stories for these systems. Except for the most primitive and nonspecific innate systems, there is clearly no mandate to have an immune system of any particular type or efficiency. Either the more complex systems overlaid on simpler systems initially (and perhaps still) serve some other function, or the circumstances that called for them no longer exist, or we do not understand them. The complexity of immune systems today is “good enough” but not perfect or of the same type in different species. Slow metabolism or rapid reproductive strategies, or reproduction from diverse sequestered parts (as in plants), may allow an organism to tolerate infection rather than having to fight it.

It is not hard to see how molecular defense strategies might evolve. Any ligand-binding system in an organism might be modified by mutation (perhaps after gene duplication) to produce alleles that bind things not normally part of the system. If it binds the wrong things too aggressively it is selected against (and we might call that a “disease”), but if it does not bind self-structures too aggressively it may end up binding to other things that get inside the organism, inactivating them, and being favored by selection. It can then undergo evolutionary elaboration.

A subject beyond this book is the ecological evolution by which species adapt to commensalisms or tolerance. From a selfish evolutionary point of view, a pathogen that kills its host can survive only if transmission to a new host occurs quickly and readily enough (that is, before the original host is dead and gone). The conditions for epidemic versus endemic infection are debated among immunologists and epidemiologists (Ewald 1994). Many parasites kill their hosts only slowly or if the latter are under nutritional or other stresses. They are endemic because they are always there to infect new susceptibles born or migrating into the population. Other parasites cause little or no reaction by the host and do no harm. As we have noted above, some parasites are symbiotes: we cannot do without them, and they cannot do without us. Each case is different. The important point is that immune systems and/or parasites evolve so that the immune reaction is strong enough to protect the host but not necessarily any stronger. Of course, neither the strategy of host nor that of the pathogen is always successful. Those that have overstepped their bounds are generally no longer here to be studied.

PART IV

External Awareness

Information Transfer between Environment to Organism

*The life of any organism largely consists of dealing with the outside world where things they need, like food and mates, as well as things that threaten them, are found. Organisms have evolved to receive and **perceive** whatever aspects of the environment sufficed, in its ancestry, for survival. There is no one way to do this, and the sources of “information” in the environment are diverse.*

One important characteristic of many vital aspects of the environment is that the organism cannot know what they will be like in advance. It cannot be genetically programmed or “hard-wired” to handle unpredictable stimuli or signals. Instead, a diversity of mechanisms has evolved that enable organisms to respond to the events they need to in order to survive, seek food and mates, and so on.

Animals use some aspects of the environment that are largely predictable, such as gravity or magnetism. Other aspects are more truly open-ended. They include sounds, lights, and chemical components. Many intriguing, and sometimes closely related, means have evolved to receive input from these environmental stimuli and resources, and turn it into useful information. Examples of the responses are phototropism in plants, or vision, olfaction, and hearing in animals.

Yet, not all organisms respond to all signals that might be useful to them, and we will consider why that is so.

Chapter 12

Detecting Physical Variability in the Environment

Almost 2500 years ago, in his treatise *On the Soul (De Anima)*, Aristotle wrote, “There is nothing in the intellect that was not first in the senses.” He enumerated five senses: sight, hearing, smell, taste, and touch. To Aristotle, the senses were formed of earth, water, air, and fire, as was all matter, and they came together in the “common sense,” the heart, which brings us the awareness of sensation, allows us to distinguish between the perceptions of each sense, and yet ultimately combines them into one common experience.

Aristotle was probably the most noteworthy empiricist in the history of Western thought, at least until Bacon and perhaps Descartes in the Enlightenment period. He believed that nothing could be known without first being experienced. He had very broad interests and investigated all subjects with the same intensive data-gathering approach. His teacher, Plato, by contrast, was a rationalist. To Plato, knowledge was innate; it preceded and was not dependent upon experience.

We can argue with Aristotle’s restricted list of the senses, because he should have realized that we can sense gravity, or balance, or the location of our bodies in space, or other ways in which we (and other organisms) sense our environment. Or we can disdain his primitive idea that the senses are of the four elements, because we know they are really receptors and ligands and proteins, or that the heart is the interpreter of sensation, because we know it is really the brain. We can smile at Plato’s rationalism, because we, too, are empiricists in the best scientific tradition. But if we step back from these particulars, we might find that our own view of the senses, even with our extensive knowledge of genes and molecules, is in fact more similar to, or even derived from, the thought of these ancient Greeks than we care to realize. In fact, from an evolutionary perspective we should expect more connectedness, less distinction, more repetition but in other ways less “rationalism” in the traits of life.

Does knowledge precede experience, or is it the other way around? What we know about senses and learning and behavior suggests that it is not one or the other.

The gravitational sense, the light-seeking behavior of many organisms, and sucking in newborn mammals are all considered to be instinctive or reflexive behaviors. Animals are born “knowing” these things. Biologists would even argue that it is “coded” in the DNA, which must be as platonic a view of inherent knowledge as there can be.

On the other hand, implicit in our understanding of the senses is that they are used to acquire information—*new* and *unpredictable* information—about the world. Knowledge derives from experience, Aristotle said—so our own view of the senses has also undeniably been Aristotelian. Furthermore, many biologists assign darwinian purpose to the senses: hearing is “for” the detection of predators, taste is “for” the detection of toxic or poisonous foodstuffs, smell is “for” detecting pheromones and finding mates, etc. The knowledge we take from our senses, then, is “for” survival and reproduction.

However, in both the Aristotelian and Darwinian views of the world, knowledge—the information our senses provide us with—has two teleological components. First, it does not seem a stretch to say that the current functional purpose of our eyes is to receive light (this is not the same as saying that eyes evolved with that intended objective, which would imply that before there were eyes, an organism dreamt of seeing and strove to create a way to do so). Second, our brains use the information received from our eyes to direct us to objectives that we have in mind—pursuing that deer, searching for our lost keys. We might note that once sight evolved as a distinct function, integrated with the brain to regulate motor and other behavior, it is philosophically reasonable to say that the sense was modified by evolution (e.g., to achieve greater light and color sensitivity or focusing resolution) “for” (at least, in relation to) what today is its function. A pure mechanician might argue that our awareness that we are purposive is itself molecularly programmed, but we can leave that ultimately philosophical issue aside.

As stated earlier, we generally attribute to René Descartes the view of the body as a machine. He wrote in 1664:

First of all, I want the reader to have a general notion of the entire machine which it is my task to describe. So I will say here that the heat in the heart is like the great spring or principle responsible for all the movements occurring in the machine. The veins are pipes which conduct the blood from all the parts of the body towards the heart, where it serves to fuel the heat there. The stomach and the intestines are another much larger pipe. . . .

(Descartes, *Treatise on Man*, 1664)

Intrinsic to the Western scientific method and tradition is the purposeful dissection of a system into its constituent parts so that we might analyze them in order to understand the whole. Thus we try to understand the senses individually, in a Cartesian way, even though we know that taste without smell is not as rich and that a homing pigeon relies on smell as well as sight to find its way. What does it mean to understand the parts, then? And, even after we reduce the senses to their molecular and cellular pathways in fact this tells us nothing about their “purpose,” how the brain interprets sensory signals sent to it, or how information from all the senses at once is integrated, and the like. These kinds of questions cannot be reduced, or answered sense by sense.

In this and succeeding chapters we discuss many of the ways in which organisms perceive and respond to stimuli in their external environments. The stimuli we discuss include sound, touch, movement, chemical gradients, temperature, gravity, light, taste, smell, and the electromagnetic spectrum including infrared and ultraviolet light, electricity, and magnetism. Except for the constant forces of gravity and magnetism, a key fact is that these stimuli are unpredictable and of varying intensities and ranges. This unpredictability is comparable to that of the molecular diversity confronting the immune system. An organism's ability to perceive them must be open-ended in a similar way. Light comes in many colors and levels of intensity, odors can be perceived as good or bad, seductive or noxious, sound is loud or soft, high-pitched or low.

One way or another, most if not all organisms, from bacteria to plants to invertebrates to vertebrates, have evolved receptors to at least a subset of these stimuli and are able to respond to a range of many of them. For each organism's life circumstances, the ability to discriminate among such stimuli has been at least sufficient for its survival—or, to survive, the organisms used what they had. Whether in some way it might be useful to be even more sensitive is moot. Thus whether humans would make use of better night vision or an ability to hear musical instruments producing lower sounds, or the like, these questions are irrelevant because our senses did not evolve with the *purpose* of seeing well in the dark, or capturing all vibration as sound.

A distinction is often made between the detection of internal and external stimuli, but this distinction, again in the manner of Descartes, may not be as clear-cut as it is often presented. Organisms monitor their internal as well as external state; the distension of the gastrointestinal system, blood pressure, or the levels of hormones such as insulin or glucose, salt and pH at the cellular level, and so forth. Receptors for internal events, *interoceptors*, share some principles of structure and function with receptors for external stimuli, *exteroceptors*. In all these systems, signals are received in specialized cells in the sensory organs and the information is transduced and transmitted to the central nervous system for deciphering. In addition, organisms can only respond to changes in pH and temperature and the like if they are monitoring their own internal environment so that they are able to determine when indeed there *is* a change. In fact, one could argue that the immune response is a form of monitoring and defending against threats from the external world—it just happens to be done internally. Immune detection and response is somewhat different from the usual notion of a “sense” because it happens strictly at the molecular level without being cognized; but there can be organismal response (sleeping in mammals, seeking warmth in cold-blooded species, and the like). However, for the purposes of this chapter, we will ignore these kinds of muddled distinctions and concentrate somewhat arbitrarily on responses to external physical and chemical stimuli.

WHAT ARE SENSES, AND WHY DO THEY EXIST?

If Aristotle and the other ancients defined senses in a rather restricted and perhaps anthropocentric way, here we are interested in generalities that we can draw from knowing something about these systems in very different organisms. What *is* sensation, and why do organisms have it?

As a general statement, we can characterize senses as the means by which organisms are stimulated by their external environment to react in some way. Here, the environment includes macromolecular (touch, sound), energetic (e.g., heat, magnetism, light), and chemical (pH, odors, tastes) characteristics, but signals from these kinds of stimuli may overlap. The nature of perception need not be neural and certainly not conscious, in the usual sense of awareness (but are ants “conscious” of each other when, as noted by Dante, they “touch their muzzles, each to each, perhaps to seek news of their fortunes and their journeyings” (Alighieri 1314)?). Similarly, in discussing response to environmental information we do not restrict ourselves to any particular types of response.

It is more than rhetorical to suggest that the division of sensation into discrete Cartesian categories is artificial. Such division is based on incompleteness but also to a considerable extent on how humans are structured to receive environmental signals (e.g., with distinct eyes, nose, and ears) as well the experiential aspects of how we perceive them, and thus seems “natural.” For practical reasons, the treatment below classifies sensation in broad categories, but we will show how this categorization is largely a human creation and, in particular, that the genetic mechanisms involved in different modes of sensation are often very closely related and probably ancestrally identical.

SOME BASIC SENSORY PRINCIPLES

Whether of internal or external stimuli, sensory perception begins at the level of the cell. This raises two general questions: first, what kinds of signals do cells sense and what mechanisms are used? Second, what do they do with the signal? (Here, inevitably trapped by language, we use words like “information” or “signal” without implying an intentional sender or coder nor that incoming data are neatly packaged).

Some sensory mechanisms cause cells to change gene expression. Other sensory cells respond to stimuli by transmitting an impulse to another cell, as in neural perception. Much of our own sensory perception is based on central nervous system (CNS) processing of such signals, and there are several ways they are received. There are in mammals a few senses not transmitted to the CNS by neurons (e.g., light receptors in the skin), but as discussed in the introduction to this chapter, we should not be too parochial because many organisms sense their environment in wholly nonneuronal ways.

RECEPTOR MECHANISMS

Recognition of mechanical and chemical stimuli begins with the binding of a ligand to a cell surface receptor. The ligands are in the form of chemical or energy signals, and the receptors are transmembrane proteins that bind specific signals and initiate the transformation of the signal into a neural event (in organisms with an organized nervous system, at any rate). This transduction of signal into perception can take either a direct or an indirect path from the receptor cell to the CNS.

In vertebrates, the organization of the sensory cranial nerves is essentially the same for all senses. The receptor cells are either true neurons or neuronlike cells. Receptor cells that are true sensory neurons have either free sensory terminal

endings or specialized sensory terminals. The axons from these cells terminate in the CNS. The senses of touch, pain, and stretch—which are perceived by the skin, muscle, and joints—generally involve this type of receptor, as does the sense of smell. Receptor cells that are not neurons are specialized cells with no axons. They relay receipt of their specific signal to the CNS by contact with neurons that transmit along their own axon. Hair cells of the vertebrate ear and lateral line of aquatic vertebrates as well as photoreceptors for light transduction are examples of this type of receptor. The basic cellular structures of different types of neurons are shown in Figure 12-1.

As with all neurons, when a sensory receptor is at rest—not activated—a *resting potential* is maintained across the cell membrane. That is, there is a difference in electrical potential across the membrane of the cell, which is generally negatively charged, and it is maintained by the sodium-potassium pump, which discharges more positive charge from the cell than it allows in. When the cell is activated by the specific stimulus to which it responds, this resting potential is disturbed and *receptor potential* results—a nerve impulse is generated. The electrochemical process that converts signals to receptor potentials is called *sensory transduction*, and generally begins with the activation of ion channels in the membrane of the receptor cell, controlling the passage of cations (positively charged ions such as sodium or potassium) into the cell.

When the flow, or *flux*, of ions into or out of the cell reaches sufficient magnitude in neuronal receptors, an *action potential* results in the axon. That is, the electric potential on the surface of the cell membrane changes, creating an electrochemical impulse that sweeps along the axon of the neuron to transmit the signal to terminals located in the brain or spinal cord. The terminals release their neuroactive chemical, which triggers *sensation*.

In a receptor cell that is not a true neuron the flow of receptor current is indirect, that is, it must be passed from the receptor to a neuron through a synapse. This often requires the intervention of a cascade of events, relayed by the second messengers, that leads to the opening of ion channels. The signal is then transmitted to a neuron to be carried along the axon to the nervous system. *Photoreceptors* and *chemoreceptors* are examples of this type of receptor. The use of secondary messengers produces a slower response than in mechanosensory transduction. Perception of sound is almost instantaneous, for example, whereas there is a delay between the stimulus and response in photo- or chemoreception.

MECHANORECEPTORS

Mechanosensory transduction is the process by which mechanical forces are converted into electrical signals. In this case, the signal is not molecular in nature, but must be translated by genetically based response mechanisms. Mechanoreceptor cells are the basis of a diversity of senses in multicelled organisms, including proprioception (the body's ability to orient itself in space and sense the movement of its own parts), balance, hearing, touch, which involves the detection of pressure on the surface of the body, the detection of stretching or twisting such as in joints or muscles or the digestive system, or the displacement of feathers or hairs or whiskers. Although they differ widely in the range of sensations they govern and the receptor that initiates the detection, the basis of each of these senses is a response to mechanical deformation—stretching, bending, displacement by air, and the like.

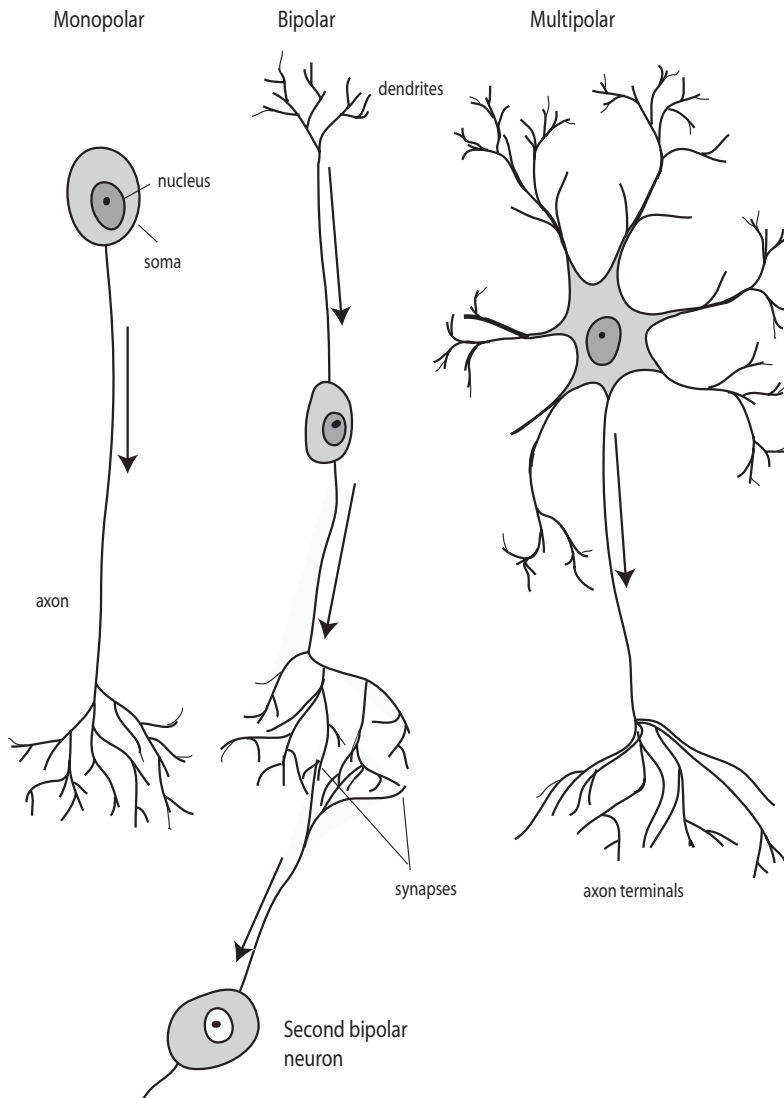


Figure 12-1. Basic cellular structure of different types of neurons. The signal travels *to* the cell body from the dendrites and *away* from the cell body, or soma, along the axon. The nucleus plays no neurological role, but functions in growth and metabolism of the cell. Multipolar neurons are the most common type in vertebrates; bipolar neurons are much less common and are found as olfactory receptors, receptors in the retina of the eye, and several cranial nerves, e.g. unipolar neurons are found in the spinal ganglia of the spinal nerves and some of the cranial nerve nuclei, where they transmit signal to the CNS.

They all depend on a mechanically gated ion channel for transduction of the mechanical force into the sensation that is ultimately perceived.

The molecular basis of the mechanical gating of ion channels is not yet well understood (Walker et al. 2000). Homologies found in this class of transduction mechanism, such as the fact that they send their message nearly instantaneously,

never requiring the action of a second messenger such as a G protein cascade, suggest to some a common evolutionary origin and/or a single molecular mechanism underlying these diverse responses.

HEARING AND BALANCE

Hearing is the transduction of vibratory energy into electrical energy, which in some species is decoded in the brain into information about the pitch and magnitude of the vibration. The stimuli can come through the air for terrestrial species or through the water for aquatic species or can come from direct contact with solid objects. Because animals with hearing have bilateral symmetry, they can typically also sense the direction from which the energy comes, relative to their midline. Hearing is mainly a feature of vertebrates and arthropods, although there are suggestions that some nonarthropod invertebrates can also hear (Budelmann 1992).

Here we use the single term “hearing,” but there is no one way to hear: different organisms use different aspects of sound in their own ways. Some simply need to respond to the presence of sound or, say, large, sudden sounds. Others respond to very specific patterns, frequencies, or even periodicities of sounds, such as the diverse ways that vertebrates and invertebrates respond to mating, territorial, or other such sounds from conspecifics. Organisms may need to recognize specific predator sounds. Or they may need to recognize sounds such as that of fire, moving water, and the like. Finally, of course, is the highly elaborate processing that humans do to interpret language.

Each of these uses of sound detection involves not just different levels of integration of sound impulses but different ways to detect sound. One cannot interpret elements of sound that one cannot detect and separate into its relevant elements. Nature has evolved particular mechanisms, especially perhaps in vertebrates, for discriminating sound. The variation in sound frequencies audible to a number of hearing organisms is shown in Figure 12-2.

HEARING IN VERTEBRATES

The Mammalian Model

There is great diversity in the shape of the ear in vertebrates, but the mechanism by which it transduces vibration into sound is similar across species (with similar mechanisms in some invertebrates). The classic system is that of mammals, and it illustrates the various modes of transmission. The outer ear collects the airborne sound waves and transfers them to the middle ear by vibrating the *tympanic membrane* stretched across the inner end of the ear canal; the middle ear contains three small, linked bones, the ear *ossicles*, that transfer the mechanical vibrations from the tympanic membrane to a thin, pliable membrane on the cochlea called the *oval window*, which communicates with the fluid-filled inner ear. Vibrations of the oval window are transmitted in the fluid and detected by ciliated hair cells in the spiral-shaped cochlea. Movements in the cilia are transduced into electrical energy and transmitted to the auditory portion of the brain via the acoustic nerve, which connects at the anchored end of the hair cells. The balance system, or labyrinth, is part of the same fluid-filled system and will be described below.

In all hearing vertebrates the ubiquitous and central player in this mechanosensory pathway, the sensory cell itself, is the ciliated *hair cell*. This cell type is ancient

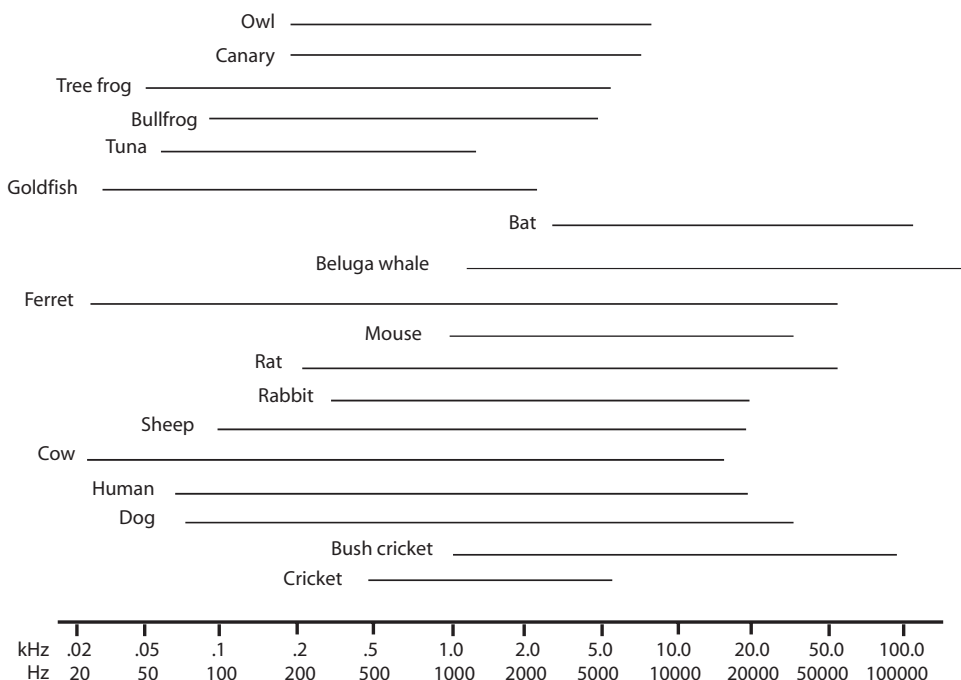


Figure 12-2. Sound frequencies (in Hz) audible to various species.

and found diversely in the living world. Cilia are tiny hairlike appendages, consisting of *microtubules* containing a protein called *tubulin*, that protrude from the surface of the cell (Alberts 1994). Ciliated cells are found in most animal species and some lower plants. Many protozoa are ciliated as well; they use the cilia to find food by washing fluid over the cell or for locomotion. Ciliated cells line the respiratory tract of many animals to move contaminants up and out of the lungs, for example, or they move eggs through the oviduct. Ciliated cells are integral to at least three sensory systems; taste buds, odorant receptors, and auditory hair cells and similar structures are used in photoreceptors.

In the hair cells of the *cochlea* and *semicircular canals* (also called the *labyrinth*) of higher vertebrates, groups of 50 or more cilia form bundles at the top of each cell and project into the fluid in the basilar membrane of the cochlea, which divides the cochlea lengthwise into an upper and a lower chamber, and in a gelatinous material, the *cupula*, in the bulging *crista ampullaris* at one end of each semicircular canal (see Figure 12-3). The cilia increase in length from one side of the bundle to the other. (See Figure 12-4 for a photograph of hair cells taken with a scanning electron microscope.) Deflection of these thin hairs by the fluid into which they protrude sends the signal to the sensory cells that connect from the basal end of the hair cell to the auditory regions of the brain. In this mechanically gated signal transduction system, if the cilia are deflected in the direction of the longest cilium, the action potential of the cell decreases, depolarizing it, and this induces the cell to release excitatory neurotransmitter at the synapse between the hair cell and the sensory neuron. Deflection away from the longest cilium, in contrast, results in hyperpolarization of the cell and decrease in the release of neurotransmitter.

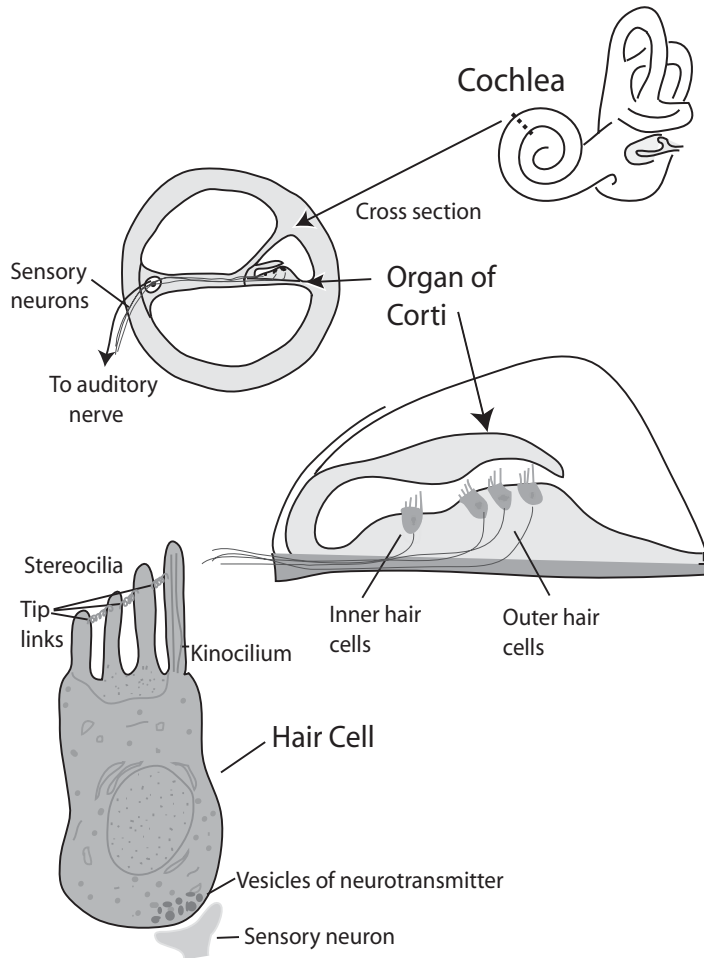


Figure 12-3. Structure of human cochlea and hair cells. Modified after (Matthews 2001).

The ear is one of the most intricately patterned structures in a vertebrate body. However, its development does not involve new principles or basically any new genes. It is beyond our scope to present ear development, but each region shown in Figure 12-3 develops through the agency of a set of signaling, transcription, and cytostructural factors (Kiernan et al. 2002; Petit 1996; Petit et al. 2001).

TFs, SFs, and signal receptor genes involved in the hind-brain region of the developing nervous system are involved in early otic patterning. A lateral otic placode invaginates to form an otic cup which closes to become the otocyst. The semicircular canals and cochlea derive from outpocketing from the otocyst. Genes from ectoderm and mesoderm are involved. Several genes related to the semicircular canals have been identified, but less is known about cochlea-specific developmental mechanisms. The hair and supporting cells become arrayed along the cochlea, probably by an activation and lateral inhibition system such as has been seen earlier, including that of the Delta/Notch system.

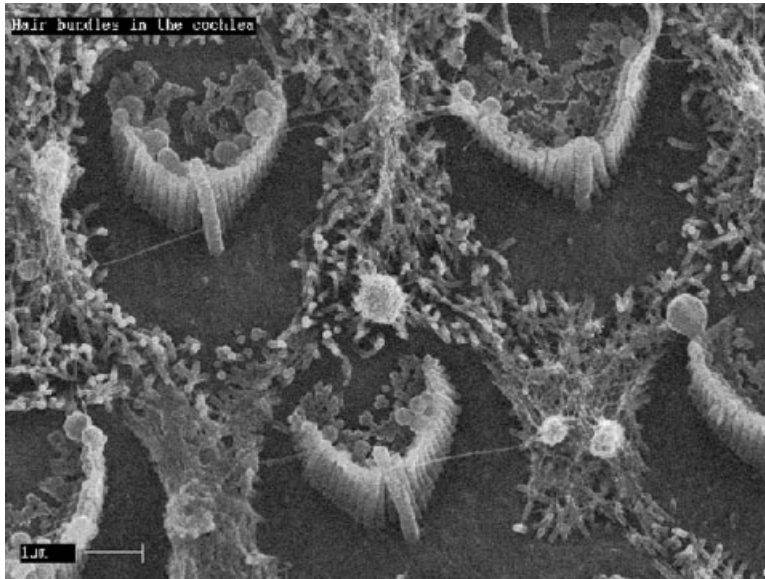


Figure 12-4. Scanning electron micrograph of hair cells. Reprinted with kind permission of Julian Thorpe and Guy Richardson, University of Sussex; see www.biols.susx.ac.uk/Home/Julian_Thorpe/coch8.htm for a series of SEM photos of hair cells at increasing magnification.

Although hair cells are not true neurons, sound detection is nearly instantaneous, as is signal transduction in all mechanically gated systems. Indeed, it is much quicker than photoreception, because the ion channel gating involved is a purely mechanical process. Each cilium in a bundle is connected to an ion channel on its neighboring cilium by a *tip link*, which is like a tiny spring (see Figure 12-3). When the bundle of cilia is deflected in the direction of the longest cilium, the tip link on the longer cilia pulls open the gate on the ion channel of the adjacent, shorter cilium. When the bundle is deflected in the opposite direction, toward the shortest cilia, the tip link relaxes, closing the ion channel of the neighboring cilium. The signal is thus transmitted through the ion channel to adjacent sensory neurons, located within the *spiral ganglion* of the cochlea. An axon extends from each spiral ganglion cell to the CNS via the auditory nerve. With the nerves that come from the vestibular apparatus of the inner ear, the semicircular canals, these axons make up cranial nerve VIII.

Frequency detection is graded along the cochlea, with the hair cells at the base nearest to the oval window responding to the highest frequencies and those at the apical end responding to the lowest frequencies. The nerve fibers that synapse with the hair cells at specific locations are “tuned” to specific frequencies. There are two competing theories as to how this happens. The first is the “place theory,” which proposes that frequency is determined by the physical properties of the basilar membrane, in which the hair cells are embedded; the membrane is rigid and narrow at the basal end and wider and more flexible at the apex. The ratio of stiffness to mass determines how far sound waves will travel down the cochlea, and thus which hair cells will be excited and send their signal to the brain. A different, “temporal” or

“frequency theory” is based on the idea that sound is periodic in nature; the auditory nerves fire at a rate determined by the periodicity of sound waves and the rate at which the basilar membrane vibrates, and the brain determines tone by the rate at which auditory nerves discharge their signal. Neither theory alone adequately explains all aspects of sound perception, so more research on this question is needed before we have a sufficient understanding.

Large animals (including humans) and miniscule insects detect the directionality of sound in two ways. The body itself diffracts sound waves, and the brain can detect and interpret the interaural difference in sound pressure caused by this diffraction as directionality. Second, the brain can also interpret the difference in the time of arrival of the sound at each of the ears.

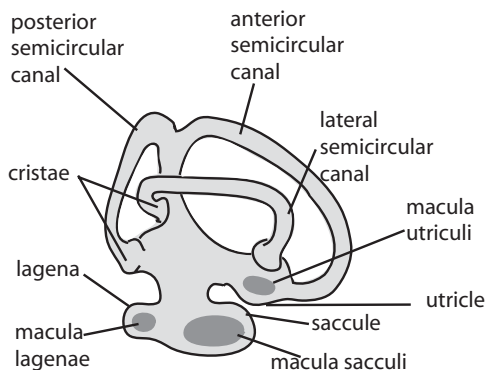
In terms of genetics, the known genes involved in sending hearing and balance information to the brain are similar to those used in other membrane potential systems. Most of what we know comes from human or mouse mutations. Two classes of genetic effects are observed. One class involves syndromic hearing loss, which means that hearing is lost in association with other craniofacial developmental anomalies. In nonsyndromic hearing loss, the latter anomalies do not occur and there may be no other phenotypic effects. Over 75 genes or chromosomal regions have been identified in association with nonsyndromic hearing loss (Van Camp and Smith 2003), and this is probably an incomplete list of the many pathways that are actually involved and hence mutable.

As with so many structures, catalogs of expressed genes are being assembled by many investigators, and the lists include members of most types of genes. Perhaps the most general characteristic is that mutations in genes specifically expressed in normal cells or structures of the mature inner ear, or expressed during its development, lead to defects generally consistent with their expression pattern. The genes affect hair cell structure, cytoarchitecture, mechano- and neurotransmission, transcription factors, and genes for gap junction communication between cells, among others (see, e.g., Petit 1996; Petit, Levilliers et al. 2001); tabulated in (Van Camp and Smith 2003). Some of these genes are specific to hearing apparatus, although most if not all are members of larger gene families and some have additional sites of expression. For example, one cell-junction gene, *Connexin26*, is frequently found to affect nonsyndromic hearing loss. Many relevant genes have been discovered in surveys of cDNA (expressed genes) from embryonic ear tissue. Although mouse homologs can typically be identified, as so often happens, when comparison has been possible between human disease and effects of mutation in the mouse, the two do not always correspond closely.

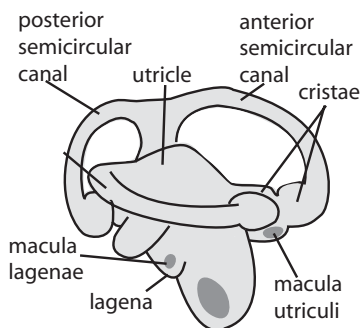
The complex transmission of mechanical signal from outer to middle to inner ear might seem rather more elaborate than necessary. Indeed, some hearing is transmitted directly to the cochlear fluid through vibrations induced by sound in the skull, and other animals have more direct transduction (see below). As with so many traits, hearing evolved its elaborations in connection with a diverse set of needs (chance, developmental or genetic connectedness, or response to selection) in a way that had to be consistent with other aspects of craniofacial development.

The sensory organs that allow vertebrates to maintain their sense of balance and equilibrium are also located in the ear, next to the hearing apparatus. (See Figure 12-5 for a comparison of the anatomy of the inner ears of fish, amphibians, birds and mammals.) These are the three mutually perpendicular semicircular canals that comprise the labyrinth and detect and convey rotational movement and the two *otolith*

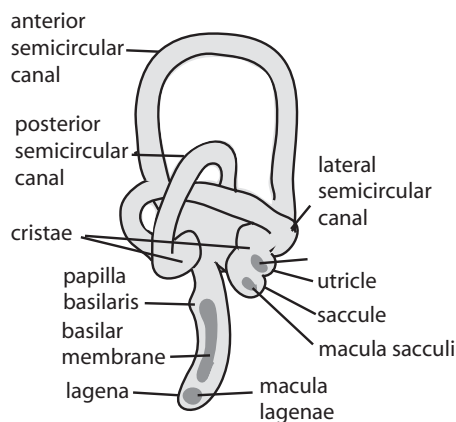
FISH



FROG



BIRD



MAMMAL

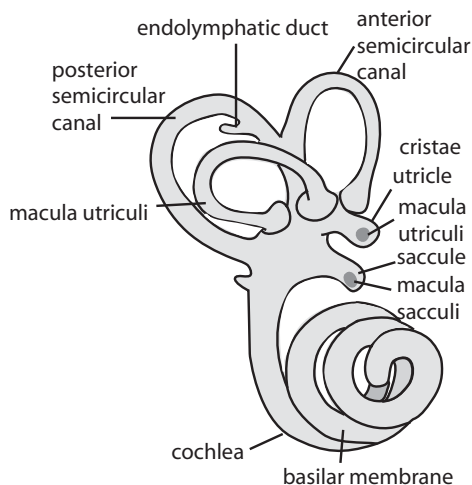


Figure 12-5. Comparative diagram of the labyrinths and inner ears of fish, amphibians, birds, and mammals. Redrawn based on version in *Encyclopedia Britannica* (Britannica 2003).

organs, the *sacculus*, which detects motion in the vertical plane, and the *utricle*, which senses motion in the horizontal plane. These sacs and canals are all filled with a fluid called *endolymph*. As with hearing, the detection of motion is based on the deflection of the sensory hair cells, which occurs when the animal moves its head, causing the endolymph to lag behind. At one end of each of the semicircular canals is the knobular structure called the *ampulla* and within that, the *crista*. The upper surface of the crista contains the hair cells, which are embedded in the gelatinous cupula. As in sound reception, deflection of the hair cells stimulates the transmission of infor-

mation about angular movement to the brain. When the animal moves its head the lymph within the canals moves, and this deflects the hair cells in the crista and either depolarizes or hyperpolarizes the cells to release or inhibit the release of neurotransmitter and thus transmit signal, also via the VIII nerve, to the brain.

The utricle and saccule are membranous sacs. The floor of the utricle and the wall of the saccule contain hair cells covered with gelatinous substance, as well as tiny grainlike crystals, or otoliths. At rest, the otoliths press straight down on the hair cells, but with motion, depending on the direction, the otoliths press less or more, and at different angles, and the cilia of the hair cells respond to the changes. The brain puts together signals from all of these organs to produce the sensation of motion in a three-dimensional space. People tend not to think of balance and equilibrium as one of the real “senses”—until they lose it and realize how pervasively it is used. The orientation of the semicircular canals is related to the habitual posture of a species and frequently detects motion in all three spatial planes.

Bilateral symmetry allows discrimination of direction, but many animals, especially higher vertebrates, do much more precise location of distant objects or phenomena via their sense of hearing. By turning the head or moving the outer ears, animals can intensify sound to help locate it. To do this, the brain must not only compare neural impulses from left and right cochleae but also take into account the orientation of the ears or head, thus combining macro- and microscale phenomena. There is also an ability to judge the distance of an object from the intensity as well as frequency spectrum of sound, based on experience stored in memory or in other ways. Organisms can also sometimes account for the effects of wind, echoes, and so on (but the difficulties this presents illustrate in interesting ways the limitations of the system).

Birds

The auditory structure in birds shares many of the features of mammalian ears—phylogenetic analysis suggests that hearing had its origins in reptiles in the Paleozoic with subsequent divergent evolution of the hearing apparatus in the distinct amniote lineages that then arose (Manley and Koppl 1998) (see Figure 12-6).

The auditory mechanism in birds is the *basilar papilla* in the cochlea. As in tetrapods and some insects, hearing is tympanic (based on the vibration of a membrane), and the tympanic membrane bulges outward at the surface of the head. Sound vibration is conducted to the cochlea along a single bony structure called the *columella*, which communicates with the cochlear oval window. The cochlea is a very short bony structure, although the length varies by species, and it encloses the basilar membrane and ends in the *macula lagenae* and the *lagena*, which do not have a hearing function (see Figure 12-5). In birds, hair cells are found throughout the inner ear, along the entire width of the auditory basilar papillae. Birds are capable of very acute frequency analysis, and their hearing is organized tonotopically, as is hearing in mammals.

The labyrinths of the avian inner ear are similar to those of higher vertebrates, with three canals arranged at angles that detect motion in all directions, in the same manner as the balance system of mammals.

Fish and Amphibians

The fish's ear is a membranous labyrinth that as in tetrapods serves two functions, hearing and maintenance of the fish's equilibrium. Although in the head, unlike in

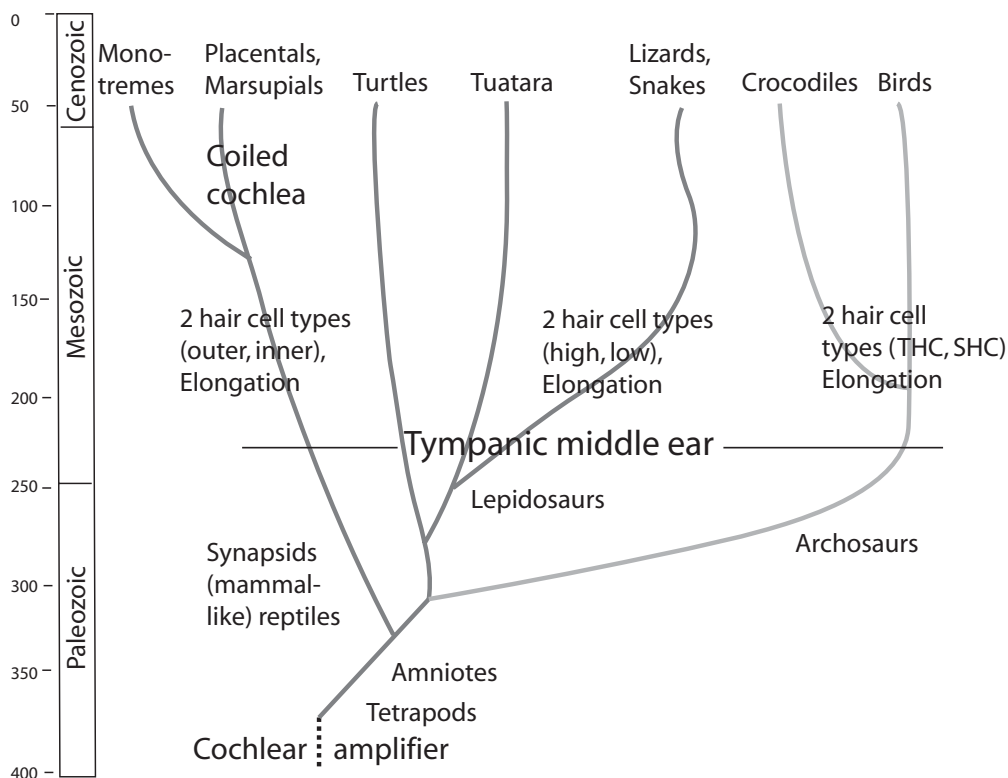


Figure 12-6. Phylogeny of cochlear types among amniote species. Redrawn from (Manley and Koppl 1998) with permission.

tetrapods, fish ears are internal organs, not open to the external environment, and there is no tympanic membrane. Sound is conducted to the hearing organs through tissue and bone. The labyrinth is made up of a series of sacs and canals, as in mammals. The semicircular canals (of which there are three in most fishes, except hagfish, which have only one, and lampreys, which have two) are oriented at right angles to each other in three different planes and are fluid filled. Each canal has an ampulla and crista, and the crista is composed of sensory cells called *neuromasts*. Neuromasts are covered by the cupula, a gelatinous membrane as in tetrapod cupulae. Fish also have the saccule and utricle and a lagena, an appendage of the saccule in fish, amphibians, and birds. In most fishes, the saccule is most likely primarily an auditory organ and the utricle primarily serves a vestibular function (Fay and Popper 1999).

Hearing in some fish also relies on an additional pathway; the swim bladder. This buoyancy organ in the body cavity of bony fishes is a gas-filled out-pocketing of the digestive tube that helps a fish maintain its depth and adjust its buoyancy. The swim bladder pathway to hearing is indirect: the gas-filled organ, or other gas bubbles near the ears, expand and contract in response to sound pressure, and this motion is transmitted to the otoliths.

Most amphibians share all the organs of the fish ear: the semicircular canals and their cristae, the saccule, lagena, and utricle. In terrestrial amphibians, reptiles, and birds, sound is conducted from the tympanic membrane to the inner ear by a single bone, rather than the three in higher vertebrates. In fish and aquatic amphibians, hair cells are found in the semicircular canals of the ear and along the lateral line. This raises the point that no one system is required for transmission of sound to the cochlea; the bony ossicles of the middle ear of mammals evolved as part of the evolution of jaws differently hinged than those of reptiles, for example. Mammalian ossicles transmit sound from the tympanic membrane to the cochlea, but this is not the only way, and it is not particularly clear what special advantage this system had. Perhaps it was just better than relying on the duller direct through-the-skull perception pathway, as bones were remodeled and recruited for the evolution of jaws and face. Maybe it did not matter, relative to the developmental constraints entailed by the importance of making jaws, so long as sound was transmitted somehow.

Fishes have evolved a large repertoire of sound-generating mechanisms that they use for attracting mates and during spawning. These include muscular vibrations of the swim bladder, pectoral girdle, and pectoral spines rubbing in the grooves of the pectoral girdle, plucking of enhanced pectoral fin tendons, or grinding of pharyngeal teeth (Ladich 2000).

Fish are probably poor at localizing sound sources; because of the way sound waves reach the hearing apparatus, interaural time and intensity differences are effectively nonexistent (Fay and Popper 1999). In some species the hearing organs are connected by perilymphatic spaces, and sound conducted by the swim bladder reaches both ears at the same time.

Echolocation

A few animals, including dolphins and bats, “see” their environment by emitting sound waves and monitoring the reflected echoes as the sound bounces off whatever it hits. The suborder of bat, *Microchiroptera* of the order *Chiroptera*, both navigates and finds its prey by *echolocating* in this way. Approximately 800 species of bats belong to this suborder, and the sound they emit ranges from biosonar pulses to clicks and other calls. These bats are able to monitor the speed of flying objects—their prey—and their size, range, and elevation among other characteristics. Bats that do not echolocate (fruit-eating bats, for example) find their prey by vision. Bats are highly speciose (richly diversified) animals as reflected in the exotic variation in their craniofacial morphology (Figure 12-6); how much of this was specifically selected for echolocation is somewhat less clear.

Bats that echolocate make laryngeal or nasal emissions (Gobbel 2002; Springer et al. 2001; Teeling et al. 2002; Teeling et al. 2000). Teeling et al. have shown that echolocating bats are probably paraphyletic, with laryngeal echolocation probably evolving first and being lost later in different lines that have modified the details of their apparatus in various ways (Teeling, Madsen et al. 2002). Dolphins have sonar, but it evolved independently of bat echolocation and uses different means. Dolphins emit clicks, receive the echo with their jawbone (*panbone*), and apparently begin to interpret it with the “melon” in their forebrain. They may actually stun their prey with sonar.

Bats, including those that echolocate, share the same auditory apparatus found in most mammals; the outer ear that captures sound waves and directs them to the

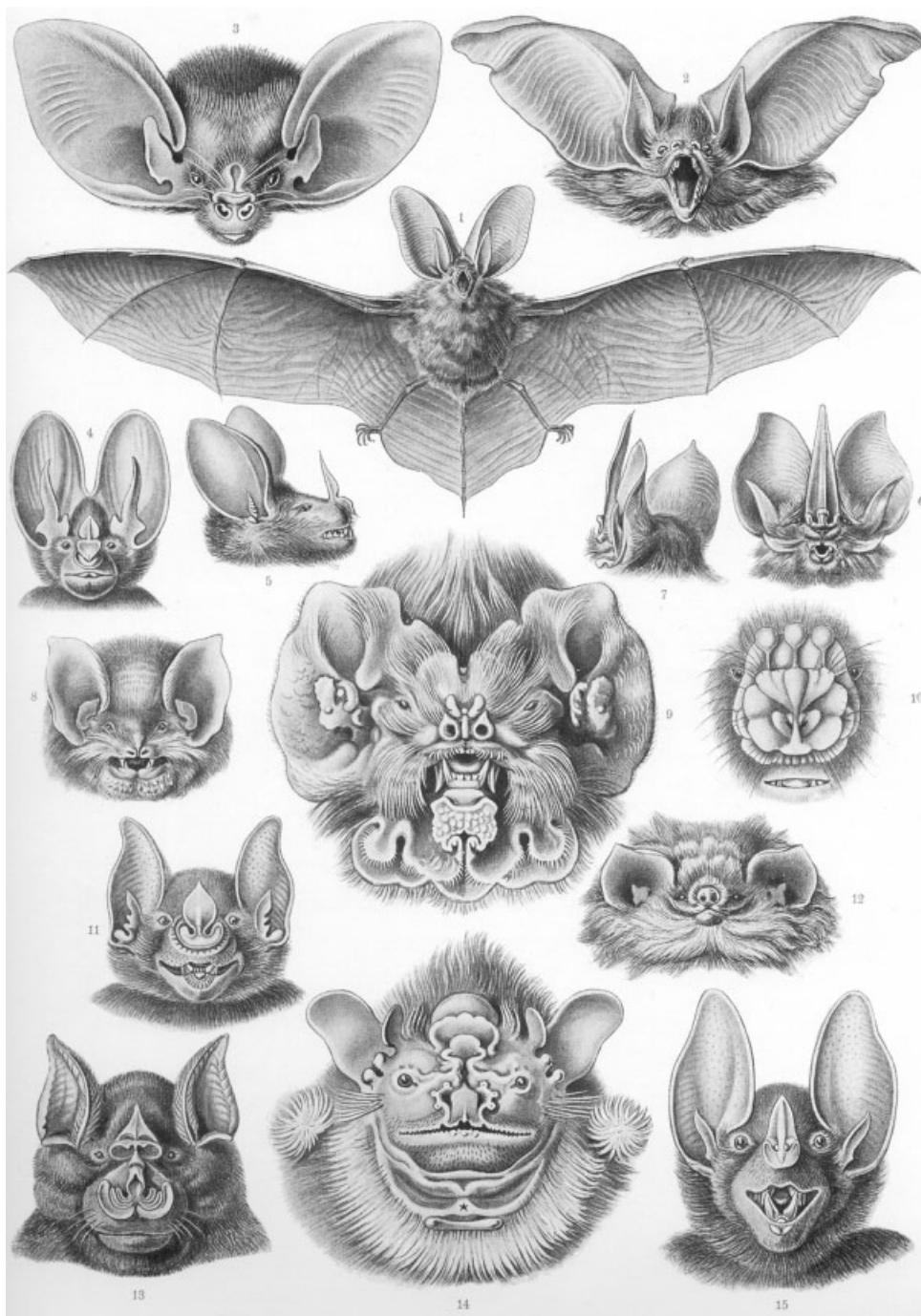


Figure 12-7. Natural diversity in bat head morphology, as artistically rendered by Ernst Haeckel in his *Art Forms in Nature* (available in reprint as Haeckel 1899).

middle ear and then the inner ear, where the sound waves are transduced from mechanical energy into electrical energy and transmitted to the brain for interpretation. However, a feature specific to echolocating bats is that the portion of the basilar membrane in the cochlea that receives sonar sound frequencies is especially sensitive and spiral ganglion cells of the brain that receive these frequencies are overrepresented relative to others.

That said, this exquisite ability to detect prey with echolocation may outrank the bat's ability to actually capture the prey once it has been detected. One study found that only about 40 percent of attempts by red bats are successful on the first try (Obrist and Wenstrup 1998). This is in part because many of the moths and other insects that bats prey upon can hear the sounds that bats emit in their echolocation calls, and in fact their hearing is most sensitive to the frequencies emitted by their most common predator and so can initiate evasive maneuvers while the bat is still fairly distant. Thus, although we think of bats as having a rather refined and exotic sensory capability (perhaps because we do not have it and hence romanticize it), echolocation is, like other systems, imperfect and highly variable among bat species, especially when the responses of moths and other prey have kept pace. If it were not so, there would be no prey—and no echolocating bats.

HEARING IN ARTHROPODS

Hearing in insects—the detection of air- or substrate-borne vibration—is used for detecting predators and for signaling to and locating mates. The mechanoreception of air- or water-borne vibration is found in seven of 27 orders of insects (Hoy et al. 1998). However, many more insects produce vibration through the branch or leaf upon which they perch by knocking against it with a leg or other body part and detect it coming from the same source with their hearing organ. Insects with body lengths less than 1 cm are generally restricted to ultrasound emissions, which are useful only at short range or in free space. Ultrasound is distorted and attenuated by most habitats; thus small insects that rely on ultrasound probably do not use sound for social communication but to detect prey or predators. Larger insects can emit sounds above about 1 kHz (Michelsen 1992), which can penetrate vegetation. Most insects that use hearing for social communication have frequency analyzers, whereas those that use it only to detect prey or predators do not seem able to detect frequency.

Interestingly, despite the bat problem, not all Lepidoptera (moths and butterflies) are able to detect sound, and the ability of those that can is highly variable, both in range and in the complexity of the actual hearing apparatus itself and its location on the body, which can be anywhere from the wings to abdomen to thorax to legs to head. Some nocturnal butterflies have “ears” on their wings that are sensitive to ultrasound (Yack and Fullard 2000); some moths hear with their mouthparts (Gopfert and Wasserthal 1999), but in all moths that can hear, the ear is tympanal and specialized to the frequency of bat ultrasonic emissions.

Insect mechanosensory organs are classified as type I or type II (Eberl et al. 2000). Type I organs have bipolar neurons with an axon extending to the CNS and a dendrite on the opposite end (see Figure 12-1). These organs are surrounded by specialized supporting cells. Type II organ cells are single multidendritic neurons with no obvious ciliary structure. Most insect sensory organs are type I (Eberl, Hardy et al. 2000).

The insects that are able to detect airborne sound have either tympanal organs, which as noted above can be almost anywhere on the body (thorax, abdomen, sternum, legs, wings, antennae), or flagellar organs. Which type of organ an insect has determines how far afield the sound it detects comes from. Tympanal organs are sound pressure detectors and as such are for detecting sound from far away. Near a sound source, on the other hand, most sound energy may be detected particle movement, and flagellar organs, which are protruding structures such as hair or antennae that function as particle velocity detectors are sufficient (Eberl 1999).

An insect's body is covered with a hard protective coating called the cuticle. Poking through at many spots are sensory bristles. These are of various types and are regularly spaced along the body, perhaps produced by reaction-diffusion types of patterning mechanisms (see, e.g., Gerhart and Kirschner 1997). Each appears to develop from a single precursor cell, and the different types of final structure are determined by combinatorial expression of numerous transcription factors (TFs) and activation-inhibition signaling not yet understood.

One of the bristle types is a sensillum known as a *chordotonal organ* (sometimes referred to as a *stretch receptor*), shown in Figure 12-8. Chordotonal organs can be specialized to perform various functions including stretch reception (of the outer chitinous shell) and a related sense of hearing. Tympanal chordotonal organs are internal structures that span two cuticle plates, where there is a thinning of the cuticle, a thin membrane that covers an air-filled sac. A four-cell structure senses

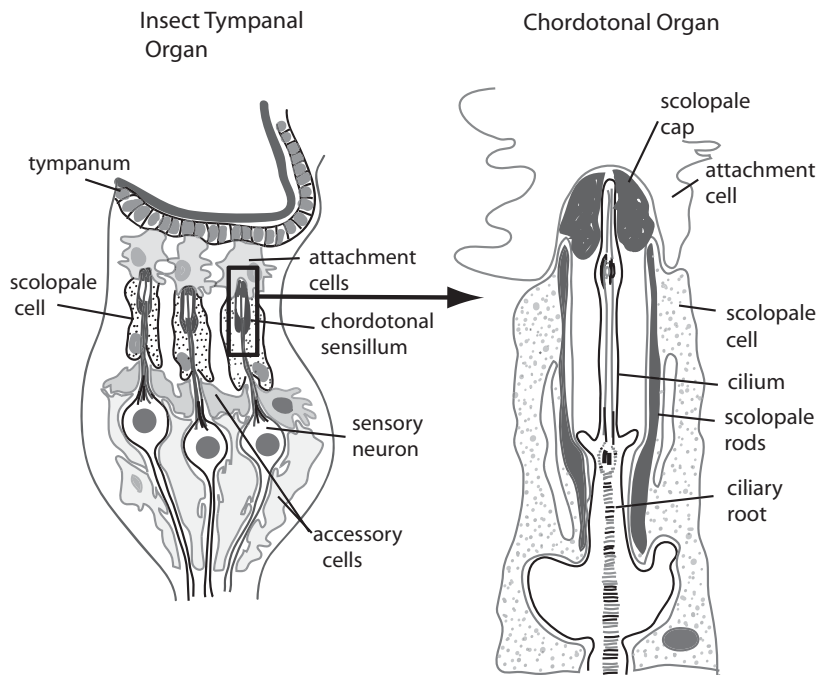


Figure 12-8. Basic structure of insect chordotonal organs. Redrawn from (Gray 1960) with permission.

vibratory movement at this location and is innervated by a single neuron. The sensory organ is sometimes called the *scolopidial organ*, in reference to the spindle-like sheaf of ciliary origin, the *scolopale cell*, into which the dendrite of the nerve cell extends. The scolopale cells and the neuron are associated with glial and support cells. These are located at basically fixed sites in a given species. Auditory chordotonal neurons project to similar sites in the CNS but what happens when the signal reaches the CNS varies among insect species, as acoustic information is extracted from the sensory input in a way specific to the sounds relevant to each insect. Although there is a wide variety of “acoustomechanical transformers” among insects, the way the mechanical signal is converted to electrochemical response in the nervous system is quite consistent across taxa (Eberl 1999).

Some of the genetics of chordotonal organs in *Drosophila* is known. The tympanal chordotonal organs are similar to the vertebrate inner ear (Eatock and Newsome 1999; Fritzsche and Beisel 2001). Both have ciliated mechanoreceptive cells and accessory or supporting cells. Homologous genes and mechanisms are involved in the development of mechanosensors in both insect chordotonal organs and vertebrate inner ears (Eberl 1999), and all ciliated mechanoreceptors share a common transduction system (Kernan and Zuker 1995).

A neurogenic TF, *Atonal*, is expressed early in embryonic development in all chordotonal organ progenitor cells (this gene is also expressed in photoreceptors). In *Atonal* mutants, all chordotonal organs are absent except for one scolopidium in the abdominal linear array of five pentascolopidial organs called *Ich5* (Lage et al. 1997). Other genes are known to be required, including *Egf receptor* signaling in precursor cells in most but not all of the eight scolopidia in each abdominal segment. Detailed studies have been done of specific gene expression in these developing organs, which essentially have to do with their patterning, number, and location.

Although most insect auditory organs are chordotonal, some also use other means of detecting air vibration. Insects such as cockroaches and crickets have, either in addition or alone, specialized bristle organs on their *cercae*, or antenna-like sensory appendages projecting from their tails, that are deflected by wind currents and can be very sensitive to sound (Eberl 1999). Bees have flagellar antennae on their heads that are thought to decipher the acoustic components of the waggle dance that is used for communicating the location of pollen sources and to detect song within the hive. *Drosophila* discriminate species-specific courtship songs at close range with their *Johnston's organ*, a collection of chordotonal organs in the antennae.

Crickets and cicadas call for mates over long distances by *stridulation*, the rubbing of ridged surfaces on their legs or wing margins, and detect sound with tympanal organs located on their front legs. Cricket songs also attract parasites (Ormiine tachinid flies, which have sternal tympanal ears for directional hearing to locate their hosts). In a defensive mechanism referred to earlier, some moths use their tympanal organs to detect the ultrasonic echolocating calls of predatory bats and respond by altering their flight patterns. Indeed, other arthropods have an even more diverse repertoire of sound-making and sensing devices. Lobsters have ridged surfaces called *plectra* on the base of their antennae that they rub against a file organ to generate sound. This seems to be for warning off predators, because the lobsters do not appear to use this for mating or defensive reactions.

LATERAL LINE

All the primarily aquatic vertebrates, cyclostomes (agnaths), fish and amphibians, have hair cells in the “touch organs,” or mechanosensors, in their outer skin. This organ is the *lateral line*, and it is sensitive to minute water displacements from vibration as well as changes in pressure, caused both by the fish itself and by other nearby animals or fish, and so is used for various behaviors including finding prey and escaping approaching predators, social behavior such as schooling, shoaling, and avoidance of stationary objects, and hearing, among others.

The morphology of a particular lateral line system determines just what an animal is going to detect—the spatial distribution of the lateral line’s mechanoreceptive organs, the neuromasts, determines the receptive field of a particular animal, the innervation patterns determine how sensitive the system will be, and the morphology of the neuromasts themselves determines which aspect of water movement the organ will respond to, velocity or acceleration (Maruska and Tricas 1998).

The lateral line and the labyrinth arise from the same embryonic placode. Together, they comprise the *acousticolateralis system*. The mechanoreceptors of the lateral line, the neuromasts, contain hair cell clusters embedded in the cupula. The organ has two kinds of sensory receptors; the *superficial neuromasts* on the skin and *canal neuromasts* recessed in fluid-filled canals beneath the skin. Superficial neuromasts detect the motion of water flow, and canal neuromasts detect its acceleration. Superficial neuromasts predominate in still water fish and canal neuromasts in fish that live in moving water (Engelmann et al. 2000).

Water enters the lateral line organ through numerous pores on the surface of the skin and flows past the neuromasts. Pressure of the water bends the cupula, and this creates an action potential in the hair cells. (See Figure 16-2.)

EVOLUTION OF THE EAR AND HEARING

In 1882, Mayser proposed that the modern ear was derived from the fish lateral line, the *acousticolateralis hypothesis*, a view that was held for most of a century (Mayser 1882). Mayser described the lateral line as an accessory hearing organ. In 1987, observing embryonic development in salmon, Wilson and Mattocks (Wilson and Mattocks 1887) suggested that the inner ear and lateral line were derived from the same embryonic placode and that the inner ear originated as part of the lateral line. Additional supporting evidence was considered to be the similarity of the hair cells in the lateral line and the inner ear.

In 1974, E. G. Wever wrote an influential paper on the evolution of vertebrate hearing, pointing out that none of the animals thought to be ancestral to vertebrates has an inner ear, so that the inner ear must not be derived from lateral line, but that the lateral line and inner ear share a common ancestor. He believed that bony fishes had the first “real ear.” He suggested that the *acousticolateralis hypothesis* be abandoned in favor of the idea that the inner ear and the lateral line evolved from a common mechanosensory system using hair cells (Wever 1974).

Similarities in the receptor cell structure and the basic function of the ear and auditory system among vertebrate groups, and in the corresponding gene usages, suggest that the ear arose early in the evolution of vertebrates (Popper and Fay 1997). There is a consensus that the earliest inner ear structures were equilibrium receptors rather than auditory receptors. It has been proposed that cochlear ampli-

fication is ancient (Manley and Koppl 1998) and that the first auditory receptor, the basilar papilla, pinched off from the vestibular organ with the transition of vertebrates from water to land.

Manley and Koppl suggest that this first appeared almost 400 million years ago in lobe-finned fish (Manley and Koppl 1998). However, a basilar papilla seems to have evolved independently several times (Manley and Koppl 1998), after, and perhaps because of, the development of the tympanic ear in the Triassic. This may be one of many examples of the conservation of a basic common ancestral mechanism, like a mechanoreceptor cell or pathway, that has repeatedly been modified for similar purposes. An ability to detect and respond to air or physical vibration apparently evolved many times in invertebrates as well (Michelsen 1992). Animals share common ancestry more closely than was thought in the decades before genetic homologies were identifiable. The distinction between “hearing” and the detection of other forms of environmental vibration is after all a human invention.

Some but not all of these systems involve similar hair cell mechanosensory mechanisms. The near ubiquity of the sensory hair cell in vertebrate ears and the lateral line suggests that it arose early in the evolution of hearing, pressure, or vibration detection. Specialization in form and function, however, suggests that the hair cell has adapted to the specific needs of a variety of vertebrates (Fay and Popper 2000). If precedent (and the results of mouse deafness studies) is a guide, different genetic mechanisms will be found even among animals having very similar physical phenotypes.

Little is known about the evolution of hearing in insects, but it probably evolved from extant mechanoreceptors. Insect auditory receptors are all based on chordotonal sensilla—regardless of where the auditory locus is on the body—and the chordotonal system is present ontogenetically everywhere. A different part of the chordotonal system has evolved for hearing in different insect lineages (Eberl 1999). Chordotonal organs serve as proprioceptors at the appendicular joints, and at intersegmental “joints” between the thorax and abdomen and in the abdomen itself. Thus the opportunity for an ear to arise from the chordotonal primordium occurs frequently over the body of a typical insect, given its many joints and appendages.

THE SOMATOSENSORY SYSTEM

Neural pathways that process signals from receptors on the skin, muscles, and joints are called *somatosensory* pathways. There are a number of different receptor types in the somatosensory system, specific to different sensations. Some receptors perceive touch, some pressure, and others pain. There are structural differences between these receptors; some are very sensitive to touch and pressure and adapt rapidly to sustained stimulus, whereas others are less sensitive and adapt slowly.

A number of different vertebrate receptors perceive pressure, vibration, and texture. These are known as *Meissner's corpuscles*, *Pacinian corpuscles*, *Merkel's disks*, and *Ruffini endings*. The first two are rapidly *adapting*, whereas the second two adapt slowly. The act of picking up a stick, for example, is quickly perceived by Meissner's and Pacinian corpuscles, but they quickly adapt and stop sending the signal to the brain. Merkel's disks and Ruffini endings, however, continue to alert the brain to the presence of the stick in the hand. The *hair follicle receptor* is found in hairy skin, where nerve endings wrap around single hair follicles and transduce signal on deflection of the hair from that follicle.

The vertebrate pain and temperature system does not have specialized receptor organs, but instead free nerve endings throughout the skin, bone, muscle, and connective tissue perceive changes in temperature and pain peptides. Pain is sometimes the result of damage to tissue but more often is the response to substances released by damaged tissue for which free nerve endings have receptors. *Nociceptors*, receptors for pain, have axons that are either myelinated or not. Myelin is the covering of the neuron, produced by a variety of neural supporting cells, that sequesters a neuron electrically from its environment much as insulation protects electrical wires from signal leakage. Myelinated nociceptors are faster conducting and produce the immediate sensation of pain on receipt of signal. Nociceptors with nonmyelinated axons are responsible for the long-lasting pain that may follow a wound or injury.

Mechanisms for the detection of some stimuli are similar across vertebrate classes, which suggests shared evolutionary history. The somatosensory systems that originate in the spinal cord and caudal brain stem, as well as the vestibular and lateral line mechanodetection systems, are very similar across the vertebrate classes in which they are found (Hodos and Butler 1997). Other sensory systems, however, like electroreception or color vision, involve mechanisms that are different enough across species to suggest that they evolved several times. Nonetheless, although receptors for various stimuli vary across vertebrates, the basic organization of the sensory cranial nerves is remarkably alike (Hodos and Butler 1997). These nerve cells are either monopolar or pseudomonopolar neurons, such as the olfactory receptors, or bipolar neurons that innervate specialized receptor cells, such as in the octavolateralis system in fishes or the vertebrate visual system.

Invertebrates have sensory bristles on their external surface, which respond to pressure or movement. They are embedded in the cuticle in insects and are innervated by a sensory neuron that transmits the signal to the insect's central nervous system.

THE STAR-NOSED MOLE

The star-nosed mole provides an interesting—and essentially novel—tactile sense. Its nose is splayed out (Figure 16-7) and contains physical receptors so that it can obtain a spatially arranged “map” of its underground environment. This will be described in more detail in Chapter 16.

OTHER

There is a variety of other physical sensing in the animal and plant world, probably not even easily broken into specific categories. Some plants respond quickly to touch or other physical disturbance. Temperature is a very important variable to many species. Two types of vertebrate *thermoreceptor neurons* have been described. One is the cold receptor, which reacts when the skin is colder than resting body temperature. Warm receptors respond when the skin is warmer than resting temperature. Arthropods that are blood feeders have thermoreceptors and are attracted to prey and induced to feed by heat. These receptors are usually on the antennae but can be on the legs. Many arthropods have humidity detectors. How they work is unknown.

Cold-blooded species seek places in their environment in which the temperature is lower or higher, for various reasons. The seeking of higher temperature to help

fight infection in species that cannot mount a mammallike febrile (fever) response has been reported. Homeothermic (warm blooded) species have internal and external receptors that monitor temperature via the hypothalamus in the brain (see Chapter 15) and regulate metabolic responses to maintain body temperature homeostasis (see, e.g., Turner 2000).

As with the other systems we discuss in this book, clearly not all species require the same kinds of thermal sensing, because there is no one way to deal with body temperature and/or its relation to the environment. Some aspects of thermal stress seem to be more fundamental, however. Heat- and cold-shock elicit a response in many organisms, including plants, bacteria, vertebrates and invertebrates. Thermal shock can alter protein structure and function in cells, for example. The response is in the form of the production of heat- or cold-shock proteins that counteract the effect of rapid temperature change, for example, by binding to proteins to prevent their mature shape from denaturing. Heat and cold shock proteins have been found in all known cells and have many functions apart from induction of cellular response to temperature change. They can activate the immune system and generally maintain cellular homeostasis and protect cells in times of stress. The genes, the chaperonins or heat shock proteins that we have mentioned in several contexts, are an ancient family, showing the importance of this kind of protective response to thermal or other traumatic conditions.

PROPRIOCEPTION

In vertebrates, information about muscle length and tension is collected by sensory receptors in the joints and muscles. In invertebrates, this information is collected by stretch receptors on the outsides of muscles and chordotonal organs in the joints, which, among other things, measure tension changes. Stimulation of bristles at the joints can also be part of the proprioceptive system in arthropods.

Proprioception requires the perception of gravity for spatial orientation. Phylogenetically, structures that allow organisms to equilibrate themselves with respect to gravity and acceleration are perhaps the oldest sensory organs. As described earlier, the semicircular canals of vertebrates are based on hair cells, the same kind of sensory cell integral to the vertebrate sense of hearing. The membrane potential of hair cells changes in response to deflection of cilia that project from one end of the cell caused by vibration or the motion of fluid surrounding the cell.

In many invertebrates, the detection of motion and gravity takes place via *statocysts*, the fluid-filled chambers lined with hair cells that transduce the motion of a small granule called a *statolith*, which is free moving in some animals and loosely anchored in the hair cells in others. Statocysts are found in all major invertebrate groups, including arthropods, jellyfish, sandworms, higher crustaceans, sea cucumbers, tunicate larvae, and mollusks. In vertebrates, it is the saccule and utricle of the vestibular system of the inner ear that detect gravity.

Vertebrates and some invertebrates are also able to detect body rotation. The vertebrate ear conveys this information to the brain: when the animal moves, the inertial lag of the fluid in the semicircular canals deflects the hair cells, which transduce that information to the brain.

Many insects are sensitive to gravity, and the receptors involved tend to be found on the body surface, not internally as in statocyst-like structures. The receptors are

tactile hairs, found in a variety of locations. In the honeybee, they are found between the head and thorax and between the thorax and abdomen.

The ability of plants to perceive gravity has long been recognized; in fact, Darwin wrote about the fact that roots always grow downwards. The receptor is a calcium granule, a statolith, found in statocysts, as in invertebrate animals. The statolith is pulled by gravity to the bottom of a cell in the root, and this stimulates the root to grow down.

ELECTRORECEPTORS

The ability to detect electric fields is common among aquatic vertebrates, particularly fishes. Few mammals are known to have this ability, but among them are platypuses and the spiny anteater, both of which have electroreceptors in the skin of their snouts. The electroreceptors of some fish are so sensitive that they can detect the electric fields produced by the contraction of the muscles of their prey as they swim. Many fishes are also able to produce electric fields with *electric organs*, for defense as well as to stun or kill their prey.

The electroreceptors in fish are in the lateral line system. They are found on the head of all elasmobranchs and many bony fishes, electric eels, and others. They are called *ampullary lateral line organs* and differ from other sensory receptors of the lateral line system in some respects: water does not come into contact with the receptor cells, which lie within the ampulla, a vesicle that opens to the surface of the fish's body through a duct. The ampulla and the duct contain a gelatinous substance that is a good conductor of electricity, and perhaps a pH sensor as well. The environmental sequestration that was the key factor in the evolution of the earliest cells probably involved an ability to monitor and adjust to the electrical conditions in their fluid environment.

MAGNETORECEPTORS

Many kinds of animals are known to respond to the Earth's magnetic field, including honeybees, planarians (flatworms), mice, birds, salamanders, and many bacteria. A magnetic material called *magnetite* (iron oxide) has been found in many of these animals and may be the receptor for this sense. Magnetite crystals are found, for example, in the nose of the trout, and their interaction with the earth's magnetic field is detectable by the nervous system. However, the molecular mechanism that transduces this signal to an electrical impulse perceived by the nervous system is still not known (Diebel et al. 2000).

CHEMORECEPTORS

Chemoreceptors are cells in the olfactory and gustatory systems that initiate a neural response to molecules in the air or dissolved in liquid that come into contact with the receptors of these particular sensory systems. They function to detect the chemical structure of individual molecules, which in a logical sense is similar to immune detection but functions differently and is specialized for externally derived, nonliving molecules. These are discussed in detail in Chapter 13.

CHEMESTHESIS

Organisms from bacteria to protists, insects, and vertebrates are able to detect chemical stimuli, and a major use is detection of noxious substances. Nerve endings responsive to irritating chemicals are a part of the general somatic sensory system, as a subset of the pain- and temperature-sensitive fibers on the skin and generally found throughout mucous membranes. The term *chemesthesis* refers to the sensations elicited by the chemical stimulation of free nerve endings (Finger and Simon 2000).

BACTERIA

Motility is an important determinant of bacterial survival. Although some only glide, many bacteria can swim by rotating their flagellum, an ability that allows them to move toward a more favorable environment when they detect toxic conditions. Bacteria sense changes in the levels of surrounding nutrients and toxic chemicals, pH, temperature, light, the magnetic field of the Earth, electron transport conditions, and the like, and they respond with *osmoregulation*, altering internal conditions to protect against the changing external environment, or by moving to a more favorable environment, *chemotaxis*. Osmoregulation is a complex “two-component system” for regulating cellular response to external conditions. Chemotaxis is also a two-component sensory system, and it is controlled by more than 40 genes and sets of genes that control the flagella, transmembrane receptors, and signal transduction involved in bacterial responses to stress. Bacterial cell membranes are loaded with *porins*, proteins that allow the diffusion of important molecules in and out to maintain homeostasis and internalize nutrients.

In a two-component system a transmembrane receptor protein transduces environmental signal into metabolic changes by phosphorylating a second protein, as described in earlier general discussions of signaling processes. The rate of phosphorylation is under the control of the receptor proteins, and in chemotaxis, for example, the second protein affects the flagella to result in a change in the direction in which the bacterium swims. Ciliated protozoa use chemical sensing to find potential mates and avoid predators. Plants and fungi also use similar two-component response systems.

QUORUM SENSING

Bacteria use the information they gather from the environment for *quorum sensing*, the ability that allows them to determine the density of other bacteria around them and, depending on population size, form into biofilm. This then leads to the expression of genes that aid in the survival of the biofilm, the colonization of higher organisms, such as the roots of legumes, or finding the sites of adhesion or invasion of higher organisms to initiate infection (catheters are a common site). The environmental sensing is done by transmembrane sensory proteins, often coupled with cytoplasmic receptors, a system that has been well-characterized at the molecular level. Bacterial transmembrane receptors have common structural features: an extracellular binding domain for one or more ligand, a transmembrane region, a

cytoplasmic linker, and methylation and signaling regions. The receptors form homodimers and cluster together on the cell membrane.

SLIME MOLD

Dictyostelium discoideum are amoeboid cells that live a well-characterized life cycle, many stages of which are triggered by environmental cues including depletion of the food supply (see, e.g., Bonner 2000). (See Figure 12-9.) Initially, the cells live in soil, where they feed on bacteria. While food is plentiful the cells multiply by mitotic division, but as the population grows the food supply can become depleted and the cells begin to starve. The cells are constantly monitoring population density by sensing and secreting a protein called *conditioned medium factor (CMF)*, and when population density is large enough to threaten starvation, that is, when the concentration of CMF is high enough, cells signal other cells to aggregate. It is time to search for greener pastures.

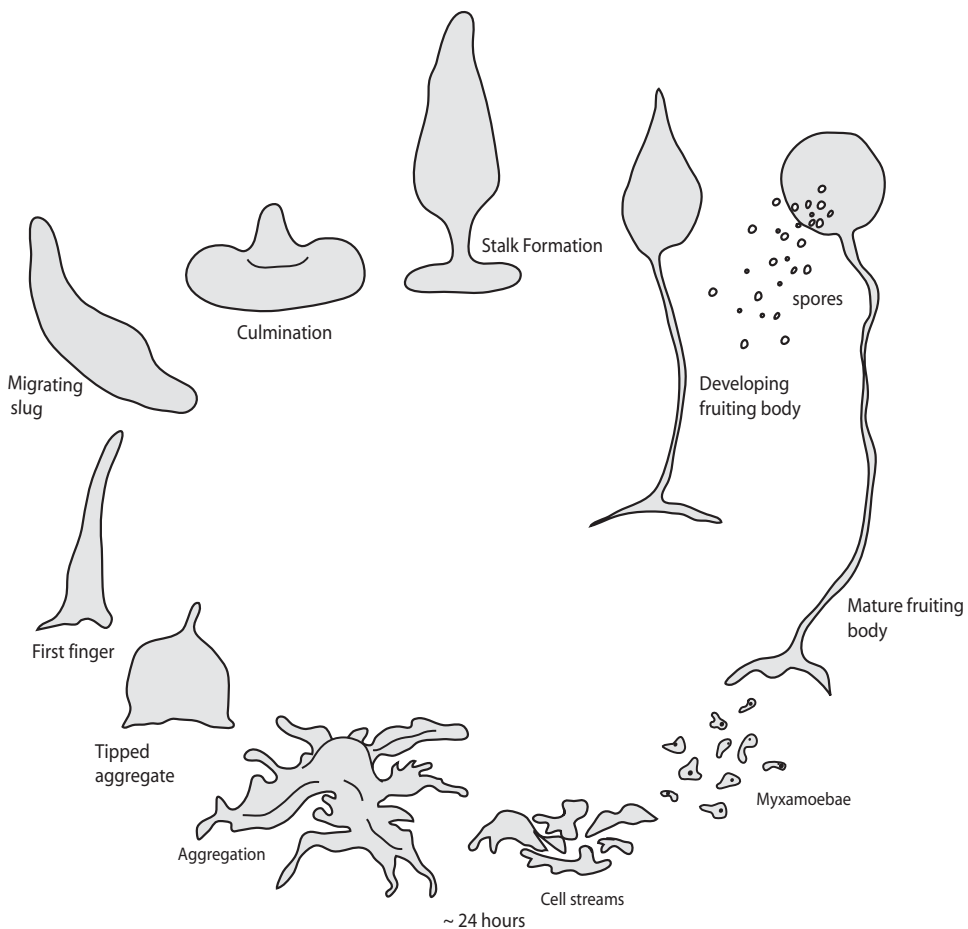


Figure 12-9. Life cycle of the slime mold *Dictyostelium discoideum*.

The signaling molecule is a substance called *acrasin*, which is cAMP. Cells adjacent to the signaling cell receive the signal via cell surface G protein-coupled receptors (GPCRs) for cAMP. The GPCRs transmit signal to the cytoplasm that initiates the pathway to the release of cAMP, which is then received by neighboring cells that, in turn, send their own signal, and so on until a colony is formed. The patterns of colony formation that result vary and are a function of the spatiotemporal pattern of signal emission. The colony is mobile, which allows the cells to migrate to a richer environment in which to produce the next generation. Within about 24 hours, the colony forms a *fruiting body* that disperses spores and begins the life cycle over again.

The colony, like a bacterial biofilm, or in fact any collection of aggregated cells, becomes more than the sum of its parts. Genes are expressed at this stage that are not expressed in independent cells. The *slug* of aggregated cells is about 1 mm long and can stay together for up to several days. The slug is polar and mobile, with its movement mediated by waves of cAMP that travel from the tip to the tail, inducing individual cells to migrate toward the tip, thus moving the slug in the direction of cAMP concentrations. The slug is extremely sensitive to environmental clues, so that small differences in light, temperature, or the concentration of ammonia gas produced by the cell mass itself will affect the direction of the slug's movement. Presumably this optimizes the location of the final fruiting body with respect to food supply. Interestingly, formation of the fruiting body depends on programmed cell death to take the slug through the stages of culmination to formation of the stalk and then to maturation, so in a very real sense this collection of single celled organisms becomes a multicellular being, with once free-standing cells sacrificing themselves for the good of the group.

A SIDE-COMMENT ON QUORUM SENSING, COOPERATION, AND GROUP BEHAVIOR

Quorum sensing is an interesting phenomenon, if it is being properly understood. It is one of many instances in which animals aggregate, swarm, or give display behaviors in a way that appears to reflect their population size or density and that has been interpreted as being a mutual signal among individuals that they interpret to alter their behavior. Controversy has been particularly heated over whether this kind of behavior could lead to altruistic self-sacrifice, individuals restricting their reproduction so that the group does not overpopulate relative to available resources. In a phrase, the issue is *group selection* as compared to the classic individual selection described in Chapter 3. The problem is the need to explain how alleles leading to self-sacrifice could increase in frequency. Formally, this becomes a mathematical problem for population genetics theory; informally, one can see many ways in which behavior can evolve that is good for the group so long as the sacrifice of individuals initially responsible for the behavior is not too immediate and complete.

CONCLUSION

One possible reading of the many homologies we find among the wide array of senses in all organisms alive today is that mechanoreceptors and chemoreceptors were the earliest sensory receptors, with all others being derivative, duplication of sensory modules having led to new receptors (Hodos and Butler 1997). For example,

the infrared detectors of the pit organs of crotalid snakes are modified thermal detectors and electroreception has apparently evolved several times in ray-finned fishes and in sharks and monotremes (Hodos and Butler 1997). We know, however, that many of the components (or modules) of the sensory mechanisms, such as those involving transmembrane G protein-linked information transduction pathways, are used for all sorts of other processes, including developmental patterning, cellular differentiation, ion transport to maintain proper chemical conditions in the cell, and so on, as well as for signal transduction in single-celled organisms. Many others have to do with cell architecture and the moving around of materials inside the cell via the cytoskeletal tubules and molecules like myosin that use the tubules. Mechanoreception in widespread contexts and species uses these very fundamental elements of cell biology. This is also another example of at least some connections or homologies among similar systems in very dissimilar species.

Thus the general repertoire of information receipt by cells probably antedates the senses that we see as organized organ systems today. At the same time, it becomes easier to see how rudimentary systems can evolve initially and then be elaborated in diverse ways—easier than having to reinvent a suitable mechanism whenever it might do some good.

Chapter 13

Chemical Signaling and Sensation from the Outside World

The light, sound, and chemical environs of an organism are often detected at a distance from their source and, to a great extent, are varied enough that they cannot be specifically anticipated. The properties of the sensory stimuli also vary. Light is fastest, but cannot penetrate solid objects, and an object cannot be seen if the requisite energy is not available (e.g., many animals do not see well at night). Because light travels in a coherent linear way, the information it carries is precisely patterned relative to the source. Sound travels more slowly but in all directions, sometimes through solid objects, bouncing and reflecting in a less orderly way, and sounds from multiple sources mix freely. To the receiving organism, sound is less spatially patterned than a light image, but, like light, it is organized in a coherent frequency spectrum, which some organisms can detect.

Chemosensation is the ability to detect and discriminate among specific individual chemical molecules. In previous chapters we have discussed ways in which cells respond to chemical stimuli in their immediate surroundings, whether as hormones produced by neighboring cells, signaling factors that regulate gene expression, bacterial responses to chemical attractants or repellants and the like. The distinctions are clearly arbitrary and based on ways humans think of our own functional diversity, but in this chapter we focus primarily on the particular subset of chemosensory phenomena that we traditionally refer to as smell and taste, by which organisms sample their environment.

Galileo is mainly known for his physics and astronomy but in other ways he was the first modern scientist, advancing from the greats like Aristotle by the use of controlled experiments, newly invented instruments and a more critical approach to relating theory to observation. In his treatise on the philosophy of knowledge, *The Assayer* (Galilei 1622), Galileo gave a rather perceptive explanation of taste and smell: particles from the detected object wafted on air to the tongue and nose, where, after mixing with water and mucus on the tongue and nasal membranes, they were detected. The outer part of the body was assumed to be insensitive to such small

particles. The smell or taste of the particles depended on their concentration, the vigor of their motion, *and on their shape*. Galileo's notions of taste and smell were to some extent rooted in ancient humoral theory, but many of the basic ideas were rather prescient surmises. We could argue that our better understanding of these systems is in the detail.

Generally, *olfaction* (smell) refers to the detection of air- or water-borne chemicals that diffuse from a source to the detecting organism, whereas *gustation* (taste) and other forms of chemosensation involve direct contact with the source. Immune detection and chemosensation are molecular recognition systems, with the former generally detecting internal invaders such as living parasites and the latter detecting external factors. One might expect similar systems to be used in these two molecular detection systems, and the similarities are interesting. Genetically, however, probably the closest parallel is, perhaps surprisingly, between chemosensation and vision.

Odorants waft through the air but do not generally have a precise spatial component at their source; certainly, the mixing of odorants in the air, and their variable diffusion depending on the vagaries of air currents, mean that there is less need for a precise spatial element in odorant perception compared with that needed for light images. But as in sound perception, organisms use both direction and intensity gradients to locate olfactory sources. Organisms vary greatly in the precision with which they detect or characterize the source, as well as with which they can compensate for the effects of movement of the air.

Olfaction is used to detect many aspects of the environment, including dangers, food sources, and conspecifics. Some species also have olfactory mechanisms thought to enable them to identify specific individuals such as family or group members. Although the chemical environment can be open-ended in general, organisms may be programmed to detect some specific chemical signals, such as *pheromones* that elicit species-specific behaviors such as mating and male-male competition. Olfaction works with the endocrine system in some of these contexts, and responses may be highly ritualized, such as visual or vocal display. Recognition of pheromones requires very specific molecular detection, and both signal and receptor must coevolve in some way. Such specificity of response occurs in both vertebrates and invertebrates.

Similar to visual and sound perception, olfactory processes in the animal world share some genetic mechanisms but have evolved morphological and neurological similarities apparently independently and have evolved different ways to use similar genes for chemosensation (e.g., Strausfeld and Hildebrand 1999). Indeed, olfaction and vision have evolved from a common ancestral receptor system and rely on some similar neural mechanisms. Transcription factors, including *Pax6*, are used in the development of both systems, and the receptors are in the same family as well.

Every species is surrounded by information in its environment that it cannot or does not use. Species are able to detect what they need to detect, and vary in how they take advantage of the chemical environment. But it is worth remembering that, as with other senses or abilities, a relatively poor sense of smell generally only reflects less reliance on chemosensation, not inferior adaptation—essentially, such judgments are purely human overlays on nature.

For convenience, this chapter will discuss olfaction and gustation as separate systems. At the molecular, genetic, and neurological levels, however, these divisions are arbitrary.

OLFACTION

The ability to detect odors varies widely across the animal world. The numbers game can be misleading, but there is a general correspondence between the number of *olfactory neurons* (ONs)—cells specialized for odorant detection—that an animal has and the probability that any given odorant will come in contact with an appropriate receptor. Humans have approximately 5–10 million olfactory neurons in the *olfactory epithelium*, enabling us to detect an estimated 10,000 or more different odorants, but this is a pittance when compared with dogs, who have more than 200 million ONs. By contrast, *Drosophila* have about 1,300 ONs. For many years, the conventional wisdom was that birds do not have much of a sense of smell, but recent work suggests that the acuity of the avian sense of smell is related to the size of a bird's olfactory bulb relative to its cerebellum (Malakoff 1999). In some species, such as songbirds, it is very small, on the order of 3 percent of the size of the cerebellum, whereas in others, such as many seabirds, it is closer to 40 percent. Although olfaction in aquatic species is of water-borne chemicals, the process works in a largely similar way. It is thought that salmon, a fish that returns to its birthplace every year to spawn, may find their way home with a well-developed sense of smell and a form of imprinting (Barinaga 1999).

Mammalian odor detection begins with olfactory receptors embedded in the olfactory membrane on the roof of the nasal cavity (see Figure 13-1). The olfactory membrane is composed of three layers of cells: (1) *supporting cells*, in which (2) *olfactory receptors cells* are embedded, and (3) *basal cells*, which produce mucus and which are the source of new olfactory receptors. One end extends from the layer of supporting cells to the central nervous system (CNS) as an afferent neuron, and the other end extends to the epithelium where it forms a knob, with cilia projecting from it. This makes the olfactory epithelium the most direct connection from the outside world to the CNS. Another distinction is that, unlike most neurons, olfactory receptors are continuously replaced during life. Olfactory receptor cells are bipolar.

Odorant molecules are detected by transmembrane receptor proteins in the cilia, which are covered by mucus produced by the supporting cells. Odorants in the air or (for aquatic animals) in the water diffuse through the mucus layer to reach the receptor, although hydrophobic odorants must be transported. Among the soluble proteins found in the aqueous medium around the olfactory receptor neurons are *odorant-binding proteins* (OBPs), to be discussed below. The detection of the presence of an odorant is based on ligand-receptor binding and depends on a class of diverse receptor genes as well as the number of sensory ONs.

THE GENERALIZED VERTEBRATE OLFACTORY SYSTEM GENES

How can an organism detect (have the right ligand for) smells it could not have been preprogrammed to detect? The discovery of *odorant* (or *olfactory*) *receptor* (*OR*) genes, made first in rats in 1991 by Buck and Axel (Buck and Axel 1991), opened the way to the understanding of the molecular basis of olfaction, which has become a model system for many aspects of sensory neurobiology. *ORs* are found in the cell surface of ONs in the olfactory epithelium and are encoded by members of the 7TMR G protein-coupled receptor (*GPCR*) gene family, which also includes opsin genes for photoreception and various hormones and many other receptors. The familiar general structure of a typical olfactory receptor is shown in Figure 13-2A.

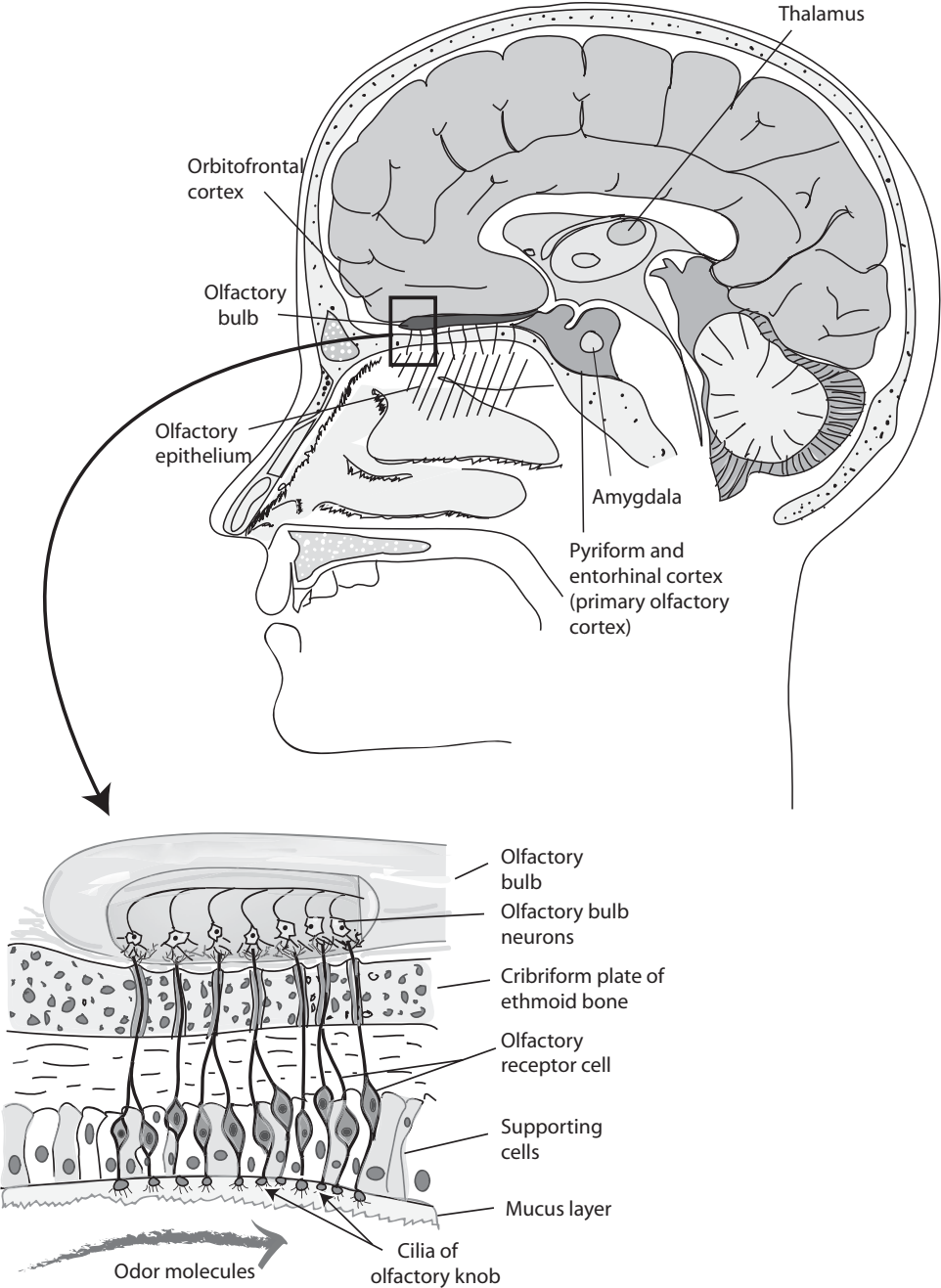


Figure 13-1. Vertebrate (human) olfactory system showing olfactory epithelium, olfactory bulb of the brain and the pathway of olfactory neurons to the olfactory regions of the brain.

OR genes are the largest subset of this gene family and comprise about 3–4 percent of the entire genomes of some species (see Crasto et al. 2003; Crasto et al. 2002; Crasto et al. 2001; Skoufos et al. 1999). Like the number of ONs, the number of *OR* genes reflects the importance of olfaction and varies greatly among species. For example, there are about 1,000 in the rat, which exceeds their number of immunoglobulin and T cell receptor genes combined (Mombaerts 2001; 2001). Among vertebrates, the number of *OR* genes varies considerably and in interesting ways that can be related to function. Some of the genes have been mutated to pseudogenes, which may relate to sloppy copying during meiosis of tandemly repeated genes—the same process that generates the useful variety of *OR*s—but may also be affected by selection. Birds typically have relatively few *OR* genes, but this varies among species. Fish have about 100 *OR* genes. Primates have roughly the same number of *OR* genes as other mammals (about 1000 in humans); however, among species that have been studied, about 60 percent of primate *OR* genes are pseudogenes; a typical human may have around 350 functional *OR* genes (Crasto, Singer et al. 2001; Glusman et al. 2001; Mombaerts 2001; Sosinsky et al. 2000). Interestingly, New World monkeys have few *OR* pseudogenes (Rouquier et al. 2000).

As shown below in Figure 13-4, *OR*s are distributed on almost all chromosomes in both mouse and human, and the pseudogenes have roughly similar proportional distribution. The high fraction of pseudogenes in humans thus is likely to have to do with selection or function, since the majority of mouse genes are still functional yet presumably the duplication and mutation processes are similar (but see some further thoughts at the end of this chapter).

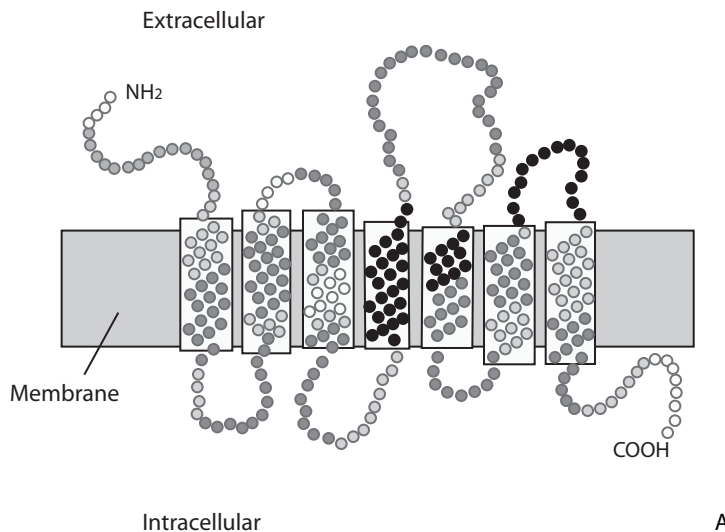
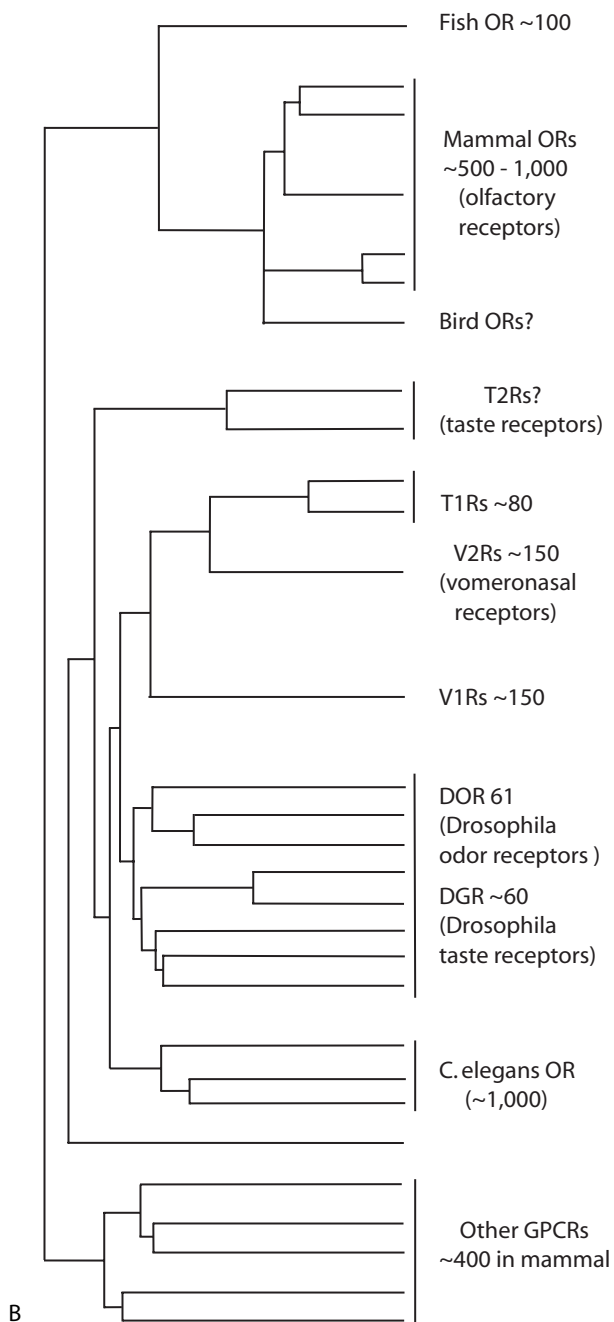


Figure 13-2. Odorant receptors and related genes. (A) Schematic of a seven transmembrane (7TMR) olfactory receptor molecule showing amino acids, specifically for receptor M71 in the mouse. The most highly conserved residues are shown in white and black, and the most variable are in shades of gray, presumably indicating the most important sites for odorant specificity. (See Firestein 2001); (B) General phylogenetic (gene sequence) relationships among major classes of chemoreceptors. The approximate numbers of genes in these classes are indicated (modified after Firestein 2001).

Figure 13-2. *Continued*

Vertebrate *ORs* have a roughly 1-kb intronless coding region that codes for a polypeptide with a little more than 300 amino acids. As a group, *ORs* can be distinguished from other seven transmembrane receptors (7TMRs) by a few conserved amino acid motifs and some conserved single residues. Seventeen hypervariable

amino acid residues in transmembrane domains 3–5 are thought to define an odorant binding pocket; these domains have high diversity and little conservation among *OR* subfamilies (Mombaerts 2001). *OR* genes have variable noncoding exons and splice sites, suggesting that they are alternatively spliced (e.g., Sosinsky, Glusman et al. 2000).

The range of odorants detectable by this variety of receptors includes molecules with very different chemical structures, implying that OR ligand-binding domains must be diverse as well, and there is corresponding variability in OR structure. An OR seems to respond optimally to a particular molecular structure—for example, only one of a series of specific odorants used in experimental tests. It is often assumed that the OR has been “tuned” for this specificity by evolution (Dreyer 1998), but the use of this adjective by sensory biologists suggests an implicitly strong darwinian view of precise selection. Whether, when, or how the OR-specific response pattern was shaped by (odorant-specific?) natural selection or whether it is simply the empirical property of any given binding pocket is not clear. As a rule, odorants are not detected in nature by single receptors (a feature shared with antigen recognition by antibodies), and, as noted earlier, odorants do not have a clear or cohesive chemical “spectrum” as do light and sound.

Transduction of an olfactory signal is initiated by the binding of an odorant molecule to the receptor on the cilia of an olfactory neuron dendrite. This triggers the intracellular G protein cascade and ultimately results in the delivery of the signal to the olfactory centers in the brain where the signal is decoded. G protein linked ORs activate signal transduction in one of two ways. Olfaction-specific G proteins convert abundant ATP into cAMP to generate an action potential in the ON. Some ORs increase intracellular cAMP concentration, opening cAMP-gated cation channels and allowing an influx of sodium ions, which depolarizes the cell and initiates a nerve impulse that travels along the axon to the brain. Others activate the inositol phospholipid pathway, IP₃-gated Ca²⁺ channels, in the plasma membrane, to activate second-messenger signal transduction molecules. ONs typically produce multiple OR copies and a given cell’s response may be thought of as the effect of the binding of odorants to a sufficient number of receptors on its surface over a suitably short time period, which, in aggregate, generates a threshold action potential.

ONs are the only sensory neurons whose axons connect directly to the brain (see Figure 13-3 for a schematic drawing of basic olfactory wiring to the brain). The mapping between specific odorant receptors and specific locations in the brain seems to be quite precise within an individual, as will be discussed in Chapters 15 and 16. The essential “wiring” characteristic may be that different ONs that express the same *OR* gene send axons to the same *glomerulus* (neuronal cluster), in the olfactory bulb (e.g., Kauer 2002; Mombaerts 2001; Zou et al. 2001). Replacement ONs are generated mitotically during life from cells at the base of the olfactory epithelium and express the same *OR* gene and send their axons to the same glomerulus as their predecessor, which seems to be an important means by which odor perception remains more or less constant throughout an organism’s life. How ON-glomerular conservation occurs is not yet completely clear, but it appears to be ON-guided; an ON could either switch which *OR* gene it expresses or use OR-specific axonal redirection to the glomerulus appropriate for the *OR* gene it is already expressing (P. Mombaerts, personal communication).

DNA sequence phylogenies suggest that there are two main classes of vertebrate OR genes (Kratz et al. 2002; Mombaerts 2001) (see Figure 13-2B). Class I appears

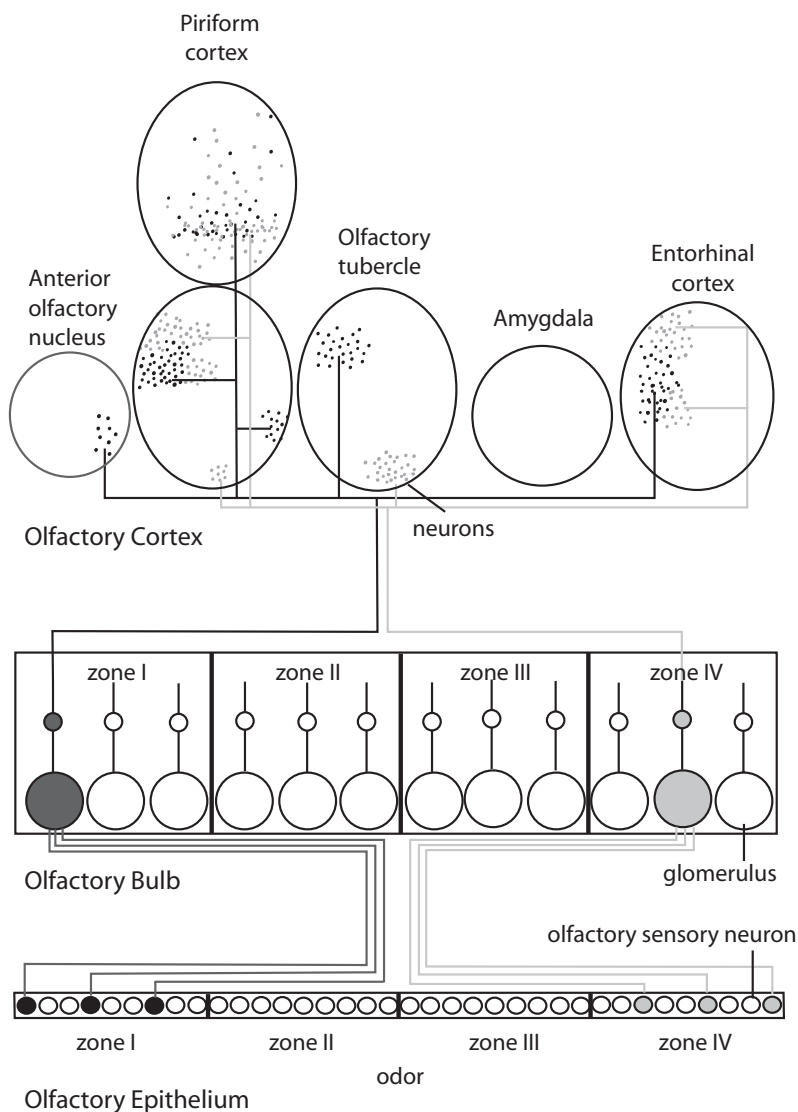


Figure 13-3. Basic olfactory wiring diagram showing pathways of two different odorant receptors. In the olfactory epithelium, sensory neurons expressing a single *OR* gene are located in the same zone. In the olfactory bulb their axons synapse with mitral cells (small circles) in the same few glomeruli (large circles). The mitral cells carrying signal of specific ORs synapse with clusters of neurons (differently shaded dots in large ovals) at stereotypical sites in olfactory areas of the cortex, creating a sensory map. Figure redrawn from (Zou, Horowitz et al. 2001) with permission. Original figure copyright 2001 by Nature Publishing.

to be the oldest and is the only group found in fish. Amphibians have class I and class II genes; at least in frogs, the class I genes are only expressed in a water-sensitive chamber and class II genes in an air-sensitive chamber. The genomic cluster arrangement of mammalian *ORs* suggests that class I genes duplicated to form the ancestor of class II genes, followed by the dispersal of duplicate *ORs*, mainly in class

II, to other chromosomes. These facts suggest that class I genes specialize for water-borne and class II for volatile odorants. However, mammals retain many class I genes scattered in their genome, and a higher fraction of human class I genes are more functional than our class II genes; thus the former seem unlikely simply to be a relic of an early aquatic life.

As with other tandemly repeated genes, *OR* genes appear to have evolved by unequal crossing over, gene conversion, and occasional chromosomal translocations or duplications, followed by subsequent additional local duplication. In mammals, genes from different subclasses are dispersed within single chromosomal clusters. There are at least two *OR* gene clusters in zebra fish, 12 in mice, and more than 25 in humans (Kratz, Dugas et al. 2002; Mombaerts 2001), in whom *ORs* are found on all chromosomes except the short chromosome 20 and the Y (Glusman, Yanai et al. 2001) (Figures 13-4 and 13-5). These genes are distributed in clusters of various sizes. Each cluster largely comprises genes from the same *OR* subfamily. There are two major clusters of class I *ORs* on human chromosome 11, comprising over 40 percent of our entire repertoire, pointing to the importance of this cluster in the evolution of the vertebrate *OR* genes. Another cluster is linked to the major histocompatibility complex (MHC) on chromosome 6, and it has been speculated that these *ORs* could be involved with mate identification, at least in some species like rodents (Younger et al. 2001).

There is at least some coherence in olfactory systems that may represent phylogeny as well as development. Frequent duplication, mutation, and perhaps gene conversion means that homology between pairs of *OR* genes is highly variable; however, as might be expected, the cluster organization for identifiable orthologs generally appears to be retained among mammals. *ORs* closely related in sequence are closely related in chromosomal location and regulation and with a few exceptions tend to be expressed in similar regions in the olfactory epithelium (Lane et al. 2001; Mombaerts 1999); these in turn project to localized regions in the olfactory bulb (Kratz, Dugas et al. 2002; Tsuboi et al. 1999). Thus, each zone has its own expression mechanism, and its *ONs* project roughly to similarly distinct regions in the olfactory bulb.

Because these epithelial regions are similar among individual animals in the same species, they may reflect temporally coordinated aspects of gene expression and embryonic neural development. Interestingly, however, little regulatory sequence sharing has been detected among paralogous genes that are expressed in the same zone of the olfactory bulb (Lane, Cutforth et al. 2001), although some potential motif sharing has been suggested (Hoppe et al. 2000; Sosinsky, Glusman et al. 2000). One problem is that recently duplicated genes may share flanking sequence, which may or may not imply that that sequence is specifically regulatory; recall that the genome is nearly saturated with potential regulatory sequences; therefore, sequence analysis alone is usually not a definitive way to identify response elements. Experiments with different *ORs* have reported differing results regarding how near the gene the relevant regulatory regions are, whether a few kilobases or hundreds (e.g., Mombaerts 2001); this may mean either that the experiments have missed something important about *OR* regulation or that that regulation varies greatly from gene to gene.

Although these spatial and chromosomal clustering aspects of organization are relevant for developmental coherence and gene expression, they may be of little relevance for odorant detection itself. The zonal organization of the olfactory

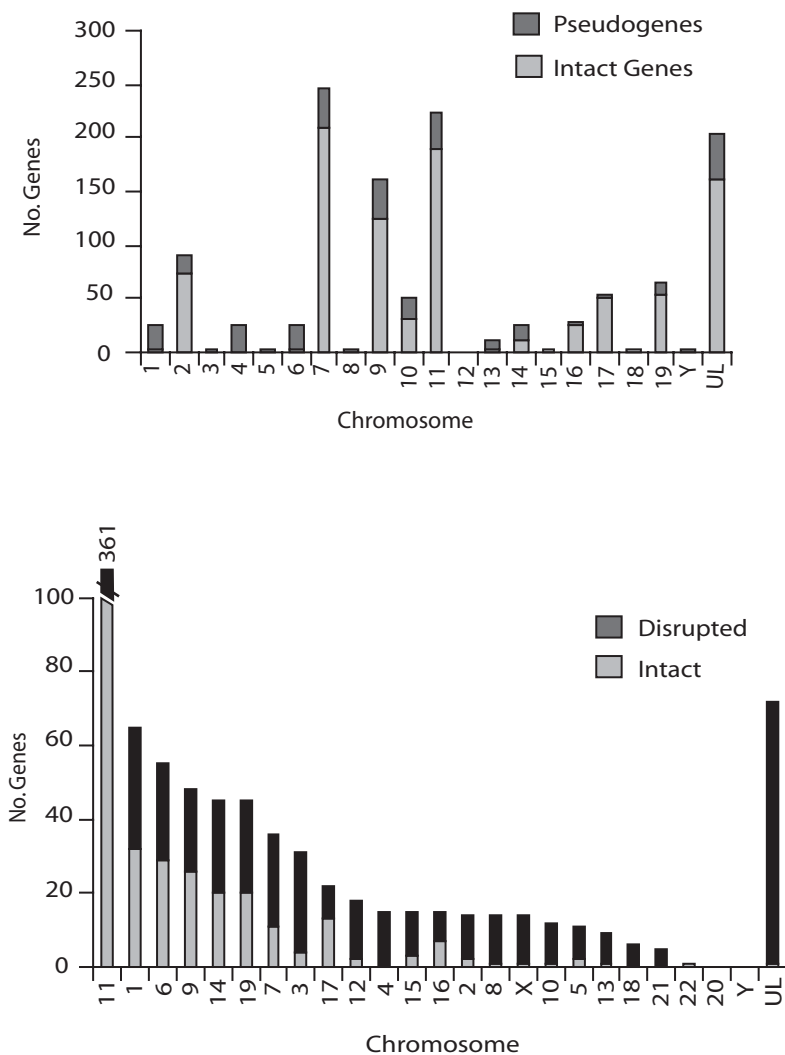


Figure 13-4. Number of functional and disrupted (pseudogene) ORs in the mouse (top), and human (bottom) genome, by chromosome. See text. Redrawn after (Zhang and Firestein 2002) and (Glusman, Yanai et al. 2001), respectively.

epithelium is widespread in general, although varying in detail, in vertebrates with very different olfactory behavior; thus, the developmental organization may relate to *OR* gene cluster and its vertebrate evolutionary history rather than clustering of ORs coding for functionally similar receptor proteins. However, an analysis of conservation of amino acid sequence motifs showed that some motifs were highly clustered within *OR* class and may have some relation to ligand-binding properties (Liu et al. 2003). While the individual motifs did not seem to these authors to be correlated with particular odorant classes or properties, some combinations of motifs did. If this interpretation bears further scrutiny, it may suggest that over evolutionary time the combinatorial shuffling of motifs among the duplicating, evolving *OR* genes is an additional means of generating olfactory diversity.

Other animals have different organizational patterns. For example, nematodes appear to have at least one chemosensory receptor that is not symmetrically expressed on left and right sides, and invertebrate receptors are located in various places, including the antennae and the separate maxillary palps (Dreyer 1998).

The regulation of *OR* expression is quite interesting, even though we do not yet understand its details. A given *OR* is expressed mainly in a single region of the olfactory epithelium. As a rule, a given vertebrate ON expresses only one *OR* gene and, at that, only one of the two alleles of that gene in the diploid individual. Similar to X-inactivation and antibody/TCR allelic exclusion, there is allelic exclusion in olfactory gene expression. Because *OR* genes are on many chromosomes (as well as the homolog of each cluster in the diploid cell) (see Figure 13-5), there must be a form of *trans-OR* expression control. Recent experiments suggest that chromosomal

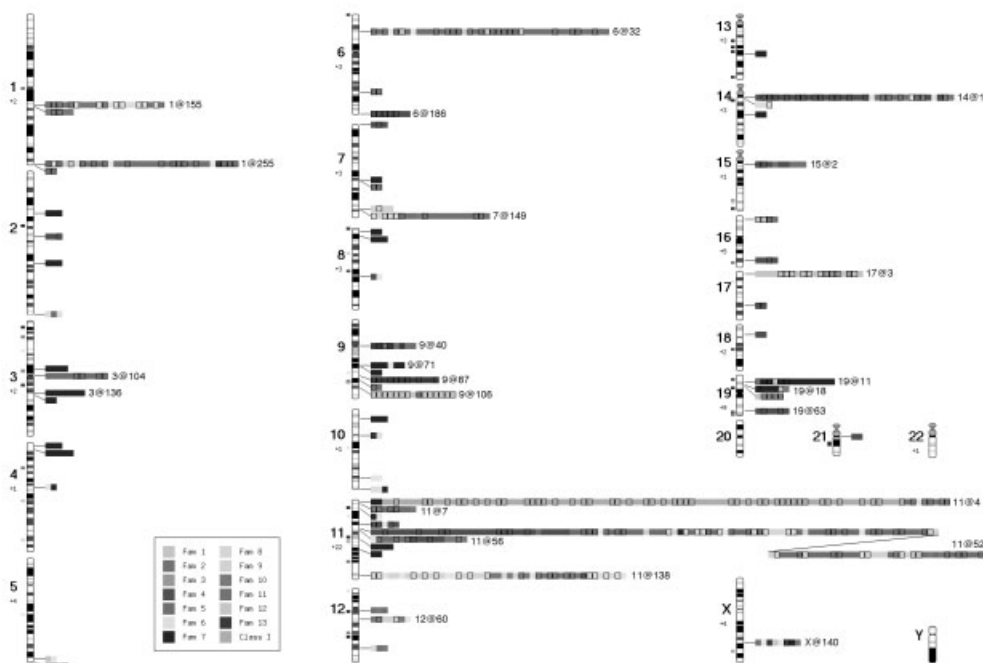


Figure 13-5. Genomic (chromosomal) distribution of *OR* classes in the human genome. This is a gray-scale version of a color original figure. Generally, the darkness shades correspond to *OR* gene subfamilies. The outlined squares are functional *OR*s (i.e., not pseudogenes). Genes shown to the left of each chromosome are singletons (isolated, not in a cluster), genes to the right are in clusters of two or more; relative order is approximate, and the large cluster on chromosome 11 is split for convenience. Even in gray-scale, the clustering of related genes and their distribution across the genome can be seen. The numbers by clusters (e.g., 3@136) refer to the chromosome and number of genes in the cluster. (Modified and reprinted with permission from Glusman, Yanai et al. 2001, and see this for details and color resolution).

exclusion may occur via asynchronous replication during mitosis, resembling a possible element of X-inactivation (Singh et al. 2003) discussed earlier. In principle, this could account for the expression of *OR* genes from only one (maternal or paternal) chromosome but does not account for the additional exclusion of expression of *OR* genes from all but the one cluster—on that same or any other chromosome. The cluster-related regionalization of expression suggests that perhaps *OR* clusters are made available for transcription (that is, the chromatin opened, and so forth) in a developmentally orderly way as the epithelium is patterned. This would reduce the exclusion problem to that of genes within the cluster(s) that are open in cells in a given epithelial region.

Within a cluster there must be some form of *cis*-exclusion. Other genes including globins and color-vision *opsins* have *cis*-based exclusion, in which only one gene within a cluster is expressed at a given time or in a given cell. Control of this type can involve enhancer competition among the genes in a cluster on the same chromosome, but *trans*-regulatory factors may also be involved (Kratz, Dugas et al. 2002; Serizawa et al. 2000). The latter possibility is suggested because single-*OR* gene expression also occurs for transgenes experimentally inserted in various places in the genome (i.e., not just in *OR* clusters). The local chromosome seems to determine the exclusion of its homolog (Singh, Ebrahimi et al. 2003), but, again, some more general *trans* mechanism(s) than asynchronous replication—perhaps the concentration of some limiting factor(s)—must exist in a cell to suppress all other *OR* expression once a first, single *OR* has been expressed anywhere in the genome. As always, there are exceptions: there is evidence that at least some rat ONs express specific pairs of *ORs* (Rawson et al. 2000) rather than only a single gene. Therefore, the system must be escapable (perhaps because the multiply expressed *OR* genes in the rat have lost their repression-control sequences).

THE VOMERONASAL ORGAN: PHEROMONES

We have described how unpredictable olfactory signals are detected. However, as we know, there are many important, indeed vital, ways in which organisms are pre-programmed for *specific* chemical signals; this is the kind of specificity found in internal communications via hormones and with developmental and regulatory signaling. How is specific *interorganismal* chemical communication achieved? In fact, many vertebrates communicate via interorganismal “hormone” systems. No new processes or mechanisms have been needed.

The nasal chemosensory organs of tetrapod vertebrates are divided into the olfactory system and the *vomer*nasal system (Johnston and Peng 2000; Wysocki 1979). The *vomer*nasal organ (VNO) is present in amphibians and reptiles which suggests that it probably emerged in early tetrapod lineages. The VNO is thought to be primarily a sensory organ for the detection of pheromones, produced in bodily secretions such as sweat, urine, and vaginal fluids, that induce stereotypical behavior (Holy et al. 2000; Keverne 1999). For example, the VNO has been shown experimentally to be vital for mating recognition and male-male aggression in mice (Stowers et al. 2002). However, at least in some animals, such as snakes, the VNO also has a role in other functions such as the detection of prey. The VNO is well developed in reptiles but less so in many mammals and has been reduced or lost in some lineages such as old-world monkeys, probably apes, and some lizards. The nature of the human VNO is still unclear. As fetuses, humans have a VNO with

apparent neural connections to the olfactory bulb (Døving and Trotier 1998). Some authors suggest that there is a depression just inside the nasal opening in adults where the requisite epithelial cells can be found, but others suggest that the whole system degenerates soon after birth (Meredith 2001). In fact, evidence for an active human pheromone system is essentially nonexistent (see below).

The VNO in nonhuman vertebrates is located in a pouch off the nasal cavity, on both sides of the nasal septum dividing the nose into its right and left halves. The VNO is fluid-filled and typically not directly open to the air as is the olfactory epithelium. Pheromones must reach the VNO receptors in a way that is different from how odors reach olfactory receptor cells. Various pumplike mechanisms draw the molecules into contact with the VNO receptors. The pump can be activated by the autonomic nervous system, perhaps triggered by conventional olfactory cues (e.g., Keverne 2002). When snakes and other reptiles flick their tongues, for example, they draw in molecules from the air. When the tongue is pulled back into the mouth after each flick, the molecules pass over the duct openings that lead to the VNO, where they are detected by pheromone receptors.

Like the olfactory epithelium, the VNO employs members of the *7TMR* family. Sequence analysis has identified two *vomeroneasal receptor* (*VR*) classes of 100 or more genes each, with a phylogeny separate from that of the *OR* genes. Each *VR* class uses (or at least is coincidentally expressed with) different message transduction G protein types (Bargmann 1997; Firestein 2001; Herrada and Dulac 1997; Matsunami and Buck 1997). From studies of rodents, the roughly 150 *VIR* class genes are found to be phylogenetically closer to *OR* genes, whereas the similarly large class of *V2Rs* are found to be closer to a different class of *7TMRs* (Figure 13-2B, from Firestein 2001), the metabotropic glutamate receptors, the most ubiquitous neurotransmitters in the CNS and a distinct class of receptors that when activated affects internal neuronal conditions, as distinct from the direct ion-channel ionotropic receptors. The class difference has to do with the length of the extracellular N-terminus of the *VR* protein, that may have to do with ligand-binding (Firestein 2001). The genes in each *VR* class are expressed in a discrete part, either in the apical or basal region, of the VNO.

The VNO and olfactory epithelium appear to have a common evolutionary origin, but there are many differences between vomeronasal and olfactory neurons. The two are innervated separately. *VRs* are structurally somewhat different and share little sequence homology with *ORs* (Matsunami and Buck 1997). A vomeronasal neuron (*VNN*) may express multiple receptors, which converge only imprecisely to their respective region in the accessory olfactory bulb of the brain, and a given *VNN* sends axons to multiple glomeruli. Interestingly, although the VNO is intrinsically involved in sexual behavior, males and females express the same receptors (Holy, Dulac et al. 2000).

Because of their use in inducing stereotypical behavior, one would expect a high degree of preprogrammed sensitivity in *VRs*; evidence presently suggests that *VRs* do have more sharply restricted binding properties than *ORs* (e.g., Leinders-Zufall et al. 2000). Consistent with this is that specific parts of rodent olfactory bulbs are activated by mating pheromones (e.g., Keverne 1999). One might predict that at least some *VRs* will not be polymorphic because polymorphism might lead a fraction of members of a population to be unable to detect the appropriate signal. We might expect that to be at least one good example of variation easily purged by natural selection. However 200+ genes in two separate receptor groups can gener-

ate much variation in detection, and one can ask why such diversity is required if pheromones are so specific.

Important aspects of behavior in some species involve individual recognition. The highly polymorphic MHC genes play some role in individual recognition (Penn and Potts 1999). The MHC is involved in the preference of mice for mates genetically dissimilar to themselves, and there is evidence for similar mate-choice effects in other vertebrates; however, the evidence is not entirely consistent and some behaviors in some species or contexts suggest preference for MHC-similar individuals. One problem is the difficulty in evaluating behavior accurately compared with the relative ease of identifying genotypes in mates and offspring. Distinguishing among an animal's two parents by MHC mediated odors may affect mating and other behaviors. Some recent evidence suggests that, rather than simply learning by experience, this discrimination is made possible by specific genetic mechanisms that, for example, compare an organism's own alleles with those of another member of its species.

Mouse major urinary proteins (MUPs), which release small volatile pheromones, have been shown experimentally to mediate individual recognition (Hurst et al. 2001; Keverne 2002). MUPs secreted by male mice have strain specificity and pregnancy-blocking activity. MUPs also have other functions. They are produced by a diverse family of polymorphic genes with multiple tissue expression patterns, largely clustered on one chromosome in the mouse (Cavaggioni and Mucignat-Caretta 2000). The relative roles of MUPs and the MHC in individual recognition are not yet clear or consistently understood (e.g., Brennan 2001).

Adaptive evolutionary reasons have been suggested for the importance of genetically determined individual recognition. There may be competition as well as cooperation between mother and fetus in placental mammals. There are various reasons why mating behavior that favors the generation of diversity in offspring may have been selected for (e.g., Penn and Potts 1999). Heterozygosity in the immune-related MHC can improve the odds of resistance to rapidly evolving infectious organisms, and heterozygosity can protect against negative consequences of inbreeding. The allocation of cooperative behavior may also require individual identification, to distinguish relatives from nonrelatives. How important this may be in regard to natural selection is unclear, especially because many natural populations consist mainly of relatives anyway. But at least, as with most examples in this book, the mechanism to serve such functions is part of the normal repertoire of genetic mechanisms.

There is some indirect evidence for pheromonal action in humans, such as the oft-reported synchronization of menstrual cycles in women who live together (e.g., at school). However, the known human *VR* genes all appear to be pseudogenes, the corresponding accessory olfactory bulb seems to be absent, and no relevant gene expression has been detected in our VNO (Giorgi et al. 2000; Keverne 1999; 2002; Meredith 2001). Further, an ion channel gene, *Trpc2*, expressed only in VNO and thought to be required for VNO function (Liman et al. 1999) is a pseudogene in Old World monkeys and apes (including humans) (Liman and Innan 2003).

Most of the MHC evidence is equivocal in humans and has mainly been based on observed vs. expected genotype frequencies in mates or between parent and offspring. These data have generally suggested MHC-dissimilar mating preference (e.g., a reduction in homozygosity relative to expectation from Hardy-Weinberg equilibrium). Several T-shirt smelling tests have shown that humans discriminate among MHC genotypes, although sometimes preferring those like themselves and

sometimes preferring those unlike themselves (e.g., Penn and Potts 1999; Potts 2002) or perhaps preferring types resembling their fathers (Jacob et al. 2002). At present, this kind of evidence is rather on the quaint side and has not been rigorously interpreted. We can, however, draw a basic conclusion: we humans seem to find our sexual way by more facultative and diverse mechanisms.

ODORANT-BINDING PROTEINS

A part of the olfactory process is getting the odorant to the detection system. A number of *odorant binding proteins* (OBPs) are found in the moist vertebrate olfactory mucosa, where they appear to bind volatile hydrophobic compounds to make them available to the ORs embedded in the hydrophobic mucosa (Finger and Simon 2000). However, the interpretation of OBPs is a complex story.

OBP genes are in the *Lipocalin* gene family (Akerstrom et al. 2000; Paine and Flower 2000; Tegoni et al. 2000). They are not as diverse as *ORs* and may be more directly relevant to the VNO than the olfactory epithelium. Lipocalins are among the MUPs that stimulate the VNOs and act as pheromones (e.g., Cavaggioni and Mucignat-Caretta 2000), but their complex expression patterns and functions go well beyond such signaling. Although lipocalins are expressed in the vertebrate nasal cavity and can bind volatile compounds, this and other lipocalin functions exist elsewhere in nature: the genes are found in invertebrates, plants, and bacteria, and there is as yet no specific sequence-based evidence that a subclass has specifically evolved to serve as OBPs (Tegoni, Pelosi et al. 2000).

INVERTEBRATE CHEMOSENSATION

Presently, only limited data are available on chemosensation in our invertebrate relatives, and it is likely that a number of different systems remain to be identified. In this regard, invertebrates are more diverse than vertebrates, and, unlike their restriction to vertebrate respiratory intake sites, invertebrate chemosensory organs are found in various structures in different parts of the body. For example, chemosensation occurs in the *osphradium*, a sensory epithelium associated with the respiratory apparatus, in some mollusks. Nematodes rely on chemosensation as a primary sensory system, with two chemo-thermo-sensing *amphid* organs in the head. *C. elegans* has 11 pairs of individually identified chemosensory neurons, about 7 percent of their total 302 neurons; each is sensitive to different types of molecules, including at least one pheromone (Troemel 1999).

Chemosensation occurs in *sensilla* on the antennae and other loci of insects and crustaceans. Sensilla are hairlike or peglike clusters of receptor cells covered in chitinous cuticle, with dendrites that extend inside a sheath and hairshaft. They can protrude from the surface of the cuticle or they may be embedded in it.

The shape and type of olfactory sensilla varies by species, but all have a number of pores or slits in the walls of the hair that allow odorants to pass through. Inside the fluid-filled lumen of the sensilla are housed one or more bipolar olfactory receptor neurons, with an axon that extends to the CNS and a dendrite that reaches upward through the hair. The odorant passes through the pores in the wall to reach the dendrite. The number of ONs varies by species, ranging from one to 50, most commonly two to six. Like those in vertebrates, invertebrate ONs are primary receptors, each sending an axon into the CNS. They connect via glomeruli in the anten-

nal lobes. As in fish and rodents, these can be specific, for example, for sex pheromones (see Strausfeld and Hildebrand 1999).

Also like vertebrates, and not unexpectedly, invertebrate *ORs* are in the *7TMR* gene family. Insect (*Drosophila*) *OR* genes (*DORs*) number around 50–100, and sequence relationships show that they form a family of their own (Figure 13-2B) (Clyne et al. 1999; Vosshall 2000; 2001; Vosshall et al. 1999). About 60 of these genes are used in adults and others in larvae (Firestein 2001). Unlike vertebrates, however, insect *OR* coding regions are interrupted by introns and have no specific homology to the *OR* subclasses in their vertebrate or nematode olfactory counterparts (Strausfeld and Hildebrand 1999; Vosshall, Amrein et al. 1999).

Current evidence suggests that each *OR* is expressed in a nonoverlapping set of neurons; that is, neurons express one or distinct subsets of a few members of the *OR* repertoire. This is bilaterally symmetric and conserved among flies (of the same species tested); the pattern appears to be more stereotypical than the vertebrate pattern in which, although restricted to a general region of the olfactory epithelium, receptors are more or less randomly expressed among the ONs. The pattern is shared among individuals. As in vertebrates, however, neurons expressing similar *OR* genes are wired together through the same glomerulus. The location of olfactory and gustatory receptors in a stereotypical insect is shown in Figure 13-6.

Insects also appear to have a diversity of odorant binding proteins to transport odorants within the chemosensory sensilla on their antenna, and these include pheromone binding proteins. However, insect OBPs appear to have no homology to the lipocalins used as vertebrate OBPs (Galindo and Smith 2001; Robertson et al. 1999). As with lipocalins in vertebrates, it may be that insect OBPs function to protect against sensory overload or to speed up sensory recovery by removing odorants from the organism.

Nematodes have about 500 active *OR* genes plus many pseudogenes, but it is not clear whether all the “active” genes are actually used for chemosensation. Nematode *ORs* are *7TMRs* unrelated to the *OR* subfamilies found in other species. The presence of so many genes and pseudogenes suggests that sloppy meiosis generates diversity, and one can predict a certain amount of point-mutational variation as well. Unlike insects and vertebrates, each nematode chemosensory neuron expresses multiple (15–25) receptor genes rather randomly distributed among the subclasses. These elicit two major responses, either attraction or repulsion, a restricted discriminatory power relative to the other systems that we have discussed. This innervation pattern raises questions about how discriminating their perception is (Troemel 1999); obviously, it’s good enough.

Perhaps rather surprising to our vertebrate perspective, insects can be trained by classical punishment-reward experiments to respond positively or negatively to odors (e.g., see Waddell and Quinn 2001). Mutations have been identified that affect the ability of flies to respond to odorants, as evaluated by training experiments, and this fact has enabled some of the neural signaling mechanisms involved in olfaction to be understood. For instance, training involves interpreting coincident stimuli, and the detection of temporal signal pairing appears to occur in structures called *mushroom bodies* in the fly brain. Perhaps more important than neurological details, these studies have shown clearly that “flies are not automata. Their tiny brains are capable of much more than hard-wired reactions” (Waddell and Quinn 2001). This is a lesson for our human-centric world that conceives of thought or mind or information